21

Oxygenation

Learning Objectives

On completion of this chapter, the reader should be able to:

1. Explain the difference between ventilation and respiration.
2. Differentiate between external and internal respiration.
3. Name two methods for assessing the oxygenation status of clients at the bedside.
4. List at least five signs of inadequate oxygenation.
5. Name two nursing interventions that can be used to improve ventilation and oxygenation.
6. Identify four items that may be needed when providing oxygen therapy.
7. Name four sources for supplemental oxygen.
8. List five common oxygen delivery devices.
9. Discuss two hazards related to the administration of oxygen.
10. Describe three additional therapeutic techniques that relate to oxygenation.
11. Discuss at least two facts concerning oxygenation that affect the care of older adults.

Words to Know

arterial blood gas
BiPAP mask
CPAP mask
diaphragmatic breathing
expiration
external respiration
face tent
flowmeter
Fowler’s position
fraction of inspired oxygen
humidifier
hyperbaric oxygen therapy

hypercarbia
hypoxemia
hypoxia
incentive spirometry
inspiration
internal respiration
liquid oxygen unit
nasal cannula
nasal catheter
non-rebreather mask
orthopneic position
oxygen analyzer
oxygen concentrator
oxygen tent
oxygen therapy
oxygen toxicity
partial rebreather mask
positive airway pressure machines
pulse oximetry
pursed-lip breathing
respiration
simple mask
stent
surfactant
tension pneumothorax
tidaling
T-piece
tracheostomy collar
transtracheal catheter
tripod position
ventilation
venturi mask
water-seal chest tube drainage

Oxygen, which measures approximately 21% in the Earth's atmosphere, is essential for sustaining life. Each cell of the human body uses oxygen to metabolize nutrients and produce energy. Without oxygen, cell death occurs rapidly.

This chapter describes the anatomic and physiologic aspects of breathing,

techniques for assessing and monitoring oxygenation, types of equipment used in oxygen therapy, and skills needed to maintain respiratory function. Techniques for airway management, such as suctioning and other methods for maintaining a patent airway, are presented in Chapter 36.

ANATOMY AND PHYSIOLOGY OF BREATHING

The elasticity of lung tissue allows the lungs to stretch and fill with air during **inspiration** (breathing in) and return to a resting position after **expiration** (breathing out). **Ventilation** (the movement of air in and out of the lungs) facilitates **respiration** (the exchange of oxygen and carbon dioxide). **External respiration** takes place at the most distal point in the airway between the alveolar and capillary membranes (Fig. 21-1). **Internal respiration** occurs at the cellular level by means of hemoglobin and body cells. For people without respiratory disease, increased blood levels of carbon dioxide and hydrogen ions trigger the stimulus to breathe, both chemically and neurologically.

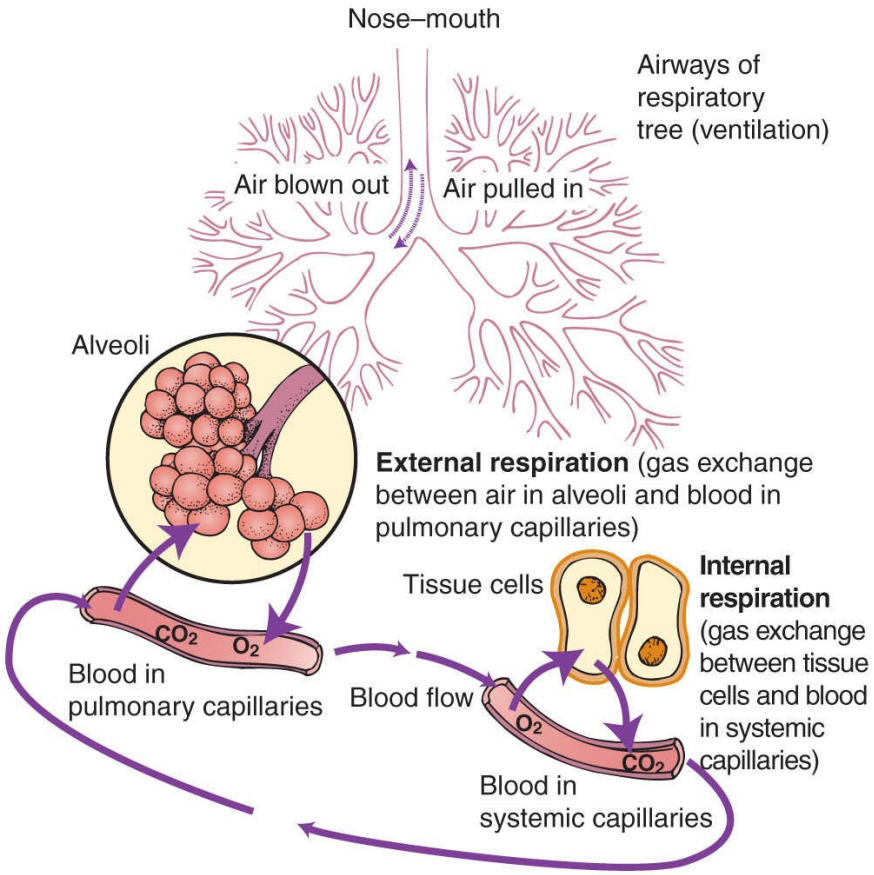


FIGURE 21-1 External and internal respiration.

Ventilation results from pressure changes within the thoracic cavity produced by the contraction and relaxation of respiratory muscles (Fig. 21-2). During inspiration, the dome-shaped diaphragm contracts and moves downward in the thorax. The intercostal muscles move the chest outward by elevating the ribs and sternum. This combination expands the thoracic cavity. Expansion creates more chest space, causing the pressure within the lungs to fall below that in the atmosphere. Because air flows from an area of higher pressure to one of lower pressure, air is pulled in through the nose, filling the lungs. When there is an acute need for oxygen, additional muscles known as accessory muscles of respiration (the pectoralis minor and sternocleidomastoid) contract to assist with even greater chest expansion.

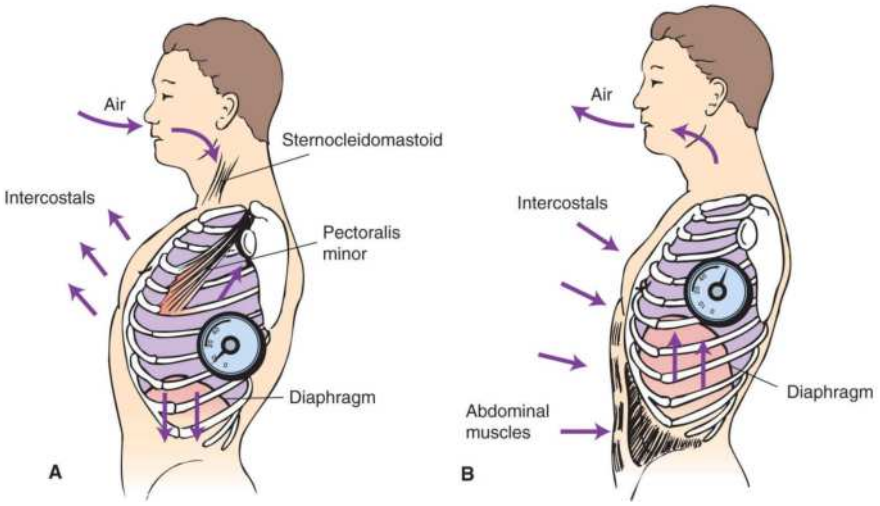


FIGURE 21-2 Ventilation and thoracic pressure changes. (A) Inspiration. (B) Expiration.

Concept Mastery Alert

Remember, during inspiration, the chest cavity increases but the chest pressure decreases which allows air to move into the lungs.

During expiration, the respiratory muscles relax, the thoracic cavity decreases, the stretched elastic lung tissue recoils, intrathoracic pressure increases as a result of the compressed pulmonary space, and air moves out of the respiratory tract. A person can forcibly exhale additional air by contracting abdominal muscles such as the rectus abdominis, transverse abdominis, and external and internal obliques.

Gerontologic Considerations

■ Reduced gas exchange and efficiency in ventilation are the age-related changes that can affect respiratory function, even in healthy older adults.
■ Functional changes to the respiratory system include diminished coughing and gag reflexes, increased use of accessory muscles for breathing, diminished efficiency of gas exchange in the lungs, and increased mouth breathing and snoring.
■ Some changes in lung volumes occur, resulting in a slight decrease in overall efficiency and increased energy expenditure by older adults.

BOX 21-1 Common Signs of Inadequate Oxygenation

- Decreased energy
- Restlessness
- Rapid, shallow breathing
- Rapid heart rate
- Sitting up to breathe
- Nasal flaring
- Use of accessory muscles
- Hypertension
- Sleepiness, confusion, stupor, coma

ASSESSING OXYGENATION

The nurse can determine the quality of a client's oxygenation by collecting physical assessment data, monitoring arterial blood gases (ABGs), and using pulse oximetry. A combination of these helps to identify signs of **hypoxemia** (insufficient oxygen within arterial blood) and **hypoxia** (inadequate oxygen at the cellular level).

Physical Assessment

The nurse physically assesses oxygenation by monitoring the client's respiratory rate, observing the breathing pattern and effort, checking chest symmetry, and auscultating lung sounds (see Chap. 13). Additional assessments include recording the heart rate and blood pressure, determining the client's level of consciousness, and observing the color of the skin, mucous membranes, lips, and nail beds (Box 21-1).

Pharmacologic Considerations

■ All clients under 5 years of age, over 65 or anyone with a compromising chronic health condition should be assessed for and offered Prevnar 13 a

pneumococcal vaccine. This vaccine is effective in reducing 13 strains of *Streptococcus pneumoniae*.

Gerontologic Considerations

- Careful assessment of older adults who demonstrate restlessness or confusion is imperative to differentiate signs of inadequate oxygenation accurately from signs of delirium or dementia.

Arterial Blood Gases

An **arterial blood gas** (ABG) assessment is a laboratory test using arterial blood to evaluate or assess oxygenation, ventilation, and acid–base balance. It measures the partial pressure of oxygen dissolved in plasma (PaO_2), the percentage of hemoglobin saturated with oxygen (SaO_2), the partial pressure of carbon dioxide in plasma ($PaCO_2$), the pH of blood, and the level of bicarbonate (HCO_3) ions (Table 21-1). Arterial blood is preferred for sampling because arteries have greater oxygen content than veins and are responsible for carrying oxygen to all cells. Initial and subsequent ABGs are ordered to assess the client in acute respiratory distress or to monitor the progress of a client receiving medical treatment.

TABLE 21-1 Values for Arterial Blood Gases

COMPONENT	NORMAL RANGE	ABNORMAL FINDINGS	INDICATION OF ABNORMAL FINDINGS
pH	7.35–7.45	<7.35	Acidosis
		>7.45	Alkalosis
Pao_2	80–100 mmHg	60–80 mmHg	Mild hypoxemia
		40–60 mmHg	Moderate hypoxemia
		<40 mmHg	Severe hypoxemia
		>100 mmHg	Hyperoxygenation
$Paco_2$	35–45 mmHg	<35 mmHg	Hyperventilation
		>45 mmHg	Hypoventilation
Sao_2	95%–100%	<95%	Hypoventilation
			Anemia
HCO_3	22–26 mEq	<22 or >26 mEq	Compensation for acid–base imbalance

In most situations, a laboratory technician and the nurse collaboratively collect arterial blood. The nurse notifies the laboratory of the need for the blood test, records pertinent assessments on the laboratory request form and in the client's medical record, prepares the client, assists the laboratory technician who obtains the specimen, and implements measures for preventing complications after the arterial puncture. In emergencies, a nurse who is trained in performing arterial punctures may obtain the specimen (see Nursing Guidelines 21-1).

Pulse Oximetry

Pulse oximetry is a noninvasive, transcutaneous technique for periodically or continuously monitoring the oxygen saturation of blood (Skill 21-1). A pulse oximeter is composed of a photodetector sensor, a red and infrared light emitter, and a microprocessor. The sensor is attached to a finger, toe, earlobe, or the bridge of the nose using spring-tension or adhesive. The sensor detects the amount of light absorbed by hemoglobin. The microprocessor then computes the information and displays it on a monitor at the bedside. The measurement of oxygen saturation when obtained by pulse oximetry is abbreviated and recorded as SpO_2 to distinguish it from the SaO_2 measurement obtained from arterial blood.

NURSING GUIDELINES 21-1

Assisting With an ABG

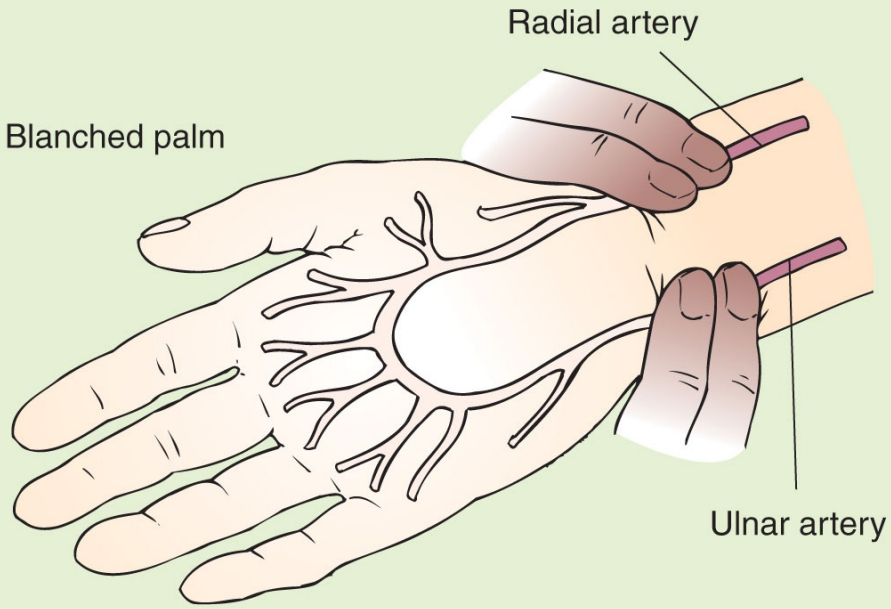
- Perform the Allen test before the arterial puncture by doing the following:
  - Flex the client's elbow and elevate the forearm where the arterial puncture will be made.
  - Compress the radial and ulnar arteries simultaneously (Fig. A).
  - Instruct the client to open and close the fist until the palm of the hand appears blanched.
  - Release pressure from the ulnar artery while maintaining pressure on the radial artery (Fig. B).
  - Observe whether the skin flushes or remains blanched.
  - Release pressure on the radial artery.

The Allen test determines if the hand has an adequate ulnar arterial blood supply should the radial artery become damaged or occluded. The radial artery should not be punctured if the Allen test shows absent or poor collateral arterial blood flow as evidenced by continued blanching after pressure on the ulnar artery has been released. Alternative sites include the brachial, femoral, or dorsalis pedis arteries.

- Keep the client at rest for at least 30 minutes before obtaining the specimen unless the procedure is an emergency. *Because an ABG reflects the client's status at the moment of blood sampling, activity can transiently lower oxygen levels in the blood and lead to an incorrect interpretation of the test results.*
- Record the client's current temperature, respiratory rate, and level of activity if other than resting. *Increased metabolism and activity affect cellular oxygen demands. Therefore, the data help in interpreting the results of laboratory findings.*
- Record the amount of oxygen the client is receiving at the time of the test (either room air or prescribed amount) and ventilator settings. *This information helps to determine if oxygen therapy is necessary or aids in evaluating its current effectiveness.*
- Hyperextend the wrist over a rolled towel. *Hyperextension brings the radial artery nearer the skin surface to facilitate penetration.*
- Comfort the client during the puncture. *An arterial puncture tends to be painful unless a local anesthetic is used.*
- After obtaining the specimen, expel all air bubbles from it. *Doing so ensures that the only gas in the specimen is that contained in the blood.*
- Rotate the collected specimen. *Rotation mixes the blood with the anticoagulant in the specimen tube, ensuring that the blood sample will not clot before it can be examined.*
- Place the specimen on ice immediately. *Blood cells deteriorate outside the body, causing changes in the oxygen content of the sample. Cooling the sample slows cellular metabolism and ensures more accurate test results.*
- Apply direct manual pressure to the arterial puncture site for 5–10 minutes. *Arterial blood flows under*

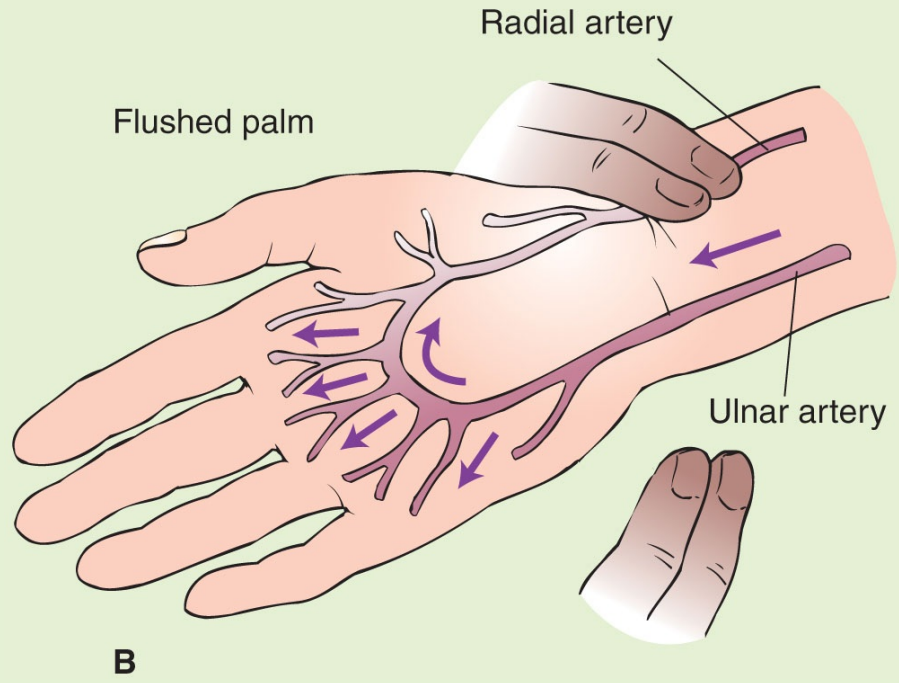
higher pressure than venous blood. Therefore, prolonged manual pressure is necessary to control bleeding.

- Cover the puncture site with a pressure dressing composed of several 4 × 4 inch gauze squares and tape. *Tight mechanical compression provides continued pressure to reduce the potential for arterial bleeding.*
- Assess the puncture site periodically for bleeding or formation of a hematoma (collection of trapped blood) beneath the skin. *Periodic inspection aids in the early identification of arterial bleeding, which can lead to substantial blood loss and discomfort.*
- Report the laboratory findings to the prescribing physician as soon as they are available. *Collaboration with the physician assists in making changes in the treatment plan to improve the client's condition.*



A

Simultaneous compression of radial and ulnar arteries.



Pressure on the ulnar artery is released.

Based on the oxygen–hemoglobin dissociation curve (Fig. 21-3), it is possible to infer the Pa_{O_2} from the pulse oximetry measurement. The normal Sp_{O_2} is 95% to 100%. A sustained level of less than 90% is cause for concern. If the Sp_{O_2} remains low, the client needs oxygen therapy. Various factors, however, affect the accuracy of the displayed information (Table 21-2). Troubleshooting the equipment, performing current physical assessments, and obtaining an ABG help to confirm the significance of the displayed findings.

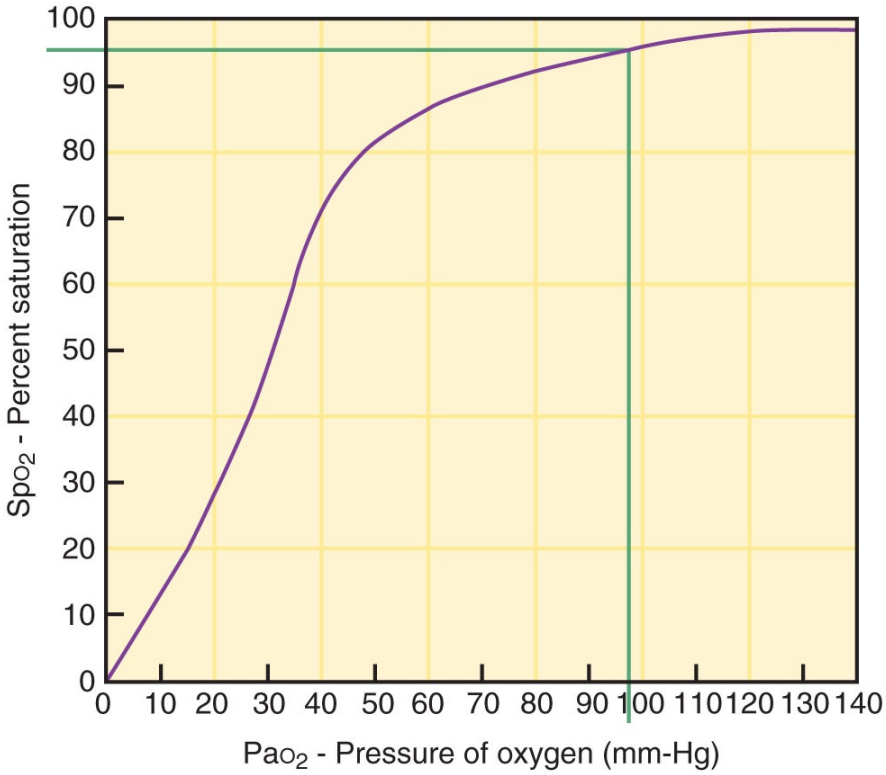


FIGURE 21-3 Draw a line from the Spo_2 in the left column across the graph to the point at which it intersects the curve. Use the numeric scale at the bottom to calculate the Pao_2. In this example, with an Spo_2 of 95%, the Pao_2 is approximately 98 mmHg.

Stop, Think, and Respond 21-1

What actions are appropriate if a client appears to be hypoxemic, but the pulse oximeter indicates a normal Spo_2? What action(s) are appropriate if the opposite occurs—that is, the client appears normal but the pulse oximeter reading gives you cause for concern?

PROMOTING OXYGENATION

Many factors affect ventilation and, subsequently, respiration (Table 21-3). Positioning and teaching breathing techniques are two nursing interventions frequently used to promote oxygenation. Adhesive nasal strips can be used to improve oxygenation by reducing nasal airway resistance and improving ventilation.

Positioning

Unless contraindicated by their condition, clients with hypoxia are placed in high **Fowler's position** (an upright seated position; see Chap. 23). This position eases breathing by allowing the abdominal organs to descend away from the

diaphragm. As a result, the lungs have the potential to fill with a greater volume of air.

As an alternative, clients who find breathing difficult may benefit from variations of Fowler's position. One option is a **tripod position** in which the client is in a seated position with the arms supported on pillows or the arm rests of a chair. The tripod position increases a client's breathing capacity by using the arms to lift the chest upward (Fig. 21-4A). Another option is an **orthopneic position** in bed. In the orthopneic position, the client leans forward over the bedside table or a chair back (Fig. 21-4B). The orthopneic position allows room for maximum vertical and lateral chest expansion and provides comfort while resting or sleeping.

TABLE 21-2 Factors That Interfere With Accurate Pulse Oximetry

FACTOR	CAUSE	REMEDY
Movement of the sensor	Tremor	Relocate sensor to another site.
	Restlessness	Replace sensor or tape in place.
	Loss of adhesion	
Poor circulation at the sensor site	Peripheral vascular disease	Change the sensor location or type of sensor.
	Edema	Loosen or change sensor location.
	Tourniquet effect from taped sensor	
	Vasoconstrictive drug effects	
Barrier to light	Nail polish	Discontinue use temporarily.
	Thick toenails	Remove polish.
	Acrylic nails	Relocate sensor.
		Remove acrylic nails.
Extraneous light	Direct sunlight	Cover sensor with a towel.
	Treatment lights	
Hemoglobin saturation with other substances	Carbon monoxide poisoning	Discontinue use temporarily.

TABLE 21-3 Factors Affecting Oxygenation

FACT	NURSING IMPLICATION
Adequate respiration depends on a minimum of 21% oxygen in the environment and normal function of the cardiopulmonary system.	Clients with cardiopulmonary disorders require more than 21% oxygen to maintain adequate oxygenation of blood and cells.
Breathing can be voluntarily controlled.	Assist clients who are hyperventilating to slow the rate of breathing; teach clients to perform pursed-lip breathing to exhale more completely.
Clients with chronic lung diseases are stimulated to breathe by low blood levels of oxygen, called "the hypoxic drive to breathe."	Giving high percentages of oxygen can depress breathing in clients with chronic lung disease. No more than 2–3 L oxygen is safe unless the client is mechanically ventilated.
Smoking causes increased amounts of inhaled carbon monoxide that compete and bond more easily than oxygen to the hemoglobin.	Clients who smoke have a greater potential for compromised gas exchange and acquiring chronic pulmonary and cardiac diseases.
Nicotine increases the heart rate and constricts arteries.	Teach people who do not smoke never to start. Identify products that are available, such as nicotine skin patches and gum, which can help with smoking cessation.
Pregnant women who smoke have a risk for low-birth-weight infants because low blood oxygenation affects fetal metabolism and growth.	Promote smoking cessation for pregnant women who are addicted to nicotine.
Pulmonary secretions within the airway and fluid within the interstitial space between the alveoli and capillaries interfere with gas exchange.	Encourage coughing, deep breathing, turning, and ambulating to keep alveoli inflated and the airway clear. Antibiotics, diuretics, and drugs that improve heart contraction reduce fluid within the lungs.
Gas exchange is increased by maximum lung expansion and compromised by any condition that compresses the diaphragm, such as obesity, intestinal gas, pregnancy, and an enlarged liver.	Assist clients to sit up to lower abdominal organs away from the diaphragm. Encourage weight loss, expulsion of gas via ambulation and bowel elimination, and assist with removing abdominal fluid by paracentesis (see Chap. 14) to improve breathing.
Activity and emotional stress increase the metabolic need for greater amounts of oxygen.	Provide rest periods and teach stress reduction techniques such as muscle relaxation to promote maintenance of blood oxygen levels.
Pain associated with muscle movement around abdominal and flank surgical incisions decreases the incentive to breathe deeply and cough forcefully.	Teach and supervise deep breathing before surgery. Support the incision with a pillow and administer drugs that relieve pain to facilitate ventilation.

Breathing Techniques

Breathing techniques such as deep breathing with or without an incentive spirometer, pursed-lip breathing, and diaphragmatic breathing help clients to breathe more efficiently.

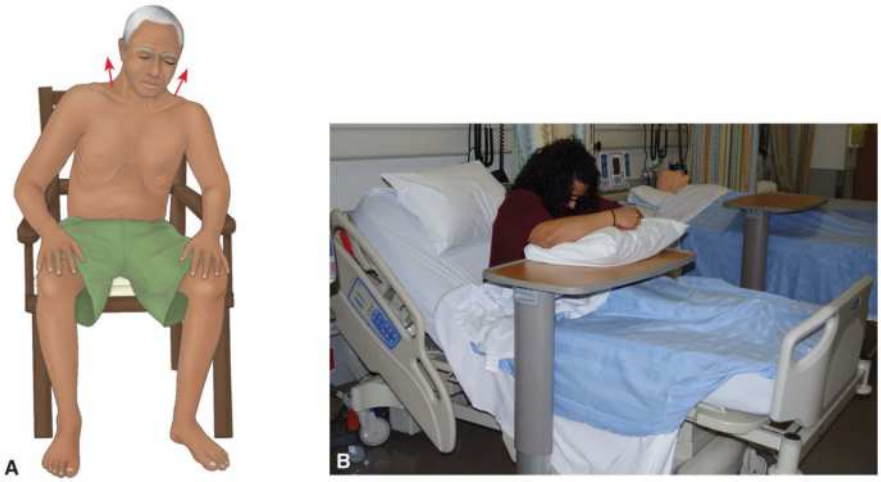


FIGURE 21-4 (A) The tripod position. (B) The orthopneic position. (From Gregory, D., Raymond-Seniuk, C., Patrick, L., & Stephen, M. N. [2014]. *Fundamentals: Perspectives on the art and science of Canadian nursing*. Philadelphia, PA: Lippincott Williams & Wilkins.)

Deep Breathing

Deep breathing is a technique for maximizing ventilation. Taking in a large volume of air fills alveoli to a greater capacity, thus improving gas exchange. Deep breathing is therapeutic for clients who tend to breathe shallowly, such as those who are inactive or in pain. To encourage deep breathing, the client learns to take in as much air as possible, hold the breath briefly, and exhale slowly. In some cases, it is helpful to use an incentive spirometer; however, deep breathing alone, if performed effectively, is sufficiently beneficial.

FIGURE 21-5 During deep inhalation, a ball rises in an incentive spirometer. (Courtesy of Swedish Hospital Medical Center.)

Incentive Spirometry

Incentive spirometry, a technique for deep breathing using a calibrated device, encourages clients to reach a goal-directed volume of inspired air. Although spirometers are constructed in different ways, all are marked in at least 100 mL increments and include some visual cue, such as elevation of lightweight balls, to show how much air the client has inhaled (Fig. 21-5). The calibrated measurement also helps the nurse to evaluate the effectiveness of the client's breathing efforts (see Client and Family Teaching 21-1).

Pursed-Lip Breathing

Pursed-lip breathing is a form of controlled ventilation in which the client consciously prolongs the expiration phase of breathing. This is another technique for improving gas exchange, which, if done correctly, helps clients eliminate more than the usual amount of carbon dioxide from the lungs. Pursed-lip breathing and diaphragmatic breathing are especially helpful for clients who

have chronic obstructive pulmonary disorders (COPD), such as emphysema, which are characterized by chronic hypoxemia and **hypercarbia** (excessive levels of carbon dioxide in the blood). The client performs pursed-lip breathing as follows:

- Inhale slowly through the nose while counting to three.
- Purse the lips as though to whistle.
- Contract the abdominal muscles.
- Exhale through pursed lips for a count of six or more.

Client and Family Teaching 21-1 Using an Incentive Spirometer

The nurse teaches the client and the family as follows:
- Sit upright unless contraindicated.
- Identify the mark indicating the goal for inhalation.
- Exhale normally.
- Insert the mouthpiece, sealing it between the lips.
- Inhale slowly and deeply until the predetermined volume has been reached.
- Hold the breath for 3–6 seconds.
- Remove the mouthpiece and exhale normally.
- Relax and breathe normally before the next breath with the spirometer.
- Repeat the exercise 10–20 times per hour while awake or as prescribed by the physician.

Expiration should be two to three times longer than inspiration. Not all clients can achieve this goal initially, but with practice the length of expiration can increase.

Diaphragmatic Breathing

Diaphragmatic breathing is breathing that promotes the use of the diaphragm rather than the upper chest muscles. It is used to increase the volume of air exchanged during inspiration and expiration. With practice, diaphragmatic breathing reduces respiratory effort and relieves rapid, ineffective breathing (see Client and Family Teaching 21-2).

Client and Family Teaching 21-2 Diaphragmatic Breathing

The nurse teaches the client and the family as follows:
- Lie down with knees slightly bent.
- Place one hand on the abdomen and the other on the chest.
- Inhale slowly and deeply through the nose while letting the abdomen rise more than the chest.
- Purse the lips.
- Contract the abdominal muscles and begin to exhale.
- Press inward and upward with the hand on the abdomen while continuing to exhale.
- Repeat the exercise for 1 full minute; rest for at least 2 minutes.

- Practice the breathing exercises at least twice a day for a period of 5–10 minutes.
- Progress to doing diaphragmatic breathing while upright and active.

Nasal Strips

Adhesive nasal strips, available commercially in drug stores, are used to reduce airflow resistance by widening the breathing passageways of the nose. Increasing the nasal diameter promotes easier breathing. Common users of nasal strips are people with ineffective breathing as well as athletes, whose oxygen requirements increase during sustained exercise. Another use for nasal strips is to reduce or eliminate snoring.

OXYGEN THERAPY

When positioning and breathing techniques are inadequate for keeping the blood adequately saturated with oxygen, oxygen therapy is necessary. **Oxygen therapy** is an intervention for administering more oxygen than is present in the atmosphere to prevent or relieve hypoxemia. It requires an oxygen source, a flowmeter, in some cases an oxygen analyzer or humidifier, and an oxygen delivery device.

Oxygen Sources

Oxygen is supplied from any one of four sources: wall outlet, portable tank, liquid oxygen unit, or oxygen concentrator.

Wall Outlet

Most modern health care facilities supply oxygen through a wall outlet in the client's room (Fig. 21-6). The outlet is connected to a large central reservoir filled with oxygen on a routine basis.

FIGURE 21-6 Wall outlet for oxygen. (From McGreer, M. A., & Carter, P. J. [2011]. *Lippincott's textbook for personal support workers*. Philadelphia, PA: Lippincott Williams & Wilkins.)

Portable Tanks

When oxygen is not piped into individual rooms or if the client needs to leave the room temporarily, oxygen is provided in portable tanks (Fig. 21-7) that hold various volumes under extreme pressure. A large tank of oxygen contains 2,000 lb of pressure per square inch. Therefore, tanks are delivered with a protective cap to prevent accidental force against the tank outlet. Any accidental force

applied to a partially opened outlet could cause the tank to take flight like a rocket, with disastrous results. Therefore, oxygen tanks are transported and stored while strapped to a wheeled carrier.

FIGURE 21-7 A portable oxygen tank. (From McGreer, M. A., & Carter, P. J. [2011]. *Lippincott's textbook for personal support workers*. Philadelphia, PA: Lippincott Williams & Wilkins.)

Before oxygen is administered from a portable tank, the tank is "cracked," a technique for clearing the outlet of dust and debris. Cracking is done by turning the tank valve slightly with a wrench to allow a brief release of pressurized oxygen (Fig. 21-8). The force causes a loud hissing noise, which may be frightening. Therefore, it is best to crack the tank away from the client's bedside.

Liquid Oxygen Unit

A **liquid oxygen unit** is a device that converts cooled liquid oxygen to a gas by passing it through heated coils (Fig. 21-9). Ambulatory clients at home primarily use these small, lightweight, portable units because they allow greater mobility inside and outside the house. Each unit holds approximately 4 to 8 hours worth of oxygen. Potential problems include that liquid oxygen is more expensive, the unit may leak during warm weather, and frozen moisture may occlude the outlet.

Oxygen Concentrator

An **oxygen concentrator** is a machine that collects and concentrates oxygen from room air and stores it for client use. To do so, the concentrator uses a substance called "zeolite" within two absorbing chambers. The machine compresses atmospheric air and diverts it into a chamber containing zeolite. The

zeolite absorbs nitrogen from the air, leaving nearly pure oxygen, which is stored in the second chamber. When the nitrogen-absorbing chamber becomes saturated, the machine releases nitrogen back into the atmosphere, and the process repeats itself, providing a constant supply of oxygen (Fig. 21-10). Oxygen concentrators typically provide oxygen flows of 1 to 5 L/min with some that are capable of flows up to 10 L/min (American Association for Respiratory Care, 2007).

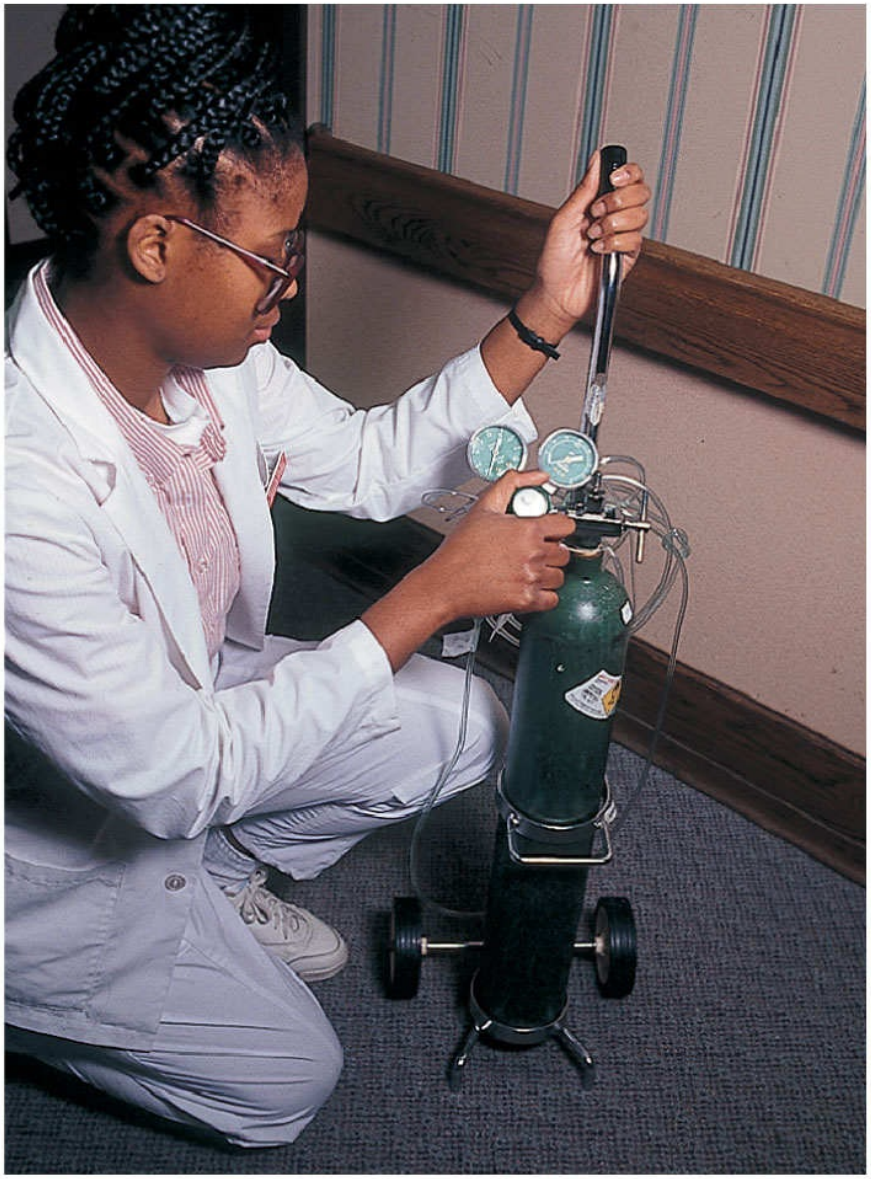


FIGURE 21-8 A nurse opens a portable oxygen tank with a wrench.

Gerontologic Considerations

- Older adults who require home oxygen need encouragement to continue socializing with others outside the home to prevent feelings of isolation and depression. It is important to teach older adults and their caregivers about portable oxygen equipment.

FIGURE 21-9 A liquid oxygen unit. (From Carter, P. J. [2015]. *Lippincott textbook for nursing assistants* (4th ed.). Philadelphia, PA: Lippincott Williams & Wilkins.)

FIGURE 21-10 A portable oxygen concentrator extracts nitrogen and concentrates oxygen to enable clients who require oxygen therapy to travel about or maintain their lifestyle without the need for multiple tanks of oxygen. (From McGreer, M. A., & Carter, P. J. [2011]. *Lippincott's textbook for personal support workers*. Philadelphia, PA: Lippincott Williams & Wilkins.)

An oxygen concentrator eliminates the need for a central reservoir of piped oxygen or the use of bulky tanks that must be constantly replaced. This type of oxygen source is used in home health care and long-term care facilities primarily because of its convenience and economy.

Although it is more economical than oxygen supplied in portable tanks, the device increases the client's electric bill. Other disadvantages are that it generates heat from its motor and that it produces an unpleasant odor or taste if the filter is not cleaned weekly. Also, it is best that clients have a secondary source of oxygen available in case of a power failure.

Equipment Used in Oxygen Administration

In addition to an oxygen source, other pieces of equipment used during the administration of oxygen are a flowmeter, oxygen analyzer, and humidifier.

Flowmeter

The flow of oxygen is measured in liters per minute (L/min). A **flowmeter** is a gauge used to regulate the amount of oxygen delivered to the client and is attached to the oxygen source (Fig. 21-11). To adjust the rate of flow, the nurse turns the dial until the indicator is directly beside the prescribed amount.

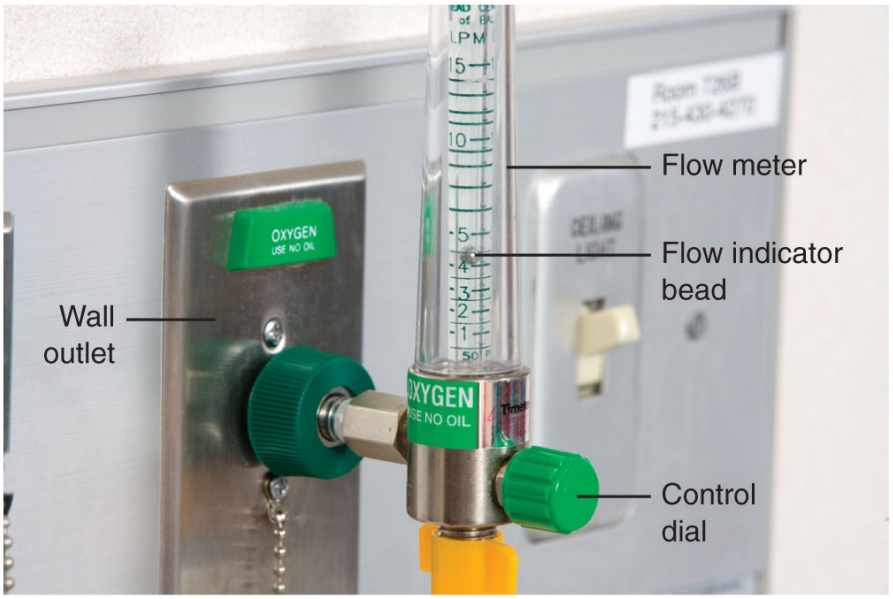


FIGURE 21-11 A flowmeter attached to a wall outlet for oxygen administration.

The physician prescribes the concentration of oxygen, also called the **fraction of inspired oxygen** (FIO_2; the portion of oxygen in relation to total inspired gas), as a percentage or as a decimal (eg, 40% or 0.40). The prescription is based on the client's condition. The Joint Commission recommends that oxygen be prescribed as a percentage rather than in liters per minute (L/min) because, depending on the oxygen delivery device, liters per minute may provide different percentages of oxygen.

Oxygen Analyzer

An **oxygen analyzer** is a device that measures the percentage of delivered oxygen to determine whether the client is receiving the amount prescribed by the physician (Fig. 21-12). The nurse or respiratory therapist first checks the percentage of oxygen in the room air with the analyzer. If there is a normal mixture of oxygen and other gases in the environment, the analyzer indicates 0.21 (21%). When the analyzer is positioned near or within the device used to deliver oxygen, the reading should register at the prescribed amount (>0.21). If there is a discrepancy, the nurse adjusts the flowmeter to reach the desired amount. Oxygen analyzers are used most often when caring for newborns in isolettes, children in croup tents, and clients who are mechanically ventilated.

FIGURE 21-12 An oxygen analyzer. (Photo by B. Proud.)

Humidifier

A **humidifier** is a device that produces small water droplets and may be used during oxygen administration because oxygen is drying to the mucous membranes. In most cases, oxygen is humidified only when more than 4 L/min is administered for an extended period. When humidification is desired, a bottle is filled with distilled water and attached to the flowmeter (Fig. 21-13). A respiratory therapist or nurse checks the water level daily and refills the bottle as needed.

FIGURE 21-13 An oxygen humidifier attached to a flowmeter.

Common Delivery Devices

Common oxygen delivery devices include a nasal cannula, masks, a face tent, a tracheostomy collar, or a T-piece (Table 21-4). The device prescribed depends on the client's oxygenation status, physical condition, and amount of oxygen needed. Skill 21-2 describes how to administer oxygen by common delivery methods.

Nasal Cannula

A **nasal cannula**, the most common oxygen delivery device, is a hollow tube with ½-inch prongs placed into the client's nostrils. It is held in place by wrapping the tubing around the ears and adjusting the fit beneath the chin (Fig. 21-14). The skin around the ears and cheeks should be assessed for pressure. Padding the area around the ears may be required to keep the skin intact. Frequent mouth and nasal care are indicated.

Stop, Think, and Respond 21-2

Explain the difference between a flowmeter and an oxygen analyzer.

TABLE 21-4 Comparison of Oxygen Delivery Devices

DEVICE	COMMON RANGE OF ADMINISTRATION	ADVANTAGES	DISADVANTAGES
Nasal cannula	2–6 L/min F_{IO_2} 24%–40%[a]	Is easy to apply; promotes comfort Does not interfere with eating or talking Is less likely to create a feeling of suffocation	Dries nasal mucosa at higher flows May irritate the skin at cheeks, nose, and behind ears Is less effective in some clients who tend to mouth breathe Does not facilitate administering high F_{IO_2} to hypoxic clients
Masks			
Simple	5–8 L/min F_{IO_2} 35%–50%[a]	Provides higher concentrations than possible with a cannula Is effective for mouth breathers or clients with nasal disorders Allows a client to inhale a mixture of 75% room air and 25% oxygen	Requires humidification Interferes with eating and talking Can cause anxiety among those who are claustrophobic Creates a risk for rebreathing CO_2 retained within mask when less than 6 L/min is administered
Partial rebreather	6–10 L/min F_{IO_2} 35%–60%[a]	Increases the amount of oxygen with lower liter flows	Requires a minimum of 6 L/min Creates a risk for suffocation Requires monitoring to verify that reservoir bag remains inflated at all times
Non-rebreather	6–10 L/min F_{IO_2} 60%–90%[a]	Delivers highest F_{IO_2} possible with a mask Prevents breathing room air Exhaled air is not rebreathed.	See partial rebreather mask. Creates a risk for oxygen toxicity
Venturi	4–8 L/min F_{IO_2} 24%–40%[a]	Delivers F_{IO_2} precisely through adjustable ports Buildup of carbon dioxide is minimal.	Permits condensation to form in tubing, which diminishes the flow of oxygen
Face tent	8–12 L/min F_{IO_2} 30%–55%[a]	Provides a comfortable fit Is useful for clients with facial trauma, facial burns, or who are claustrophobic when an oxygen mask is applied Facilitates humidification	Interferes with eating May result in inconsistent F_{IO_2}, depending on environmental loss
Tracheostomy collar	4–10 L/min F_{IO_2} 24%–100%[a]	Facilitates humidifying and warming oxygen	Allows water vapor to collect in tubing, which may drain into airway
T-piece	4–10 L/min F_{IO_2} 24%–100%[a]	Delivers any desired F_{IO_2} with high humidity	May pull on tracheostomy tube Allows humidity to collect and moisten gauze dressing

[a]Source: American Association for Respiratory Care (AARC).

A nasal cannula provides a means of administering low concentrations of oxygen. Therefore, it is ideal for clients who are not extremely hypoxic or who have chronic lung diseases. High percentages of oxygen are contraindicated for clients with chronic lung disease because they have adapted to excessive levels of retained carbon dioxide and low blood oxygen levels to stimulate their drive to breathe. Consequently, if clients with chronic lung disease receive more than 2 to 3 L of oxygen over a sustained period, their respiratory rate slows or even stops.

FIGURE 21-14 Oxygen delivered with a nasal cannula.

Gerontologic Considerations

- The skin behind the ears of older adults as well as others should be assessed for breakdown if oxygen administration equipment is secured by tubing or elastic.

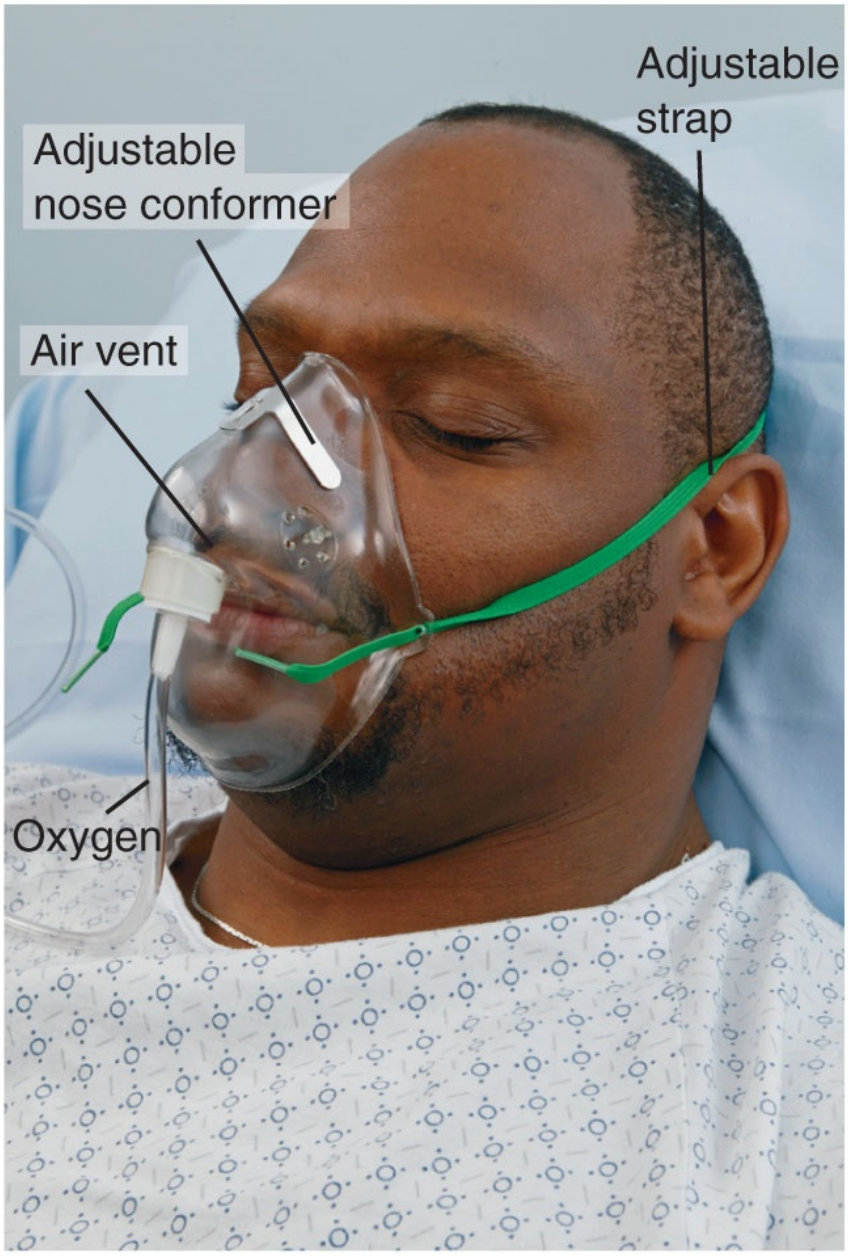


FIGURE 21-15 A simple mask is used to administer oxygen.

FIGURE 21-16 A partial rebreather mask delivers a higher percentage of oxygen than a simple mask.

Masks

Oxygen can be delivered using a simple mask, a partial rebreather mask, a non-rebreather mask, or a venturi mask.

Simple Mask

A **simple mask** fits over the nose and mouth and allows atmospheric air to enter and exit through side ports (Fig. 21-15). An elastic strap holds it in place. The

simple mask, like other types of masks, allows for the administration of higher levels of oxygen than are possible with a cannula. A simple mask is sometimes substituted for a cannula when a client has nasal trauma or breathes through the mouth. When a simple mask is used, oxygen is delivered at no less than 5 L/min.

The efficiency of any mask is affected by how well it fits the face. Without a good seal, the oxygen leaks from the mask, thus diminishing its concentration. Other problems are associated with masks as well. All oxygen masks interfere with eating and make verbal communication difficult to understand. Also, some clients become anxious when their nose and mouth are covered because it creates a feeling of being suffocated. Skin care also becomes a priority because masks create pressure and trap moisture.

Partial Rebreather Mask

A **partial rebreather mask** is an oxygen delivery device through which a client inhales a mixture of atmospheric air, oxygen from its source, and oxygen contained within a reservoir bag (Fig. 21-16). It provides a means for recycling oxygen and venting all the carbon dioxide during expiration from the mask. During expiration, the first third of exhaled air enters the reservoir bag. The portion of exhaled air in the reservoir bag contains a high proportion of oxygen because it comes directly from the upper airways; the gas in this area has not been involved in gas exchange at the alveolar level. Once the reservoir bag is filled, the remainder of exhaled air is forced from the mask through small ports. With a simple mask, some carbon dioxide always remains within the mask and is reinhaled.

Gerontologic Considerations

- Older adults who have lost weight and subcutaneous fat in their cheeks or who are not wearing their dentures may not receive the prescribed amounts of oxygen by mask because of an inadequate facial seal.

Non-Rebreather Mask

A **non-rebreather mask** is an oxygen delivery device in which all the exhaled air leaves the mask rather than partially entering the reservoir bag (Fig. 21-17). It is designed to deliver an F_{IO_2} of 90% to 100%. This type of mask contains one-way valves that allow only oxygen from its source, as well as the oxygen in the reservoir bag, to be inhaled. No air from the atmosphere is inhaled. All the air that is exhaled is vented from the mask. None enters the reservoir bag.

Obviously, clients for whom non-rebreather masks are used are those who require high concentrations of oxygen. They are usually critically ill and may eventually need mechanical ventilation. Despite the high concentration of oxygen, humidification is not employed when a mask with a reservoir bag is used. Also, clients with partial and non-rebreather masks are monitored closely to ensure that the reservoir bag remains partially inflated at all times.

Oxygen toxicity, a condition caused by breathing a high concentration of oxygen, can develop among clients who require the prolonged use of a non-rebreather mask, is discussed later in this chapter. Oxygen toxicity can result in irreversible pulmonary damage as well as decreases the stimulus to breathe that naturally occurs when carbon dioxide levels rise.

FIGURE 21-17 A non-rebreathing mask can deliver the highest amount of oxygen without the use of a mechanical ventilator.

Venturi Mask

A **venturi mask** mixes a precise amount of oxygen and atmospheric air. Sometimes called a Venti mask, this mask has a large ringed tube extending from it (Fig. 21-18). Adapters within the tube, which are color-coded or regulated by a dial system, permit only specific amounts of room air to mix with the oxygen. This feature ensures that the venturi mask delivers the exact amount of

prescribed oxygen. Unlike masks with reservoir bags, humidification can be added when a venturi mask is used.

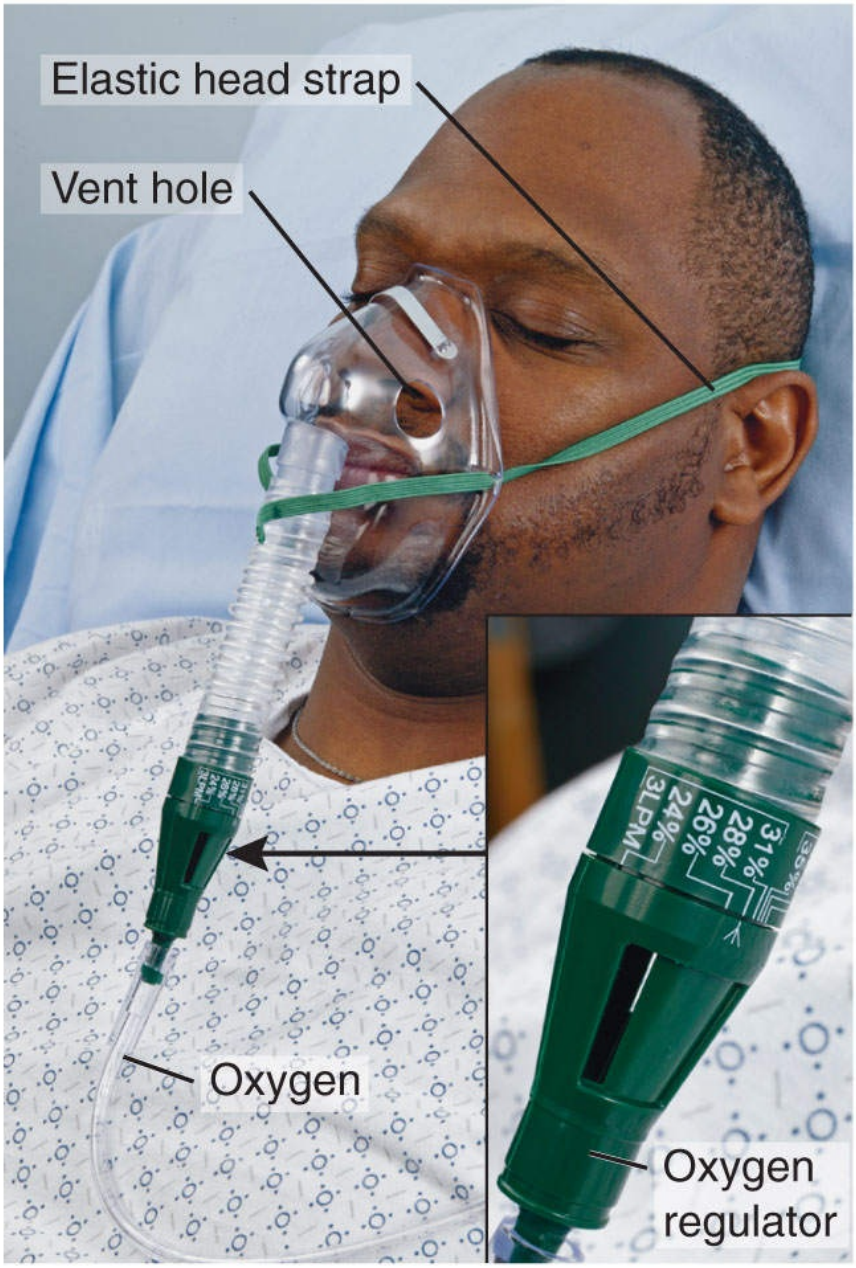


FIGURE 21-18 A venturi mask can be adjusted to deliver a variety of oxygen percentages.

Face Tent

A **face tent** provides oxygen to the nose and mouth without the discomfort of a mask (Fig. 21-19). Because the face tent is open and loose around the face, clients are less likely to feel claustrophobic. An added advantage is that a face

mask can be used for clients with facial trauma or burns. A disadvantage is that the amount of oxygen clients actually receive may be inconsistent with what is prescribed because of environmental losses.

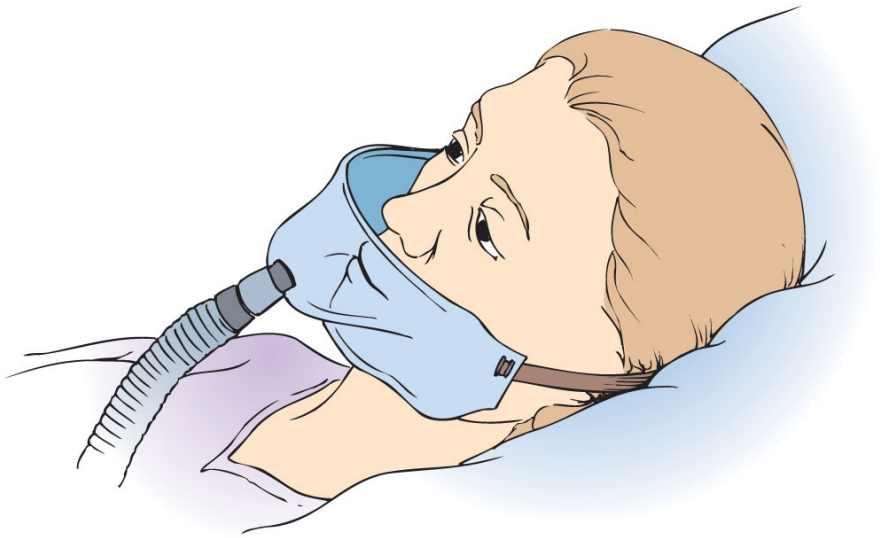


FIGURE 21-19 A face tent supplies oxygen toward the nose and mouth of a client.

Tracheostomy Collar

A **tracheostomy collar** delivers oxygen near an artificial opening in the neck (Fig. 21-20). It is applied over a tracheostomy—an opening into the trachea through which a client breathes (see Chap. 36). Because it bypasses the warming and moisturizing functions of the nose, a tracheostomy collar provides a means for both oxygenation and humidification. The moisture that collects, however, tends to saturate the gauze dressing around the tracheostomy, making it necessary to change it frequently.

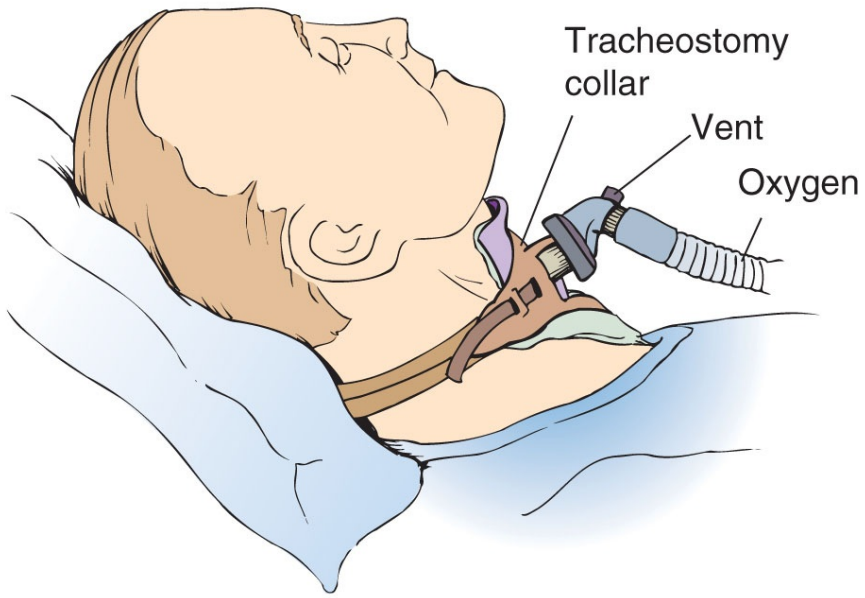


FIGURE 21-20 A tracheostomy collar is used to deliver oxygen via an artificial airway when the client does not breathe through the nose or mouth.

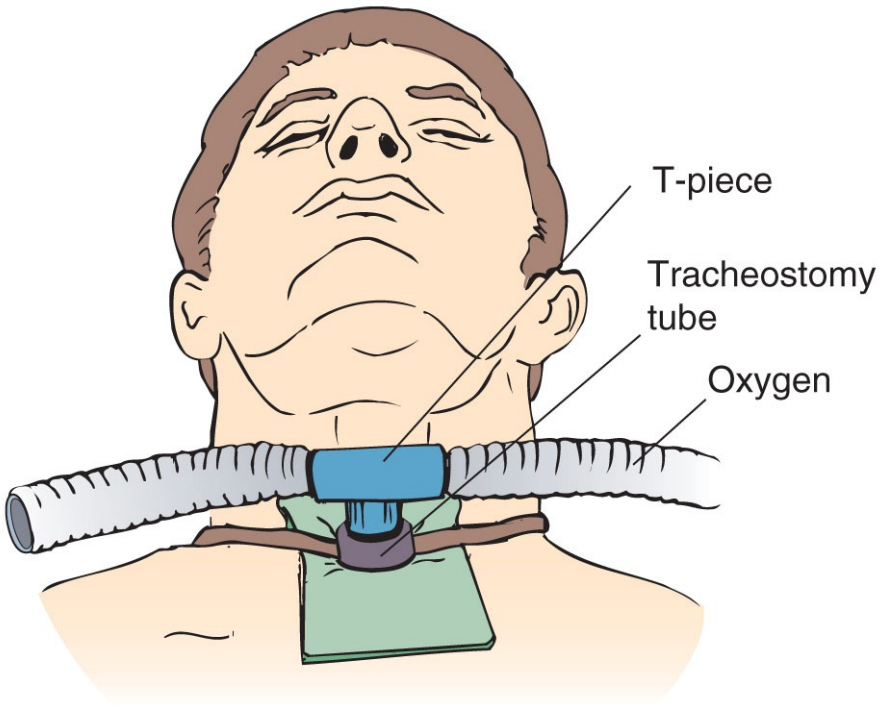


FIGURE 21-21 A T-piece attached to an airway.

T-Piece

A **T-piece** fits securely onto a tracheostomy tube or endotracheal tube (Fig. 21-21). It is similar to a tracheostomy collar but is attached directly to the artificial

airway. Although the gauze around the tracheostomy usually remains dry, the moisture that collects within the tubing tends to condense and may enter the airway during position changes if it is not drained periodically. Another disadvantage is that the weight of the T-piece, or its manipulation, may pull on the tracheostomy tube, causing the client to cough or experience discomfort. To avoid displacing the tracheostomy tube, the T-piece may need to be supported with a rolled towel.

Additional Delivery Devices

Other methods for delivering oxygen are used less commonly. Occasionally, oxygen is delivered by means of a nasal catheter, oxygen tent, transtracheal catheter, or positive airway pressure machines.

Nasal Catheter

A **nasal catheter** is a method for delivering oxygen with a tube inserted through the nose into the posterior nasal pharynx (Fig. 21-22). It is used for clients who tend to breathe through the mouth or experience claustrophobia when a mask covers their face. The catheter tends to irritate the nasopharynx; therefore, some clients find it uncomfortable. If a catheter is prescribed, the nurse secures it to the nose to avoid displacement and regularly cleans the nostril with a cotton applicator to remove dried mucus.

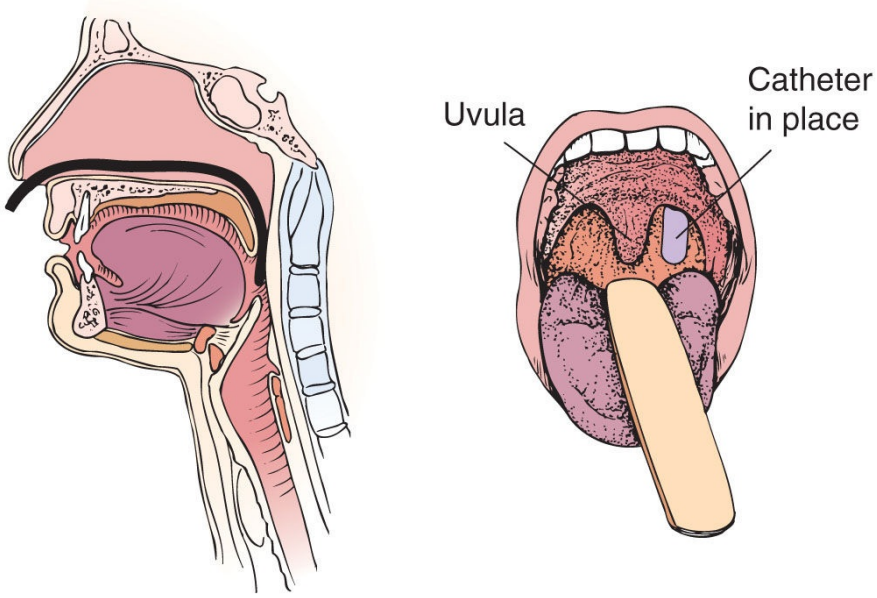


FIGURE 21-22 Nasal catheter placement.

Transtracheal Oxygen

Some clients who require long-term oxygen therapy may prefer its administration through a **transtracheal catheter** (a hollow tube inserted within the trachea to deliver oxygen; Fig. 21-23). This device is less noticeable than a nasal cannula. The client is adequately oxygenated with lower flows, decreasing the costs of replenishing the oxygen source.

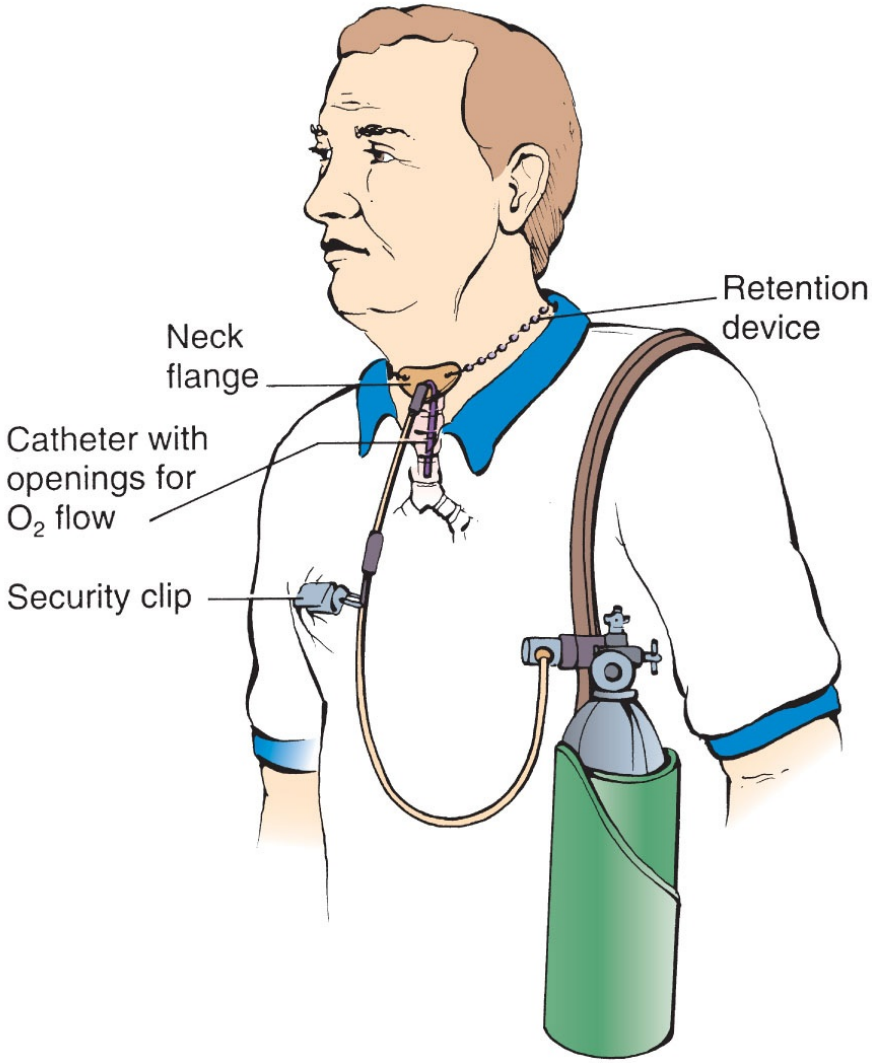


FIGURE 21-23 Transtracheal oxygen administration.

Before transtracheal oxygen is used, a **stent** (tube that keeps a channel open) is inserted into a surgically created opening and remains there until the wound heals. Thereafter, the stent is removed, and the catheter is inserted and held in place with a necklace-type chain. Clients learn how to clean the tracheal opening and catheter, a procedure performed several times a day. During cleaning, clients administer oxygen with a nasal cannula.

Oxygen Tent

An **oxygen tent** is a clear plastic enclosure that provides cooled, humidified oxygen. It is most often used in the care of active toddlers. Children this age are less likely to keep a mask or cannula in place but may require oxygenation and humidification for respiratory conditions such as croup or bronchitis. A face hood may be used for less-active infants.

Oxygen concentrations are difficult to control when an oxygen tent is used. Therefore, when caring for a child in an oxygen tent, the edges of the tent must be tucked securely beneath the mattress; limit opening the zippered access ports so that oxygen does not escape too freely. Oxygen levels must be monitored with an oxygen analyzer.

Stop, Think, and Respond 21-3

What evidence indicates that a client is well oxygenated?

Oxygen Hazards

Regardless of which device is used, oxygen administration involves potential hazards: first and foremost, oxygen's capacity to support fires, and second, the potential for oxygen toxicity.

Fire Potential

Oxygen itself does not burn, but it does support combustion; in other words, it contributes to the burning process. Therefore, it is necessary to control all possible sources of open flames or ungrounded electricity (see Nursing Guidelines 21-2).

Oxygen Toxicity

Oxygen toxicity refers to lung damage that develops when oxygen concentrations of more than 50% are administered for longer than 48 to 72 hours. The exact mechanism by which hyperoxygenation damages the lungs is not definitely known. One theory is that it reduces **surfactant**, which is a lipoprotein produced by cells in the alveoli that promotes elasticity of the lungs and enhances gas diffusion. Once oxygen toxicity develops, it is difficult to reverse. Unfortunately, early symptoms are quite subtle (Box 21-2). The best prevention is to administer the lowest F_{IO_2} possible for the shortest amount of time.

BOX 21-2 Signs and Symptoms of Oxygen Toxicity

- Nonproductive cough
- Substernal chest pain
- Nasal stuffiness
- Nausea and vomiting
- Fatigue
- Headache
- Sore throat
- Hypoventilation

RELATED OXYGENATION TECHNIQUES

Additional techniques that relate to oxygenation include positive airway pressure machines, water-seal chest tube drainage systems, and hyperbaric oxygen therapy (HBOT).

Positive Airway Pressure Machines

Positive airway pressure machines are devices that help to relieve impaired oxygen levels caused by apnea or hypopnea during sleep. The underlying cause of sleep apnea/hypopnea is a temporary relaxation of muscles that should support structures in the soft palate and tongue. As a result of the muscle relaxation, the airway becomes obstructed, breathing becomes difficult or temporarily ceases. Those affected awaken briefly multiple times during the night in an effort to restore breathing which interferes with normal sleep cycles. Consequently, they experience daytime sleepiness, difficulty concentrating on work-related tasks, emotional irritability, headaches, and even chest pain. Falling oxygen saturation levels may precipitate cardiac arrest and death.

Sleep apnea can be relieved with one of several types of positive airway pressure machines that keep the airways open with controlled ventilation. Some examples include full face or nasal masks that maintain continuous positive airway pressure (CPAP), bilevel positive airway pressure (BiPAP), and other variations or mode adjustments of the two.

CPAP Mask

A **CPAP mask** which is attached to a portable ventilator maintains continuous positive airway pressure keeping the alveoli partially inflated even during periods of expiration (Fig. 21-24). The positive pressure prevents the airway from collapsing allowing inflated alveoli to diffuse oxygen into the blood during apneic episodes that may last 10 or more seconds as frequently as 10 to 15 times an hour.

NURSING GUIDELINES 21-2

Administering Oxygen Safely

- Post "Oxygen in Use" signs wherever oxygen is stored or in use. *The sign warns others of a potential fire hazard.*
- Prohibit the burning of candles during religious rites. *Doing so eliminates a source of open flames.*
- Check that electrical devices have a three-pronged plug (see Chap. 19). *This type of plug provides a ground for leaking electricity.*
- Inspect electrical equipment for frayed wires or loose connections. *Inspection helps to prevent sparks or an uncontrolled pathway for electricity.*
- Avoid using petroleum products, aerosol products (such as hair spray), and products containing acetone (such as nail polish remover) where oxygen is used. *This measure prevents ignition of flammable substances.*
- Secure portable oxygen cylinders to rigid stands. *Doing so prevents the tank from rupturing.*

However, some find it difficult to adjust to wearing the mask during the night as well as overcoming an effort to exhale against a continuous fixed level of pressure. Some abandon treatment due to their discomfort. An alternative is to switch to a BiPAP mask.

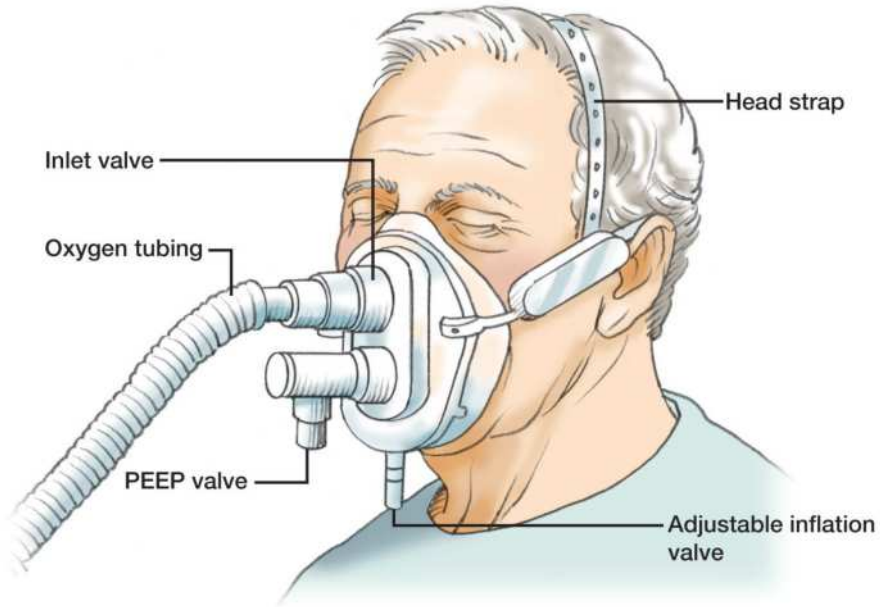


FIGURE 21-24 A CPAP mask. (From *Lippincott's Nursing Procedures and Skills 2013 Q1*. Philadelphia: Lippincott Williams & Wilkins.)

BiPAP Mask

A **BiPAP mask,** which looks similar to a CPAP device, provides two different levels of airway pressure: *inspiratory positive airway pressure*, which is higher during inhalation, and *expiratory positive airway pressure*, which is lower during

expiration. The variation promotes better tolerance among those requiring positive airway pressure therapy.

Further modifications are available. They include systems known as *automatically adjusting positive airway pressure* (APAP) and *variable positive airway pressure* (VPAP) machines. The APAP machine adjusts the pressure according to the sleeper's position in bed and stages of sleep. A VPAP device normalizes breathing on a breath-by-breath basis thus suppressing apnea and promoting the time spent in REM sleep (see Chap.18).

Water-Seal Chest Tube Drainage

Water-seal chest tube drainage is a technique for evacuating air or blood from the pleural cavity, which helps to restore negative intrapleural pressure and reinflate the lung. Clients who require water-seal drainage have one or two chest tubes connected to the drainage system.

Several companies provide equipment for water-seal drainage. All of these products consist of a three-chamber system (Fig. 21-25):

- One chamber collects blood or acts as an exit route for pleural air.
- A second compartment holds water that prevents atmospheric air from reentering the pleural space (hence the term "water seal").
- A third chamber, if used, facilitates the use of suction, which may speed the evacuation of blood or air.

One of the most important principles when caring for clients with water-seal drainage is that the chest tube must never be separated from the drainage system unless it is clamped. Even then, the tube is clamped for only a brief time. Additional nursing responsibilities are included in Skill 21-3.

Stop, Think, and Respond 21-4

Discuss how a collapsed lung affects oxygenation.

Hyperbaric Oxygen Therapy

Hyperbaric oxygen therapy (HBOT) consists of the delivery of 100% oxygen at three times the normal atmospheric pressure within an airtight chamber (Fig. 21-26). Treatments, which last approximately 90 minutes, are repeated over days, weeks, or months of therapy. Providing pressurized oxygen can deliver 15 times as much oxygen to tissues as can be obtained by breathing room air (Mayo Clinic Staff, 2014). Providing clients with brief periods of breathing room air

helps to prevent oxygen toxicity.

HBOT helps to regenerate new tissue at a faster rate; thus, its most popular use is for promoting wound healing. It also is used to treat carbon monoxide poisoning, gangrene associated with diabetes or other conditions of vascular insufficiency, decompression sickness experienced by deep-sea divers, anaerobic infections (especially in burn clients), and several other medical conditions.

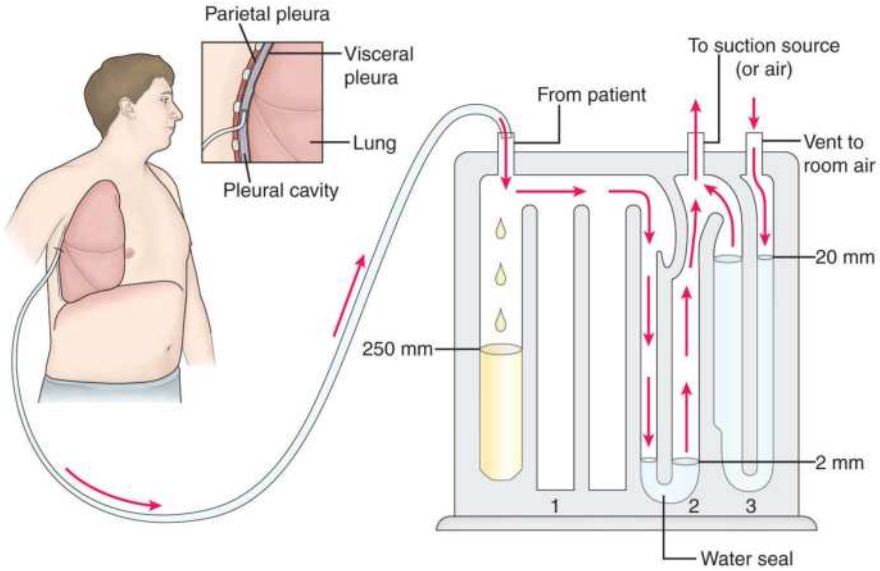


FIGURE 21-25 The three-chambered water-seal drainage system: (1) drainage collection chamber from the client, (2) the water-seal chamber, and (3) the suction control chamber attached to a source of suction and vented to room air.

FIGURE 21-26 A hyperbaric oxygen chamber. (Photo courtesy of Moose Jaw Union Hospital, Saskatchewan, Canada.)

NURSING IMPLICATIONS

Nurses assess the oxygenation status of clients on a day-by-day and a shift-by-shift basis. Therefore, it is not unusual to identify any one or several of the following nursing diagnoses among clients experiencing hypoxemia or hypoxia:

- Ineffective Breathing Pattern
- Impaired Gas Exchange
- Activity Intolerance
- Anxiety
- Risk for Injury (Related to Oxygen Hazards)

Abnormal assessment findings often lead to collaboration with the physician and the prescription for oxygen therapy.

Gerontologic Considerations

Advise older adults to receive annual influenza immunizations and a pneumonia immunization after 65 years of age or earlier if there is a history of chronic illness. Current guidelines recommend a booster dose for older adults who received their initial pneumonia immunization 5 or more years ago.

Clinical Scenario An older adult client is assessed by her physician because she has developed a frequent productive cough. She has a history of smoking one to two packs of cigarettes daily for 30 years. She tells her physician, “It seems so hard for me to get my breath. I cannot work in my flower garden because I get winded when I try to do any gardening. I cannot sleep lying down because I cannot breathe except sleeping in a chair.” Her physical examination reveals a barrel chest and diminished lung sounds bilaterally. The physician arranges a hospital admission.

Nursing Care Plan 21-1 is one example of how the nursing process applies to a client with the nursing diagnosis of Ineffective Breathing Pattern. This diagnostic category is defined in the NANDA-I taxonomy (2015, p. 227) as “inspiration and/or expiration that does not provide adequate ventilation.”

NURSING CARE PLAN 21-1 Ineffective Breathing Pattern

Assessment

- Determine the client’s respiratory rate and effort.
- Check the radial or apical pulse rate.
- Measure the client’s blood pressure.
- Note the client’s level of consciousness and mental status.
- Assess for the evidence of a cough and its characteristics.
- Observe the use of accessory thoracic and abdominal muscles for breathing.
- Observe the client’s chest contour.
- Inspect the skin, oral mucous membranes, and nail beds for signs of cyanosis.
- Palpate the client’s abdomen for evidence of distention that could crowd the diaphragm.
- Note the client’s body position, which may or may not facilitate breathing.
- Measure the client’s SpO_2 with a pulse oximeter.
- Review the results of arterial blood gas measurements.

- Auscultate anterior, posterior, and lateral lung sounds.
- Ask the client to describe his or her current status of oxygenation.
- Perform a pain assessment.
- Inquire as to the client's medical history of respiratory disorders or other conditions that can affect ventilation.
- Identify the client's smoking history.
- Review the client's current medication history for drugs that can impair oxygenation.

Nursing Diagnosis. **Ineffective Breathing Pattern** related to the retention of carbon dioxide secondary to chronic pulmonary damage from long-term cigarette smoking as manifested by rapid, shallow breathing at 40 breaths per minute accompanied by the use of accessory muscles to breathe.

Expected Outcome. The client will demonstrate an effective breathing pattern by 5/10 as evidenced by a respiratory rate no greater than 32 while performing mild activity such as bathing face, arms, and chest.

Interventions	Rationales
Provide periods of rest between activities.	Rest decreases oxygen demand and facilitates maintenance or restoration of oxygen within blood.
Elevate the head of the bed up to 90 degrees.	Head elevation lowers abdominal organs by gravity and provides an increased area for chest expansion when the diaphragm contracts.
Teach how to perform diaphragmatic and pursed-lip breathing and practice same at least b.i.d.	Pursed-lip breathing decreases respiratory rate, increases tidal volume, decreases arterial CO_2, increases arterial oxygen, and improves exercise performance.
Provide a minimum of 2,000 mL of oral fluid per 24 hour.	Adequate hydration liquefies respiratory secretions and facilitates expectoration. Expectoration of sputum clears the airway and promotes ventilation.
Ensure a daily dietary intake of approximately 2,000–2,500 calories.	The act of breathing creates additional caloric demands for energy.
Administer oxygen per nasal cannula at 2 L/min as prescribed by the physician if SpO_2 falls below 90% and is sustained there.	Supplemental oxygen relieves hypoxemia. Administering 2–3 L/min prevents suppressing the hypoxic drive to breathe, experienced by clients with chronic respiratory diseases.
Explore nicotine cessation therapy with transdermal skin patches.	Transdermal nicotine skin patches reduce symptoms associated with nicotine withdrawal. The dose of nicotine can be reduced gradually to promote nicotine cessation.

Evaluation of Expected Outcomes

- Respiratory rate decreases from 34 to 26 when placed in high Fowler's position.
- SpO_2 increases from 86% to 90% with 2 L of oxygen per minute.
- The client demonstrates and performs pursed-lip breathing.
- The client consumes three cans of supplemental liquid nourishment, each of which has 350 cal, three times a day, to facilitate reaching a minimum caloric goal of 2,000 cal.
- Fluid intake for 24 hours is between 1,800 and 2,200 mL.
- Client expectorates copious volume of sputum.

CRITICAL THINKING EXERCISES

1. What levels of oxygen saturation and pulse rates are a cause for nursing concern and indicate a need for further assessment?
2. Discuss some differences between oxygen therapy in a health care setting and that in a home environment.
3. What health teaching would you provide to reduce potential problems with oxygenation?
4. What nursing actions may be appropriate if the alarm on a pulse oximeter sounds frequently because the sensor does not stay on a client's finger?

NCLEX-STYLE REVIEW QUESTIONS

1. When a nurse assesses a client returning from surgery, which sign is an early indication that the client's oxygenation status is compromised?
   1. The client's dressing is bloody.
   2. The client becomes restless.
   3. The client's heart rate is irregular.
   4. The client reports being thirsty.

   Test-Taking Strategy: Note the key words, "early indication." Select a sign that correlates with hypoxemia.
2. When the nurse uses a pulse oximeter, what range in the SpO_2 measurement indicates that the client is adequately oxygenated?
   1. 80 to 100 mmHg
   2. 95 to 100 mmHg
   3. 80% to 100%
   4. 95% to 100%

   Test-Taking Strategy: Use the process of elimination to select the measurement and range for normal SpO_2.
3. When administering oxygen with a partial rebreather mask, which of the following observations is most important for the nurse to report to the respiratory therapy department?
   1. Moisture accumulates inside the mask.
   2. The reservoir bag collapses during inspiration.
   3. The mask covers the mouth and nose.
   4. The strap around the head is snug.

   Test-Taking Strategy: Note the key word and modifier, "most important." Select the option that requires immediate action more so than any of the others.

4. When the physician orders oxygen by nasal cannula at 5 L/min for a client with chronic pulmonary disease, what is the most appropriate nursing action to take initially?
   1. Add an oxygen humidifier to the flowmeter.
   2. Apply a pulse oximeter sensor to a finger.
   3. Question the prescribed oxygen flow rate.
   4. Substitute a mask for the nasal cannula.

   Test-Taking Strategy: Note the key words and modifier, "most appropriate" and "initially." Analyze the options and select the one that promotes the safety of the client.
5. When the nurse assesses a client with a chest tube connected to a water-seal drainage system, what are indications that the nurse should take corrective action? Select all that apply.
   1. Fluid in the water-seal chamber rises and falls with respirations.
   2. The fluid in the water seal is at the 2 cm mark.
   3. There is dark red blood in the drainage compartment.
   4. Fluid vigorously bubbles in the suction chamber.
   5. There is 10 cm of water in the suction chamber.

   Test-Taking Strategy: Use the process of elimination to select options that do not correlate with the safe use of water-seal drainage system.

SKILL 21-1 Using a Pulse Oximeter

23

Body Mechanics, Positioning, and Moving

Learning Objectives

On completion of this chapter, the reader should be able to:

1. Identify characteristics of good posture in a standing, sitting, or lying position.
2. Describe three principles of correct body mechanics.
3. Explain the purpose of ergonomics.
4. Give at least two examples of ergonomic recommendations in the workplace.
5. Describe at least 10 signs or symptoms associated with the disuse syndrome.
6. Describe six common client positions.
7. Explain the purpose of five different positioning devices used for safety and comfort.
8. Name one advantage for each of three different pressure-relieving devices.
9. Discuss four types of transfer devices.
10. Give at least five general guidelines that apply to transferring clients.

Words to Know

alignment
anatomic position
balance
bariatric client
base of support
bed board
body mechanics
center of gravity
contractures
disuse syndrome
energy
ergonomics
foot drop

Fowler's position
functional mobility
functional position
gravity
lateral oblique position
lateral position
line of gravity
muscle spasms
neutral position
posture
prone position
repetitive strain injuries
shearing
Sims' position
supine position
transfer

Inactivity leads to deterioration of health. Multiple complications can occur among people with limited activity and movement (Table 23-1).

Gerontologic Considerations

■ By the seventh or eighth decade of life, muscle strength, endurance, and coordination decline. Engaging in physical activity is an important health promotion intervention for preventing mobility limitations. The risk for social isolation among older adults increases as mobility is limited.

The consequences of inactivity are collectively referred to as **disuse syndrome** (signs and symptoms that result from inactivity). Nursing care activities such as positioning and moving clients reduce the potential for disuse syndrome. Nurses can become injured, however, if they fail to use good posture and body mechanics while performing these activities.

This chapter describes how to position and move clients to prevent complications associated with inactivity. It also discusses methods for protecting nurses from work-related injuries. Basic terms are defined in Box 23-1.

TABLE 23-1 Dangers of Inactivity

SYSTEMS	EFFECTS
Muscular	Weakness
	Decreased tone/strength
	Decreased size (atrophy)
Skeletal	Poor posture
	Contractures
	Foot drop
Cardiovascular	Impaired circulation
	Thrombus (clot) formation
	Dependent edema
Respiratory	Pooling of secretions
	Shallow respirations
	Atelectasis (collapsed alveoli)
Urinary	Oliguria (scanty urine)
	Urinary tract infections
	Calculi (stone) formation
	Incontinence (inability to control elimination)
Gastrointestinal	Anorexia (loss of appetite)
	Constipation
	Fecal impaction
Integumentary	Pressure sores
Endocrine	Decreased metabolic rate
	Decreased hormonal secretions
Central nervous	Sleep pattern disturbances
	Psychosocial changes

BOX 23-1 Basic Terminology Used in Body Mechanics

Gravity: Force that pulls objects toward the center of the earth. The pull of gravity causes objects, such as an item dropped from the hand, to fall to the ground. It causes water to drain to its lowest level.
Energy: Capacity to do work. Energy is used to move the body from place to place. Energy is required to overcome the force of gravity.
Balance: Steady position with weight. A person falls when off balance.
Center of gravity: Point at which the mass of an object is centered. The center of gravity for a standing position is the center of the pelvis and about halfway between the umbilicus and the pubic bone.
Line of gravity: Imaginary vertical line that passes through the center of gravity. The line of gravity in a standing person is a straight line from the head to the feet through the center of the body.
Base of support: Area on which an object rests. The feet are the base of support when a person is in a standing position.
Alignment: Parts of an object being in proper relationship to one another. The body is in good alignment in a position of good posture.
Neutral position: The position of a limb that is turned neither toward nor away from the body's midline.
Anatomic position: Frontal and back views with arms at the sides and palms forward.
Functional position: Position in which an activity is performed properly and normally. In the hand, the

wrists are slightly dorsiflexed between 20 and 35 degrees and the proximal finger joints are flexed between 45 and 60 degrees, with the thumb in opposition and in alignment with the pads of the fingers.

MAINTAINING GOOD POSTURE

Posture (the position of the body, or the way in which it is held) affects a person's appearance, stamina, and ability to use the musculoskeletal system efficiently. Good posture, whether in a standing, sitting, or lying position, distributes gravity through the center of the body over a wide base of support (Fig. 23-1). Good posture is important for both clients and nurses.

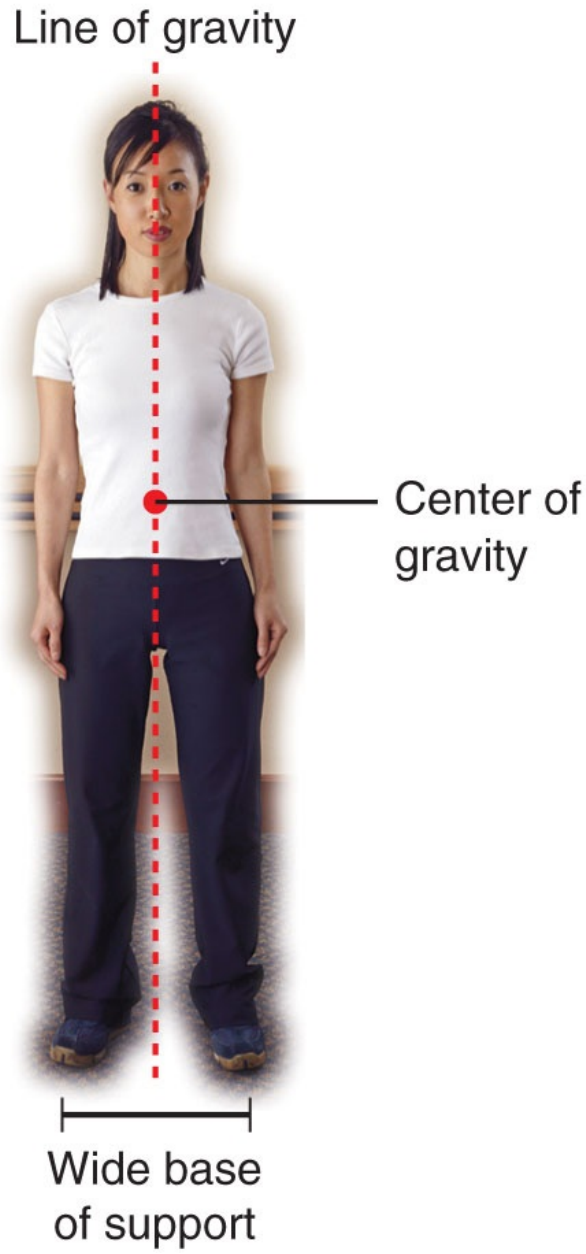


FIGURE 23-1 Good posture helps to align gravity through the center of the body. A wide stance provides a stable base for support.

When a person performs work while using poor posture, **muscle spasms** (sudden, forceful, and involuntary muscle contractions) often result. They occur more often when muscles are strained and forced to work beyond their capacity.

Pharmacologic Considerations

- Older women may be prescribed bisphosphonate drugs (bone resorption

inhibitors), such as alendronate (Fosamax), to maintain bone density and prevent bone fractures. Specific administration guidelines must be followed for best results and to prevent gastrointestinal side effects.

Standing

To maintain good posture in a standing position (Fig. 23-2): Keep the feet parallel, at right angles to the lower legs, and about 4 to 8 inches (10 to 20 cm) apart.

- Distribute weight equally on both feet to provide a broad base of support.
- Bend the knees slightly to avoid straining the joints.
- Maintain the hips at an even level.
- Pull in the buttocks and hold the abdomen up and in to keep the spine properly aligned. This position supports the abdominal organs and reduces strain on both back and abdominal muscles.
- Hold the chest up and slightly forward and extend or stretch the waist to give internal organs more space and maintain good alignment of the spine.
- Keep the shoulders even and centered above the hips.
- Hold the head erect with the face forward and the chin slightly tucked.

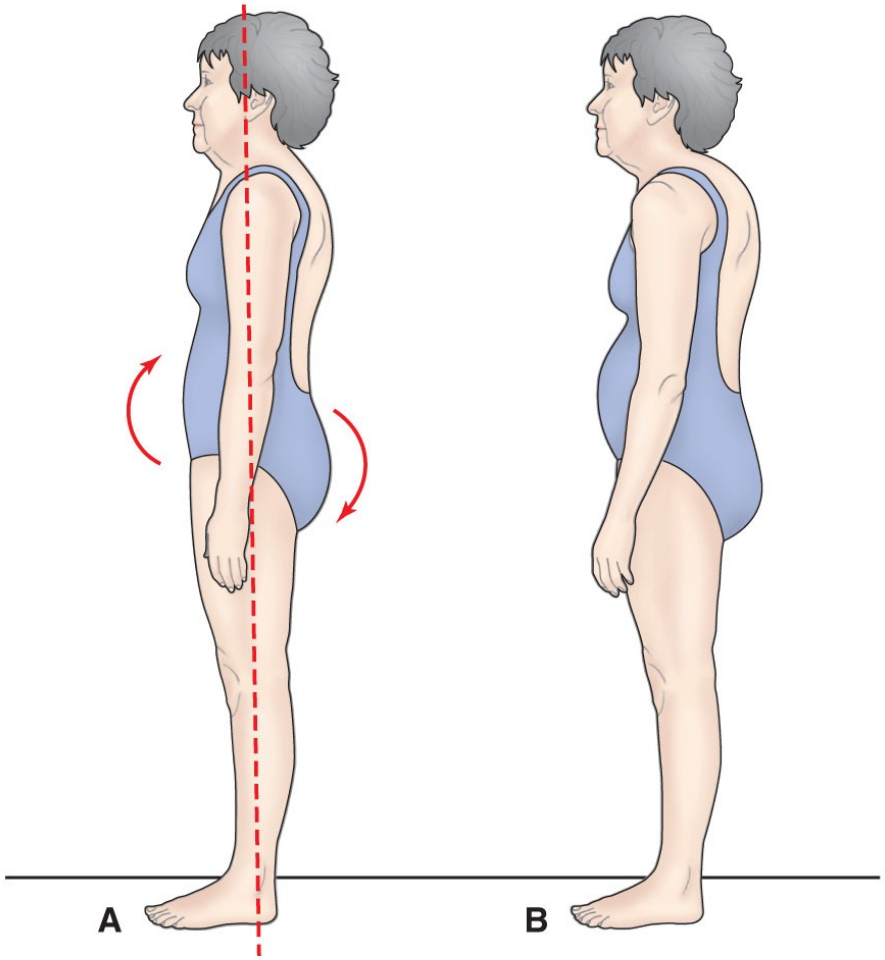


FIGURE 23-2 (A) A good standing posture results when abdominal and gluteal muscles are contracted. (B) A poor standing posture results when abdominal muscles are relaxed, causing altered body alignment.

Sitting

In a good sitting position (Fig. 23-3), the buttocks and upper thighs become the base of support. Both feet rest on the floor. The knees are bent, with the posterior of the knee free from the edge of the chair to avoid interfering with distal circulation.

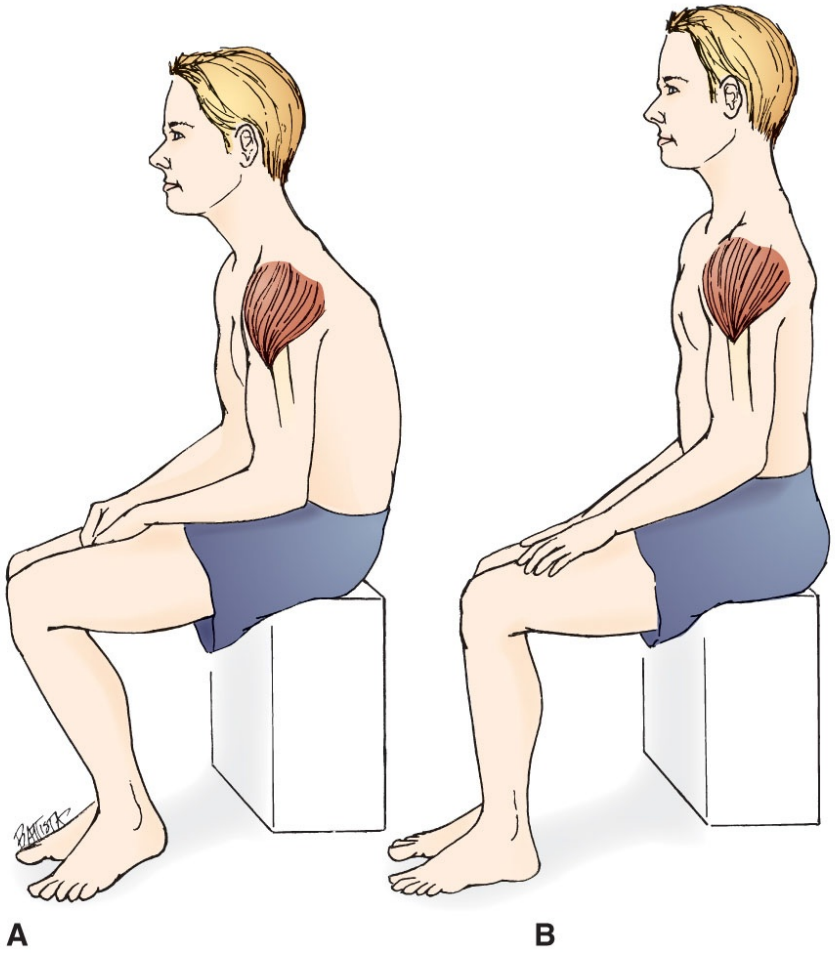


FIGURE 23-3 (A) A correct sitting posture. (B) An incorrect sitting posture. (From Hendrickson, T. [2010]. *Massage and manual therapy for orthopedic conditions* [2nd ed.]. Philadelphia, PA: Wolters Kluwer Health and Pharma.)

Lying Down

Good posture in a lying position looks the same as in a standing position, except the person is horizontal (Fig. 23-4). The head and neck muscles are in a neutral position, centered between the shoulders. The shoulders are level, whereas the arms, hips, and knees are slightly flexed with no compression of the arms or legs under the body. The trunk is straight, and the hips are level. The legs are parallel to each other with the feet at right angles to the leg.

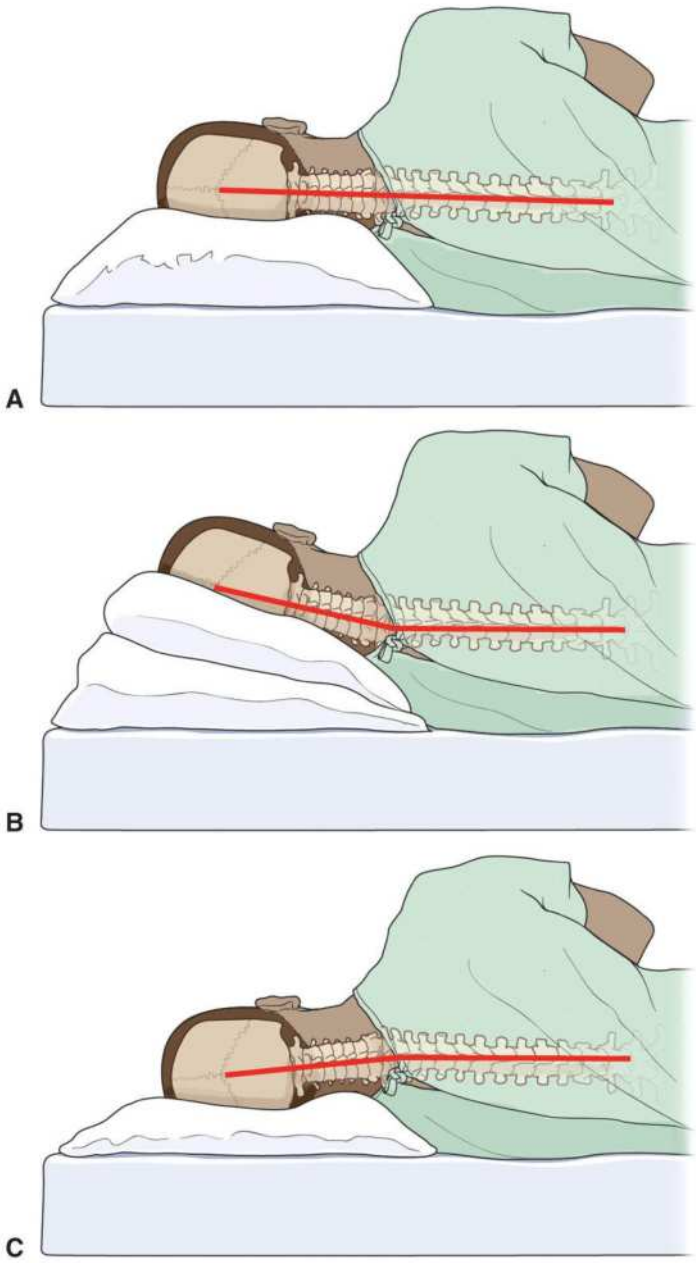


FIGURE 23-4 (A) **Maintaining the spine in neutral position (top) facilitates the best lying posture.** (B) **Positions that create neck flexion (middle) or** (C) **hyperextension (bottom) are undesirable.**

BODY MECHANICS

The use of proper **body mechanics** (the efficient use of the musculoskeletal system) increases muscle effectiveness, reduces fatigue, and helps to avoid **repetitive strain injuries** (disorders that result from cumulative trauma to musculoskeletal structures). Basic principles of body mechanics are important regardless of a person's occupation or daily activities, but body mechanics alone

will not necessarily reduce musculoskeletal injuries (see Nursing Guidelines 23-1).

Gerontologic Considerations

- Skeletal changes such as kyphosis, lordosis, or scoliosis change the older person's center of gravity.
- Kyphotic changes can cause pressure on cervical vertebrae when someone is in a supine position. Effects of this can be minimized by placing a small towel or cervical pillow under the neck.

NURSING GUIDELINES 23-1

Using Good Body Mechanics

- Use the longest and strongest muscles of the arms and legs. *Use of these muscles provides the greatest strength and potential for performing work.*
- When lifting a heavy load, center it over the feet. *Such positioning creates a base of support.*
- Hold objects close to the body. *Doing so increases balance.*
- Bend the knees. *Bending the knees prepares the spine to accept the weight of the load.*
- Contract the abdominal muscles and make a long midriff. *Doing so protects the muscles of the abdomen and pelvis and prevents strain and injury to the abdominal wall.*
- Push, pull, or roll objects whenever possible rather than lifting them. *Lifting requires more effort.*
- Use body weight as a lever to assist with pushing or pulling an object. *This reduces muscle strain.*
- Keep feet apart for a broad base of support. *This stance lowers the center of gravity, which promotes stability.*
- Bend the knees and keep the back straight when lifting an object, rather than bending over from the waist with straight knees. *This stance makes best use of the longest and strongest body muscles and improves balance by keeping the weight of the object close to the center of gravity.*
- Avoid twisting and stretching muscles during work. *Twisting can strain muscles because the line of gravity is outside the body's base of support.*
- Rest between periods of exertion. *Resting promotes work endurance.*

Gerontologic Considerations

- Teach older adults to use appropriate body mechanics, such as sitting in a chair when lifting an object. Also teach about the importance of positioning oneself directly in front of objects when lifting to prevent pulling of lateral back muscles or vertebral disk compression.

ERGONOMICS

Using proper body mechanics is one component of preserving the integrity of the body. Another component is applying and implementing **ergonomics** (a

specialty field of engineering science devoted to promoting comfort, performance, and health in the workplace). Ergonomics is used to improve the design of the work environment and equipment. The National Institute for Occupational Safety and Health (NIOSH), a division of the Centers for Disease Control and Prevention, requires employers to comply with many ergonomic recommendations. Examples include the following:

- Using assistive devices to lift or transport heavy items or clients
- Using alternative equipment for tasks that require repetitive motions—for instance, telephone headsets or automatic staplers
- Positioning equipment no more than 20 to 30 degrees away—about an arm's length—to avoid reaching or twisting the trunk or neck
- Using a chair with good back support. A chair should be high enough so the user can place his or her feet firmly on the floor. There should be room for two fingers between the edge of the seat and the back of the knees. Arm rests should allow a relaxed shoulder position.
- Keeping the elbows flexed no more than 100 to 110 degrees, or resting the wrists in neutral position when working at a computer
- Working under nonglare lighting

Despite being taught principles of good body mechanics, health care providers, particularly nurses, are vulnerable to ergonomic hazards in the workplace as a direct consequence of (1) lifting heavy loads (ie, clients), (2) reaching and lifting with loads far from the body, (3) twisting while lifting, (4) unexpected changes in load demand during the lift, (5) reaching low or high to begin a lift, and (6) moving or carrying a load a significant distance. In 2011, US health care providers experienced seven times the national rate of musculoskeletal disorders compared with all other private-sector workers (CDC, 2015b). Because of the pervasiveness of the problem and its direct link to a shortage of employed nurses, the American Nurses Association (ANA) took the initiative in 2003 (still ongoing) to reduce injuries to nurses and their clients by recommending a "no lift policy," known as the *Handle With Care Campaign* (deCastro, 2004). The campaign's goal is to reduce injuries through the use of assistive equipment and devices, which would result in many advantages (Box 23-2). In addition, the ANA has developed a framework for establishing a comprehensive program titled *Safe Patient Handling and Mobility: Interprofessional National Standards* to eliminate the manual handling of patients. In 2009 and again in 2013, the principles are incorporated in a US congressional bill (H.R.2480) known as the *Nurse and Health Care Worker Protection Act* (U.S. House, 2013). The proposed

legislation mandates safe patient/client movement for direct-care licensed nurses and other health care providers as a critical component in protecting health care providers and increasing patient/client safety. It supports methods to reduce risks associated with moving clients and evaluating alternatives or restricting manual lifting to emergency, life-threatening, or exceptional circumstances. Despite the support of nurses and other health care providers for H.R.2480, to date there has been no roll call vote on this bill. However, health care agencies are already implementing the ANA guidelines. Voluntary changes in nursing practices have not been delayed while waiting for federal legislative action. As of 2014, eleven states enacted safe client handling legislation (ANA, 2015).

BOX 23-2 Advantages of Assistive Devices

Nurses

- Lessens physical exertion during positioning, moving, and transferring clients
- Reduces musculoskeletal injuries
- Decreases sick or absentee time
- Lowers medical costs, pain, and suffering
- Decreases workman's compensation claims
- Maintains workforce of employed nurses

Clients

- Provides more security during repositioning and transfers from bed, chairs, toilets, stretchers
- Results in fewer handling mishaps and secondary injuries
- Relieves anxiety concerning safety
- Promotes comfort by reducing awkward or forceful manual handling
- Maintains dignity and self-esteem
- Promotes faster recovery

Adapted from American Nurses Association. (2003). *Handle with care campaign*. From http://www.nursingworld.org/handlewithcare.

POSITIONING CLIENTS

Good posture and body mechanics and ergonomically designed assistive devices are necessary when inactive clients require positioning and moving. An inactive client's position is changed to relieve pressure on bony areas of the body, promote **functional mobility** (an alignment that maintains the potential for movement and ambulation), and provide for therapeutic needs. General principles for positioning are as follows:

- Change the inactive client's position at least every 2 hours.
- Enlist the assistance of at least one other caregiver.

- Raise the bed to the height of the caregiver's elbow.
- Remove pillows and positioning devices.
- Unfasten drainage tubes from the bed linen.
- Use a low-friction fabric or gel-filled plastic sheet, roller sheet with handles, or a repositioning sling to slide, rather than to drag or lift, the client while turning or transferring from bed to a stretcher (Fig. 23-5).
- Turn the client as a complete unit to avoid twisting the spine.
- Place the client in good alignment with joints slightly flexed.
- Replace pillows and positioning devices.
- Support limbs in a functional position.
- Use elevation to relieve swelling or promote comfort.
- Provide skin care after repositioning.

FIGURE 23-5 These nurses are using the Phil-e-slide Patient Handling System to transfer a client. (Photo courtesy of ErgoSafe Products, LLC. St. Louis, MO).

Gerontologic Considerations

■ Older adults with cognitive impairment may have difficulty following directions regarding positioning and transferring. Instructions should be given

using clear, simple words to make one request at a time. Demonstrations and illustrations can be used to help convey the message.

Common Positions

Nurses commonly use six body positions when caring for bedridden clients: supine, lateral, lateral oblique, prone, Sims', and Fowler's.

Supine Position

In the **supine position**, the person lies on his or her back (Fig. 23-6).

FIGURE 23-6 Supine position.

There are two primary concerns associated with the supine position: prolonged pressure, especially at the end of the spine, leads to skin breakdown; and gravity, combined with pressure on the toes from bed linen, creates a potential for **foot drop** (a permanent dysfunctional position caused by a shortening of the calf muscles and a lengthening of the opposing muscles on the anterior leg; Fig. 23-7). Foot drop hinders ambulation because it interferes with a person's ability to place the heel on the floor. The supine position, however, is recommended as a way to reduce the incidence of sudden infant death syndrome among newborns (American Academy of Pediatrics, 2011).

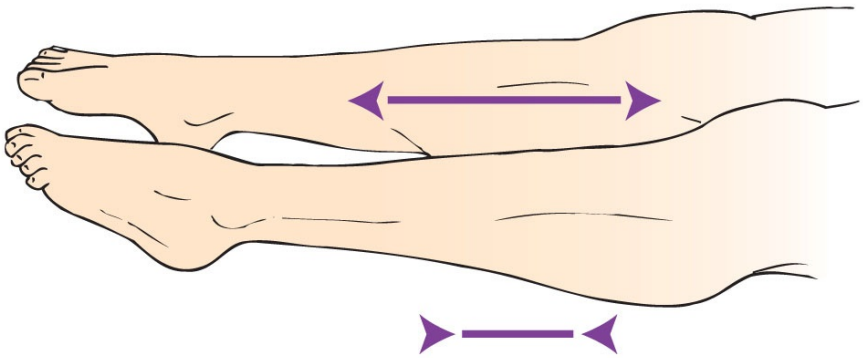


FIGURE 23-7 Foot drop is a consequence of weakened muscles for dorsiflexion resulting in permanent plantar flexion.

Lateral Position

With the **lateral position** (a side-lying position), foot drop is of less concern because gravity does not pull the feet down as happens when clients are supine (Fig. 23-8). Nevertheless, unless the upper shoulder and arm are supported, they may rotate forward and interfere with breathing.

FIGURE 23-8 Lateral position.

Lateral Oblique Position

In the **lateral oblique position** (a variation of the side-lying position), the client lies on the side with the top leg placed in 30 degrees of hip flexion and 35 degrees of knee flexion (Fig. 23-9). The calf of the top leg is placed behind the midline of the body on a support such as a pillow. The back is supported, and the

bottom leg is in neutral position. This position produces less pressure on the hip than a strictly lateral position and reduces the potential for skin breakdown.

FIGURE 23-9 Lateral oblique position.

Prone Position

The **prone position** (one in which the client lies on the abdomen, Fig. 23-10) is an alternative position for the person with skin breakdown from pressure ulcers (see Chap. 28). The prone position also provides good drainage from bronchioles, stretches the trunk and extremities, and keeps the hips in an extended position. The prone position has been found to improve arterial oxygenation in critically ill clients with adult respiratory distress syndrome who are mechanically ventilated, but does not necessarily result in higher rates of survival (Schwartz et al., 2015). The prone position poses a nursing challenge for assessing and communicating with clients, however, and it is uncomfortable for clients with recent abdominal surgery or back pain, and interferes with eating.

FIGURE 23-10 Prone position.

Sims' Position

In **Sims' position** (a semiprone position), the client lies on the left side with the right knee drawn up toward the chest (Fig. 23-11). An arm is positioned along the client's back, and the chest and abdomen are allowed to lean forward. The Sims' position is also used for the examination of and procedures involving the rectum and vagina (see Chap. 14).

Fowler's Position

The **Fowler's position** (a semi-sitting position) makes it easier for the client to eat, talk, and look around. Three variations are common. In a *low Fowler's position*, the head and torso are elevated to 30 degrees (Fig. 23-12). A *mid-Fowler's* or *semi-Fowler's position* refers to an elevation of up to 45 degrees. A *high Fowler's position* is an elevation of 60 to 90 degrees (Fig. 23-13). The knees may not be elevated, but doing so relieves strain on the lower spine.

FIGURE 23-11 Sims' position.

FIGURE 23-12 Low Fowler's position.

FIGURE 23-13 High Fowler's position.

The Fowler's position is especially helpful for clients with dyspnea because it causes the abdominal organs to drop away from the diaphragm. Relieving pressure on the diaphragm allows the exchange of a greater volume of air. Sitting for a prolonged period, however, decreases blood flow to tissues in the coccyx area and increases the risk for pressure ulcers in that area.

››› *Stop, Think, and Respond 23-1*

Give one advantage and one disadvantage for the supine, lateral and lateral oblique, prone, Sims', and Fowler's positions.

Positioning Devices

Many devices are available to help maintain good body alignment in bed and to prevent discomfort or pressure. Any position, no matter how comfortable or anatomically correct, must be changed frequently.

Adjustable Bed

The adjustable bed (see Chap. 18) can be raised or lowered and allows the position of the head and knees to be changed. The high position facilitates the performance of nursing care. Raising the head of the bed helps the client to look around without twisting and bending. It also promotes drainage of the upper lobes of the lungs and prepares the client for eventually standing and walking. The low position enables an independent client to get in and out of the bed safely (Fig. 23-14). Placing a bed in a slight Trendelenburg position may help keep the client from sliding down toward the foot of the bed (Fig. 23-15).

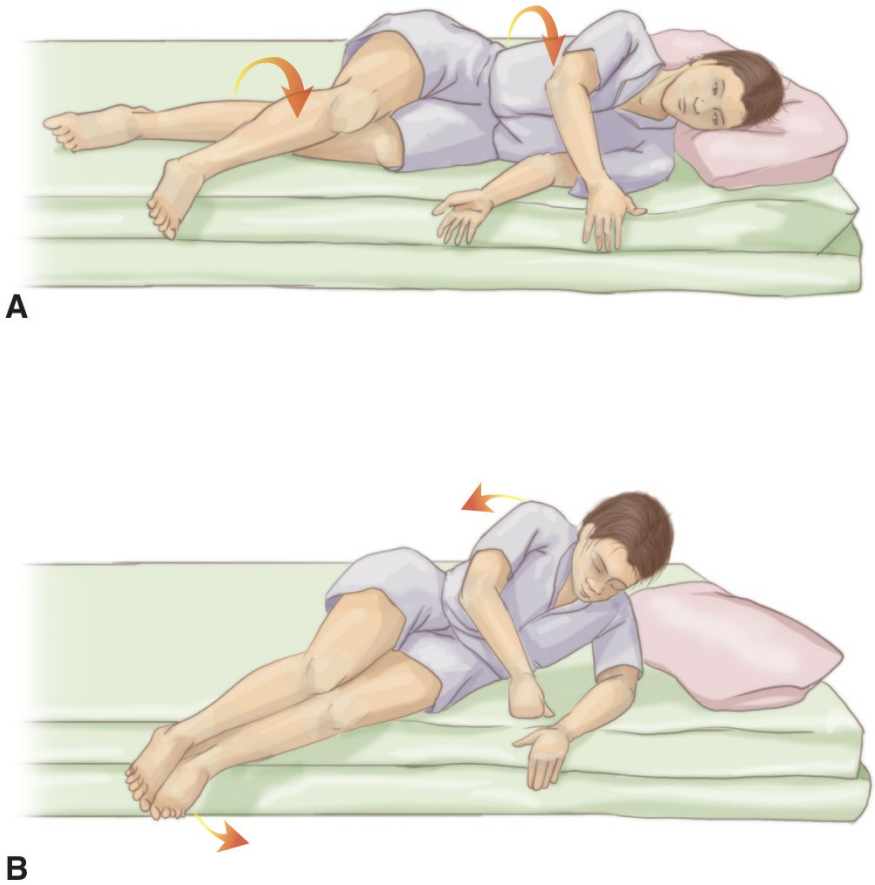


FIGURE 23-14 Grasping the mattress and pushing down with the other hand is an independent technique for sitting on the edge of the bed in preparation for ambulating. (From Craven, R. F., Hirnle, C. J., & Jensen, S. [2012]. *Fundamentals of nursing* [7th ed.]. Philadelphia, PA: Lippincott Williams & Wilkins)

Mattress

A comfortable, supportive mattress is firm but flexible enough to permit good body alignment. An unsupportive mattress promotes an unnatural curvature of the spine.

Bed Board

A **bed board** (a rigid structure placed under a mattress) provides additional skeletal support. Bed boards usually are made of plywood or some other firm material. The size varies with the situation. If sections of the bed (the head and foot) can be raised, the board must be divided into hinged sections. For home use, full bed boards can be purchased or made from sheets of plywood.

FIGURE 23-15 In the Trendelenburg position, the head is lower than the feet.

Pillows

Pillows are used to support and elevate a body part. Small pillows, such as contour pillows, triangular wedges, and bolsters, are ideal for supporting and elevating the head, extremities, and shoulders. For home use, oversized pillows are useful for elevating the upper part of the body if an adjustable bed is not available.

Roller Sheet

A roller sheet (also known as a slider sheet) that extends from the upper back to mid-thighs is a helpful positioning device. Some are designed with handles on either side. When made of substances that reduce friction, the roller sheets diminish the work of turning a client and avoid the potential for skin injuries. They are used to slide and roll, rather than to lift, the client. They help clients to move up in bed from a supine position in the center of the bed to the side of the bed, to turn clients to a lateral position, or to transfer clients from bed to a stretcher. Working as a team, nurses use the roller sheet to change the client to an alternate position while avoiding any stooping, reaching, or twisting. The sheet is removed after being used or kept dry and free of wrinkles to prevent skin breakdown. A mechanical lift, which is discussed later, or a repositioning sling is recommended when major repositioning is required. The roller sheet is placed close to the sides of the client's body during repositioning (Fig. 23-16).

Turning and Moving Clients

In some cases the client may be fully capable of assisting with turning or moving. The amount of client assistance depends on factors such as size, weight, mental status, and strength (Fig. 23-17).

FIGURE 23-16 A roller sheet is used for turning, moving, and repositioning.

Gerontologic Considerations

- Elevated toilet seats with handrails may help older adults use their arm muscles during transfers.
- Older adults require extra time and assistance during positioning, transferring, and ambulating. Positions

should be modified to address pain or functional limitations.

- Allow extra time when older adults are changing their positions, such as from supine to sitting or standing, to prevent orthostatic hypotension. Teach the client to wait until any dizziness has resolved before moving, thus decreasing the risk for falls.
- Older adults may limit their mobility due to fear of falling. Handrails may be strategically placed to promote confidence in ambulation. In addition, placing chairs along usual walking pathways will provide a "rest stop," thus increasing confidence in ambulation.

If all criteria suggest that the nurse and client can accomplish the task at hand, the nurse enlists the client's cooperation by explaining the plan and how the client can help. Assistive devices and additional caregivers are needed when turning or moving a client who cannot change from one position to another independently or who needs help doing so. Good turning and moving skills are important to prevent injury to the nurse and the client. Skill 23-1 describes the process of repositioning and moving clients.

Trochanter Rolls

Trochanter rolls (Fig. 23-18) prevent the legs from turning outward. The trochanters are the bony protrusions at the head of the femur near the hip. Placing a positioning device at the trochanters helps to prevent the leg from rotating outward (see Nursing Guidelines 23-2).

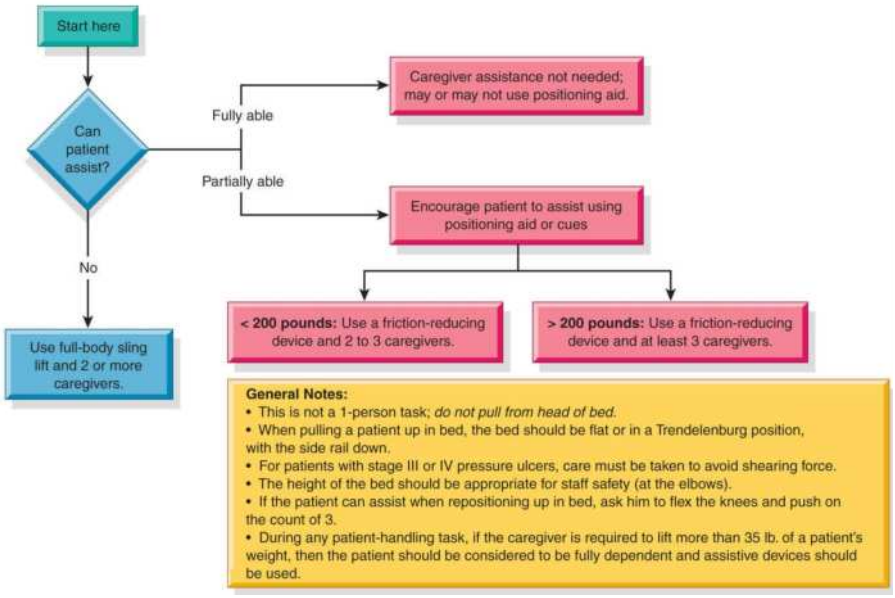


FIGURE 23-17 Repositioning algorithm for reposition in bed, from side to side or up. (From VISN 8 Patient Safety Center. [2009]. *Safe patient handling and movement algorithms.* Retrieved September 29, 2015, from http://www.visn8-med-va.gov/patientsafetycenter/safePThandling/default.asp.)

24

Fitness and Therapeutic Exercise

Learning Objectives

On completion of this chapter, the reader should be able to:

1. List at least five benefits of regular exercise.
2. Define fitness.
3. Identify seven factors that interfere with fitness.
4. Name at least two methods of fitness testing.
5. Describe how to calculate a person's target heart rate.
6. Define metabolic energy equivalent.
7. Differentiate fitness exercise from therapeutic exercise.
8. Differentiate isotonic exercise from isometric exercise.
9. Give at least one example of isotonic and isometric and isokinetic exercises.
10. Differentiate between active exercise and passive exercise.
11. Discuss how and why range-of-motion exercises are performed.
12. Provide at least two suggestions for helping older adults become or stay physically active.

Words to Know

active exercise
aerobic exercise
ambulatory electrocardiogram
ankylosis
artifacts
body composition
cardiac ischemia
continuous passive motion machine
exercise
fitness
fitness exercise
isometric exercise
isotonic exercise

isokinetic exercise
maximum heart rate
metabolic energy equivalent
passive exercise
pedometer
range-of-motion exercises
recovery index
step test
stress electrocardiogram
submaximal fitness test
target heart rate
therapeutic exercise
walk-a-mile test

Exercise (purposeful physical activity) is beneficial to people of all age groups (Box 24-1), and the health risks of a sedentary lifestyle are well documented. This chapter addresses techniques for improving health and maintaining or restoring muscle and joint function by promoting exercise. Because exercise must be individualized, nurses are responsible for assessing each person's fitness level before initiating an exercise program with a client.

FITNESS ASSESSMENT

Fitness means the capacity to exercise. Factors such as a sedentary lifestyle, health problems, compromised muscle and skeletal function, obesity, advanced age, smoking, and high blood pressure can impair a client's fitness and stamina. They could even result in injury during exercise. Therefore, before a client begins an exercise program, assessment of his or her fitness level is necessary. Some assessment techniques include measuring body composition, evaluating trends in vital signs, and performing fitness tests.

Body Composition

Body composition is the amount of body tissue that is lean versus the amount that is fat. Determining factors include anthropometric measurements such as height, weight, body-mass index, skin-fold thickness, and mid-arm muscle circumference (see Chap. 13). Inactivity without reduced food intake tends to promote obesity. Overweight or obese people are less fit than their leaner counterparts and need to proceed gradually when initiating an exercise program.

BOX 24-1 Benefits of Physical Exercise

- Improved cardiopulmonary function
- Reduced blood pressure
- Increased muscle tone and strength
- Greater physical endurance
- Increased lean mass and weight loss
- Reduced blood glucose level
- Decreased low-density blood lipids
- Improved physical appearance
- Increased bone density
- Regularity of bowel elimination
- Promotion of sleep
- Reduced tension and depression

Vital Signs

Vital signs—temperature, pulse rate, respiratory rate, and blood pressure—reflect a person's physical status (see Chap. 12). Elevated pulse rate, respiratory rate, and blood pressure while resting are signs that the person may have life-threatening cardiovascular symptoms during exercise. After a period of modified exercise, vital signs may decrease, thus reducing the potential for heart-related complications.

Fitness Tests

Fitness tests provide an objective measure of a person's current fitness level and his or her potential for safe exercise. They also help to establish safe parameters for the level and duration of exercise. Two methods of fitness testing are a stress electrocardiogram (ECG) and an ambulatory ECG. Another is a **submaximal fitness test**, which is an exercise test that does not stress a person to exhaustion. Examples of submaximal fitness tests include a step test and a walk-a-mile test. Because submaximal tests are less demanding, the validity of their results is less reliable than results obtained through ECG testing.

Stress Electrocardiogram

A **stress electrocardiogram** tests electrical conduction through the heart during maximal activity and is performed in an acute care facility or an outpatient clinic (Fig. 24-1). The client first walks slowly on a flat treadmill. As the test progresses, the speed and incline of the treadmill are increased. The examiner notes the client's heart rate and rhythm, blood pressure, breathing, and symptoms such as dizziness and chest pain. A pulse oximeter (see Chap. 21) is used to measure peripheral oxygenation. The examiner stops the test if the client develops an abnormal heart rhythm, **cardiac ischemia** (impaired blood flow to

the heart), elevated blood pressure, or exhaustion.

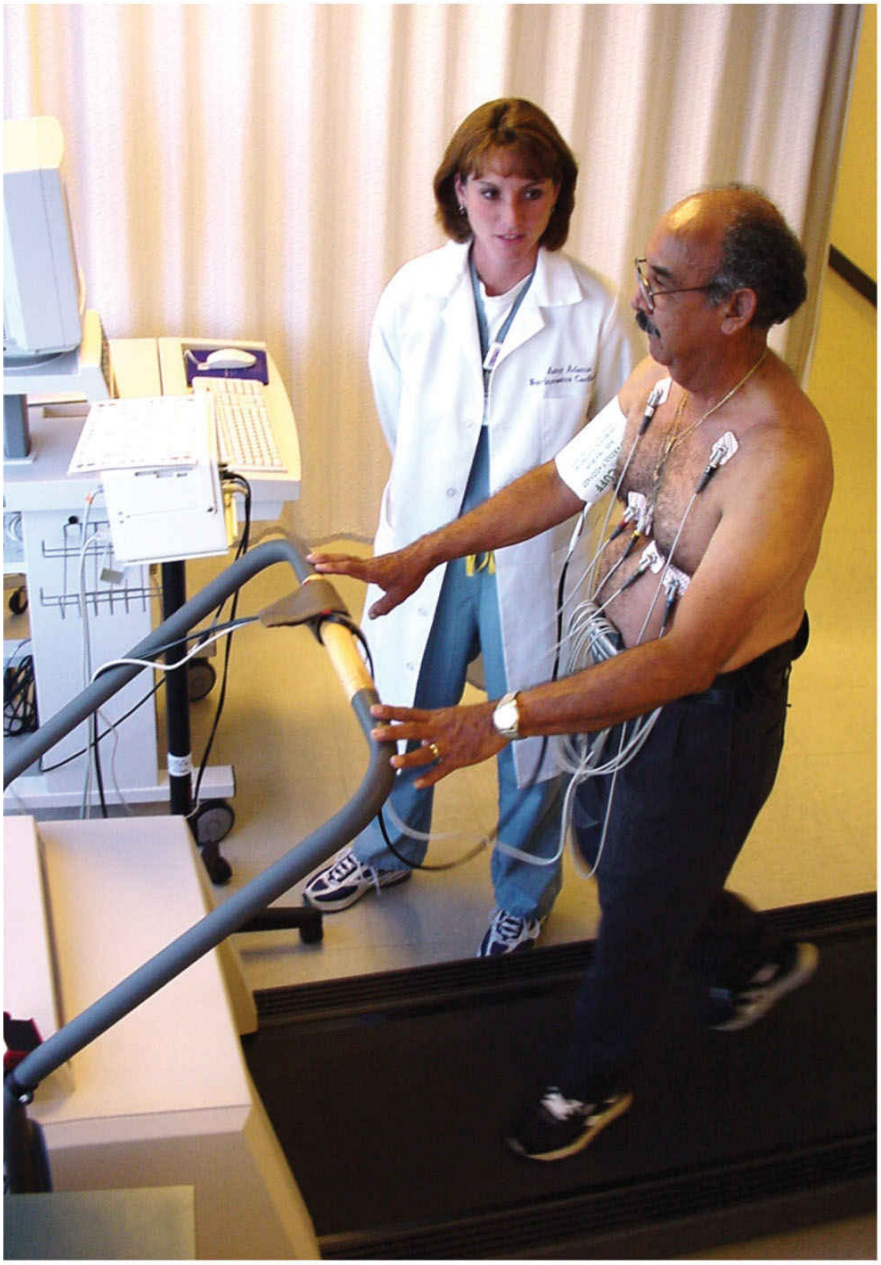


FIGURE 24-1 A stress electrocardiogram test. (Image Texas Heart Institute, www.texasheart.org)

Ambulatory Electrocardiogram

An **ambulatory electrocardiogram** is a continuous recording of heart rate and rhythm during normal activity. It requires the client to wear a device called a Holter monitor for 24 hours. This less-taxing version of a stress ECG is used when the person has had prior cardiac-related symptoms, such as chest pain, or

has major health risks that contraindicate a stress ECG.

Ambulatory electrocardiography helps to assess the heart's response to normal activity rather than activity imposed during a stress ECG. It also helps to evaluate how a person is responding to cardiac rehabilitation and medical therapy.

The Holter monitor, which is connected to chest leads, is attached to a belt or shoulder strap or carried in a pocket (Fig. 24-2). During ambulatory electrocardiography, the client should not shower or swim; a sponge bath is permitted as long as the monitor does not get wet. The client also should avoid magnets, metal detectors, electric blankets, and high-voltage areas that may cause **artifacts** (abnormal appearance due to external distortion) on the recordings that interfere with an accurate interpretation of the test results.

The client keeps a diary of the time and type of activities performed, when he or she took medications and when symptoms, if any, were experienced. After the test period, the client returns the monitor, and then a computer and the physician check the electrically recorded information. The physician compares the client's diary with the ECG. The assessment results help to determine whether oxygenation to the heart muscle was temporarily impaired during an activity or if an abnormal heart rhythm developed. Either finding indicates that exercise should be delayed until further treatment or started at a very low intensity and for a short duration.

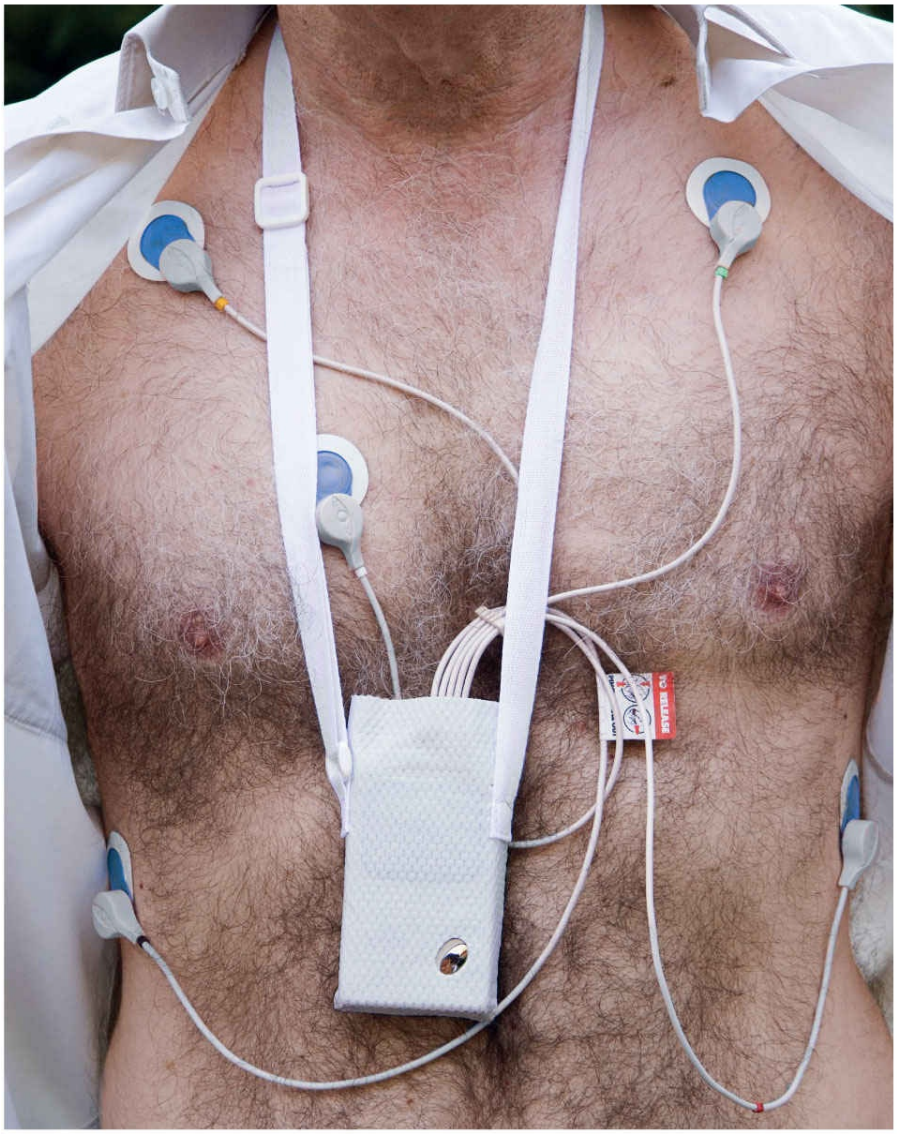


FIGURE 24-2 Ambulatory electrocardiography.

Step Test

A **step test** is a submaximal fitness test involving a timed stepping activity. Several variations include the Harvard Step Test, the Queens College Step Test, and the Chester Step Test. A person undergoing this type of fitness analysis will step up and down on a platform of a prescribed height (20 inches for men, 16 inches for women) for 3 to 5 minutes at a rate of at least 76 steps/minute. A step up or down is considered one step. The time is shortened when the client can no longer sustain the prescribed rate or develops discomfort. The examiner uses a metronome and a stopwatch to keep track of the rate and the time.

Examiners calculate the client's **recovery index** (a guide for determining a

person's fitness level) by taking a 30-second pulse rate at 1, 2, and 3 minutes after the test using the following formula:

$$\text{Recovery index} = \frac{(100 \times \text{test duration in seconds})}{2 \times \text{total of the 30-second pulse assessments}}$$

The examiner compares results with standardized fitness levels (Table 24-1). A fit person has a smaller decline in heart rate at each assessment. Another fitness indicator is how close the pulse rate at the end of recovery compares with the pretest pulse rate. The more similar the pretest and posttest pulse rates, the more fit is the person.

The step test must be used with caution. Personnel certified in cardiopulmonary resuscitation and use of an automated electronic defibrillator (AED) should be available to assist if there is an adverse cardiac event (see Chap. 37).

TABLE 24-1 Cardiovascular Endurance Fitness Levels

SCORE	FITNESS CLASSIFICATION
≥96	Excellent
83–96	Good
68–82	Average
54–67	Low average
≤54	Poor

From Wood, R. (2008). *Harvard step test*. Retrieved October, 2015, from http://www.topendsports.com/testing/tests/step-harvard.htm

Walk-a-Mile Test

The **walk-a-mile test**, devised by the American College of Sports Medicine (2011), measures the time it takes a person to walk 1 mile. The person is instructed to walk 1 mile on a flat surface as fast as possible. The examiner calculates the time from start to finish and interprets results using the guidelines in Table 24-2. If a person walks at a brisk pace, approximately 3 to 4 miles/hour, it will take approximately 30 minutes to complete 1 to 2 miles.

Activity Monitoring

A person's activity level can be monitored and evaluated with a **pedometer**, a self-monitoring motion sensing device that tracks total steps and distance

walked. Another option is a wristband monitor such as Fitbit. A daily goal is 10,000 steps/day, or approximately five miles. An individual's activity level can be compared with the levels in Table 24-3.

EXERCISE PRESCRIPTIONS

The prescription for an exercise program involves determining the person's target heart rate and the metabolic energy equivalents (METs) of particular activities based on the person's fitness level.

Target Heart Rate

Target heart rate refers to the goal for heart rate during exercise. It is determined by first calculating the person's **maximum heart rate** (the highest limit for heart rate during exercise). Maximum heart rate is determined by subtracting a person's age from 220. Thus, a 20 year old has a maximum heart rate of 200 beats per minute (bpm), whereas a 50 year old has a maximum heart rate of 170 bpm. The target heart rate for moderate intensity is 50% to 70% of the maximum heart rate (Centers for Disease Control and Prevention, 2015). Beginners should not exceed 50%, intermediates can exercise up to 70%, and competitive athletes may tolerate 70% to 85% of their maximum heart rate during vigorous intensive activity.

TABLE 24-2 Evaluation Criteria for the Walk-a-Mile Test

PERFORMANCE TIME FOR MEN (MINUTES)	PERFORMANCE TIME FOR WOMEN (MINUTES)	FITNESS LEVEL[a]
≥15:3	≥17:3	Poor
14:01–14:42	15:07–16:06	Average
12:54–14:00	14:12–15:06	Good
<12:54	<14:12	Excellent

[a]Based on adults age 40–49.

TABLE 24-3 Pedometer or Fitbit Assessment

NUMBER OF STEPS	ACTIVITY LEVEL
0–5,000	Sedentary
5,000–7,499	Low active
7,500–9,999	Somewhat active
10,000–12,500	Active
12,500 or more	Highly active

From American College of Sports Medicine. (2011). *Starting a walking program.* Retrieved October, 2015, from https:www.acsm.org/docs/brochures/starting-a-walking-program.pdf

»» *Stop, Think, and Respond 24-1*

What is the maximum heart rate and minimum target heart rate for a person who is 25 years old and a competitive athlete who is 32 years old?

Exercising at the target rate for 15 minutes (excluding the warm-up and cool-down periods) three or more times per week strengthens the heart muscle and promotes the use of fat reserves for energy. Exercising beyond the target heart rate reduces endurance by increasing fatigue.

Metabolic Energy Equivalent

Because fitness levels vary, exercises also are prescribed according to their **metabolic energy equivalent** (MET; the measure of energy and oxygen consumption during exercise). This is the prescribed amount that a person's cardiovascular system can safely support. Low-to-vigorous physical activities and their approximate METs are listed in Table 24-4.

TYPES OF EXERCISE

Exercise is performed to promote fitness or to achieve therapeutic outcomes. The two major types of exercise are fitness exercise and therapeutic exercise.

Fitness Exercise

Fitness exercise means physical activity performed by healthy adults. Fitness exercise develops and maintains cardiorespiratory function, muscular strength, and endurance (Fig. 24-3). The three categories of fitness exercise are isotonic, isometric, and isokinetic.

FIGURE 24-3 Stationary cycling.

Isotonic exercise is activity that involves movement and work. The muscles being exercised change length; they shorten and lengthen during contraction (Fig. 24-4).

Aerobic exercise is an example of isotonic exercise. It involves rhythmically moving all parts of the body at a moderate to slow speed without hindering the ability to breathe (Fig. 24-5). In other words, the person can talk comfortably if the exercise is within his or her level of fitness. To promote cardiorespiratory conditioning and increase lean muscle mass, a person should perform isotonic exercise at his or her target heart rate.

TABLE 24-4 Levels of Physical Activity

METABOLIC ENERGY EQUIVALENT (MET)	EXAMPLES OF ACTIVITIES
1	Sewing
	Watching television
	Dressing
1–2	Walking 1 mph on level ground
	Bowling
2–3	Golfing with a cart
	Mowing lawn with a power mower
3–4	Playing badminton (doubles)
	Raking leaves
4–5	Slow swimming
	Lifting 50 lb
5–6	Square dancing
	Shoveling snow
6–7	Water skiing
	Moving heavy furniture
7–8	Playing basketball
	Playing noncompetitive handball
8–9	Cross-country skiing
	Playing contact football
≥10	Running 6 mph or faster

Isometric exercise consists of stationary exercises during which there is no change in the length of the contracting muscle. The concept involves working to hold a position still. Examples include holding weights in a steady position, using an elastic band or stationary ball to achieve a static position and holding the position, or less intense activities such as simply contracting and relaxing muscle groups while sitting or standing (Fig. 24-6). Isometric exercises increase muscle mass and strength, and tone and define muscle groups. Although they improve blood circulation, they do *not* promote cardiorespiratory function. In fact, strenuous isometric exercises elevate blood pressure temporarily.

Isokinetic exercise combines movement at a constant speed with a form of resistance. The speed and resistance is preprogrammed and controlled with a machine. For example, a stationary bicycle like those in therapeutic settings or fitness centers (Fig. 24-7) can be set at a level that allows a certain number of revolutions per minute. Isokinetic exercises are often used for the purpose of rehabilitation after an athlete experiences an injury (Fig. 24-8) or to strengthen weakened muscles when a person is recovering from a stroke or accident.

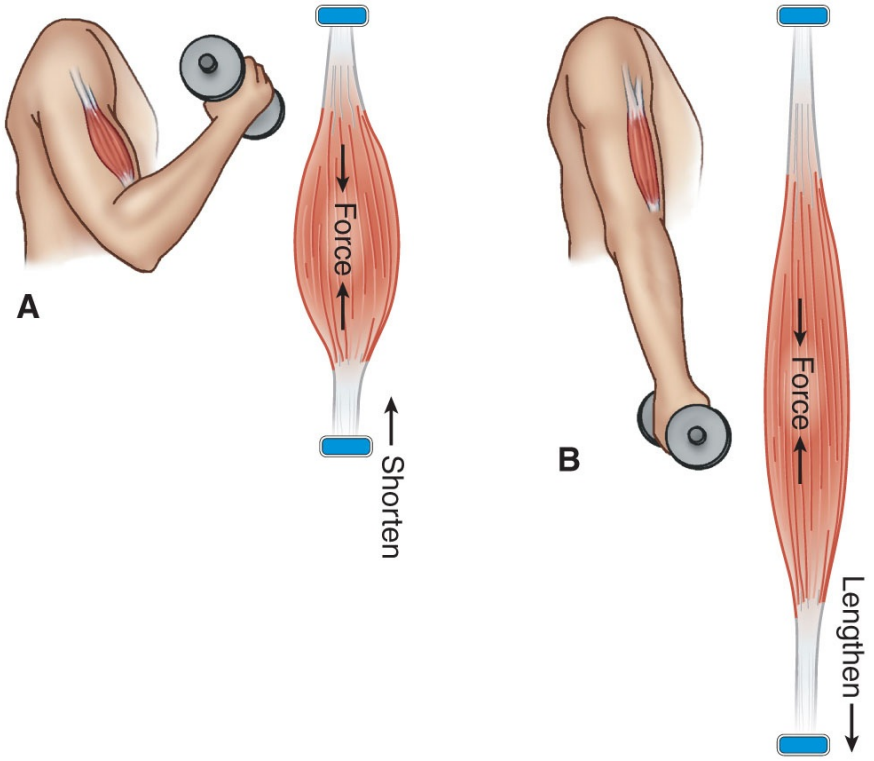


FIGURE 24-4 Example of an isotonic exercise with shortening **(A)** and lengthening **(B)** of muscles. (**A, Left** from McArdle, W. D., Katch, F. I., & Katch, V. L. [2011]. *Essentials of exercise physiology* [4th ed.]. Philadelphia, PA: Lippincott Williams & Wilkins.; **Right**, from Kraemer, W. J., Fleck, S. J., & Deschenes, M. R. [2015]. *Exercise physiology* [2nd ed.]. Philadelphia, PA: Lippincott Williams & Wilkins.; **B, Left** from McArdle, W. D., Katch, F. I., & Katch, V. L. [2011]. *Essentials of exercise physiology* [4th ed.]. Philadelphia, PA: Lippincott Williams & Wilkins.; **Right**, from Kraemer, W. J., Fleck, S. J., & Deschenes, M. R. [2015]. *Exercise physiology* [2nd ed.]. Philadelphia, PA: Wolters Kluwer Health and Pharma solutions.)

Before beginning any exercise program, the nurse can offer information on how to promote safety and prevent injuries (Client and Family Teaching 24-1).

FIGURE 24-5 Example of aerobic exercise. (From Anatomical Chart Company. [2002] *Understanding menopause* [2nd ed.]. Philadelphia, PA: Lippincott Williams & Wilkins.)

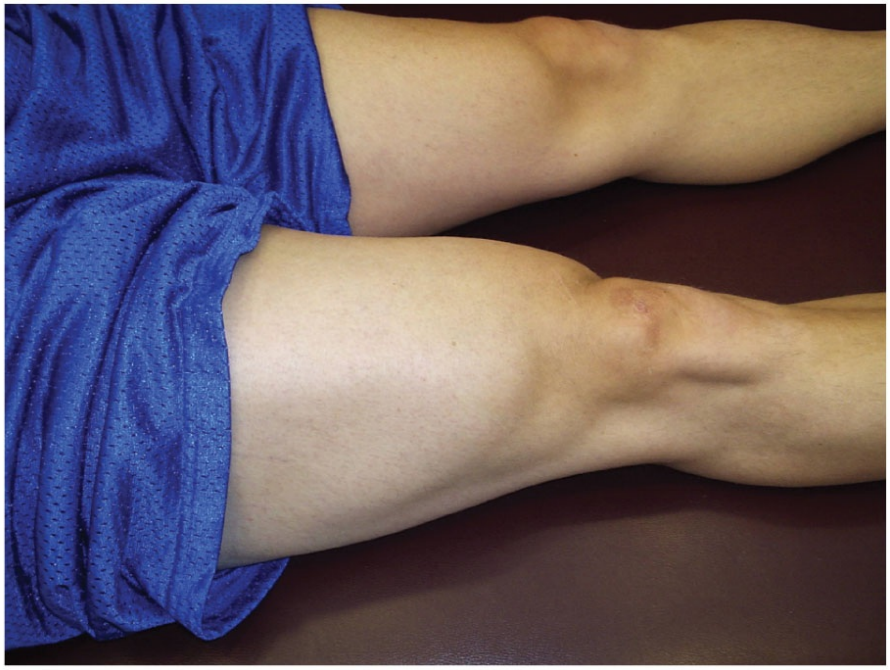


FIGURE 24-6 Isometric exercise occurs by repeatedly tensing and relaxing muscles such as the quadriceps with little or no movement of the exercised body part. (From Lotke, P. A., Abboud, J. A., & Ende, J. [2008]. *Lippincott's primary care orthopaedics*. Philadelphia, PA: Lippincott Williams & Wilkins.)

Nutrition Notes

- Because muscle is composed of protein, many people believe that it is necessary to eat more protein to build muscle. While protein is vital to the process, the amount needed to stimulate exercise-related muscle growth is relatively small.
- Nutritionally, the most important factor for increasing muscle mass is calories. If calorie intake is adequate, muscle growth can occur over a range of protein intakes. Protein and amino acid supplements are not more effective than protein in food for increasing muscle mass.

FIGURE 24-7 Isokinetic exercise using fitness equipment. (From American College of Sports Medicine. [2007]. *ACSM's health-related physical fitness assessment manual* [2nd ed.]. Philadelphia, PA: Lippincott Williams & Wilkins.)

FIGURE 24-8 Isokinetic exercise to recover strength following an injury. (From Anderson, M. K. [2013]. *Foundations of athletic training* [5th ed.]. Philadelphia, PA: Lippincott Williams & Wilkins.)

Therapeutic Exercise

Therapeutic exercise is activity performed by people with health risks or those being treated for an existing health problem such as following cardiac surgery. Clients perform therapeutic exercise to prevent health-related complications or to restore lost functions (see Performing Leg Exercises in Chap. 27 and Strengthening Pelvic Floor Muscles in Chap. 30). Therapeutic exercise may be isotonic or isometric; isotonic exercises are performed actively or passively.

Active Exercise

Active exercise is therapeutic activity that the client performs independently after proper instruction. For example, clients who have undergone a mastectomy learn to exercise the arm on the surgical side by combing their hair, squeezing a soft ball, finger-climbing the vertical surface of a wall, and swinging a rope attached to a doorknob.

Active therapeutic exercise often is limited to a particular part of the body that is in a weakened condition. It is assumed that clients will use their unaffected muscle groups while performing activities of daily living such as bathing and dressing.

Passive Exercise

Passive exercise is therapeutic activity that the client performs with assistance and is provided when a client cannot move one or more parts of the body. For example, for clients who are comatose or paralyzed from a stroke or spinal injury, nurses perform exercises that maintain muscle tone and flexible joints. One form of frequently provided passive therapeutic exercise is range-of-motion (ROM) exercise. Another form is delivered with a continuous passive motion (CPM) machine, which is discussed later.

Client and Family Teaching 24-1 A Safe Exercise Program

The nurse teaches the client and the family as follows:

- Seek a pre-exercise fitness evaluation from a health care provider or a certified sports trainer.
- Determine the target heart rate according to fitness level.
- Determine the appropriate level of METs.
- Choose a form of exercise that seems pleasurable and involves as many muscle groups as possible.
- Plan at least 150 minutes of moderate-intensity exercise per week that can be divided in no less than 10 minutes or longer over multiple days to achieve target amount (American College of Sports Medicine, 2011).
- Build up to 30 minutes or more of moderate-intensity physical activity on most (preferably all) days of the week to prevent cardiovascular disease (Douglas, 2014).
- Exercise with a partner for safety and motivation.
- Avoid exercising in extreme weather conditions (high humidity, smog).
- Dress in layers according to the temperature and weather conditions.
- Wear supportive shoes.
- Wear reflective clothing after dark.
- Walk or jog against traffic; cycle in the same direction as traffic.
- Eat complex carbohydrates (pasta, rice, cooked cereal) rather than fasting or eating simple sugars (cookies, chocolate, sweetened drinks) before exercising.
- Avoid drinking alcohol, which dilates the blood vessels, promotes heat loss, and interferes with good judgment.
- Warm up for 5 minutes by stretching muscle groups or doing light calisthenics.
- Measure the heart rate two or three times while exercising.
- Slow down if the heart rate exceeds the preestablished target.

- Try to sustain the target heart rate for at least 12 to 15 minutes.
- Never stop exercising abruptly.
- Cool down for at least 5 minutes in a manner similar to the warm-up.

Range-of-Motion Exercises

Range-of-motion (ROM) exercises are therapeutic activities that move the joints. They are performed for the following reasons:

- To assess joint flexibility before initiating an exercise program
- To maintain joint mobility and flexibility in inactive clients
- To prevent **ankylosis** (the permanent loss of joint movement)
- To stretch joints before performing more strenuous activities
- To evaluate the client's response to a therapeutic exercise program

TABLE 24-5 Joint Positions

POSITION	DESCRIPTION
Flexion	Bending so as to decrease the angle between two adjoining bones
Extension	Straightening so as to increase the angle between two adjoining bones up to 180 degrees
Hyperextension	Increasing the angle between two adjoining bones more than 180 degrees
Abduction	Moving away from the midline
Adduction	Moving toward the midline
Rotation	Turning from side to side as in an arc
External rotation	Turning outward, away from the midline of the body
Internal rotation	Turning inward, toward the midline of the body
Circumduction	Forming a circle
Pronation	Turning downward
Supination	Turning upward
Plantar flexion	Bending toward the sole of the foot
Dorsiflexion	Bending the foot toward the dorsum or anterior side
Inversion	Turning the sole of the foot toward the midline
Eversion	Turning the sole of the foot away from the midline

During ROM exercises, the client moves or is assisted to move joints in the positions that the joint normally permits (Table 24-5). Whenever possible, the client actively exercises as many joints as possible while the nurse assists with those that are compromised (see Nursing Guidelines 24-1).

Nurses perform ROM exercises whenever they care for inactive clients (Skill 24-1).

Concept Mastery Alert

When performing passive ROM exercises, the nurse evaluates the client's tolerance for the exercises, noting any verbal and nonverbal expressions of discomfort or pain. Extra caution is needed when performing passive ROM exercises with nonresponsive clients because of their inability to communicate discomfort or pain.

Stop, Think, and Respond 24-2

Why would a nurse promote active ROM exercises in the upper body for a client who is paralyzed below the waist after a motor vehicle collision?

Continuous Passive Motion Machine

A **continuous passive motion (CPM) machine** is a mechanical device powered electrically as a supplement or substitute for manual ROM exercise (Fig. 24-9). A machine-assisted ROM sometimes is preferred during the rehabilitation of clients who have experienced burns or have had knee or hip replacement surgery because the machine precisely controls the degree of joint movement and can increase it in specific increments throughout recovery.

NURSING GUIDELINES 24-1

Performing Range-of-Motion Exercises

- Use good body mechanics (see Chap. 23). *Doing so conserves energy and avoids muscle strain and injury*.
- Remove pillows and other positioning devices. *Such items can interfere with the exercises.*
- Position the client to facilitate movement of the joint through all its usual positions. *This positioning makes it easier to perform a comprehensive exercise program.*
- Follow a systematic, repetitive pattern—such as beginning at the head and moving down. *A routine prevents overlooking a joint.*
- Perform similar movements with each extremity. *Doing so exercises the joints bilaterally.*
- Support the joint being exercised. *Support reduces discomfort.*
- Move each joint until there is resistance but not pain. *This method exercises each joint to its point of limitation*.
- Watch for nonverbal communication. *Nonverbal signs may indicate the client's response*.
- Avoid exercising a painful joint. *Doing so can contribute to injury.*
- Stop if spasticity develops, as manifested by a sudden, continuous muscle contraction. *Taking a break gives muscles time to relax and recover.*
- Apply gentle pressure to the muscle or move the spastic limb more slowly. *These actions relieve spasticity.*
- Expect the client's respiratory and heart rates to increase during exercise but to return to a resting rate later. *This is a normal cardiopulmonary response to activity.*
- Teach the family to perform ROM exercises. *A regular exercise program improves the potential for regaining function.*

In addition to restoring and increasing joint ROM, the movement created by the machine prevents the pooling of venous blood, thus decreasing the risk of blood clots. Also, it accelerates wound healing because the synovial fluid circulates around the joint. Most machines produce 0 to 110 degrees of motion 2 to 10 times per minute. Initially, the nurse sets the machine at very low speeds and degrees of movement—it is common to begin with 5 or 10 degrees of flexion cycling twice a minute for at least six times a day. The nurse increases the settings as the client's tolerance builds. The nurse positions the client's

extremity in the machine and programs the speed and the degree of desired joint flexion according to the physician's exercise prescription (Skill 24-2).

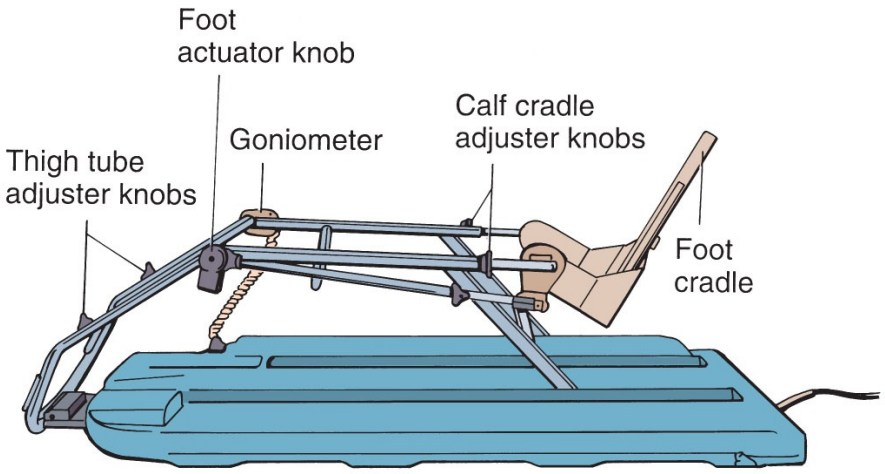


FIGURE 24-9 A continuous passive motion machine.

Stop, Think, and Respond 24-3

List examples of assessment findings that indicate a positive response to the use of a CPM machine.

NURSING IMPLICATIONS

Few people exercise sufficiently to promote optimal health. With this in mind, the Department of Health and Human Services has established *Physical Activity Guidelines for Americans* for improving the health of United States citizens (Table 24-6). Nurses can set an example for others in the community by improving their own physical fitness and encouraging others to do so.

Gerontologic Considerations

- Older adults, especially those who are disabled, need to balance periods of physical activity with periods of rest. Shortness of breath or an increased heart rate indicates that the level of activity is beyond the client's tolerance.
- Older adults need to eliminate their intake of caffeinated and alcoholic beverages before and during physical activity to avoid depleting fluid volume. Water is the preferred drink for fluid replacement.
- Encourage families and caregivers of older adults with cognitive impairment to help older persons participate in physical activities such as walking and ball throwing. If the older person has difficulty with balance, exercises may be done while sitting or lying down. Active ROM exercises should be scheduled daily

and may be divided into short sessions. If the older adult cannot participate actively in an exercise program, the caregivers can perform passive ROM exercises, at least daily, to prevent muscle atrophy and disuse syndrome.

- Many shopping malls permit, and even encourage, people to walk through the mall before stores open for business.
- Swimming or exercising in water puts less stress on joints and is beneficial for older adults.
- Many physically challenging sports, such as bowling, golfing, walking in marathons, and weight lifting, have competition categories for older adults.
- Precautions, such as wearing safe shoes with nonskid soles, are necessary to prevent falls when older adults exercise. Complications from falls contribute to morbidity and mortality among older people.

For people with medical disorders, nurses may identify one or more of the following nursing diagnoses that are treated with activity or an exercise regimen:

- Impaired Physical Mobility
- Impaired Bed Mobility
- Risk for Disuse Syndrome
- Unilateral Neglect
- Risk for Delayed Surgical Recovery
- Activity Intolerance

Clinical Scenario An 88-year-old female is a resident in a nursing home, following a stroke. The

client has lost motor skills including range of motion on the right side of her body. She does not notice people or objects on her affected side. Her muscle mass and strength on the right side have diminished due to neglect of her arm and leg as well as its position. She depends on others for turning and positioning.

Nursing Care Plan 24-1 illustrates how a nurse can incorporate exercise into the care of a client with a stroke using the nursing diagnosis of Unilateral Neglect. The NANDA-I taxonomy (2015, p. 251) defines this diagnosis as "impairment in sensory and motor response, mental representation, and spatial attention of the body, and the corresponding environment, characterized by inattention to one side (of the body) and overattention to the opposite side."

TABLE 24-6 Physical Activity Guidelines for Americans

GOAL	RECOMMENDATION	STRATEGIES	EXAMPLES
Increase aerobic physical activities.	Do at least 2½ hours of moderate level activities or 1¼ hours of vigorous activities per week.	Build up time slowly. Do at least 10 minutes at a time. Combine moderate and vigorous activities.	*Moderate* (can talk, but not sing during performance): Walking briskly Ballroom and line dancing Biking on level ground or with few hills General gardening (raking, trimming shrubs) Sports in which you catch and throw (baseball, softball, volleyball) Water aerobics *Vigorous* (can say a few words without stopping for a breath) Jogging Fast or aerobic dancing Biking faster than 10 mph Heavy gardening (digging, hoeing) Jumping rope Swimming fast or swimming laps Sports with a lot of running (basketball, soccer, hockey)
Increase muscle strengthening activities.	Do at least 2 days/week.	Include all the major muscle groups (legs, hips, back, chest, stomach, shoulders, and arms). Repeat 8–12 times for each muscle group per session.	

From US Department of Health and Human Services. (2008). *Be active your way: Physical activity guidelines for Americans.* Retrieved October, 2015, from http://health.gov/paguidelines/pdf/adultguide.pdf

NURSING CARE PLAN 24-1 Unilateral Neglect

Assessment

- Observe the client's bilateral movement or unilateral lack of movement.
- Note whether the client uses both sides of the body in an integrated and coordinated manner.
- Determine whether the client omits, ignores, or favors activities or objects consistently on one side.
- Check the client's vision and sensation bilaterally.

Nursing Diagnosis. **Unilateral Neglect** related to a lack of awareness of objects in the right visual field secondary to stroke as manifested by a lack of attention to food on the right side of the plate and tray, an inability to see objects placed on the right side, combing only the left side of hair, no response to touch or pain stimuli on the right side, and an inability to differentiate between warm and cold on the right.

Expected Outcome. The client will identify his or her own body parts on the right side, attend to his or her care, and incorporate objects within his or her extrapersonal environment located to the client's right side by 4/21.

Interventions	Rationales
Approach the client always from the left side.	The client's perception and attention are limited

	to the unaffected side.
Place items for safety, such as the signal cord, and those for self-care, such as a glass of water, on the client's left side.	The neurologic deficit predisposes the client to ignore objects on the affected side.
Suggest that the client turn his or her head from side to side for a panoramic view of the environment.	Directing the client to scan the environment uses the visual areas in the unaffected structures of the brain.
Show the client three objects on the left side of the visual field each shift; then relocate objects to the right side and encourage the client to turn his or her head and identify where they are located.	Repetition in scanning both sides helps the client to develop awareness skills.
Have the client locate and touch the right arm and other body structures on the right side.	Attending to the affected side helps to retrain the client's brain to recognize and integrate parts of the self.
Add one self-care task at a time such as bathing the affected arm, inserting the arm into a gown or shirt, and grasping and exercising the affected hand with the unaffected hand as the client's awareness and competence develop.	Practice and repetition facilitate progress in reaching goals.

Evaluation of Expected Outcomes

- The nurse transfers the client to a room with a door on the left side of the client to facilitate awareness.
- The client locates and identifies one of three objects such as pen, watch, and banana after looking at them in the left visual field and then in the right.
- The client states "That's my arm and leg" when instructed to look to the right side of his or her body.
- The client touches and moves the affected right arm with the left arm.
- The client performs ROM exercises with assistance from the nurse for affected extremities.
- The client continues to practice bathing and exercise.

CRITICAL THINKING EXERCISES

1. List at least five excuses people give for not exercising and offer counterarguments for each.
2. A client with paralysis of the lower extremities is depressed and questions the purpose for performing passive ROM exercises on the lower body. Assuming paralysis is permanent and the client will never walk again, how would you respond?
3. What advantages would you offer to a friend who is physically inactive and could benefit from exercise?
4. What are some reasons the federal government sets goals and objectives for physical activity and fitness in its *Healthy People* campaigns?

NCLEX-STYLE REVIEW QUESTIONS

1. When the nurse teaches a client how to perform isometric exercises of the quadriceps muscles, which of the following instructions is correct?
   1. Move your toes toward and away from your head.
   2. Tighten and relax the muscles on the front of your thigh.
   3. Lift your lower leg up and down from the bed.
   4. Bend your knee and pull your lower leg upward.

   Test-Taking Strategy: Analyze the choices and select the option that describes keeping the exercised limb stationary.
2. When the nursing team develops a plan of care for a client with a stroke, which area of nursing management is most important to the client's rehabilitation?
   1. Regulating bowel and bladder elimination
   2. Dealing with problems of disturbed body image
   3. Preventing contractures and joint deformities
   4. Facilitating positive outcomes from grieving

   Test-Taking Strategy: Note the key word and modifier, "most important." Analyze the choices and select one that promotes restoration of the client's future mobility and self-care.
3. When a client asks of what benefit a stress electrocardiography (ECG) will be, what is the most accurate answer the nurse can give?
   1. A stress ECG shows how the heart performs during progressive exercise.
   2. A stress ECG helps determine the client's potential target heart rate.
   3. A stress ECG verifies how much exercise is needed to improve fitness.
   4. A stress ECG can predict whether the client will have a heart attack soon.

   Test-Taking Strategy: Note the key word and modifier, "most accurate." Analyze the choices and select the option that describes the primary purpose of a stress ECG better than any of the others.
4. When the nurse performs range of motion (ROM) exercises on a sedentary client, which of the following indicates the exercise should be stopped immediately? Select all that apply.
   1. The joint movement is accompanied by pain.
   2. The joint movement causes a sudden muscle spasm.
   3. The joint movement meets with slight resistance.
   4. The joint movement increases the client's respirations.
   5. The joint movement results in a grimaced expression.

   Test-Taking Strategy: Note the key word, "immediately." Analyze the choices and select options that indicate the exercise may be too excessive at the present time.
5. What information is essential for the nurse to document when caring for a

client for whom the physician has ordered the use of a continuous passive motion (CPM) machine? Select all that apply.

1. Condition of the sutures around the incision
2. Degree of joint flexion
3. Amount of time the client used the machine
4. Characteristics of drainage from the wound
5. Number of cycles per minute
6. Presence and quality of arterial pulses

Test-Taking Strategy: Note the key word “essential.” Analyze the choices and select options that provide information to team members about the current therapy using a CPM machine.

SKILL 24-1 Range-of-Motion Exercises

27

Perioperative Care

Learning Objectives

On completion of this chapter, the reader should be able to:

1. Define perioperative care.
2. Identify the three phases of perioperative care.
3. Differentiate inpatient from outpatient surgery.
4. List at least four advantages of laser surgery.
5. Discuss two methods for donating blood before surgery.
6. Identify four major activities that nurses perform for all clients immediately before surgery.
7. Name three topics to address in preoperative teaching.
8. Explain the purpose of antiembolism stockings.
9. Name two recommended methods for removing hair when preparing the skin for surgery.
10. List at least five items that are verified on the preoperative checklist.
11. Discuss the purpose and information required during a presurgical time-out according to The Joint Commission's Universal Protocol.
12. Name three areas of the surgical department used during the intraoperative period.
13. Describe the focus of nursing care during the immediate postoperative period.
14. Give four examples of common postoperative complications.
15. Discuss the purpose of a pneumatic compression device.
16. Describe at least two items of information included in discharge instructions for postsurgical clients.
17. Discuss at least two ways in which the surgical care of older adults differs from that of other age groups.

Words to Know

anesthesiologist
anesthetist

antiembolism stockings
atelectasis
autologous transfusion
conscious sedation
depilatory agent
directed donors
discharge instructions
emancipated minor
emboli
forced coughing
informed consent
inpatient surgery
intraoperative period
microabrasions
outpatient surgery
perioperative care
plume
pneumatic compression device
pneumonia
postanesthesia care unit
postoperative care
postoperative period
preoperative checklist
preoperative period
receiving room
reversal drugs
substituted judgment
surgical waiting area
thrombus
Universal Protocol

Perioperative care (care that clients receive before, during, and after surgery) is unique. The current trend is to facilitate as short a perioperative period as possible. This trend is driven by efforts to control health care costs by facilitating the client's return to the comfort and support of his or her home environment. This chapter discusses the general responsibilities nurses assume when caring for clients during the preoperative, intraoperative, and postoperative periods of perioperative care.

Gerontologic Considerations

■ Chronic health conditions may be present in older adults and may increase the complexity of both the preoperative, intraoperative, and postoperative periods.

TABLE 27-1 Types of Surgery According to Their Urgency

TYPE	DESCRIPTION	EXAMPLE
Optional	Surgery is performed at the client's request.	Surgery for cosmetic purposes
Elective	Surgery is planned at the client's convenience. Failure to have the surgery does not result in catastrophe.	Surgery for the removal of a superficial cyst
Required	Surgery is necessary and should be done relatively promptly.	Surgery for the removal of a cataract
Urgent	Surgery is required promptly, within 1 or 2 days if at all possible.	Surgery for the removal of a malignant tumor
Emergency	Surgery is required immediately for survival.	Surgery to relieve an intestinal perforation

PREOPERATIVE PERIOD

The **preoperative period** starts when clients, or their families in an emergency, learn that surgery is necessary and ends when clients are transported to the operating room. This period can be short or long; one major factor affecting its length is the urgency with which the surgery must be performed (Table 27-1).

Inpatient Surgery

Surgery is performed for various reasons (Table 27-2). **Inpatient surgery** is the term used for procedures performed on a client who is admitted to the hospital, expected to remain at least overnight, and in need of nursing care for more than 1 day after surgery. All except the sickest of clients usually are admitted the morning of the scheduled surgery.

TABLE 27-2 Reasons for Surgery

TYPE OF SURGERY	PURPOSE	EXAMPLES
Diagnostic	Removal and study of tissue to make a diagnosis	Breast biopsy Biopsy of skin lesion
Exploratory	More extensive means to diagnose a problem; usually involves exploration of a body cavity or use of scopes inserted through small incisions	Exploration of abdomen for unexplained pain Exploratory laparoscopy
Curative	Removal or replacement of defective tissue to restore function	Cholecystectomy Total hip replacement
Palliative	Relief of symptoms or enhancement of function without cure	Resection of a tumor to relieve pressure and pain
Cosmetic	Correction of defects, improvement of appearance, or change to a physical feature	Rhinoplasty Cleft lip repair Mammoplasty

Many people who have inpatient surgery undergo prior laboratory and diagnostic tests. Some have met with an **anesthesiologist** (a physician who administers chemical agents that temporarily eliminate sensation and pain; Table 27-3) or an **anesthetist** (a certified nurse specialist who administers anesthesia under the direction of a physician). Most clients will have received preoperative instructions from either the surgeon's office nurse or a hospital nurse.

Outpatient Surgery

Outpatient surgery, also called ambulatory surgery and same-day surgery, is the term used for operative procedures performed on clients who return home the same day. It generally is reserved for clients in an optimal state of health whose recovery is expected to be uneventful. Advantages and disadvantages of outpatient surgery are listed in Table 27-4.

Outpatient surgical units are located in either a hospital or a separate building that the hospital owns. Others are free-standing, privately owned facilities such as a surgeon's office that is not affiliated with a hospital. The client remains in the location of the surgery for a brief time and is able to return home by midafternoon or early evening when (1) the client is awake and alert, (2) vital signs are stable, (3) pain and nausea are controlled, (4) oral fluids are retained, (5) the client voids a sufficient quantity of urine, and (6) the client has received home care instructions. If a complication develops, the client is transferred and admitted to a hospital unit.

Pharmacologic Considerations

- Outpatients are frequently instructed to take preoperative or routine medications at home before procedures; this is especially important for those who take heart, hypertension, or diabetes medications. It is critical for the physician to know when and what dose was taken for appropriate monitoring during procedures.

Outpatient procedures have increased dramatically as a result of advances in techniques such as those using endoscopes, an instrument for performing internal procedures in lieu of those requiring an incision (see Chap. 14), and lasers; methods of anesthesia; prospective reimbursement; managed care; and changes in Medicare and Medicaid provisions (Hinkle and Cheever, 2013).

TABLE 27-3 Types of Anesthesia

TYPE	DESCRIPTION
General anesthesia	Eliminates all sensation and consciousness or memory of the event
Inhalants	Includes gas or volatile liquids
Injectables	Are given intravenously
Regional anesthesia	Blocks sensation in an area, but consciousness is unaffected
Spinal (includes epidural)	Eliminates sensation in lower extremities, lower abdomen, and pelvis
Local	Blocks sensation in a circumscribed area of skin and subcutaneous tissue
Topical	Inhibits sensation where directly applied in epithelial tissues such as skin and mucous membranes

Laser Surgery

The acronym LASER stands for **l**ight **a**mplification by the **s**timulated **e**mission of **r**adiation. Lasers convert a solid, gas, or liquid into light. When focused, the energy from the light is converted to heat, causing the vaporization of tissue and coagulation of blood vessels. Examples include the carbon dioxide laser, the argon laser, the ruby laser, and the yttrium–aluminum–garnet (YAG) laser.

Laser surgery is used as an alternative to many previously conventional surgical techniques such as reattaching the retina, removing skin tattoos, and revascularizing ischemic heart muscle (instead of coronary artery bypass graft surgery). Laser surgery offers the following advantages:

- Cost effectiveness
- Reduced need for general anesthesia
- Smaller incisions
- Minimal blood loss
- Reduced swelling
- Less pain
- Decreased incidence of wound infections
- Reduced scarring
- Less time recuperating

TABLE 27-4 Advantages and Disadvantages of Outpatient Surgery

ADVANTAGES	DISADVANTAGES
Lowers the surgical costs because of the reduced use of hospital services	Reduces the time for establishing a nurse–client relationship
Reduces the time spent away from home, school, or place of employment	Requires intensive preoperative teaching in a short amount of time
Interferes less with the client's usual daily routine	Reduces the opportunity for reinforcement of teaching and for answering questions
Provides the potential for more rest and sleep before and after surgery	Allows for fewer delays in assessing and preparing a client once he or she arrives for surgery
Allows more opportunity for family contact and support	Requires that care of the client after discharge be carried out by unskilled people

Laser technology requires unique safety precautions such as eye, fire, heat, and vapor protection. Depending on the type of laser used, everyone—including the client—wears goggles. In some cases, prescription glasses with side shields are sufficient, but contact lenses are not allowed. Because lasers produce heat, fire, and electrical discharge, safety is paramount. Volatile substances such as alcohol and acetone are not used around lasers because of their flammability. Surgical instruments are coated black to avoid absorbing scattered light that causes them to retain heat. Sometimes even the client's teeth are covered with plastic or a rubber mouth guard to shield metal dental fillings. For the same reason, no jewelry is allowed.

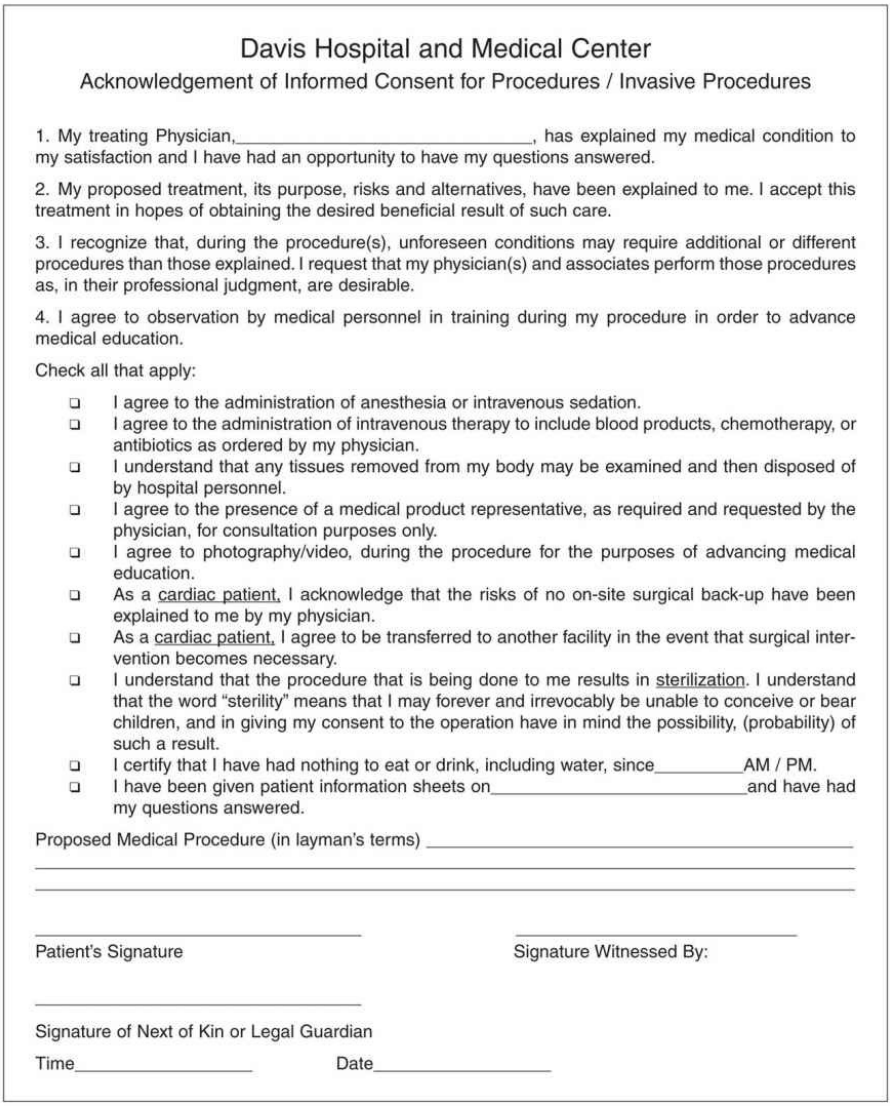
When a laser is used, it releases a **plume** (a substance composed of vaporized tissue, carbon dioxide, and water) that may contain intact cells. Plumes, sometimes referred to as surgical smoke, are accompanied by an offensive odor, and (for some) burning and itching eyes. The latter effects are not hazardous and usually can be reduced with the use of local exhaust ventilators and smoke evacuators. The greater concern involves the consequences of inhaling plumes. Airborne cells in the inhaled plume may contain viruses such as human papilloma virus (HPV) and possibly human immunodeficiency virus (HIV).

Although no cases of HIV transmission through lasers have been documented, high-efficiency respirator masks like a fit-tested surgical N95 mask (see Chap. 22) are recommended to reduce the risk for infection transmission (Association of periOperative Registered Nurses, 2014; Kumar, 2013).

Informed Consent

Regardless of whether surgery is performed conventionally, endoscopically, or with a laser, clients commonly are fearful and anxious. They often have many questions and preconceived ideas about what surgery involves. Health care providers may answer some of these questions. Nevertheless, the physician is responsible for providing information that meets the criteria for **informed consent** (permission a client gives after an explanation of the risks, benefits, and alternatives; see Chap. 14). A signed form, witnessed by a nurse, is evidence that consent has been obtained (Fig. 27-1).

If an adult client is confused, unconscious, or mentally incompetent, the client's spouse, nearest blood relative, or someone with a durable power of attorney for the client's health care must sign the consent form. If an adult client is under the influence of a mind-altering drug such as a narcotic or is alcohol intoxicated, obtaining consent must be delayed until the drug has been metabolized. In a life-threatening emergency, a court may waive the need to obtain a written or verbal consent from a client who requires immediate surgery on the basis of **substituted judgment**; that is, the court believes that if the client had the capacity to consent, he or she would have done so. Refer to Chapter 14 for the elements that constitute informed consent.



Davis Hospital and Medical Center

Acknowledgement of Informed Consent for Procedures / Invasive Procedures

1. My treating Physician,_____, has explained my medical condition to my satisfaction and I have had an opportunity to have my questions answered.

2. My proposed treatment, its purpose, risks and alternatives, have been explained to me. I accept this treatment in hopes of obtaining the desired beneficial result of such care.

3. I recognize that, during the procedure(s), unforeseen conditions may require additional or different procedures than those explained. I request that my physician(s) and associates perform those procedures as, in their professional judgment, are desirable.

4. I agree to observation by medical personnel in training during my procedure in order to advance medical education.

Check all that apply:

- ❑ I agree to the administration of anesthesia or intravenous sedation.
- ❑ I agree to the administration of intravenous therapy to include blood products, chemotherapy, or antibiotics as ordered by my physician.
- ❑ I understand that any tissues removed from my body may be examined and then disposed of by hospital personnel.
- ❑ I agree to the presence of a medical product representative, as required and requested by the physician, for consultation purposes only.
- ❑ I agree to photography/video, during the procedure for the purposes of advancing medical education.
- ❑ As a cardiac patient, I acknowledge that the risks of no on-site surgical back-up have been explained to me by my physician.
- ❑ As a cardiac patient, I agree to be transferred to another facility in the event that surgical intervention becomes necessary.
- ❑ I understand that the procedure that is being done to me results in sterilization. I understand that the word "sterility" means that I may forever and irrevocably be unable to conceive or bear children, and in giving my consent to the operation have in mind the possibility, (probability) of such a result.
- ❑ I certify that I have had nothing to eat or drink, including water, since_____AM / PM.
- ❑ I have been given patient information sheets on_____and have had my questions answered.

Proposed Medical Procedure (in layman's terms) _____

_____

_____

_____ _____

Patient's Signature Signature Witnessed By:

_____

Signature of Next of Kin or Legal Guardian

Time_____ Date_____

FIGURE 27-1 A surgical consent form. (From Carter, P. J. [2011]. *Lippincott textbook for nursing assistants* [3rd ed.]. Philadelphia, PA: Lippincott Williams & Wilkins. Courtesy of Davis Hospital and Medical Center, Layton, Utah.)

If the client is younger than 18 years, a parent or legal guardian must sign the consent form. In an emergency, health care personnel make every effort to obtain consent by telephone, telegram, or fax. Adolescents younger than 18 years, living independently, and supporting themselves are regarded as **emancipated minors** and may sign their own consent forms.

Each nurse must be familiar with agency policies and state laws regarding surgical consent forms. Clients must sign the consent form before receiving any preoperative sedatives. When the client or designated person has signed the permit, an adult witness also signs it to indicate that the client or designee signed

voluntarily. This witness usually is a member of the health care team or an employee in the admissions department. The nurse is responsible for ensuring that all necessary parties have signed the consent form and that it is in the client's chart before the client goes to the operating room.

Preoperative Blood Donation

When the need for a blood transfusion during the perioperative period is anticipated, the low risk for acquiring HIV from a blood transfusion sometimes is discussed before the surgical procedure. Although publicly donated blood is tested for several pathogens including HIV and hepatitis B, the potential, although slight, for acquiring a blood-borne disease still exists. Therefore, some clients undergoing surgery donate their own blood preoperatively if it does not jeopardize their own health. Predonated blood is held on reserve in the event that the client needs a blood transfusion during or after surgery. Receiving one's own blood is called an **autologous transfusion** (self-donated blood). Autologous transfusions also are prepared by salvaging blood lost during or immediately after surgery. The salvaged blood is suctioned, cleaned, and filtered from drainage collection devices.

TABLE 27-5 American Red Cross Criteria for Autologous and Directed Blood Donation

AUTOLOGOUS DONATION	DIRECTED DONATION
To bank one's own blood, the donor must:	**To be a directed donor, the person must:**
Have a physician's recommendation.	Be at least 17 years of age.
Have a hematocrit within a safe range.	Meet all the criteria of a public donor.
Be free of infection at the time of donation.	Have the same blood type as the potential recipient or one that is compatible.
Meet the blood collection center's minimum weight requirement.	Not have received a blood transfusion within the last 6 months.
Donate 40 days before the anticipated date of use, but not within 72 hours of the procedure.	Donate 20 to 3 days before the anticipated use.
Donate no more frequently than every 3–5 days; once per week is preferred.	Be free from blood-borne pathogens and high-risk behaviors.
Avoid acetaminophen (Tylenol), aspirin, and alcohol for 48 hours before a donation.	
Assume responsibility for costs above the usual processing fees even if blood is not used.	
Be advised that his or her blood will be discarded if unused.	

Clients who do not meet the time or health requirements for self-donation may select **directed donors** (blood donors chosen from among the client's relatives and friends). The client's siblings should not donate blood for the client. Doing so would rule them out as future organ or tissue donors for the client because antigens in the transfused blood would sensitize the recipient, increasing the risk for organ or tissue rejection. Also, a male sexual partner of a woman in her reproductive years should not be a directed donor to avoid possible antibody reactions against a fetus in any future pregnancy.

Most authorities believe that receiving blood from directed donors is no safer than receiving blood from public donors. Although predonation of blood is available in the United States, the criteria for autologous and directed donors

(Table 27-5) may vary among regions and hospitals. Because directed donors must meet the same requirements as public donors, if the intended recipient does not use the blood, it is released into the public pool and can be given to someone else.

Immediate Preoperative Care

Although some presurgical activities take place weeks in advance, others cannot be performed until just before surgery. During the immediate preoperative period —the few hours before the procedure—several major tasks must be completed: conducting a nursing assessment, providing preoperative teaching, performing methods of physical preparation, administering medications, assisting with psychosocial preparation, and completing the surgical checklist.

Nursing Assessment

Nurses share with physicians the responsibility for assessing preoperative clients. The assessment varies depending on the urgency of the surgery and if the client is admitted the same day of surgery or earlier. Although assessment of the surgical client is always necessary, the particular circumstances dictate the extent of the process. There may not be time to perform a detailed assessment.

When surgery is not an emergency, the nurse performs a thorough history and physical examination. He or she assesses the client's understanding of the surgical procedure, postoperative expectations, and ability to participate in recovery. The nurse also considers cultural needs, specifically as they relate to beliefs about surgery, personal privacy, and presence of family members during the preoperative and postoperative phases. Clients from different cultures may experience differences in postoperative pain sensitivity and analgesic requirements (Incayawar & Todd, 2013).The nurse may question the client regarding strong culturally influenced feelings about disposal of body parts and blood transfusions.

On admission, the nurse reviews preoperative instructions, such as diet and fluid restrictions, bowel and skin preparations, and the withholding or self-administration of medications, to ensure that the client has followed them. If the client has not carried out a specific portion of the instructions, the nurse immediately notifies the surgeon.

Gerontologic Considerations

- The older person should be educated about taking or omitting usual medications before surgical procedures and about resuming usual or new

medications after surgery.

Pharmacologic Considerations

- When asked some adults do not realize the anticoagulant (bleeding) effect of the medications and supplements routinely taken. They may not readily tell you about these preparations because they do not know what might constitute a postoperative risk for bleeding. Ask about the following items in addition to standard anticoagulant therapy (such as Coumadin):
  - Low-dose aspirin self-therapy (anticoagulant)
  - Cold and cough product use (contains salicylates)
  - Nonsteroidal anti-inflammatory drugs (NSAIDs) such as ibuprofen (Advil) and naproxen (Aleve) (increase gastrointestinal [GI] bleeding risk)
  - Health supplements used for memory improvement (eg, ginkgo and ginseng) (anticoagulant)

The nurse identifies the client's potential risks for complications during or after the surgery. Certain surgical risk factors increase the likelihood of perioperative complications:

- Extremes of age; very young and very old
- Dehydration
- Malnutrition
- Obesity
- Smoking
- Diabetes
- Cardiopulmonary disease
- Drug and alcohol abuse
- Bleeding tendencies
- Low hemoglobin and red cells
- Pregnancy

Some problems, such as an unexplained elevation in temperature, abnormal laboratory data, current infectious disease, or significant deviations in vital signs, are causes for postponing or canceling the surgery (Table 27-6).

Preoperative Teaching

Preoperative teaching varies with the type of surgery and the length of hospitalization. Preoperatively, clients are alert and free from pain or currently in less pain, which facilitates their participation. Knowledge of what to expect on the part of clients and family can enhance recovery from surgery.

The following are examples of information to include in preoperative

teaching:

- Preoperative medications—when they are given and their effects
- Postoperative pain control
- Explanation and description of the postanesthesia recovery room or postsurgical protocol
- Discussion of the frequency of assessing vital signs and the use of monitoring equipment

The nurse also explains and demonstrates how to perform deep breathing, coughing, and leg exercises.

Deep Breathing

Deep breathing, a form of controlled ventilation that opens and fills small air passages in the lungs (see Chap. 21), is especially advantageous for clients who receive general anesthesia or who breathe shallowly after surgery because of pain. Deep breathing reduces the postoperative risk for respiratory complications such as **atelectasis** (airless, collapsed lung areas) and **pneumonia** (lung infection), both of which can lead to hypoxemia.

The nurse practices deep breathing with clients before they undergo surgery (Fig. 27-2). Deep breathing involves inhaling deeply using the abdominal muscles, holding the breath for several seconds, and exhaling slowly. Pursing the lips may extend the period of exhalation. Incentive spirometers (see Chap. 21) also are used to promote deep breathing.

Coughing

Thickened respiratory secretions often accompany impaired ventilation. Coughing is a natural method for clearing secretions from the airways. Deep breathing alone is sometimes sufficient to produce a natural cough. **Forced coughing** (coughing that is purposely produced) may not be necessary for all postoperative clients. Forced coughing is most appropriate for clients who have diminished or moist lung sounds or who raise thick sputum. Nevertheless, all clients need to be prepared for the possibility of having to perform this technique and should receive instructions about it (see Client and Family Teaching 27-1).

Coughing is more painful for clients with abdominal or chest incisions. Administering pain medication approximately 30 minutes before coughing or splinting the incision during coughing can reduce discomfort. Methods of splinting include pressing on the incision with both hands, pressing on a pillow placed over the incision, or wrapping a bath blanket around the client (Fig. 27-

3).

FIGURE 27-2 Teaching deep breathing. (Photo by B. Proud.)

TABLE 27-6 Surgical Risk Factors and Potential Complications

VARIABLE	POTENTIAL COMPLICATION
Age	
Very young—Immaturity of organ systems and regulatory mechanisms	Respiratory obstruction, fluid overload, dehydration, hypothermia, and infection
Elderly—Multiple organ degeneration and slowed regulatory mechanisms	Decreased metabolism and excretion of anesthetics and pain medications, fluid overload, renal failure, formation of blood clots, delayed wound healing, infection, confusion, and respiratory complications
Nutritional Status	
Malnourished—Low weight and nutrient deficiencies	Fluid and electrolyte imbalances, cardiac dysrhythmias, delayed wound healing, and wound infections
Obese—Stressed cardiovascular system, decreased circulation, and decreased pulmonary function	Atelectasis, pneumonia, blood clots, delayed wound healing, wound infection, delayed metabolism, and excretion of anesthetics and pain medication
Substance Abuse	
Alcohol, tobacco, sedatives—Altered respiratory function, nutritional status, or liver function	Atelectasis, pneumonia, altered effectiveness of anesthetics and pain medications, drug interactions, and drug withdrawal
Medical Problems	
Immune—Allergies and immunosuppression secondary to corticosteroid therapy, transplants, chemotherapy, or diseases such as AIDS	Adverse reactions to medications, blood transfusions, or latex; infection
Respiratory—Acute and chronic respiratory problems and history of tobacco use	Atelectasis, bronchopneumonia, and respiratory failure
Cardiovascular—Hypertension, coronary artery disease, and peripheral vascular disease	Hypotension, hypertension, fluid overload, congestive heart failure, shock, dysrhythmias, myocardial infarction, stroke, and blood clots
Hepatic —Liver dysfunction	Delayed drug metabolism leading to drug toxicity, disrupted clotting mechanisms leading to excessive bleeding or hemorrhage, confusion, and increased risk for infection
Renal—Kidney disease, chronic renal insufficiency, and renal failure	Fluid and electrolyte imbalances, congestive heart failure, dysrhythmias, delayed excretion of drugs leading to drug toxicity
Endocrine—Diabetes	Hypoglycemia, hyperglycemia, hypokalemia, infection, and delayed wound healing

Leg Exercises

Leg exercises help to promote circulation and reduce the risk for forming a **thrombus** (a stationary blood clot) in the veins. Blood clots form when venous

circulation is sluggish and when the fluid component of blood is reduced. Surgical clients are predisposed to both. Surgical clients have reduced circulatory volume because of the preoperative restriction of food and fluids and blood loss during surgery. Also, blood tends to pool in the lower extremities because of the stationary position during surgery and the clients' reluctance to move afterward. With the use of leg exercises, efforts to reduce circulatory complications begin as soon as the client recovers from anesthesia (see Client and Family Teaching 27-2).

Client and Family Teaching 27-1 Performing Forced Coughing

The nurse teaches the client and the family as follows:

- Sit upright.
- Take a slow, deep breath through the nose.
- Make the lower abdomen rise as much as possible.
- Lean slightly forward.
- Exhale slowly through the mouth.
- Pull the abdomen inward.
- Repeat, but this time cough three times in a row while exhaling.

FIGURE 27-3 Teaching the client to splint the incision and to cough. (Photo by Ken Kasper.)

Antiembolism stockings are knee-high or thigh-high elastic stockings. They are sometimes called thromboembolic disorder (TED) hose. These stockings help to prevent thrombi and **emboli** (mobile blood clots) by compressing superficial veins and capillaries, redirecting more blood to larger and deeper veins, where it flows more effectively toward the heart. Intermittent pneumatic

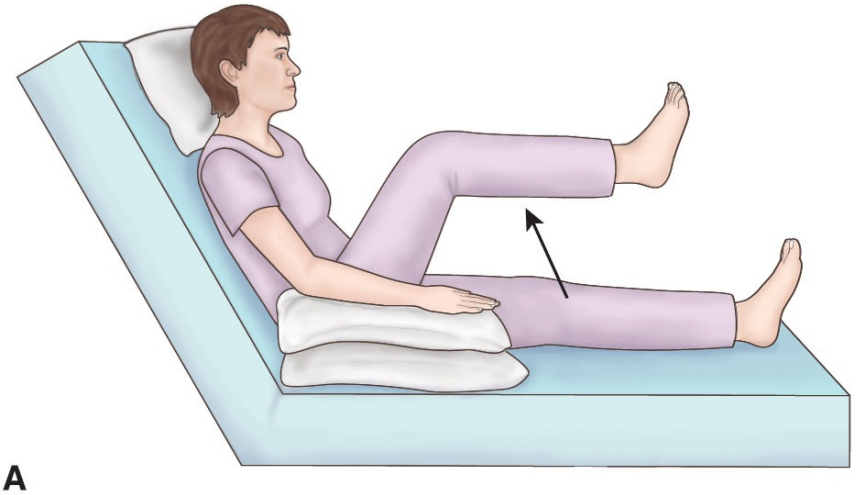
compression devices (discussed later in this chapter) are used for the same purpose but are applied postoperatively.

Antiembolism stockings must fit the client properly and must be applied correctly (Skill 27-1). Stockings that become dirty are laundered, during which a second pair is used. If washed by hand, the stockings are laid flat to dry to prevent loss of their elasticity.

Client and Family Teaching 27-2 Performing Leg Exercises

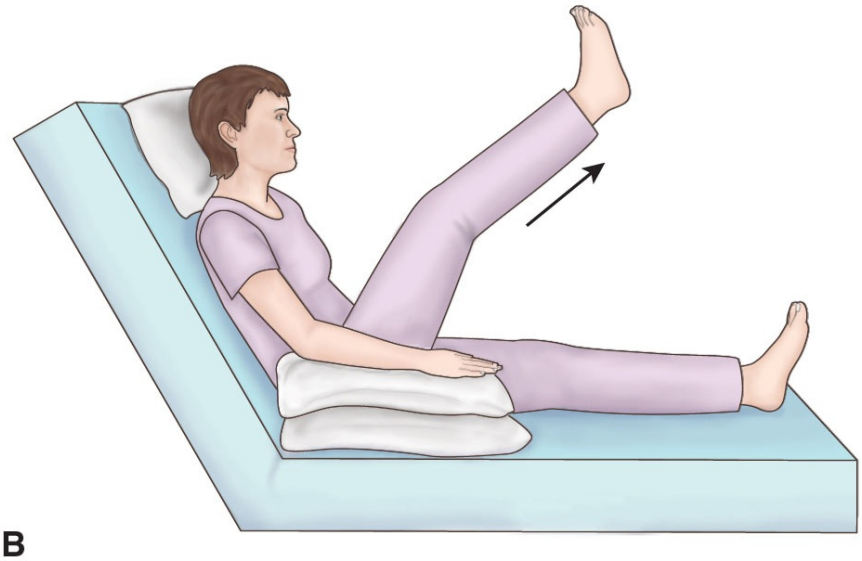
The nurse teaches the client and the family as follows:

- Sit with the head slightly raised.
- Bend one knee. Raise and hold the leg above the mattress for a few seconds (Fig. A).



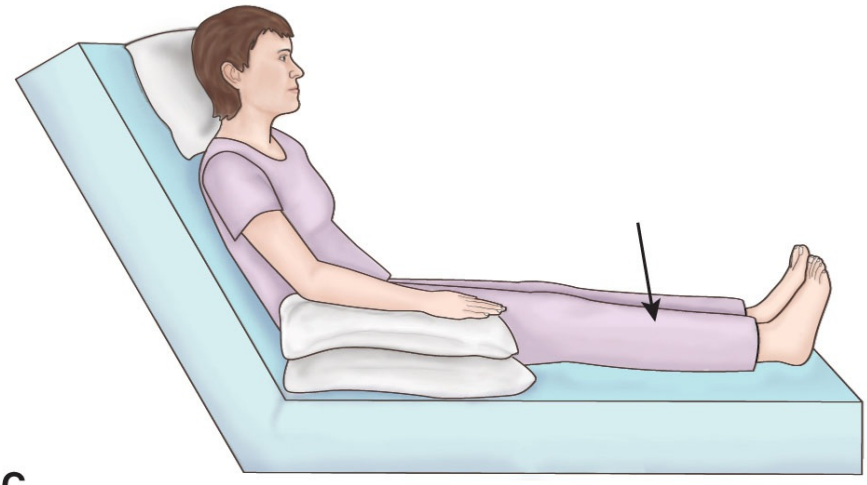
A

- Straighten the raised leg (Fig. B).



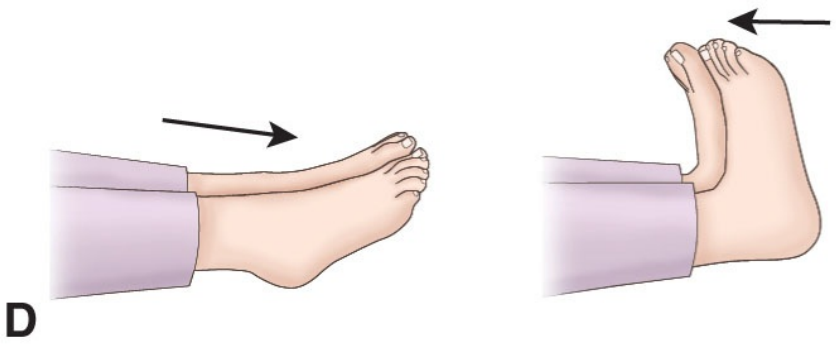
B

- Lower the leg back to the bed gradually (Fig. C).

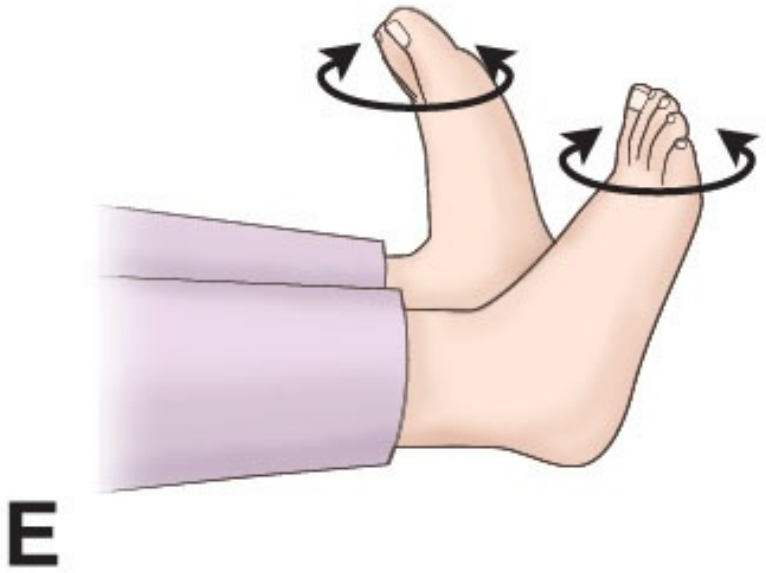


C

- Do the same with the other leg.
- Rest both legs on the bed.
- Point the toes toward the mattress and then toward the head (Fig. D).



- Move both feet in clockwise and then counterclockwise circles (Fig. E).



- Repeat the exercises five times at least every 2 hours while awake.

A through E from Taylor, C. R., Lillis, C., LeMone, P., & Lynn, P. (2011). *Fundamentals of nursing.*(7th ed.). Philadelphia, PA: Lippincott Williams & Wilkins.

Stop, Think, and Respond 27-1

Discuss reasons why surgical clients are not as active and mobile as nonsurgical clients.

Physical Preparation

Depending on the time of admission to the hospital or surgical facility, the nurse may perform some physical preparation that includes skin preparation, attention

to elimination, restriction of food and fluids, care of valuables, putting on surgical attire, and disposition of prostheses.

Skin Preparation

Skin preparation involves cleansing the skin and, in some cases, hair removal because skin and hair are reservoirs for microorganisms (Skill 27-2). The goal is to decrease transient and resident bacteria without compromising skin integrity. Reducing bacteria helps to prevent postoperative wound infections.

For planned surgery, the client may be asked to bathe or shower twice at home with 2% chlorhexidine gluconate for a minimum of 2 minutes contact time; dry with a fresh, clean dry towel; and don clean clothing afterwards (Edmiston et al., 2010, 2015; Grayling & Vasaly, 2013). The soap leaves a persistent antimicrobial film on the skin in preparation for surgery.

Hair usually is not removed before surgery unless it is likely to interfere with the incision. Shaving causes **microabrasions** (tiny cuts that provide an entrance for microorganisms). For this reason, institutions are switching from razors to electric or battery-operated clippers for hair removal. **Depilatory agents**, chemicals that remove hair, are another alternative, but their use is associated with skin irritation and allergic reactions. Some authorities believe that simply washing the skin and hair is sufficient to prevent infections.

> ***Stop, Think, and Respond 27-2***
>
> *Correlate the potential for transmitting an infection using a razor for presurgical skin preparation with the chain of infection discussed in Chapter 10.*

Elimination

The nurse may need to insert an indwelling urinary catheter (see Chap. 30) preoperatively for some surgeries, particularly of the lower abdomen. A distended bladder increases the risks for bladder trauma and difficulty in performing the procedure. The catheter keeps the bladder empty during surgery. If a catheter is not inserted, the nurse instructs the client to urinate immediately before receiving preoperative medication.

Enemas or a laxative may be ordered to clean the lower bowel (see Chap. 31) if the client is having abdominal or pelvic surgery. A clean bowel allows for improved visualization of the surgical site and prevents trauma to the intestines or accidental contamination of the abdominal cavity with feces. A cleansing enema or laxative is prescribed the evening before surgery and may be repeated the morning of surgery. If bowel surgery is scheduled, intravenous antibiotics

may be prescribed to destroy intestinal microorganisms.

Food and Fluids

The physician gives specific instructions about how long to restrict food and fluids preoperatively. Fasting from food and water from midnight onward before surgery is common, but the basis for the practice is now questionable. Fasting is used to reduce the potential for aspirating (inhaling) stomach contents while a client is anesthetized. However, aspiration is uncommon today with standard practices used by those administering general anesthesia. Consequently, the American Society of Anesthesiology (2011) recommends that healthy preoperative clients can consume clear liquids 2 hours before elective surgery, have a light breakfast 6 hours before a surgical procedure, and eat a heavier meal 6 to 8 hours beforehand. Despite these newer recommendations, old practices persist. The nurse, therefore, encourages clients to maintain good nutrition and hydration before the restricted time to promote nutrients, such as protein and ascorbic acid (vitamin C), which are needed for healing.

Concept Mastery Alert

Despite current scientific evidence and recommendations that a healthy client can have clear liquids up to 2 hours prior to surgery, the old recommendation of nothing by mouth after midnight continues to persist in practice.

Gerontologic Considerations

- Before any period of fluid restriction in older adults, it is important to assess vital signs, weight, and sternal skin turgor to establish a baseline for comparison.

Valuables

The nurse instructs the client preoperatively to leave valuables at home. If the client forgets or does not follow this instruction, he or she must entrust valuables to a family member. Otherwise, health care agency personnel itemize them, place them in an envelope, and lock them in a designated area. The client signs a receipt, and the nurse notes the items' whereabouts in the client's medical record.

If the client is reluctant to remove a wedding band, the nurse may slip a ribbon of gauze under the ring and then loop the gauze around the finger and wrist or apply adhesive tape around a plain wedding band. The client also removes

eyeglasses and contact lenses, which the nurse places in a safe location or gives to a family member.

Surgical Attire

Usually, clients wear a hospital gown and surgical cap to the operating room. The physician may order thigh-high or knee-high antiembolism stockings or order the client's legs wrapped in elastic roller bandages (see Chap. 28) before surgery to prevent venous stasis.

Hair ornaments are removed to avoid injury with equipment used to administer oxygen and inhalant anesthetics. Makeup and nail polish are omitted to facilitate assessing oxygenation. If a client has acrylic nails, one usually is removed to attach a pulse oximeter, which measures oxygen saturation (see Chap. 21).

Dentures and Prostheses

Depending on agency policy and the preference of the anesthesiologist or surgeon, the client removes full or partial dentures. Doing so prevents the dentures from causing airway obstruction during the administration of a general anesthetic. Some anesthesiologists prefer that well-fitting dentures remain in place to preserve facial contours, but that information must be communicated and well documented. When dentures are removed, they are placed in a denture container and stored at the client's bedside or with the client's belongings. Other prostheses, such as artificial limbs, also are removed unless otherwise ordered.

Gerontologic Considerations

- Older adults also are likely to be self-conscious when dentures are removed before surgery. Collaboration with operating room personnel regarding the removal of dentures, eyeglasses, and hearing aids is helpful to ensure their use as much or as long as possible.
- Older adults who rely on eyeglasses or hearing aids may experience sensory deprivation if these aids are removed before surgery or other procedures. Removal may interfere with communication or contribute to confusion and altered mental status.
- Assure that dentures, eyeglasses, hearing aids are labeled with the person's name.

Preoperative Medications

The anesthesiologist or surgeon orders preoperative parenteral medications.

Pharmacologic Considerations

Common preoperative medications include one or more of the following:

- *Antianxiety drugs*, such as lorazepam (Ativan), reduce preoperative anxiety, cause slight sedation, slow motor activity, and promote the induction of anesthesia.
- *Histamine-2 receptor antagonists*, such as cimetidine (Tagamet), decrease gastric acidity and volume.
- *Anticholinergics*, such as glycopyrrolate (Robinul), decrease respiratory secretions, dry mucous membranes, and prevent vagal nerve stimulation during endotracheal intubation.
- *Neuromuscular blocking agents*, such as succinylcholine (Anectine), promote skeletal muscle relaxation during the procedure, and allow for rapid intubation.
- *Opioids*, such as fentanyl, sedate and decrease the amount of anesthesia.
- *Sedatives*, such as midazolam (Versed), promote sleep or amnesia and decrease anxiety.
- *Antibiotics*, such as kanamycin, destroy enteric microorganisms.

Before administering preoperative medications, the nurse uses at least two methods to verify the identity of the client. An example would be checking the client's identification bracelet and asking the client to state his or her name and date of birth (see Chaps. 19, 32, and 34). The nurse asks about drug allergies, obtains vital signs, asks the client to void, and ensures that the surgical consent form has been signed.

Psychosocial Preparation

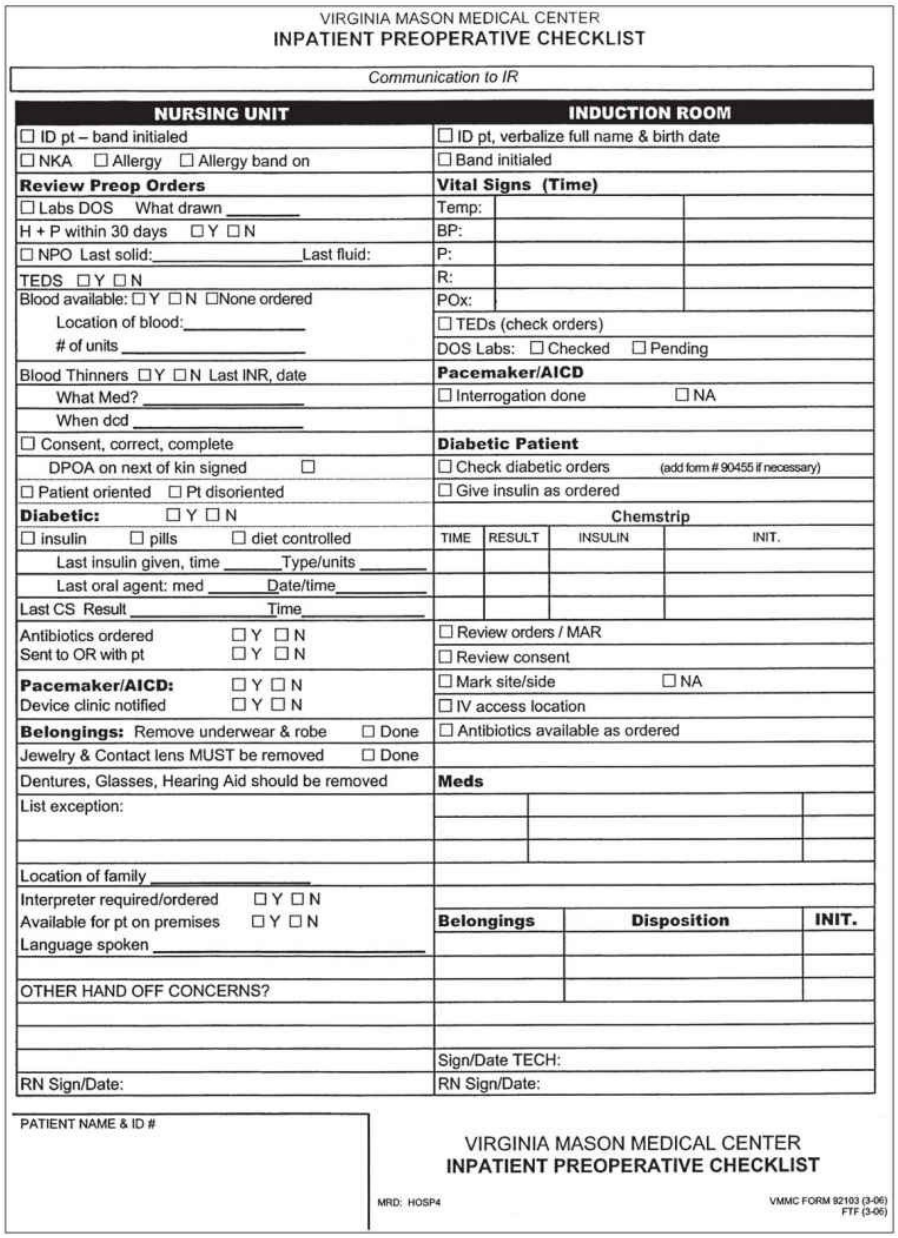
Preparing the client emotionally and spiritually is as important as doing so physically. Psychosocial preparation should begin as soon as the client is aware that surgery is necessary. Anxiety and fear, if extreme, can affect a client's condition during and after surgery. Anxious clients have a poor response to surgery and are prone to complications (Whitlock, 2014). Many clients are fearful because they know little or nothing about what will happen before, during, and after surgery. Careful listening and explaining by the nurse about what will happen and what to expect can help to allay some of these fears and anxieties. The nurse also must assess methods the client uses for coping. Religious faith is a source of strength for many clients; therefore, nurses question the client as part of the spiritual assessment, then facilitate contact with a client's clergyperson or the hospital chaplain, if requested.

Preoperative Checklist

A **preoperative checklist** (Fig. 27-4) is a form that identifies the status of essential presurgical activities and is completed before surgery. The nurse verifies the following:

- The history and physical examination have been documented.
- The name of the procedure on the surgical consent form matches that

scheduled in the operating room.
- The surgical consent form has been signed and witnessed.
- All laboratory and diagnostic test results, such as a fasting blood sugar or electrocardiogram (ECG), have been returned and reported if abnormal.
- Allergies have been identified.
- The client is wearing an identification bracelet and allergy bracelet, if any exist.
- The client has had nothing by mouth (NPO, *nil per os*) since midnight or the number of hours prescribed.
- Skin preparation, if required, has been completed.
- Vital signs have been assessed and recorded.
- Nail polish, glasses, contact lenses, and hairpins have been removed.
- Jewelry has been removed or the wedding ring has been secured.
- Dentures have been removed or left in place if requested by the person administering inhalant anesthesia.
- The client is wearing only a hospital gown and hair cover.
- The client has urinated.
- Location of intravenous site, type of solution, and rate of infusion are identified.
- The prescribed preoperative medication has been given.



VIRGINIA MASON MEDICAL CENTER
INPATIENT PREOPERATIVE CHECKLIST

Communication to IR

NURSING UNIT	INDUCTION ROOM
☐ ID pt – band initialed	☐ ID pt, verbalize full name & birth date
☐ NKA ☐ Allergy ☐ Allergy band on	☐ Band initialed
Review Preop Orders	**Vital Signs (Time)**
☐ Labs DOS What drawn _____	Temp:
H + P within 30 days ☐ Y ☐ N	BP:
☐ NPO Last solid:_____Last fluid:	P:
TEDS ☐ Y ☐ N	R:
Blood available: ☐ Y ☐ N ☐None ordered	POx:
Location of blood:_____	☐ TEDs (check orders)
# of units _____	DOS Labs: ☐ Checked ☐ Pending
Blood Thinners ☐ Y ☐ N Last INR, date	**Pacemaker/AICD**
What Med? _____	☐ Interrogation done ☐ NA
When dcd _____	
☐ Consent, correct, complete	**Diabetic Patient**
DPOA on next of kin signed ☐	☐ Check diabetic orders (add form # 90455 if necessary)
☐ Patient oriented ☐ Pt disoriented	☐ Give insulin as ordered
Diabetic: ☐ Y ☐ N	Chemstrip
☐ insulin ☐ pills ☐ diet controlled	TIME / RESULT / INSULIN / INIT.
Last insulin given, time _____Type/units _____	
Last oral agent: med _____Date/time_____	
Last CS Result _____Time_____	
Antibiotics ordered ☐ Y ☐ N Sent to OR with pt ☐ Y ☐ N	☐ Review orders / MAR ☐ Review consent
Pacemaker/AICD: ☐ Y ☐ N Device clinic notified ☐ Y ☐ N	☐ Mark site/side ☐ NA ☐ IV access location
Belongings: Remove underwear & robe ☐ Done	☐ Antibiotics available as ordered
Jewelry & Contact lens MUST be removed ☐ Done	
Dentures, Glasses, Hearing Aid should be removed	**Meds**
List exception:	
Location of family _____	
Interpreter required/ordered ☐ Y ☐ N Available for pt on premises ☐ Y ☐ N Language spoken _____	**Belongings / Disposition / INIT.**
OTHER HAND OFF CONCERNS?	
	Sign/Date TECH:
RN Sign/Date:	RN Sign/Date:

PATIENT NAME & ID #

VIRGINIA MASON MEDICAL CENTER
INPATIENT PREOPERATIVE CHECKLIST

MRD: HOSP4

VMMC FORM 92103 (3-06)
FTF (3-06)

FIGURE 27-4 Preoperative checklist. (From Craven, R. F., Hirnle, C. J., & Jensen, H. [2012]. *Fundamentals of nursing* [7th ed.]. Philadelphia, PA: Lippincott Williams & Wilkins.)

BOX 27-1 Universal Protocol for Preventing Wrong Site, Wrong Procedure, Wrong Person Surgery

Preoperative Verification Process

- *Purpose:* To ensure that all of the relevant documents and studies are available before the start of the procedure; that they have been reviewed; and that they are consistent with each other, with the client's expectations, and with the team's understanding of the intended client, procedure, site, and, as

applicable, any implants. Missing information or discrepancies must be addressed before starting the procedure.
- *Process:* An ongoing process of information gathering and verification, beginning with the determination to do the procedure, continuing through all settings and interventions involved in the preoperative preparation of the client, up to and including the "time-out" just before the start of the procedure.

Marking the Operative Site

- *Purpose:* To identify unambiguously the intended site of incision or insertion.
- *Process:* For procedures involving right/left distinction, multiple structures (such as fingers and toes), or multiple levels (as in spinal procedures), the intended site must be marked such that the mark will be visible after the client has been prepped and draped.
- The person who knows the most about the client—usually the person performing the procedure—should mark the site before the procedure and, if possible, with the client involved.

"Time Out" Immediately before Starting the Procedure

- *Purpose:* To conduct a final verification of the correct client, procedure, site and, as applicable, implants.
- *Process:* Active communication among all members of the surgical/procedure team, consistently initiated by a designated member of the team, conducted in a "fail-safe" mode; that is, the procedure is not started until any questions or concerns are resolved.

Adapted from The Joint Commission. (2015). *Introduction to the universal protocol for preventing wrong site, wrong procedure, and wrong person surgery.* Retrieved November 3, 2015, from http://www.jointcommission.org/assets/1/6/2015_NPSG_HAP.pdf.

The nurse is responsible for completing and signing the checklist. Operating room personnel review it when they arrive to transport the client. Surgery may be delayed if the checklist is incomplete.

Efforts have increased to ensure that the right client has the proper procedure at the correct site (if that applies). See Box 27-1 for the **Universal Protocol** developed by The Joint Commission (2015) to prevent errors in these categories.

INTRAOPERATIVE PERIOD

The **intraoperative period** (the time during which the client undergoes surgery) takes place in the operating suite. It involves transportation to a receiving room then to the operating room where anesthesia is administered and the procedure is performed. The family is directed to a surgical waiting area during this time.

FIGURE 27-5 A client awaiting surgery in a receiving room. (From Carter, P. J. [2011]. *Lippincott textbook for nursing assistants* [3rd ed.]. Philadelphia, PA: Lippincott Williams & Wilkins.)

Receiving Room

The **receiving room** (Fig. 27-5) is a place in the surgery department where clients are observed until the operating room and surgical team are ready. In some hospitals, preoperative medication is administered when clients reach the receiving room rather than before leaving the nursing unit. This practice coordinates the client's sedation more closely with the actual time of surgery.

Skin preparation may be delayed until this time as well. There is a direct relationship between the time the skin preparation is performed and the rate of microbial proliferation (Smith, et al, 2013).

Operating Room

Eventually, clients are taken to the operating room, where their care and safety are in the hands of a team of experts including physicians and nurses. Anesthesia is administered in the operating room. Various types of anesthesia cause partial or complete loss of sensation with or without a loss of consciousness. They include general, regional, and local anesthesia.

General Anesthesia

General anesthesia acts on the central nervous system to produce a loss of sensation, reflexes, and consciousness. General anesthetics commonly are administered via inhaled gases (Fig. 27-6) or a drug like propofol (Diprivan) intravenously.

Throughout the duration of and recovery from anesthesia, the client is monitored closely for effective breathing and oxygenation; effective circulatory status, including blood pressure and pulse within normal ranges; effective temperature regulation; and adequate fluid balance. During weaning from the anesthetic at the end of surgery, the client's consciousness will be elevated sufficiently for him or her to follow commands and breathe independently. The recovery period can be brief or long. Many effects of general anesthesia take some time for the client to eliminate completely. Usually, clients do not remember much about the initial recovery period.

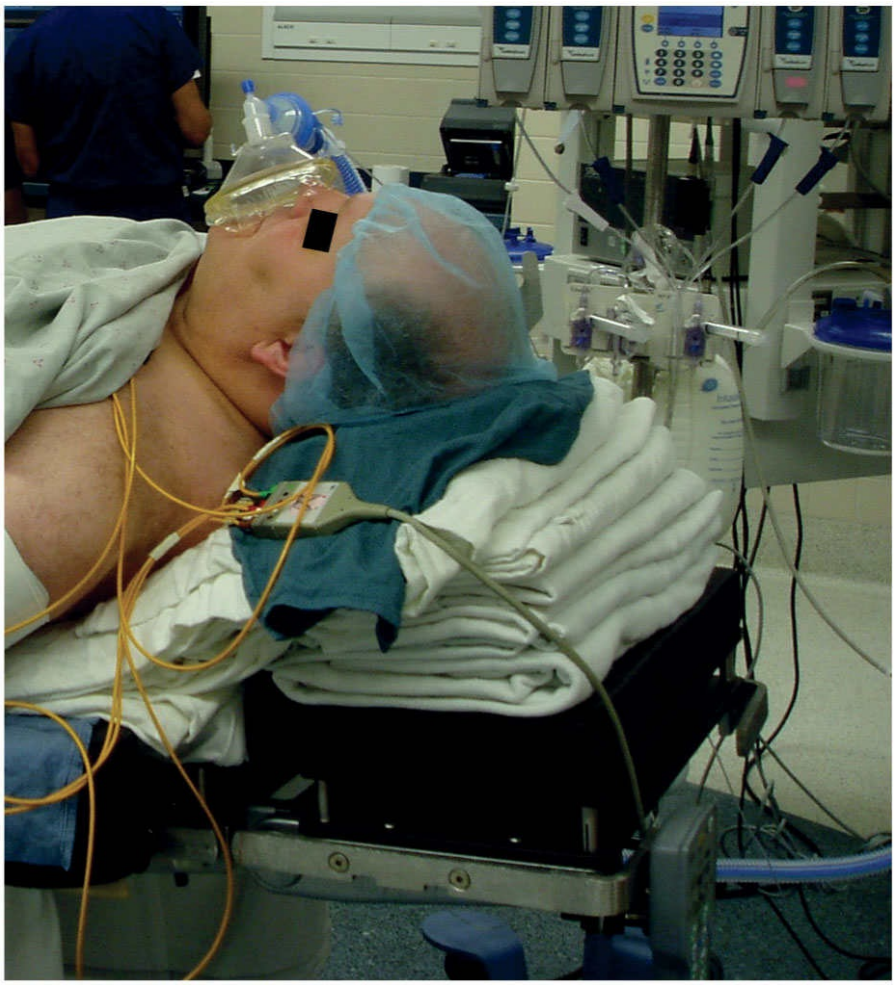


FIGURE 27-6 Client receiving inhaled general anesthesia. (From Orbaugh, S. L., & Gigeleisen, P. E. [2012]. *Atlas of airway management: techniques and tools* [2nd ed.]. Philadelphia: Lippincott Williams & Wilkins.]

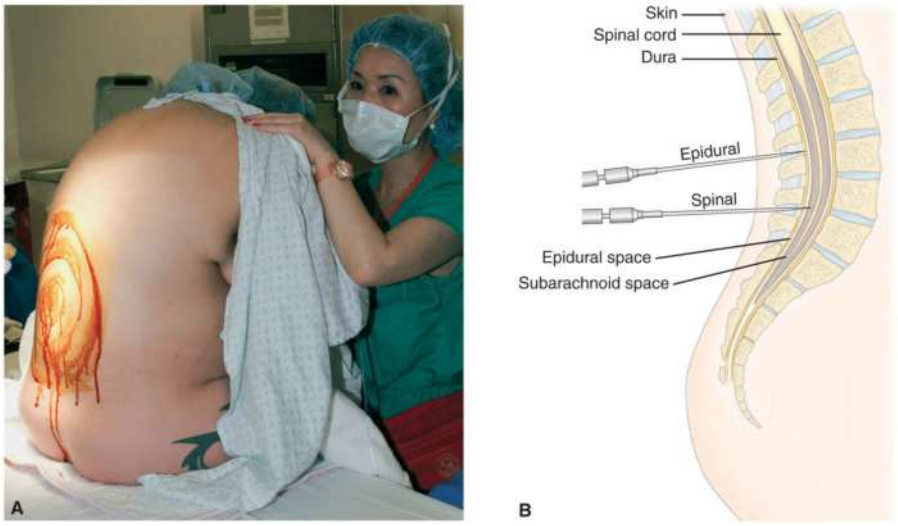


FIGURE 27-7 Administration of regional anesthesia. ((**A)**, from Evans, R. J., Brown, Y. M., & Evans, M. K. [2015]. *Canadian maternity, newborn & women's health nursing* [2nd ed.]. Philadelphia, PA: Lippincott Williams & Wilkins; (**B)**, from Hinkle, J. L., & Cheever, K. H. [2014]. *Brunner & Suddarth's textbook of medical-surgical nursing* [13th ed.]. Philadelphia, PA: Lippincott Williams & Wilkins.)

Regional Anesthesia

Regional anesthesia interferes with the conduction of sensory and motor nerve impulses to a specific area of the body. The client experiences loss of sensation and decreased mobility to the specific anesthetized area. He or she does not lose consciousness. Depending on the surgery, the client may receive a sedative to promote relaxation and comfort during the procedure. Types of regional anesthesia include local and spinal anesthesia and epidural and peripheral nerve blocks (Fig. 27-7).

The major advantage of regional anesthesia is the decreased risk for respiratory, cardiac, and GI complications. Team members must monitor the client for signs of allergic reactions, changes in vital signs, and toxic reactions. In addition, they must protect the anesthetized area if sensation is absent, because the client is at risk for injury.

Conscious Sedation

Conscious sedation refers to a state in which clients are sedated and in a state of relaxation and emotional comfort, but are not unconscious. They are free of pain, fear, and anxiety and can tolerate unpleasant diagnostic and short therapeutic surgical procedures, such as endoscopies or bone marrow aspiration, while maintaining independent cardiorespiratory function. They can respond verbally and physically.

The intravenous route is used to administer medications that create conscious sedation. If other routes are used, the client must have venous access for the

treatment of possible adverse effects such as hypoxemia and central nervous system depression. The responsibility for ensuring client safety and comfort during sedation rests with the nurse directly involved in the client's care. Although numerous types of equipment for monitoring clients are available, no equipment replaces a nurse's careful observations. Clients are discharged shortly after the procedure in which conscious sedation is used.

Pharmacologic Considerations

Reversal drugs, medications that counteract the effects of those used for conscious sedation, must be readily available in case the client becomes overly sedated. Two examples of reversal drugs are naloxone (Narcan), which is the antagonist for opiates like morphine, and flumazenil (Romazicon), which reverses antianxiety drugs like midazolam (Versed).

Surgical Waiting Area

The **surgical waiting area** is the room where family and friends await information about the client. It is staffed by volunteers who provide comfort, support, and news about how the client's surgery is progressing. Many agencies provide food and beverages, public telephones, television, and magazines in this area. Often, the surgeon comes to the waiting area immediately after the procedure to contact the family. The family and surgeon generally go to a private room where the surgeon discusses the client's status and the procedure so as to ensure confidentiality.

POSTOPERATIVE PERIOD

The **postoperative period** begins after the operative procedure is completed and the client is transported to an area to recover from the anesthesia and ends when the client is discharged. The **postanesthesia care unit** (PACU), also known as the *postanesthesia reacting* (PAR) room or the *recovery room*, is the area in the surgical department where clients are intensively monitored (Fig. 27-8). Nurses in the PACU ensure the safe recovery of surgical clients from anesthesia.

FIGURE 27-8 A postanesthesia care unit. (Photo by B. Proud.)

Immediate Postoperative Care

The focus of **postoperative care** (nursing care after surgery) is different during the immediate postoperative period than it is later, when clients are more stable. The immediate postoperative period refers to the first 24 hours after surgery. During this time, nurses monitor the client for complications as he or she recovers from anesthesia and is sufficiently stable to be transferred to a nursing unit for continued assessment.

Initial Postoperative Assessments

The circulating surgical nurse or anesthesiologist reports pertinent information regarding the surgery and the client's condition to the nurse in the PACU. Once the care of the client is transitioned to the recovery room nurse, the PACU nurse's major responsibilities are to ensure a patent airway; maintain adequate circulation; prevent or assist with the management of shock; maintain proper positions and function of drains, tubes, and intravenous infusions; and detect evidence of any complications. The nurse systematically checks the following:

- Level of consciousness
- Vital signs
- Effectiveness of respirations
- Presence or need for supplemental oxygen
- Condition of the wound and dressing
- Location of drains and drainage characteristics
- Location, type, and rate of intravenous fluid
- Level of pain and need for analgesia
- Presence of a urinary catheter and urine volume

Continuing Postoperative Care

Once the client is stable, the client is readied for transport to the general surgical unit where the client's room is prepared and assessments will continue to prevent, detect, or minimize complications.

Preparing the Room

The next stage of care begins with getting the client's bed and the environment ready. The nurses fold the top bed linen toward the foot or side of the bed. They place the bed in a high position to facilitate transferring the client from the stretcher. Often, they keep additional blankets ready for use because some clients feel cold after being quiet and inactive.

Additionally, nurses assemble bedside supplies and equipment that facilitate caring for the client. Potentially useful items include oxygen equipment (see Chap. 21), a pole or electronic infusion device for continuing the administration of intravenous fluids (see Chap. 16), an emesis basin if the client becomes nauseous, paper tissues, and a device for collecting and measuring urine (see Chap. 30). Suction canisters may be necessary for clients who have gastric tubes (see Chap. 29).

TABLE 27-7 Postoperative Complications

COMPLICATION	DESCRIPTION	TREATMENT
Airway occlusion	Obstruction of throat	Tilt the head and lift the chin. Insert an artificial airway.
Hemorrhage	Severe, rapid blood loss	Control bleeding. Administer intravenous fluid. Replace blood.
Shock	Inadequate blood flow	Place the client in a modified Trendelenburg position. Replace fluids. Administer oxygen. Give emergency drugs.
Pulmonary embolus	Obstruction of circulation through the lung as a result of a wedged blood clot that began as a thrombus	Give oxygen. Administer anticoagulant drugs.
Hypoxemia	Inadequate oxygenation of blood	Give oxygen.
Adynamic ileus	Lack of bowel motility	Treat the cause. Give nothing by mouth. Insert a nasogastric tube and connect to suction. Administer intravenous fluid.
Urinary retention	Inability to void	Insert a catheter.
Wound infection	Proliferation of pathogens at or beneath the incision	Cleanse with antimicrobial agents. Open and drain incision. Administer antibiotics.
Dehiscence	Separation of incision	Reinforce wound edges. Apply a binder.
Evisceration	Protrusion of abdominal organs through separated wound	Cover with wet dressing. Reapproximate wound.

Monitoring for Complications

Postoperative clients are at risk for many complications (Table 27-7), some of which are more likely to develop soon after surgery. Frequently focused assessments of the client and equipment facilitate a safe postoperative recovery (see Nursing Guidelines 27-1).

Gerontologic Considerations

■ The cardiac status of older adults is monitored carefully after surgery because they may not be able to tolerate or eliminate intravenous fluids given at standard rates due to compromised cardiac or renal status.

Providing Food and Oral Fluids

After surgery, the client needs to resume eating. Food and oral fluids are withheld until surgical clients are awake, are free of nausea and vomiting, and bowel sounds are active. Postoperative clients usually progress from a clear liquid diet to more solid food unless complications develop. Nurses monitor fluid intake and output to ensure that clients are adequately hydrated.

Nutrition Notes

■ The postsurgical diet order may be “Progress from clear liquids to a regular diet as tolerated.” A quick progression to self-selected regular food by the second postsurgical meal is safe for most clients, even those who have had major GI surgery, and may even hasten recovery.

Gerontologic Considerations

■ If an indwelling catheter is inserted before surgery, it is best to remove it as soon as possible after surgery to prevent urinary tract infections. Careful assessment of urination pattern and volume is indicated to ensure adequate voiding amounts and timing, especially if a bedpan will be required during a period of ambulatory restrictions.

Promoting Venous Circulation

Surgical clients ambulate with assistance as soon as possible to reduce the potential for pulmonary and vascular complications. After some surgical procedures, however, antiembolism stockings, leg exercises, ambulation, and elevation of the lower extremities may not be enough to reduce swelling of the lower extremities and the potential for thrombus formation.

NURSING GUIDELINES 27-1

Providing Postoperative Care

- Obtain a summary report from a PACU nurse. *This report provides current assessment data concerning the client's progress.*
- Check the postoperative medical orders on the chart. *The medical orders provide instructions for individualized care.*
- Assist PACU personnel to transfer the client to bed. *The client should be observed continuously at this time.*
- Observe the client's respiratory pattern and auscultate the lungs. *Maintaining breathing is a priority for care.*
- Check oxygen saturation using a pulse oximeter if the client seems hypoxic (see Chap. 21). *An oximeter indicates the quality of internal respiration.*
- Administer oxygen if the oxygen saturation is less than 90% or if prescribed by the physician. *Oxygen administration increases oxygen available for binding with hemoglobin and for becoming dissolved in the plasma.*
- Note the client's level of consciousness and response to stimulation. *Findings indicate the client's neurologic status.*
- Orient the client and instruct him or her to take several deep breaths, as taught preoperatively. *Deep breathing improves ventilation and gas exchange.*
- Check vital signs. *Findings provide data for assessing the client's current general condition.*
- Repeat vital sign assessments at least every 15 minutes until they are stable; then follow agency policy and retake them every hour to every 4 hours depending on the client's condition or medical orders. *Repeat assessment of vital signs provides comparative data.*
- Check the incisional area and the dressing for drainage. *Findings provide data concerning the status of the wound and blood loss.*
- Inspect all tubes, insertion sites, and connections. *For optimal outcomes, the equipment must function properly.*
- Check the type of intravenous fluid, rate of administration, and volume that remains. *Findings provide data regarding fluid therapy.*
- Monitor urination; report failure to void within 8 hours of surgery. *Failure to void indicates urinary*

retention.
- Auscultate bowel sounds. *Findings provide data concerning bowel motility.*
- Assess the client's level of pain, its location, and characteristics. *Pain indicates the need for analgesia.*
- Administer analgesic drugs according to prescribed medical orders, if doing so is safe. *Analgesic drugs relieve pain.*
- Remind the client to perform leg exercises or apply antiembolism stockings. *Leg exercises and antiembolism stockings promote circulation.*
- Use a side-lying position if the client is lethargic or unresponsive. *This position prevents airway obstruction by the tongue and aspiration of emesis if vomiting occurs.*
- Raise the side rails unless providing direct care. *Keeping the side rails up ensures safety.*
- Fasten the signal device within the client's reach. *The signal device is a way for the client to communicate and obtain assistance.*

Gerontologic Considerations

- Muscle atrophy occurs in older adults who have been on bed rest even for 1 or 2 days. Range of motion and muscle tone can be maintained through routine active or passive range-of-motion exercises.
- Consider requesting a referral for physical therapy for older adults who have been on bed rest.

For clients who have the potential for impaired circulation in one or both extremities, a **pneumatic compression device,** a machine that promotes the circulation of venous blood and relocation of excess fluid into the lymphatic vessels, may be medically prescribed. Various companies make pneumatic compression devices, but they all consist of an extremity sleeve with tubes that connect to an electrical air pump (Fig. 27-9). The device compresses the sleeved extremity either intermittently or sequentially from distal to proximal areas. Most devices cycle on for a few seconds and then cycle off for a longer period. Depending on the manufacturer, pumps may cycle one to four times per minute. The nurse is responsible for applying this device (Skill 27-3).

Stop, Think, and Respond 27-3

Compare the use of TED hose with a pneumatic compression device; list advantages and disadvantages for each.

Other measures to prevent thrombi include drinking plenty of fluids, avoiding long periods of sitting, keeping the legs uncrossed (especially at the knees), ambulating, and changing position frequently.

FIGURE 27-9 A pneumatic compression device.

Preventing Thrombus Formation

Postoperatively, clients are at risk for developing deep vein thromboses as a consequence of preoperative fluid restrictions, a decrease in blood volume due to fluid loss during the surgical procedure, and decreased activity following the surgery. The nurse encourages leg exercises that were taught prior to surgery, early ambulation, and administers prophylactic antithrombotic medications as prescribed by the physician.

Pharmacologic Considerations

■ Following major surgical procedures, a client typically receives an injectable anticoagulant, such as dalteparin (Fragmin) or fondaparinux (Arixtra) to prevent thrombus formation. This medication supports standard nursing postoperative care and is not a substitute for ambulation or other activities to promote venous circulation.

Performing Wound Management

Nurses assess the condition of the wound and the characteristics of drainage at least once in each shift. Dressings are reinforced or changed if they become loose or saturated. Eventually, sutures or staples are removed (see Chap. 28). Most hospitalized clients are discharged within 3 to 5 days of surgery or sooner to continue their recuperation at home.

Gerontologic Considerations

■ Wound healing in older adults may occur more slowly because of age-related skin changes and impaired circulation and oxygenation. Poor hydration and nutrition further interfere with wound healing. A registered dietitian can recommend nutritional interventions such as protein, zinc, and vitamin C to improve wound healing.

■ If older adults develop postoperative infections, the manifestations are likely to be subtle or delayed. Older adults are likely to have a lower "normal" temperature. Therefore, it is imperative to document the client's usual baseline temperature so that deviations can be assessed. A change in mental status may be an early indicator of infection.

Providing Discharge Instructions

The nurse provides **discharge instructions** (directions for managing self-care and medical follow-up) before the client leaves. Common areas to address when discharging clients who have undergone surgery include the following:

- How to care for the incision site
- Signs of complications to report
- What drugs to use to relieve pain
- How to self-administer prescribed drugs
- When presurgical activity can be resumed
- If and how much weight can be lifted
- Which foods to consume or avoid
- When and where to return for a medical appointment

The nurse gives information verbally and in written form.

Gerontologic Considerations

■ A thorough assessment of an older client's support system must be done well before discharge. It should include the ability of the support system to provide assistance once the client is discharged. Support people should be included in discharge teaching, with plenty of time to provide any return demonstration of learning regarding the needs of the older adult. Additionally, the home environment should be assessed before discharge for safety issues (eg, use of scatter rugs, lighting, rails, grab bars).

■ If the older person cannot manage his or her postoperative care independently

or with the assistance of supportive family or friends, options relative to extended or skilled nursing care should be explored and discussed. Skilled nursing or rehabilitation therapists may be available from home care agencies and covered by health insurance for home settings.

NURSING IMPLICATIONS

Surgical clients offer unique nursing care problems. Applicable nursing diagnoses include the following:

- Deficient Knowledge
- Fear
- Acute Pain
- Impaired Skin Integrity
- Risk for Infection
- Risk for Deficient Fluid Volume
- Ineffective Breathing Pattern
- Ineffective Airway Clearance
- Risk for Impaired Gas Exchange
- Disturbed Body Image
- Readiness for Enhanced Self-Care

Clinical Scenario A 73-year-old female was diagnosed with a colorectal malignancy following a routine colonoscopy. She has become depressed during the postoperative recovery.

Nursing Care Plan 27-1 shows how the nurse can use the nursing process to identify and resolve a diagnosis of Disturbed Body Image, defined in the NANDA-I taxonomy (2015, p. 275) as "confusion in (the) mental picture of one's physical self." This diagnosis is especially pertinent to clients who have had their appearance altered as a result of surgery.

NURSING CARE PLAN 27-1 Disturbed Body Image

Assessment

- Observe the client's reaction to his or her body changes.
- Note if the client refuses to touch or look at the body part that has been altered.
- Scrutinize the client's involvement, or lack of it, in learning techniques for self-care or rehabilitation.
- Observe if the client seeks others to manage care for which he or she is capable.
- Watch the quality and quantity of the client's social interactions or avoidance of others.
- Listen for self-depreciating remarks or hostility toward others.

Nursing Diagnosis. **Disturbed Body Image** related to fear of rejection based on altered elimination secondary to a colectomy with ileostomy as evidenced by asking that room freshener be sprayed frequently, applying perfume heavily, positioning herself more than 5 feet from visitors, and stating, "I hate myself for agreeing to this operation. This 'thing' fills up, it bulges, and it smells. No one will ever want to come near me again."

Expected Outcome. The client will demonstrate acceptance and less self-consciousness about changed body image by interacting with a visitor within 3 feet by 10/9.

Interventions	Rationales
Spend at least 15 minutes with the client midmorning, midafternoon, and early evening without performing direct care.	Social interaction not associated with performing a task communicates interest and acceptance of the client as a worthwhile person.
During interaction, sit within 3 feet of the client.	Sitting closely provides evidence that closeness is not a problem.
Acknowledge verbally that the ostomy and resulting change in elimination are difficult to accept.	Verbalizing what the client is implying nonverbally and actively demonstrating shows empathy.
Offer to contact another person with an ostomy through the United Ostomy Association.	Interacting with another person who is coping well with a similar change can help the client to share feelings and acquire a different perspective from an objective role model.
Offer a referral to an enterostomal nurse therapist.	An enterostomal nurse therapist has knowledge and skills for managing problems experienced by clients with ostomies, such as odor control and other wound and skin impairments.
During ostomy teaching sessions and care of the stoma, avoid facial expressions that may communicate disgust or repulsion.	Nonverbal behavior is more accurate than verbal expressions during communication.
Use terminology such as "your stoma," and avoid any depersonalized or slang names for the	Using inappropriate terms trivializes the significance of the issue with which the client is

changed body part.	coping.

Evaluation of Expected Outcomes

- Client moved away to provide more distance during close interaction.
- Client looked at stoma while skin care and changing of appliance were demonstrated.
- Client read booklet provided by the United Ostomy Association.
- Client agreed to meet with the enterostomal therapist.

CRITICAL THINKING EXERCISES

1. The following data are reviewed by a nurse preparing a client for surgery: The client is 60 years old; weighs 205 lb; has a history of chronic pulmonary disease; quit smoking 10 years ago; vital signs are BP 140/88, temperature is 101.8°F, pulse is 92, and respiratory rate is 28 breaths per minute. Which finding is most important to report to the surgeon?
2. A client reports having taken only one shower with chlorhexidine gluconate rather than two the night before surgery. What actions could the nurse take?
3. A nurse assesses a postoperative client and obtains the following data: BP 102/64, pulse rate 90, respirations 32 and shallow, responds when shaken, and experiences nausea. What finding is most serious at this time, and what nursing actions are appropriate?
4. A preoperative client who is Native American wants you to attach a dream catcher, a circular object with a woven web, to the IV pole. Based on principles of transcultural nursing, what is an appropriate way to respond to the client's request?

NCLEX-STYLE REVIEW QUESTIONS

1. Assuming a client is admitted the evening before surgery, when is it best for the nurse to perform preoperative skin antisepsis and hair removal, if the latter is necessary, on a client who is scheduled for a procedure at 1300?
   1. The night before surgery
   2. After the morning shower
   3. Before transport to the receiving area
   4. When in the operating room

   Test-Taking Strategy: Note the key word "best." Select the option that is better than any of the others for reducing the risk for a surgical infection.
2. What information is essential to verify during a presurgical time-out? Select all that apply.
   1. The client's room number

2. The client's identity
3. The name of the surgeon
4. The name of the procedure
5. The type of anesthesia
6. The site of the procedure

Test-Taking Strategy: Analyze the choices and select the options that comply with The Joint Commission's Universal Protocol that promotes the performance of safe surgery.

3. From whom is it most appropriate for the nurse to obtain consent to perform surgery on an adolescent?

1. The client
2. The client's physician
3. The client's minister
4. The client's parent

Test-Taking Strategy: Note the key word and modifier "most appropriate." Select the option that is better legally than any of the others.

4. What is the most important nursing action after giving a preoperative medication containing a narcotic?

1. Raise the side rails on the bed.
2. Help the client to use the toilet.
3. Give the client a surgical gown.
4. Apply an identification band.

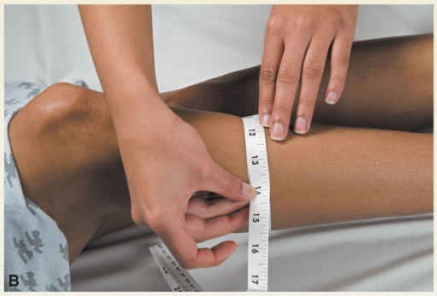
Test-Taking Strategy: Note the key word and modifier "most important." Select the option that is better than any of the others when preparing a client for surgery.

5. When the nurse assesses a client during the immediate postoperative period, which assessment is most indicative of impending shock?

1. Bounding pulse
2. Slow respirations
3. Low blood pressure
4. High body temperature

Test-Taking Strategy: Note the key word and modifier "most indicative". Eliminate choices until the option suggesting that the client is developing a postoperative complication of shock remains.

SKILL 27-1 Applying Antiembolism Stockings

Suggested Action	Reason for Action
ASSESSMENT	
Review the medical orders and the nursing plan for care.	Directs client care.
Wash your hands or perform an alcohol-based hand rub (see Chap. 10).	Reduces the transmission of microorganisms.
Measure the client's leg from the flat of the heel to the bend of the knee or to midthigh (Fig. A)	Determines the length needed for knee-high or thigh-high stockings.
A	Measuring the leg length from bottom of heel to back of knee. (From Springhouse. [2009]. *Lippincott's visual encyclopedia of clinical skills.* Philadelphia, PA: Lippincott Williams & Wilkins.)
Measure the calf or thigh circumference (Fig. B)	Determines the size needed.
 B	Measuring the circumference at its widest point. (From Springhouse. [2009]. *Lippincott's visual encyclopedia of clinical skills.* Philadelphia, PA: Lippincott Williams & Wilkins.)
Assess the client's understanding of the purpose and use of elastic stockings.	Determines the type and amount of health teaching needed.
Check the fit of stockings that the client is currently wearing.	Identifies the potential complications from tight, loose, or wrinkled stockings.
PLANNING	
Obtain the correct size of stockings before surgery or as soon as possible after they are ordered.	Facilitates early preventive treatment.
Plan to remove the stockings for 20 minutes once in each shift or at least twice a day and then reapply them.	Allows for assessment and hygiene.
Elevate the legs for at least 15 minutes before applying the stockings if the client has been sitting or standing for some time.	Promotes venous circulation and avoids trapping venous blood in the lower extremities.

IMPLEMENTATION

Wash and dry the feet.
Apply corn starch or talcum powder if desired (Fig. C).

Removes dirt, skin oil, and some microorganisms.
Reduces friction when applying the stockings.

Powdering the skin. (From Craven, R. F., Hirnle, C. J., & Jensen, S. [2012]. *Fundamentals of nursing.* [7th ed.]. Philadelphia, PA: Lippincott Williams & Wilkins.)

Avoid massaging the legs.
Turn the stockings inside out (Fig. D).

Prevents dislodging a thrombus if one is present.

Turning the stocking inside out and tucking the heel inside.

Insert the toes and pull the stocking upward a few inches until it covers the foot (Fig. E).

Facilitates threading the stockings over the foot and leg.

Easing the foot section over the toe and heel. (Photo by B. Proud.)

Gather the remaining length of the stocking and pull it upward a few inches at a time (Fig. F).	Reduces bunching and bulkiness. Eases application and avoids forming wrinkles.

F

Pulling the stocking upward over the rest of the leg. (Photo by B. Proud.)

EVALUATION

- Skin remains intact and circulation is adequate.
- Stockings are removed and reapplied at least b.i.d.

DOCUMENT

- Assessment findings
- Removal and reapplication of elastic stockings

SAMPLE DOCUMENTATION

Date and Time Toes are warm. Blood returns to nail beds within 3 seconds of compression. Skin over legs is smooth and intact. TED hose reapplied after bathing. J.Doe, LPN

SKILL 27-2 Performing Presurgical Skin Preparation

Suggested Action	Reason for Action
ASSESSMENT	
Determine that the client has followed instructions regarding showering and avoiding shaving the surgical site before coming to the facility.	Washing and rinsing with an antiseptic removes microorganisms from the skin; shaving the surgical area hours or the day before surgery significantly increases the risk for a surgical site infection.
Consult the preoperative medical orders to determine if it is necessary to remove hair in the area of the potential surgical incision.	Studies indicate that surgical site infections are reduced by omitting hair removal or only removing hair without a razor at or around the incision site if it will interfere with the procedure.
Wash your hands or perform an alcohol-based hand rub (see Chap. 10).	Reduces the transmission of microorganisms.
Assess the condition of the skin, looking especially for skin lesions.	Indicates areas that may bleed if irritated or provide a reservoir of microorganisms.
Explore how much the client understands about the purpose and extent of skin preparation.	Helps to identify the extent and level of health teaching needed.

PLANNING	
Arrange to perform the skin preparation shortly before the client is transported for surgery.	Reduces the time during which microorganisms will recolonize the skin.
Explain the procedure.	Reduces anxiety and promotes cooperation.
Provide an opportunity for the client to don a hospital gown.	Protects personal clothing and provides access for care.
Obtain electric or battery-operated clippers or depilatory agent, if ordered, a towel, a bath blanket, and gloves.	Provides essential supplies.
Braid scalp hair or use a nonflammable gel to keep hair out of the way prior to surgical procedures in which an incision will be made in the scalp.	Leaving scalp hair in place has not been shown to increase the incidence of surgical site infections and promotes a client's self-esteem postoperatively.
IMPLEMENTATION	
Wash your hands or perform an alcohol-based hand rub (see Chap. 10) and put on clean gloves.	Reduces the transmission of microorganisms.
Provide privacy.	Shows respect for dignity.
Position the client so the area to be prepared is accessible.	Facilitates performing the procedure.
Drape the client with a bath blanket.	Maintains dignity as well as warmth.
Protect the bed with towels or a disposable pad.	Contains the dispersal of loose hair.
Use a single-use hair clipper or clipper with a reusable head that can be disinfected to remove hair from the designated area.	Prevents transmission of microorganisms to other clients.
Follow the manufacturer's directions regarding skin testing in a small area if a depilatory is used.	Determines if hypersensitivity or skin irritation develops.
Keep a depilatory away from the client's eyes and genitalia.	Reduces the potential for skin and tissue irritation.
Deposit or dispose of items used for skin antisepsis and hair removal in appropriate containers.	Confines sources of infectious disease transmission, and restores comfort and orderliness.
Remove the reusable head from a nondisposable hair clipper and follow the agency's policy for disinfection.	Reduces the transmission of microorganisms.
Remove gloves and wash hands.	Reduces the transmission of microorganisms.
Return reusable clippers to their designated location and recharge the battery.	Ensures that reusable hair clippers are in working condition for future use.
EVALUATION	
• Skin has been prepared according to policy and medical orders. • Skin remains essentially intact.	
DOCUMENT	
• Assessment findings • Technique for preoperative skin antisepsis (ie, bathing, showers, hair removed with clippers, depilatory, or not removed) • Area prepared	

SAMPLE DOCUMENTATION

Date and Time Client reports taking two showers with chlorhexidine gluconate the evening before surgery. No hair removed from the potential site of the incision. Skin is intact. No evidence of lesions or body piercings. _____ J.Doe, LPN

SKILL 27-3 Applying a Pneumatic Compression Device

Suggested Action	Reason for Action
ASSESSMENT	
Review the medical orders and the nursing plan for care.	Directs client care.
Determine whether the device will be applied to one or both extremities.	Gives direction for gathering assessment data and applying the device.
Wash your hands or perform an alcohol-based hand rub (see Chap. 10).	Reduces the potential for the transmission of microorganisms.
Assess the circulation of the toes and integrity of the skin.	Provides a baseline of data for future comparison.
Measure the calf circumference and assess for pitting edema in extremities.	Provides a baseline of data for future comparisons.
Palpate the pedal pulses.	Validates arterial blood flow to the foot if present and strong.
Assess the client's understanding of the purpose and use of a pneumatic compression device.	Determines the type and amount of health teaching needed.
PLANNING	
Obtain the extremity sleeves, electric air pump, and accompanying air tubes.	Facilitates expeditious implementation of the medical order.
Assist the client with any elimination needs.	Avoids having to disconnect the equipment shortly after the device is applied.
Arrange supplies the client may need within his or her reach, including the signal device.	Promotes independence yet ensures that the client can call for assistance.
Help the client to a position of comfort such as a supine or low Fowler's position.	Fosters rest and relaxation.
IMPLEMENTATION	
Wrap the extremity sleeve snugly around the extremity (Fig. A).	Positions the sleeve where compression is desired.
A	Applying the extremity sleeve. (Photo by B. Proud.)
Secure the sleeve once it encircles the leg; most are secured with Velcro.	Ensures that the sleeve will remain in the applied position.
Secure the air pump to the bottom of the bed or a stable surface.	Protects the device from damage and prevents injury to staff or visitors.

Attach the air tubes to the ports that extend from the sleeve and to the adapter within the air pump (Fig. B).	Provides a channel through which air is delivered to the extremity sleeve.
B	**Attaching air tubes so that the arrows align. (Photo by B. Proud.)**
Check that the air tubes are unkinked and not compressed under the client or the wheels of the bed.	Ensures the unobstructed delivery of air.
Plug the air pump into an electrical outlet.	Delivers power to the air pump motor.
Set the pressure on the air pump to the amount prescribed (most medical orders range from 35 to 55 mmHg, with a common average of 40 mmHg).	Provides intermittent compression at an appropriate pressure to promote venous circulation.
Turn the power switch on and observe that the function lights illuminate during compression and turn off between compressions.	Indicates that the machine is operational.
Assess the client's circulatory status and comfort every 2–4 hours throughout the therapeutic treatment, which is continuous for some clients.	Focuses assessment on signs that indicate adverse effects.
Remove the extremity sleeve before ambulation or other out-of-bed activities.	Allows freedom of movement from the tether of the air tubes and pump.
Discontinue the compressions if serious impairment of circulation and sensation, tingling, numbness, or leg pain occurs.	Helps to avoid serious complications.
Remove the extremity sleeve and assess calf size and circulation to distal areas of the extremity at least once per day.	Provides comparative data with which to evaluate the therapeutic response.
Apply elastic stockings and reinforce the need to perform leg exercises every hour when the machine is not in use.	Promotes venous circulation.
Place equipment in a safe area where it is available for the next use.	Demonstrates regard for safety and efficient time management.

EVALUATION

- Calf size is reduced or does not increase in diameter.
- Skin in lower extremity is intact, warm, and is the appropriate color for ethnicity.
- Capillary refill is less than 2–3 seconds.
- Pedal pulses are present and strong.

DOCUMENT

- Assessment findings before and after application
- Extremity to which device was applied
- Setting and duration of application
- To whom abnormal assessment findings have been reported and the outcome of the communication

SAMPLE DOCUMENTATION

Date and Time Right calf measures 18 inches (45 cm). Left calf is 20 inches (50 cm). Toes are warm. Blood returns to nail beds within 3 seconds of compression. Skin over legs is pink, warm, and intact. Homans' sign is negative bilaterally. Pneumatic compression device applied to calves of both legs and set at a pressure of 40 mmHg. _____ J.Doe, LPN

Date and Time Pneumatic compression device removed after 2 hours of use to facilitate bathing and reapplied at 40 mmHg. _____ J. Doe, LPN

28

Wound Care

Learning Objectives

On completion of this chapter, the reader should be able to:

1. Define the term wound.
2. Name three phases of wound repair.
3. Identify five signs and symptoms classically associated with the inflammatory response.
4. Discuss the purpose of phagocytosis, including the two types of cells involved.
5. Name three ways in which the integrity of a wound is restored.
6. Explain first-, second-, and third-intention healing.
7. Name two types of wounds.
8. State at least three purposes for using a dressing.
9. Explain the rationale for keeping wounds moist.
10. Describe two types of drains, including the purpose of each.
11. Name the two major methods for securing surgical wounds together until they heal.
12. Explain three reasons for using a bandage or binder.
13. Discuss the purpose for using one type of binder.
14. Give examples of four methods used to remove nonliving tissue from a wound.
15. List three commonly irrigated structures.
16. State two uses each for applying heat and for applying cold.
17. Identify at least four methods for applying heat and cold.
18. List at least five risk factors for developing pressure ulcers.
19. Discuss three techniques for preventing pressure ulcers.

Words to Know

aquathermia pad
bandage
binder

capillary action
closed wound
collagen
compresses
debridement
dehiscence
douche
drains
dressing
evisceration
first-intention healing
granulation tissue
hydrotherapy
inflammation
irrigation
leukocytosis
lysosomes
macrophages
maggot therapy
monocytes
Montgomery straps
necrotic tissue
negative pressure wound therapy (NPWT)
neutrophils
open wound
pack
phagocytosis
polymorphonuclear leukocytes
pressure ulcer
proliferation
purulent drainage
regeneration
remodeling
resolution
scar formation
second-intention healing
sepsis
serous drainage
shearing force

sitz bath
skin tear
slough
soak
staples
sutures
therapeutic baths
third-intention healing
trauma
undermining
vacuum-assisted closure
wound

Body tissues have a remarkable ability to recover when injured. This chapter discusses several types of tissue injury, including those caused by surgical incisions and prolonged pressure. It also addresses nursing interventions to support the healing process and actions to prevent tissue injury.

WOUNDS

A **wound** (damaged skin or soft tissue) results from **trauma** (a general term referring to injury). Examples of tissue trauma include cuts, blows, poor circulation, strong chemicals, and excessive heat or cold. Such trauma produces two basic types of wounds: open and closed (Table 28-1).

An **open wound** is one in which the surface of the skin or mucous membrane is no longer intact. It may be caused accidentally or intentionally, as when a surgeon incises the tissue. In a **closed wound**, there is no opening in the skin or mucous membrane. Closed wounds occur more often from blunt trauma or pressure.

TABLE 28-1 Types of Wounds

WOUND TYPES	DESCRIPTION
Open Wounds	
Incision	A clean separation of skin and tissue with smooth, even edges
Laceration	A separation of skin and tissue in which the edges are torn and irregular
Abrasion	A wound in which the surface layers of skin are scraped away
Avulsion	Stripping away of large areas of skin and underlying tissue, leaving cartilage and bone exposed
Ulceration	A shallow crater in which the skin or the mucous membrane is missing
Puncture	An opening of skin, underlying tissue, or mucous membrane caused by a narrow, sharp, pointed object
Closed Wounds	
Contusion	Injury to soft tissue underlying the skin from the force of contact with a hard object, sometimes called a bruise

WOUND REPAIR

Regardless of the type of wound, the body immediately attempts to repair the injury and heal the wound. The process of wound repair proceeds in three sequential phases: inflammation, proliferation, and remodeling.

Inflammation

Inflammation, the physiologic process immediately after tissue injury, lasts approximately 2 to 5 days. Its purposes are to (1) limit the local damage, (2) remove injured cells and debris, and (3) prepare the wound for healing. Inflammation progresses through several stages (Fig. 28-1).

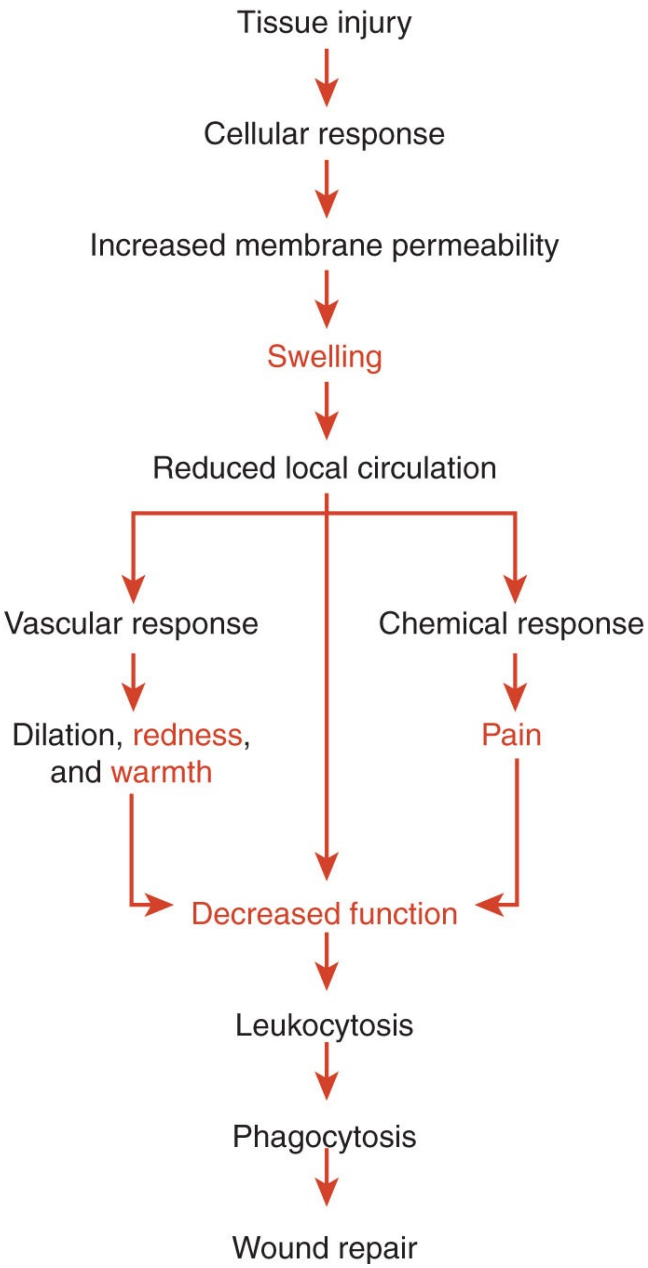


FIGURE 28-1 The inflammatory response. The words in *red* are the five classic signs and symptoms of inflammation.

During the first stage, local changes occur. Immediately following an injury, blood vessels constrict to control blood loss and confine the damage. Shortly thereafter, the blood vessels dilate to deliver platelets that form a loose clot. The membranes of the damaged cells become more permeable, causing the release of plasma and chemical substances that transmit a sensation of discomfort. The local response produces the characteristic signs and symptoms of inflammation:

swelling, redness, warmth, pain, and *decreased function.*

A second wave of defense follows the local changes when **polymorphonuclear leukocytes** (**neutrophils**) and **macrophages** (**monocytes**), types of white blood cells, migrate to the site of injury, and the body produces more and more white blood cells to take their place. **Leukocytosis** (an increased production of white blood cells) is confirmed and monitored by counting the number and type of white blood cells in a sample of the client's blood. The laboratory test is called a white blood cell count and differential count. Increased production of white blood cells, particularly neutrophils and monocytes, suggests an inflammatory and, in some cases, infectious process.

Neutrophils and monocytes are primarily responsible for **phagocytosis**, which is a process by which these cells emigrate from blood vessels to consume pathogens, coagulated blood, and cellular debris. Consumed substances are enclosed within **lysosomes**, enzymatic sacs inside the phagocytes that digest the engulfed matter. Once digestion occurs, the degraded products are released into extracellular fluid (Fig. 28-2). Collectively, neutrophils and monocytes clean the injured area and prepare the site for wound healing.

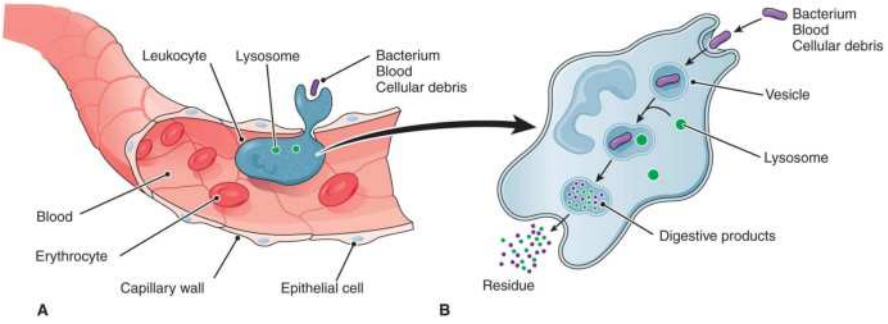


FIGURE 28-2 An example of the process of phagocytosis. (**A)** A phagocytic leukocyte exits through a capillary wall. (**B)** The debris is enclosed within a vesicle and digested by a lysosome. (From Cohen, B. J. [2012]. *Memmler's structure and function of the human body* [10th ed.]. Philadelphia, PA: Lippincott Williams & Wilkins.)

Proliferation

Proliferation (a period during which new cells fill and seal a wound) occurs from 2 days to 3 weeks after the inflammatory phase. It is characterized by the appearance of **granulation tissue** (a combination of new blood vessels, fibroblasts, and epithelial cells) that forms in the bed of an open wound. Granulation tissue has a somewhat irregular surface and looks bright pink to red because of the extensive projections of capillaries in the area (Fig. 28-3).

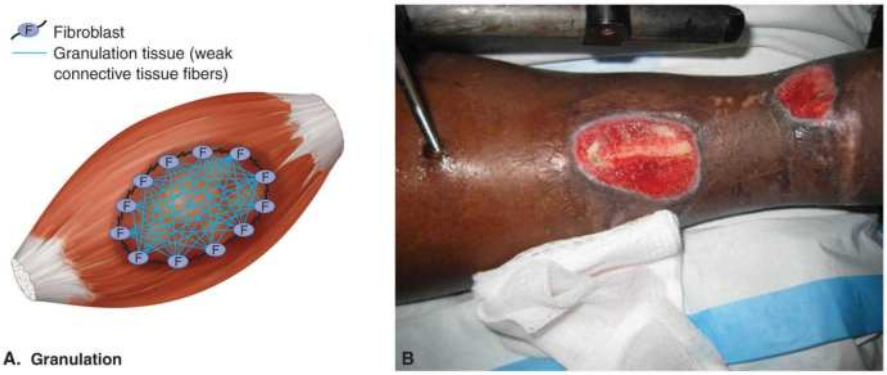


FIGURE 28-3 Wound repair. (**A)** To promote tissue healing, phagocytes decrease and fibroblasts increase. The fibroblasts migrate to the perimeter forming a web of granulation fibers. (**B)** A wound with healthy granulation tissue. (A, from Archer, P., & Nelson, L. A. [2012]. *Applied anatomy & physiology for manual therapists*. Philadelphia, PA: Lippincott Williams & Wilkins [PE]; B, from Sussman, C., & Bates-Jensen, B. [2012]. *Wound care* [4th ed.]. Philadelphia, PA: Lippincott Williams & Wilkins [PE].)

Granulation tissue grows from the wound margin toward the center. It is fragile and easily disrupted by physical or chemical means. As more and more fibroblasts produce **collagen** (a tough and inelastic protein substance), the adhesive strength of the wound increases. Toward the end of the proliferative phase, the new blood vessels degenerate, causing the previously pink color to regress. Epithelial cells use previously formed granulation tissue as a surface to move across as they re-cover the wound (Katz & Kirr, 2012).

Generally, the integrity of skin and damaged tissue is restored by (1) **resolution** (a process by which damaged cells recover and reestablish their normal function), (2) **regeneration** (cell duplication), or (3) **scar formation** (replacement of damaged cells with fibrous scar tissue). Fibrous scar tissue acts as a nonfunctioning patch. The extent of scar tissue that forms depends on the magnitude of tissue damage and the manner of wound healing (discussed later in this chapter).

Remodeling

Remodeling (a period during which the wound undergoes changes and maturation) follows the proliferative phase and may last 6 months to 2 years (Grossman & Porth, 2013). During this time, the wound contracts, and the scar shrinks.

WOUND HEALING

Several factors affect wound healing:

- Type of wound injury

- Expanse or depth of wound
- Quality of circulation
- Amount of wound debris
- Presence of infection
- Status of the client's health

The speed of wound repair and the extent of scar tissue that forms depend on whether the wound heals by first, second, or third intention (Fig. 28-4).

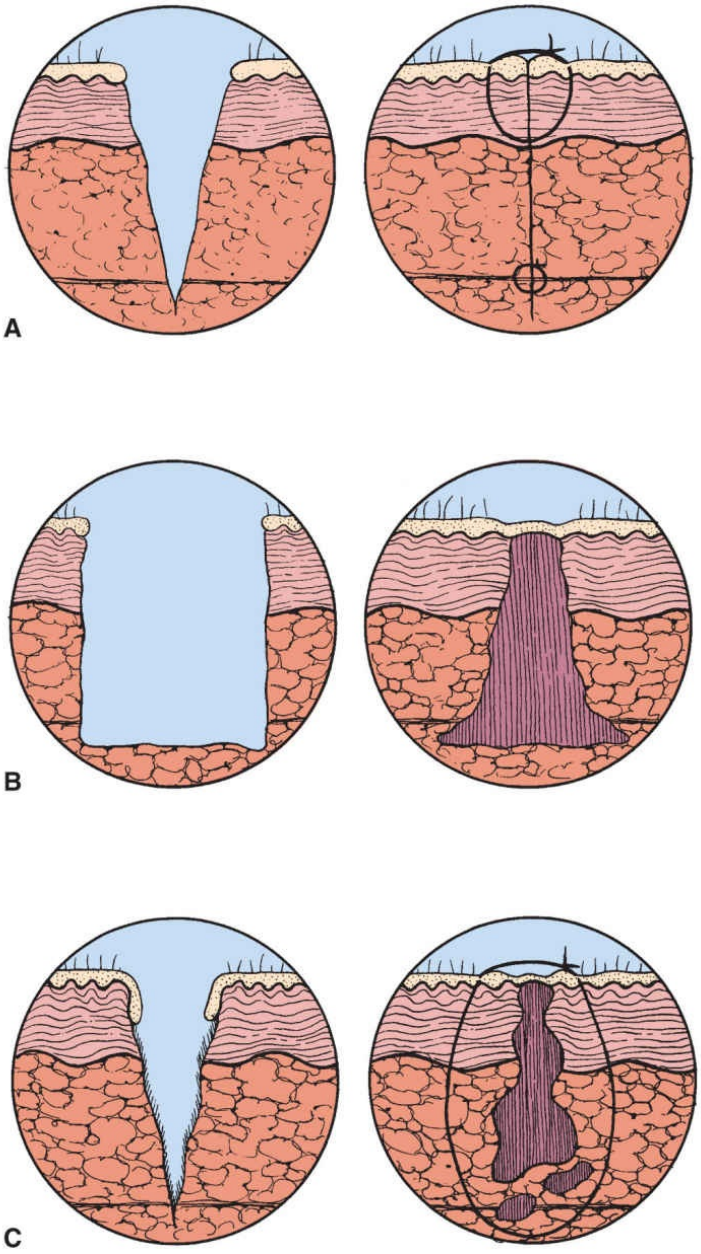


FIGURE 28-4 (**A)** First-intention healing. (**B)** Second-intention healing. (**C)** Third-intention healing.

First-intention healing, also called healing by primary intention, is a reparative process in which the wound edges are directly next to each other. Because the space between the wound is so narrow, only a small amount of scar tissue forms. Most surgical wounds that are closely approximated heal by first intention (Fig. 28-5).

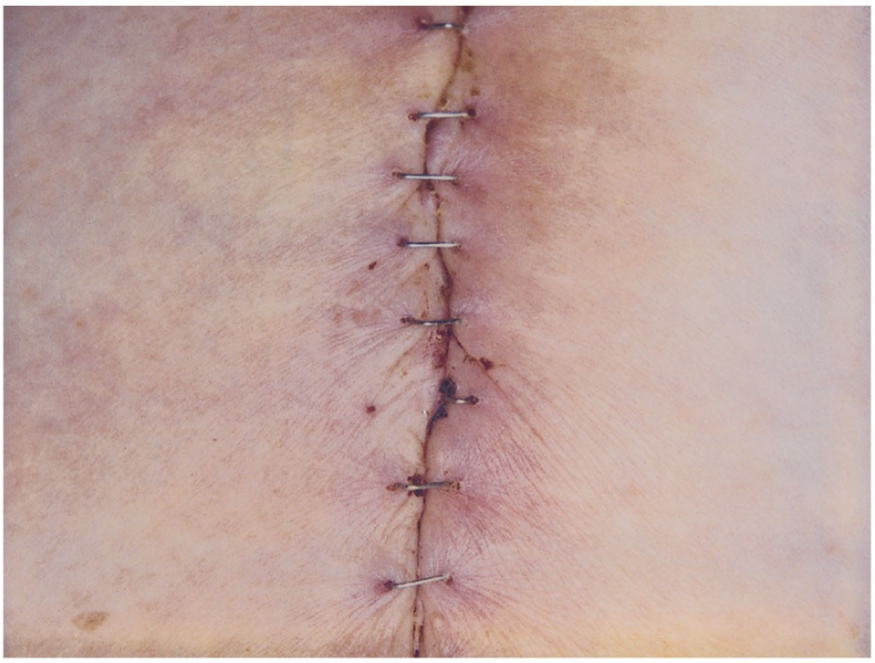


FIGURE 28-5 Example of a first-intention wound healing.

In **second-intention healing**, the wound edges are widely separated, leading to a more time-consuming and complex reparative process. Because the margins of the wound are not in direct contact, the granulation tissue needs additional time to extend across the expanse of the wound. Generally, a conspicuous scar results. Healing by second intention is prolonged when the wound contains body fluid or other wound debris. Wound care must be performed cautiously to avoid disrupting the granulation tissue and retarding the healing process.

With **third-intention healing**, the wound edges are intentionally left widely separated and are later brought together with some type of closure material. This reparative process results in a broad, deep scar. Generally, wounds that heal by third intention are deep and are likely to contain extensive drainage and tissue debris. To speed up healing, they may contain drainage devices or be packed with absorbent gauze.

Gerontologic Considerations

- Wound healing is delayed in older adults. Regeneration of healthy skin takes twice as long for an 80-year-old client as it does for a 30-year-old client.
- Age-related changes that affect wound healing include diminished collagen and blood supply and decreased quality of elastin, necessary components for wound repair. Long-term exposure to ultraviolet rays from the sun compounds these age-related changes.
- Factors such as depression, poor appetite, cognitive impairments, and physical or economic barriers that interfere with adequate nutrition in older adults may impair wound healing. These factors may be addressed by enlisting the assistance of registered dietitians, who can suggest appropriate nutritional interventions, and by making referrals to community resources such as home-delivered meals or homemaker/home health aide services.

Nutrition Notes

- Adequate protein, calories, and micronutrients are necessary for wound healing.
- The extent to which healing increases nutrient needs is dependent upon the severity of the wound and the client's nutritional status. For instance, extensive burns may increase protein needs two to three times above normal, whereas minor surgery may have no impact on nutrient needs.
- Although nutrient deficiencies are linked with poor wound healing, there is no magic bullet that will accelerate healing in an adequately nourished client.

WOUND-HEALING COMPLICATIONS

The key to wound healing is adequate blood flow to the injured tissue. Factors that may interfere include compromised circulation, infection, and purulent, bloody, or serous fluid accumulation that prevent skin and tissue approximation. In addition, excessive tension or pulling on wound edges contributes to wound disruption and delays healing. One or several of these factors may be secondary to poor nutrition, impaired inflammatory or immune responses related to drugs like corticosteroids, and obesity (see discussion on surgical risks in Chap. 27).

Gerontologic Considerations

- Diminished immune response from reduced T-lymphocyte cells predisposes

older adults to wound infections.
- Signs of inflammation may be subtle in older adults (see Chap. 27).
- Diabetes or other conditions that may interfere with circulation increase the older adult's susceptibility to delayed wound healing and wound infections.

The nurse assesses the wound to determine whether it is intact or shows evidence of unusual swelling, redness, warmth, drainage, and increasing discomfort. When assessing the wound it is important to look for **undermining**, erosion of tissue from underneath intact skin at the wound edge; **slough**, which is dead tissue on the wound surface that is moist, stringy, yellow, tan, gray, or green; and **necrotic tissue**, which is dry, brown or black devitalized tissue (Fig. 28-6). The latter two must be removed to facilitate wound healing (see the later discussion on debridement).

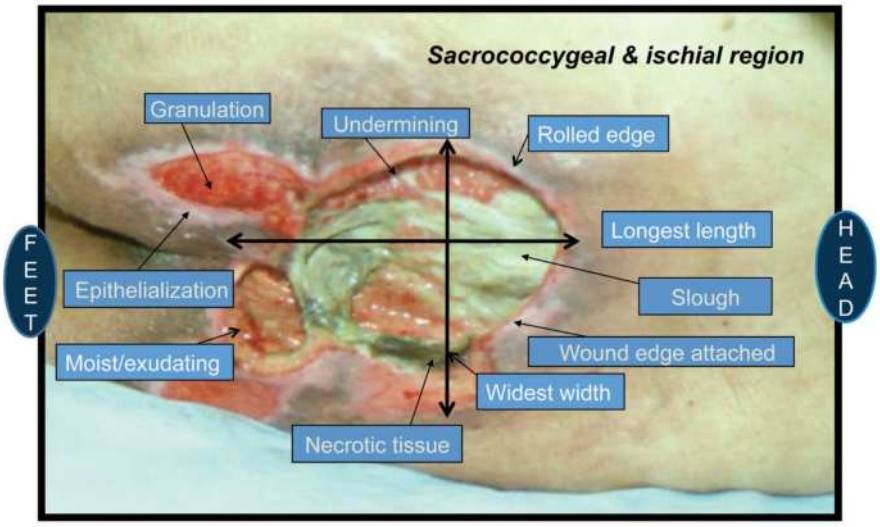


FIGURE 28-6 Components in wound assessment.

Two potentially serious surgical wound complications include **dehiscence** (the separation of wound edges) and **evisceration** (wound separation with the protrusion of organs) (Fig. 28-7). These complications are most likely within 7 to 10 days after surgery. They may be caused by insufficient dietary intake of protein and sources of vitamin C; premature removal of sutures or staples; unusual strain on the incision from severe coughing, sneezing, vomiting, dry heaves, or hiccupping; weak tissue or muscular support secondary to obesity; distention of the abdomen from accumulated intestinal gas; or compromised tissue integrity from previous surgical procedures in the same area.

The client may describe that something has "given way." Pinkish drainage

may appear suddenly on the dressing. If wound disruption is suspected, the nurse positions the client to put the least strain on the open area. If evisceration occurs, the nurse places sterile dressings moistened with normal saline over the protruding organs and tissues. For any wound disruption, the nurse notifies the physician immediately. The nurse must be alert for signs and symptoms of impaired blood flow such as swelling, localized pallor or mottled appearance, and coolness of the tissue in the area around the wound.

Stop, Think, and Respond 28-1

Discuss the signs and symptoms a person would exhibit if a wound were infected.

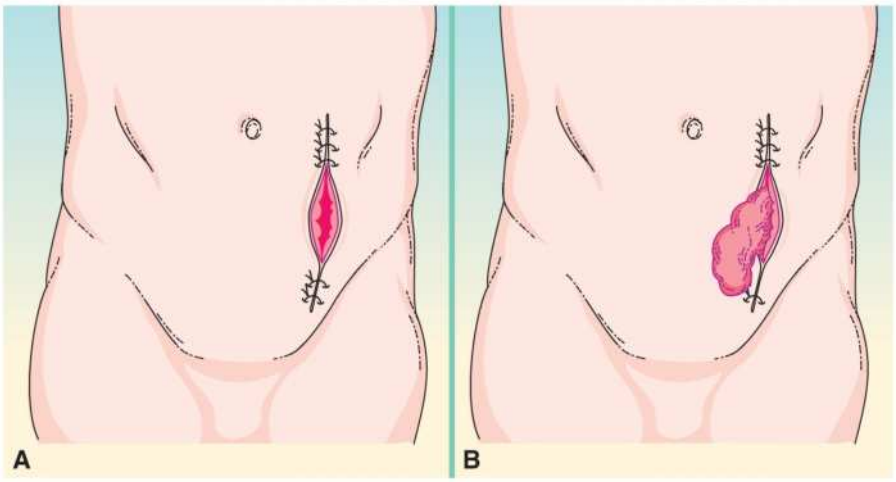


FIGURE 28-7 (**A)** Wound dehiscence. (**B)** Wound evisceration.

WOUND MANAGEMENT

Wound management involves techniques that promote wound healing. Surgical wounds result from incising tissue with a laser (see Chap. 27) or an instrument called a scalpel. The primary goal of surgical or open wound management is to reapproximate the tissue to restore its integrity. This involves changing dressings, caring for drains, removing sutures or staples when directed by the surgeon, applying bandages and binders, and performing wound irrigations.

Dressings

A **dressing** (the cover over a wound) serves one or more purposes:

- Keeping the wound clean
- Absorbing drainage
- Controlling bleeding

- Protecting the wound from further injury
- Holding medication in place
- Maintaining a moist environment

Types and sizes of dressings differ depending on their purpose. The most common wound coverings are gauze, transparent, hydrocolloid, hydrogel, and alginate dressings.

Gauze Dressings

Gauze dressings are made of woven cloth fibers. Their highly absorbent nature makes them ideal for covering fresh wounds that are likely to bleed or wounds that exude drainage. Unfortunately, gauze dressings obscure the wound and interfere with wound assessment. Unless an ointment is used on the wound or the gauze is lubricated with an ointment such as petroleum, granulation tissue may adhere to the gauze fibers and disrupt the wound when removed.

Gauze dressings usually are secured with tape. If gauze dressings need frequent changing, **Montgomery straps** (strips of tape with eyelets) may be used (Fig. 28-8). Another method may be necessary if the client is allergic to tape (see the discussion on bandages and binders later in this chapter).

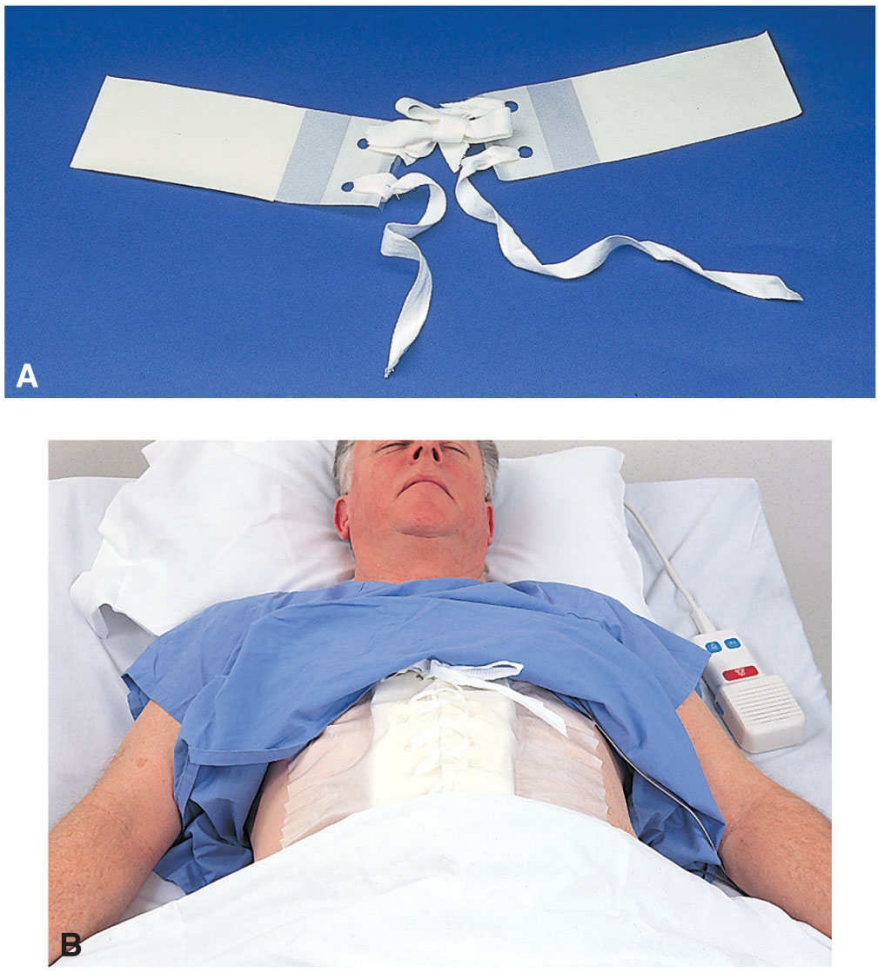


FIGURE 28-8 **(A)** The adhesive outer edge of Montgomery straps are applied to either side of a wound. **(B)** The inner edges of Montgomery straps are tied to hold a dressing over a wound. They prevent skin breakdown and wound disruption from repeated tape removal when checking or changing a dressing.

Transparent Dressings

Transparent dressings, such as OpSite and Tegaderm, are clear acrylic film wound coverings. One of their chief advantages is that they allow the nurse to assess a wound without removing the dressing. In addition, they are less bulky than gauze dressings and do not require tape because they consist of a single sheet of adhesive material (Fig. 28-9). They commonly are used to cover peripheral and central intravenous insertion sites. Transparent dressings are nonabsorbent; therefore, if wound drainage accumulates, they tend to loosen. Once a dressing is no longer intact, many of its original purposes are defeated.

FIGURE 28-9 A transparent dressing. (Photo by B. Proud.)

Hydrocolloid, Hydrogel, and Alginate Dressings

Hydrocolloid dressings, such as DuoDerm and Tegasorb, and hydrogel dressings such as DuoDerm and Intrasite are self-adhesive, opaque, air- and water-occlusive wound coverings. Hydrocolloids contain granules of gelatin or pectin in the matrix of the dressing (Fig. 28-10). The granules in hydrocolloids become gelatinous when in contact with exudate in a wound, keeping the wound moist. Hydrogels, which contain water and a network of fibers and alginates such as Algisite and Tegagel, which contain a seaweed component, have a similar function.

Moist wounds heal more quickly because new cells grow more rapidly in a wet environment. If the hydrocolloid dressing remains intact, it can be left in place for up to 1 week. Its occlusive nature also repels other body substances such as urine or stool. For proper use, these dressings must be sized generously, allowing at least a 1-inch margin of healthy skin around the wound.

Dressing Changes

Health care providers change dressings when a wound requires assessment or care and when the dressing becomes loose or saturated with drainage. In some cases, the physician may choose to assume total responsibility for changing the dressing—at least for the initial dressing change. Nurses, however, commonly *reinforce* dressings (apply additional absorbent layers), when dressings become

moist. Reinforcing a dressing prevents wicking (absorbing or drawing) microorganisms toward the wound (see Chap. 10).

Because most surgical wounds are covered with gauze dressings, this example is used when describing the technique for changing a dressing in Skill 28-1. When using dressings made of materials other than gauze, nurses can modify the technique by following the manufacturer's directions.

Drains

Drains are tubes that provide a means for removing blood and drainage from a wound. They promote wound healing by removing fluid and cellular debris. Although some drains are placed directly within a wound, the current trend is to insert them so that they exit from a separate location beside the wound. This approach keeps the wound margins approximated and avoids a direct entry site for pathogens. The physician may choose to use an open or closed drain.

FIGURE 28-10 A hydrocolloid dressing absorbs drainage into its matrix.

Open Drains

Open drains are flat, flexible tubes that provide a pathway for drainage toward the dressing. Draining occurs passively by gravity and **capillary action** (the movement of a liquid at the point of contact with a solid, which in this case is the

drain). Sometimes, a safety pin or long clip is attached to the drain as it extends from the wound. This prevents the drain from slipping within the tissue. As the drainage decreases, the physician may instruct the nurse to shorten the drain, enabling healing to take place from inside toward the outside of the wound. To shorten a drain, the nurse pulls it from the wound for the specified length. He or she then repositions the safety pin or clip near the wound to prevent the drain from sliding back internally within the wound (Fig. 28-11).

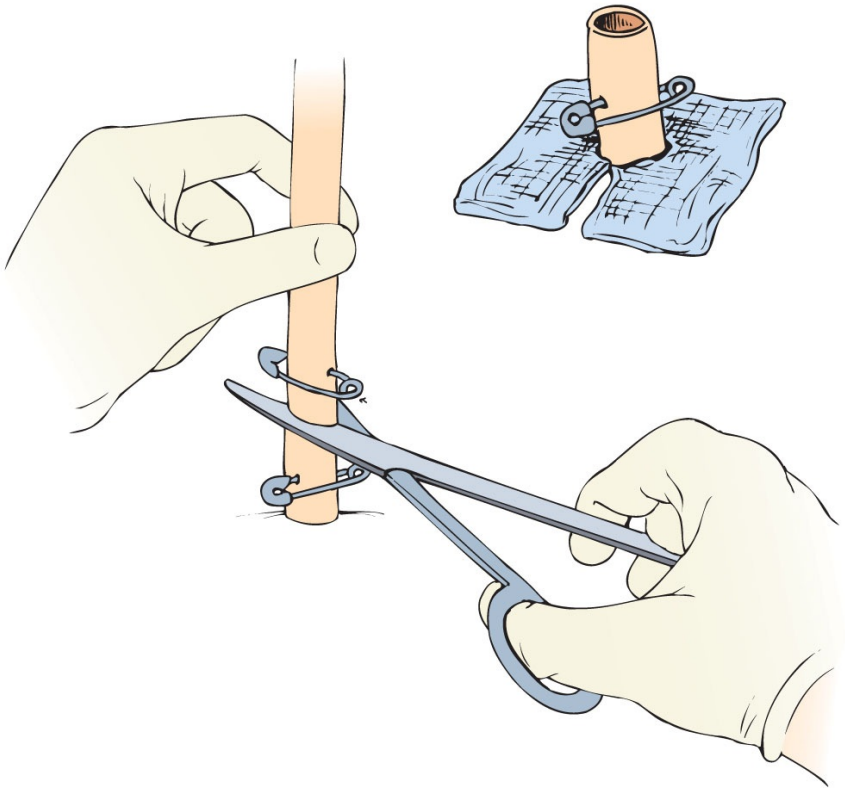


FIGURE 28-11 An open drain is pulled from the wound, and the excess portion is cut. A drain sponge is placed around the drain, and the wound is covered with a gauze dressing.

FIGURE 28-12 A Jackson-Pratt (closed) drain. (Photo by B. Proud.)

Closed Drains

Closed drains are tubes that terminate in a receptacle. Some examples of closed drainage systems are the Hemovac and the Jackson-Pratt drain (Fig. 28-12). Closed drains are more efficient than open drains because they pull fluid by creating a vacuum or negative pressure. This is done by opening the vent on the receptacle, compressing the drainage collection chamber, then capping the vent (Fig. 28-13).

FIGURE 28-13 Compressing the bulb on a Jackson-Pratt drain and capping the vent reestablishes negative

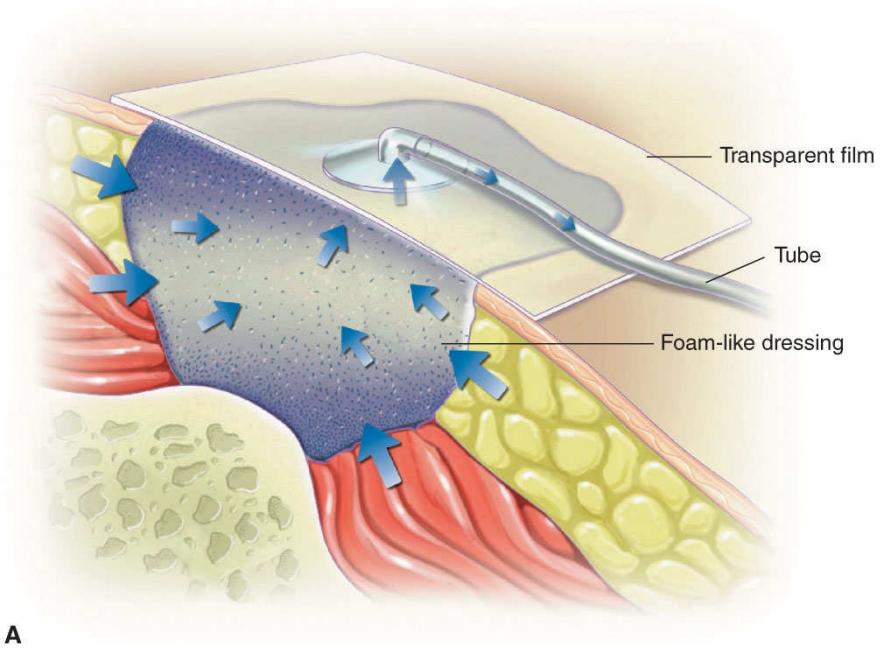
pressure that allows the collection of wound drainage.

When caring for a wound with a drain, the nurse cleans the insertion site in a circular manner from the center outward. After cleansing, he or she places a precut drain sponge or gauze, which is open to its center, around the base of the drain. An open drain may require additional layers of gauze because the drainage does not collect in a receptacle.

Vacuum-Assisted Closure

Wound healing is complicated when the edges are so widely separated they cannot be approximated with sutures or staples. Consequently, reepithelialization across the surface of the wound is delayed due to inadequate formation of new capillaries, the presence of cellular debris including an accumulation of tissue fluid, or an infectious process, any of which leads to a chronic wound.

With **vacuum-assisted closure** (VAC) also known as **negative pressure wound therapy (NPWT)**, it is possible to promote more rapid wound healing. The process involves a foam filler within the wound that is covered with a sealed occlusive dressing connected to a suction tube and pump (Fig. 28-14). When suction is applied either intermittently or continuously, fluid and debris in the entire wound bed is pulled through the foam filler into a collection canister. As a result, the wound gradually shrinks, cellular growth is promoted, blood flow increases, and healing improves. On average, the wound dressing is changed two or three times per week. Wound healing may take 6 months or longer.



A

B

FIGURE 28-14 (**A)** Schematic of the components of a vacuum-assisted closure (VAC) system. (**B)** A commercial VAC system. (Courtesy of KCI Licensing, Inc. 2010.)

Sutures and Staples

Sutures, knotted ties that hold an incision together, generally are constructed from silk or synthetic materials such as nylon. **Staples** (wide metal clips) perform a similar function. Staples do not encircle a wound like sutures; instead, they form a bridge that holds the two wound margins together. Staples are advantageous because they do not compress the tissue if the wound swells.

Sutures and staples are left in place until the wound has healed sufficiently to prevent reopening. Depending on the location of the incision, this may be a few days to as long as 2 weeks. The physician may direct the nurse to remove sutures and staples (Fig. 28-15), sometimes every other one on one day and the other

half on another day. Adhesive Steri-Strip skin closures, also known as *butterflies* because of their winged appearance, can hold a weak incision together temporarily. Sometimes Steri-Strips are used instead of sutures or staples to close superficial lacerations.

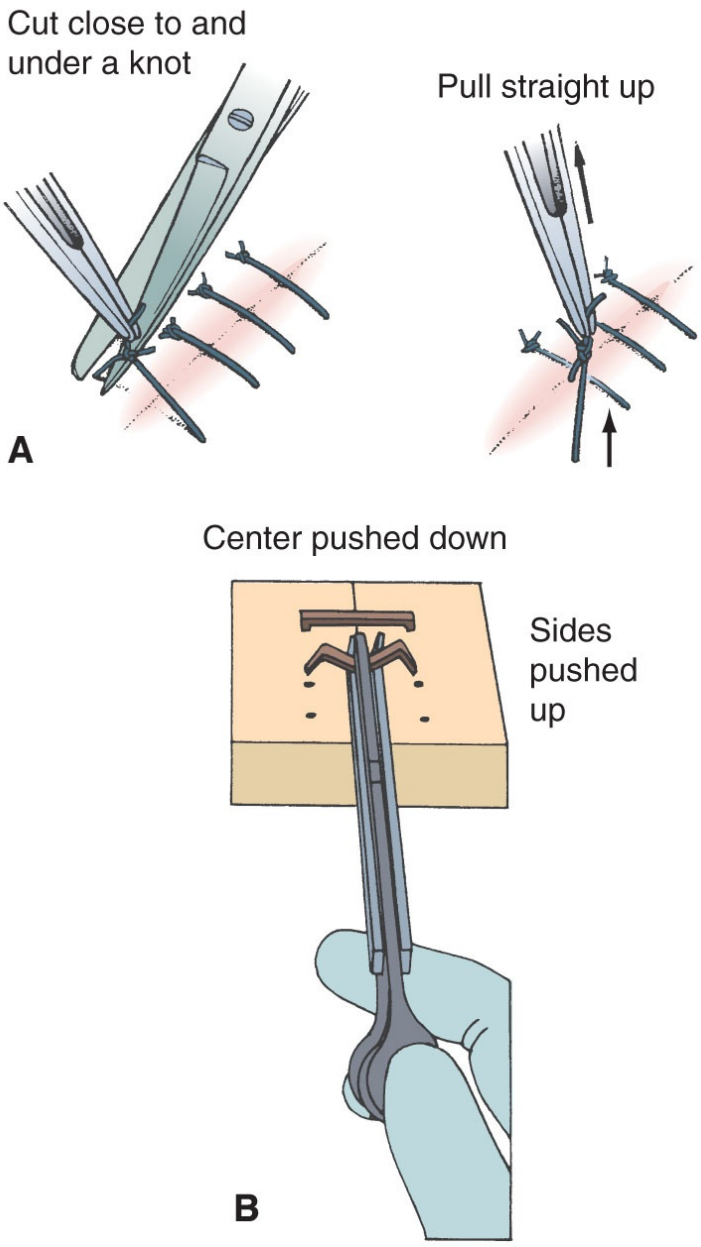


FIGURE 28-15 (**A)** A technique for suture removal. (**B)** A technique for staple removal

Bandages and Binders

A **bandage** is a strip or roll of cloth wrapped around a body part. One example is

an Ace bandage. A **binder** is a type of cloth cover generally applied to a particular body part such as the abdomen or breast. Bandages and binders are made from gauze, muslin, elastic rolls, and stockinette (see Chap. 25).

Bandages and binders serve various purposes:

- Holding dressings in place, especially when tape cannot be used or if the dressing is extremely large
- Supporting the area around a wound or injury to reduce pain
- Limiting movement in the wound area to promote healing

Concept Mastery Alert

Bandages do not help to keep a wound clean. Dressings do. Bandages hold dressings in place, support the area and limit movement.

Roller Bandage Application

Most bandages are prepared in rolls of varying widths. The nurse holds the end in one hand while passing the roll around the part being bandaged. Nurses follow several principles when applying a roller bandage:

- Elevate and support the limb.
- Wrap from a distal to proximal direction.
- Avoid gaps between each turn of the bandage.
- Exert equal, but not excessive, tension with each turn.
- Keep the bandage free of wrinkles.
- Secure the end of the roller bandage with metal clips.
- Check the color and sensation of exposed fingers or toes often.
- Remove the bandage for hygiene and replace at least twice a day.

Six basic techniques are used to wrap a roller bandage (Fig. 28-16): circular turn, spiral turn, spiral-reverse turn, figure-of-eight turn, spica turn, and recurrent turn.

A *circular turn* is used to anchor and secure a bandage where it starts and ends. It simply involves holding the free end of the rolled material in one hand and wrapping it around the area, bringing it back to the starting point.

A *spiral turn* partly overlaps a previous turn. The amount of overlapping varies from one-half to three-fourths of the width of the bandage. Spiral turns are used when wrapping cylindrical parts of the body such as the arms and legs.

A *spiral-reverse turn* is a modification of a spiral turn. The roll is reversed or turned downward halfway through the turn.

A *figure-of-eight turn* is best when bandaging a joint such as the elbow or knee. This pattern is made by making oblique turns that alternately ascend and descend, simulating the number eight.

A *spica turn* is a variation of the figure-of-eight pattern. It differs in that the wrap includes a portion of the trunk or chest (see Spica Cast, Chap. 25).

A *recurrent turn* is made by passing the roll back and forth over the tip of a body part. Once several recurrent turns are made, the bandage is anchored by completing the application with another basic turn such as the figure-of-eight turn. A recurrent turn is especially beneficial when wrapping the stump of an amputated limb or the head.

Binder Application

Binders are not used as commonly as bandages; more convenient commercial devices have largely replaced binders. For example, brassieres frequently are used instead of breast binders. Sometimes after rectal or vaginal surgery, nurses apply a T-binder which, as the name implies, looks like the letter T. T-binders are used to secure a dressing to the anus or perineum or within the groin. To apply a T-binder, the nurse fastens the crossbar of the T around the waist. Then, he or she passes the single or double tails between the client's legs and pins the tails to the belt (Fig. 28-17). Adhesive sanitary napkins worn inside underwear briefs are an alternative to a T-binder for stabilizing absorbent materials.

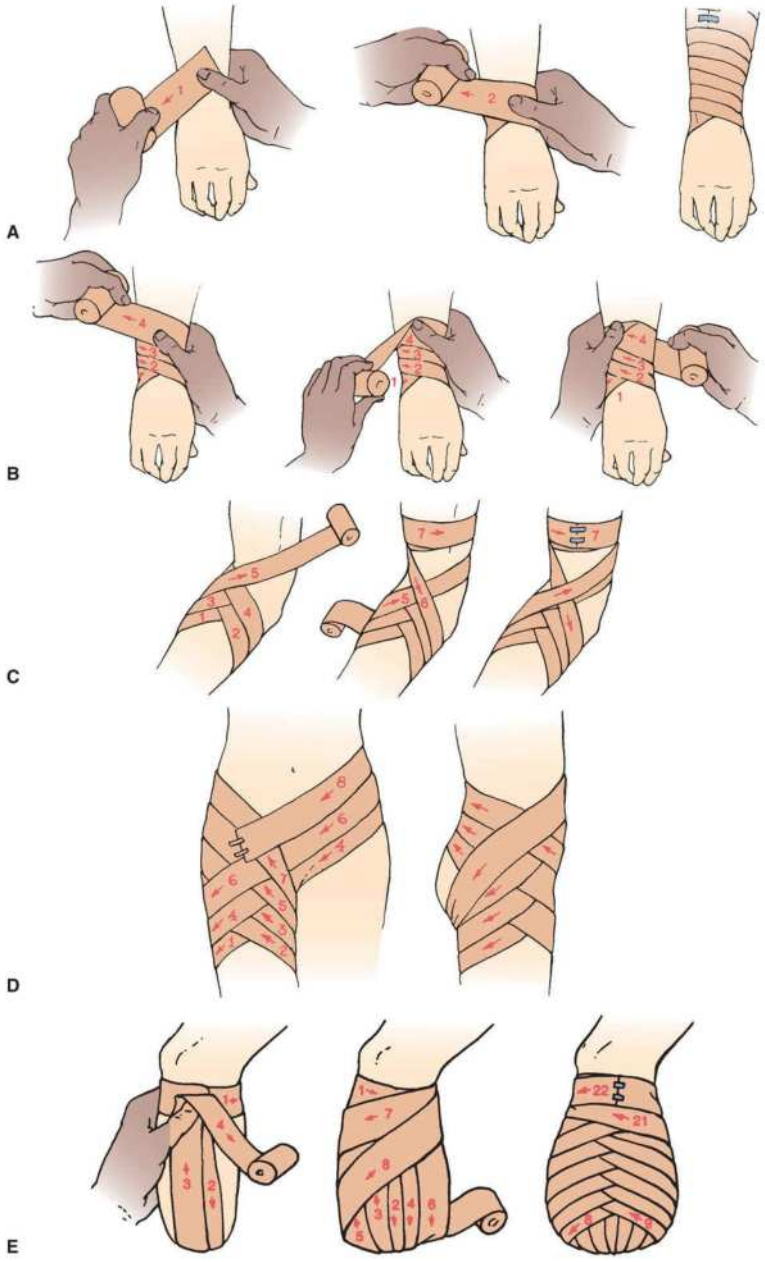


FIGURE 28-16 **(A)** A circular and spiral turn. **(B)** A spiral-reverse turn. **(C)** A figure-of-eight turn. **(D)** A spica turn. **(E)** A recurrent turn.

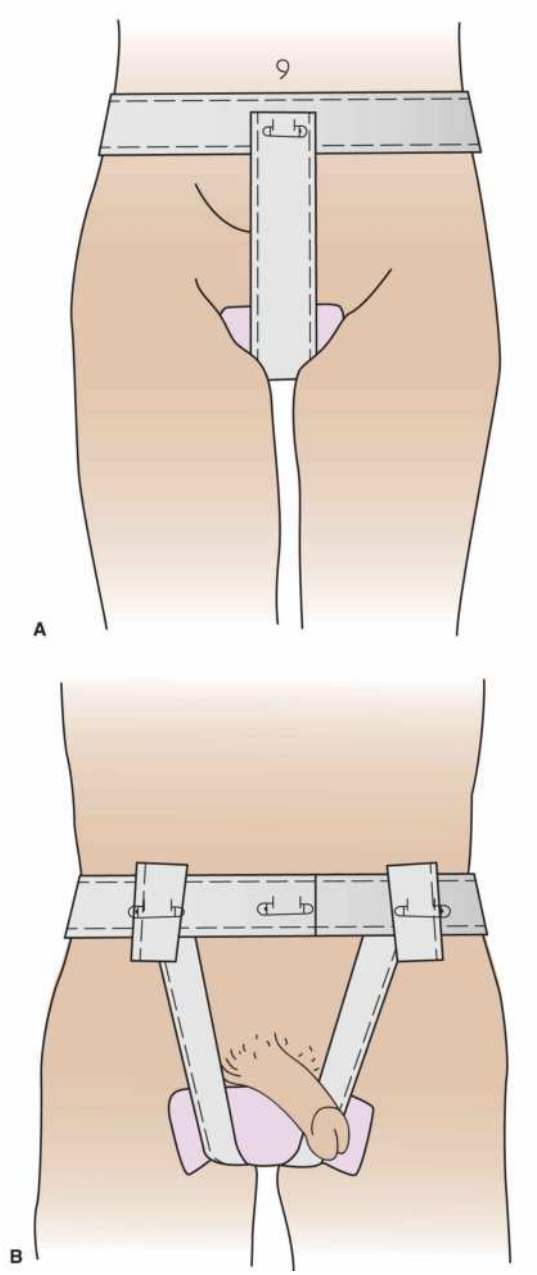


FIGURE 28-17 **(A)** A single T-binder. **(B)** A double T-binder. (From *Lippincott's nursing procedures and skills 2009*. Philadelphia, PA: Lippincott Williams & Wilkins.)

Debridement

Most wounds heal rapidly with conventional care. Nevertheless, some wounds require **debridement** (the removal of dead tissue) to promote healing. Four methods of debridement are sharp, enzymatic, autolytic, and mechanical.

Sharp Debridement

Sharp debridement is the removal of necrotic nonliving tissue from the healthy

areas of a wound with sterile scissors, forceps, or other instruments (Fig. 28-18). This method is preferred if the wound is infected because it helps the wound heal more quickly. The procedure is done at the bedside or in the operating room if the wound is extensive. Sharp debridement is painful, and the wound may bleed afterward.

FIGURE 28-18 A sharp debridement at the bedside.

Enzymatic Debridement

Enzymatic debridement involves the use of topically applied chemical substances that break down and liquefy wound debris. A dressing is used to keep the enzyme in contact with the wound and to help absorb the drainage. This form of debridement is appropriate for uninfected wounds or for clients who cannot tolerate sharp debridement.

Autolytic Debridement

Autolytic debridement, or self-dissolution, is a painless, natural physiologic process that allows the body's enzymes to soften, liquefy, and release devitalized tissue. It is used when a wound is small and free of infection. The main disadvantage in autolysis is the prolonged time it takes to achieve desired results. To accelerate autolysis, an occlusive or semiocclusive hydrocolloid, hydrogel, or calcium alginate dressing keeps the wound moist. Because removal of tissue debris is slow, the nurse monitors the client closely for signs of wound infection.

Mechanical Debridement

Mechanical debridement involves the physical removal of debris from nonhealing wounds. One technique is **maggot therapy,** which was approved by the Food and Drug Administration in 2004. Medical maggots are the larvae of a species of blow flies. Once the maggots are sterilized by a supplying laboratory, they are deposited into the wound. The maggots secrete an enzyme that dissolves dead tissue leaving healthy tissue alone. The maggots are confined beneath a dressing that provides air to the maggots and drainage of the liquefied tissue. When the debridement is completed, the maggots are transferred into a double-bagged biohazard bag.

Another approach to the mechanical removal of wound debris is **hydrotherapy** (the therapeutic use of water), in which the body part with the wound is submerged in a whirlpool tank. The agitation of the water, which contains an antiseptic, softens the dead tissue. Loose debris that remains attached is removed afterward by sharp debridement.

A third method for mechanically removing wound debris is **irrigation** (a technique for flushing debris). An irrigation is used when caring for a wound and also when cleaning an area of the body such as the eyes, ears, and vagina.

Wet-to-dry dressings were used in the past as a form of debridement in which moist gauze was packed within the wound and allowed to dry. When the dry dressing was removed, dead tissue that was embedded within the dry gauze was also removed. This type of debridement is now considered substandard practice because it disrupts the formation of new blood vessels and increases costs and time as well as the risk for infection from multiple dressing changes.

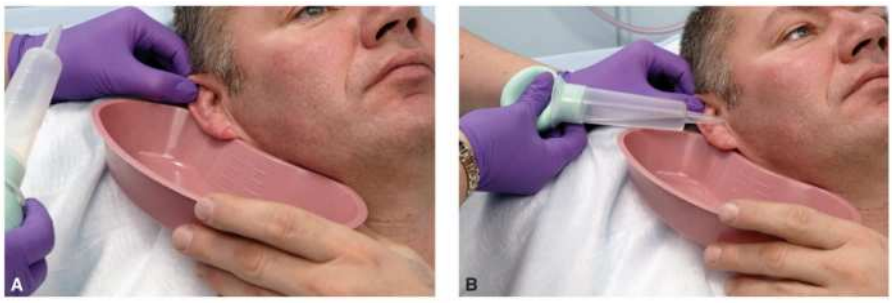


FIGURE 28-19 Ear irrigation **(A and B)**.

Stop, Think, and Respond 28-2

State an advantage and disadvantage of methods used for wound debridement.

Wound Irrigation

Wound irrigation (Skill 28-2) generally is carried out just before applying a new dressing. This technique is best used when granulation tissue has formed. Surface debris should be removed gently without disturbing the healthy proliferating cells.

Eye Irrigation

Eye irrigation flushes a toxic chemical from one or both eyes or displaces dried mucus or other drainage that accumulates from inflamed or infected eye structures (see Nursing Guidelines 28-1).

Ear Irrigation

Ear irrigation removes debris from the ear. Ear irrigation is contraindicated if the tympanic membrane (eardrum) is perforated. Performing a gross inspection of the ear is important if a foreign body is suspected because a bean, pea, or other dehydrated substance can swell if the ear is irrigated, causing it to become even more tightly fixed. Solid objects may require removal with an instrument.

If ear irrigation is not contraindicated, it is performed much like an eye irrigation except that the nurse directs the solution toward the roof of the auditory canal (Fig. 28-19). Also, the nurse takes care to avoid occluding the ear canal with the tip of the syringe because the pressure of the trapped solution could rupture the eardrum. After the irrigation, the nurse places a cotton ball loosely within the ear to absorb drainage but not to obstruct its flow.

NURSING GUIDELINES 28-1

Eye Irrigation

- Assemble supplies: a bulb syringe, an irrigating solution, gauze squares, gloves and other standard precaution apparel, absorbent pads, and at least one towel. *Assembling equipment ahead of time ensures organization and efficient time management.*
- Warm the solution to approximately body temperature by placing the container in warm water except when administering emergency first aid. *A warm solution is more comfortable for the client.*
- Position the client with the head tilted slightly toward the side. *This position facilitates drainage.*
- Place absorbent material in the area of the shoulder (Fig. A). *Use of absorbent material prevents saturating the client's gown and bed linen.*

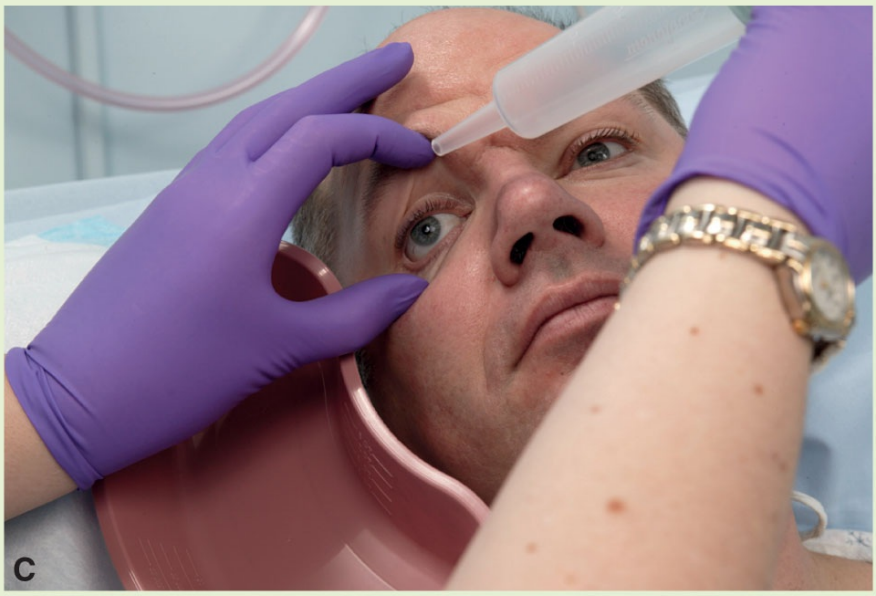
Protecting the client's gown and bed linen. (From *Lippincott's nursing procedures and skills 2007*. Philadelphia, PA: Lippincott Williams & Wilkins.)

- Give the client an emesis basin to hold beneath the cheek (Fig. B). *The basin can be used to collect the irrigating solution.*

An emesis basin collects irrigant. (From *Lippincott's nursing procedures and skills 2008*. Philadelphia, PA: Lippincott Williams & Wilkins.)

- Wash hands or use an alcohol-based hand rub and put on gloves. *Hand hygiene and glove use reduce the transmission of microorganisms.*
- Open and prepare supplies. *This enables the nurse to perform the irrigation efficiently.*
- Wipe a moistened gauze square from the nasal corner of the eye toward the temple; use additional

gauze squares, one at a time, as needed. *This removes gross debris.*
- Separate the eyelids widely with the fingers of one hand. *This action widens the exposed surface area.*
- Direct the solution onto the conjunctiva, holding the syringe or irrigating device about 1 in. (2.5 cm) above the eye (Fig. C). *Holding the syringe away from the eye prevents injury to the cornea.*



Instilling the irrigation onto the conjunctiva. (From *Lippincott's nursing procedures and skills 2007*. Philadelphia, PA: Lippincott Williams & Wilkins.)

- Instruct the client to blink periodically. *Blinking distributes solution under the eyelid and around the eye.*
- Continue irrigating until the debris is removed. *This accomplishes the desired result.*
- Dry the client's face and replace a wet gown or linen. *These actions promote client comfort.*
- Dispose of soiled materials and gloves; wash hands. *These measures reduce the transmission of microorganisms.*
- Record assessment data, the specifics of the procedure, and the outcome. *Documentation records performance of the nursing intervention and the client's response.*

Vaginal Irrigation

A vaginal irrigation, also known as a **douche** (a procedure for cleansing the vaginal canal), is sometimes necessary to treat an infection (see Client and Family Teaching 28-1).

Heat and Cold Applications

Heat and cold have various therapeutic uses (Box 28-1), and each can be used in several ways. Examples include an ice bag, collar, chemical pack, compress, and aquathermia pad. Heat is also applied with soaks, moist packs, and therapeutic baths.

The terms *hot* and *cold* are subject to wide interpretation. Table 28-2

correlates common terms with temperature ranges. Because exposing the skin to extremes of temperature can result in injuries, the nurse assesses the temperature of the application and frequently monitors the condition of the skin. Direct contact between the skin and the heating or cooling device is avoided. Hot and cold applications are used cautiously in children younger than 2 years, older adults, clients with diabetes, and clients who are comatose or neurologically impaired.

Client and Family Teaching 28-1 Douching

The nurse teaches the client or the family as follows:

- Do not douche routinely because douching removes microbes, called *Döderlein bacilli*, that help prevent vaginal infections.
- Do not douche 24–48 hours before a Pap test (see Chap. 14). Douching may wash away diagnostic cells.
- Consult a physician about symptoms such as itching, burning, or drainage rather than attempting self-diagnosis.
- Find out from the physician if sexual partners also need to be treated with medications to avoid reinfection.
- Buy douching equipment from a drugstore; prefilled disposable containers are available.
- Warm the solution to a comfortable temperature (no more than 110°F [43.3°C]).
- Clamp the tubing (on reusable equipment) and fill the reservoir bag.
- Undress and lie down in the bathtub.
- Suspend the douche bag (if used) about 18–24 in. (45–60 cm) above the hips.
- Insert the lubricated tip of the nozzle or the prefilled container downward and backward within the vagina about the distance of a tampon.
- Unclamp the tubing and rotate the nozzle as the fluid is instilled.
- Contract the perineal muscles as though trying to stop urinating and then relax the muscles. Repeat the exercise four or five times while douching.
- Sit up to facilitate drainage or shower afterward.
- Use a sanitary napkin or perineal pad to absorb residual drainage.

BOX 28-1 Common Uses for Heat and Cold Applications

USES FOR HEAT	USES FOR COLD
Provides warmth	Reduces fevers
Promotes circulation	Prevents swelling
Speeds healing	Controls bleeding
Relieves muscle spasm	Relieves pain
Reduces pain	Numbs sensation

TABLE 28-2 Temperature Ranges for Applications of Heat and Cold

LEVEL OF HEAT OR COLD	TEMPERATURE RANGE
Very hot	40.5°–46.1°C (105°–115°F)
Hot	36.6°–40.5°C (98°–105°F)
Warm and neutral	33.8°–36.6°C (93°–98°F)
Tepid	26.6°–33.8°C (80°–93°F)
Cool	18.3°–26.6°C (65°–80°F)
Cold	10°–18.3°C (50°–65°F)
Very cold	Below 10°C (below 50°F)

Gerontologic Considerations

■ The risk for thermal skin injury is increased in older adults with impaired tactile sensation or sensory nerve damage because of circulatory or neurologic disorders. Older adults who have problems with the ability to sense temperatures need to take special precautions such as using a thermometer to ensure that applications involving heat are less than 100°F (38°C) to avoid burns or injury.

Ice Bag and Ice Collar

Ice bags and ice collars are containers for holding crushed ice or small ice cubes (Fig. 28-20). Ice collars usually are applied after tonsil removal. Ice bags are applied to any small injury in the process of swelling. Although ice bags are available commercially, they can also be improvised. A rubber or plastic glove, a plastic bag with a zipper closure, or a bag of small frozen vegetables, such as peas, can be used. Client instruction minimizes the risk for injury (see Client and Family Teaching 28-2).

FIGURE 28-20 An ice bag filled with crushed ice.

Client and Family Teaching 28-2 Using an Ice Bag

The nurse teaches the client or the family as follows:

- Test the ice bag for leaks.
- Fill it one-half to two-thirds full of crushed ice or small cubes so it can be molded easily to the injured area.
- Eliminate as much air from the bag as possible.
- Pour water over the ice to provide slight melting. This tends to smooth the sharp edges from frozen ice crystals.
- Cover the ice bag with a layer of cloth before placing it on the body.
- Leave the ice bag in place no more than 20–30 minutes. Allow the skin and tissue to recover for at least 30 minutes before reapplying.
- If the skin becomes mottled or numb, remove the ice bag—it is too cold.

Chemical Packs

Commercial cold packs are struck or crushed to activate the chemicals inside, causing them to become cool. Most first-aid kits generally include this type of cold pack. Commercial cold packs can be used only once. Gel packs, designed for cold or hot application, are reusable. They are stored in the freezer until needed or heated in a microwave.

Compresses

Compresses (moist, warm, or cool cloths) are applied to the skin. Before applying the compress, the nurse soaks it in tap water or a medicated solution at the appropriate temperature and then wrings out excess moisture. To maintain

the moisture and temperature, a piece of plastic or plastic wrap is used to cover the compress, and the area is secured in a towel. As the compress material cools or warms outside the range of the intended temperature, the nurse removes it and reapplies it if necessary.

If the skin is not intact, as in the case of a draining wound, nurses wear gloves when applying a compress. They use aseptic surgical technique when applying compresses to an open wound.

Aquathermia Pad

An **aquathermia pad** (an electrical heating or cooling device) is sometimes called a K-pad. It resembles a mat but contains hollow channels through which heated or cooled distilled water circulates (Fig. 28-21). An aquathermia pad is used alone or as a cover over a compress. A thermostat is used to keep the temperature of the water at a specified setting. As with other forms of hot and cold therapeutic devices, the nurse assesses the skin frequently and removes the device periodically.

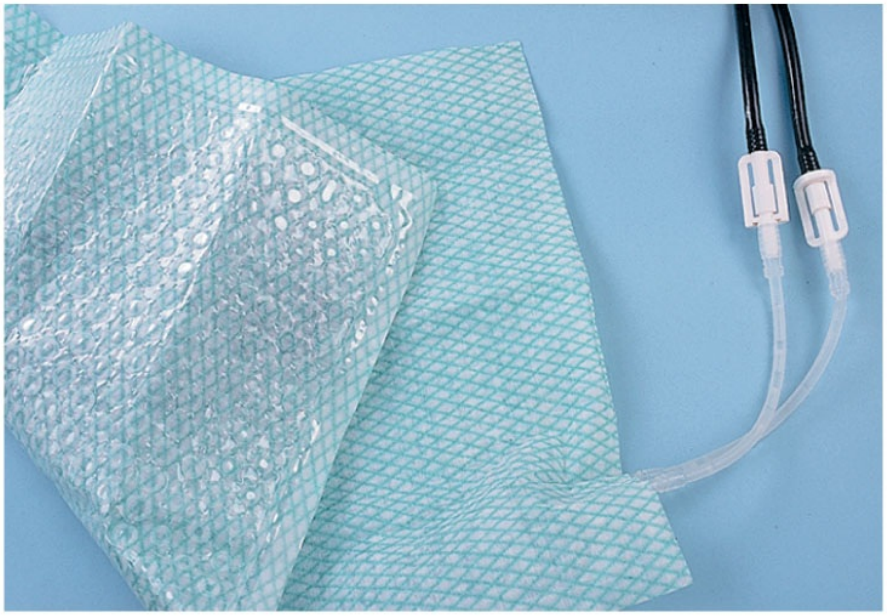


FIGURE 28-21 An aquathermia pad (K-pad). (Photo by B. Proud.)

Before placing the client on the aquathermia pad or wrapping it around a body part, the nurse covers the pad to help prevent thermal skin damage. A roller bandage may help hold the pad in place. The nurse positions the electrical unit slightly higher than the client to promote gravity circulation of the fluid.

Larger styles are used to warm clients who are hypothermic or to cool those with heat stroke. Because these clients have dangerously altered body

temperatures, the nurse must monitor vital signs continuously.

Soaks and Moist Packs

A **soak** is a technique in which a body part is submerged in fluid to provide warmth or to apply a medicated solution. A **pack** (a commercial device for applying moist heat) also can be used. Moist heat is more comforting and therapeutic than dry heat.

A soak usually lasts 15 to 20 minutes. The nurse keeps the temperature of the fluid as constant as possible, which requires frequent emptying and refilling of the basin. The newly added water should not be too hot; overly hot water causes discomfort or tissue damage.

Packs differ from soaks in two major ways: the duration of the application is usually longer, and the initial application of heat is generally more intense. Packs usually are applied at temperatures as warm as the client can tolerate. Because of the potential for causing burns, a pack never is used on a client who is unresponsive or paralyzed and cannot perceive temperatures. The nurse must make frequent assessments and remove the pack if there is any likelihood of a thermal injury.

Therapeutic Baths

Therapeutic baths (those performed for other than hygiene purposes) help reduce a high fever or apply medicated substances to the skin to treat skin disorders or discomfort. Examples are baths to which sodium bicarbonate (baking soda), cornstarch, or oatmeal paste is added.

The most common type of therapeutic bath is a **sitz bath** (a soak of the perianal area). Sitz baths reduce swelling and inflammation and promote healing of wounds after a *hemorrhoidectomy* (the surgical removal of engorged veins inside and outside the anal sphincter) or an *episiotomy* (an incision that facilitates vaginal birth). Some health care agencies have special tubs for administering sitz baths, but most provide clients with disposable equipment (Skill 28-3).

Stop, Think, and Respond 28-3

What assessment findings suggest that a sitz bath is providing a therapeutic effect?

PRESSURE ULCERS

A **pressure ulcer**, also referred to as a *decubitus ulcer*, is a wound caused by prolonged capillary compression that is sufficient to impair circulation to the

skin and underlying tissue. Pressure ulcers most often appear over bony prominences of the sacrum, hips, and heels. They also can develop in other locations such as the elbows, shoulder blades, the back of the head, and places where pressure is unrelieved because of infrequent movement (Fig. 28-22). The tissue in these areas is particularly vulnerable because body fat, which acts as a pressure-absorbing cushion, is minimal. Consequently, the tissue is compressed between the bony mass and a rigid surface such as a chair seat or a bed mattress. If the compression reduces the pressure in local capillaries to less than 32 mmHg for 1 to 2 hours without intermittent relief, the cells die from a lack of oxygen and nutrients. The primary goal in managing pressure ulcers is prevention. Once a pressure ulcer forms, however, the nurse implements measures to reduce its size and restore skin and tissue integrity.

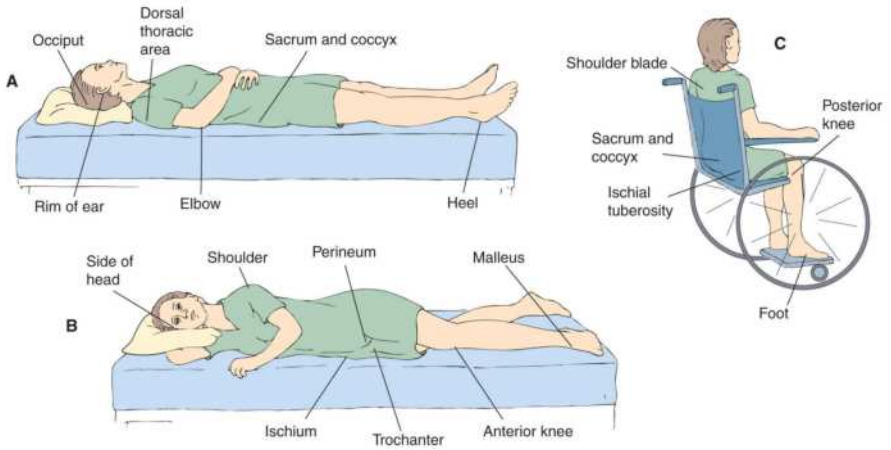


FIGURE 28-22 Locations where pressure ulcers commonly form. (**A)** The supine position. (**B)** A side-lying position. (**C)** The sitting position.

Stages of Pressure Ulcers

Pressure ulcers are grouped into four stages according to the extent of tissue injury (Fig. 28-23). Care and healing depend on the stage of injury. Without aggressive nursing care, early stage pressure ulcers can easily progress to much more serious ones.

Stage I is characterized by intact but reddened skin. The hallmark of cellular damage is skin that remains red and fails to resume its normal color when pressure is relieved.

Gerontologic Considerations

- Age-related changes (ie, a thinning dermal layer of skin, decreased subcutaneous tissue) result in increased susceptibility to pressure ulcers and

shear-type injuries in older adults. Because of the decreased blood supply to the skin, an older adult may need position changes every 60 to 90 minutes, rather than every 120 minutes. Take special care when moving older adults to avoid friction on the skin.

- Absorbent undergarments may contribute to skin breakdown because they may not allow for air circulation. Urine or feces next to the skin will cause damage and possible skin breakdown. Therefore, any incontinent older adult must be checked every 60 to 90 minutes to prevent skin damage. If urinary incontinence interferes significantly with wound healing, an indwelling catheter (see Chap. 30) may be necessary. It should be removed as soon as feasible, however, and efforts must be made to restore continence.
- Older adults with diminished mobility require aggressive skin care to prevent pressure ulcers. The elbows, heels, coccyx, shoulder blades, and hips are especially vulnerable, as are the creases above the ears if oxygen tubing is in use. Special precautions include heel and elbow protectors, pressure-relief pads, and mattresses, and a strict routine of changing the client's position at least every 2 hours or more frequently if the person's skin becomes reddened in a shorter period. Assessment of at-risk pressure point areas should be done before the 2-hour period.

A *stage II* pressure ulcer is red and accompanied by blistering or a **skin tear** (a shallow break in the skin) without slough. Impairment of the skin may lead to colonization and infection of the wound.

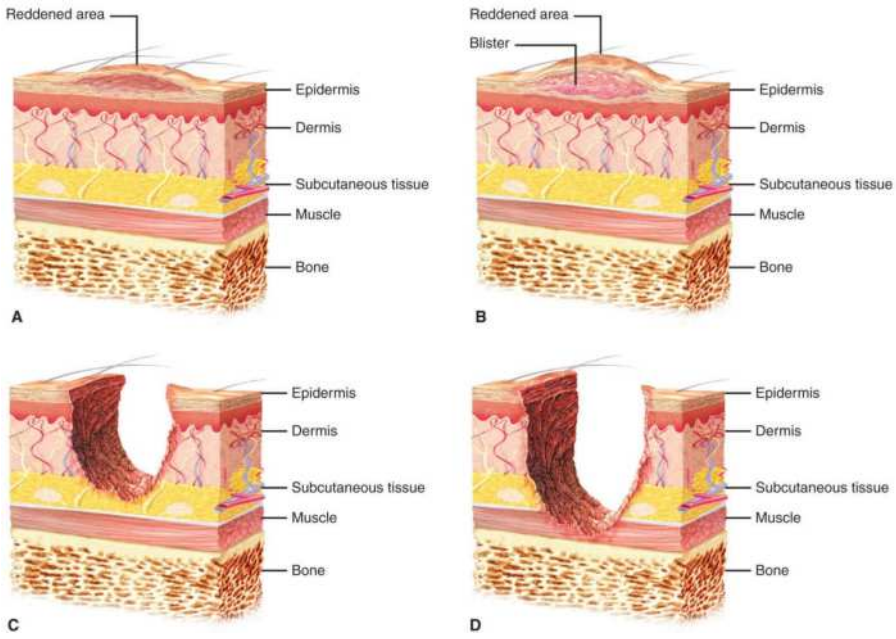


FIGURE 28-23 Pressure sore stages. (**A)** Stage I. (**B)** Stage II. (**C)** Stage III. (**D)** Stage IV.

A *stage III* pressure ulcer has a shallow skin crater that extends to the subcutaneous tissue. It may be accompanied by **serous drainage** (leaking plasma), undermining, slough, or **purulent drainage** (white or greenish fluid) caused by a wound infection. The area is relatively painless despite the severity of the ulcer.

Stage IV pressure ulcers are life-threatening. The tissue is deeply ulcerated, exposing muscle and bone (Fig. 28-24). Slough and necrotic tissue may be evident. The dead or infected tissue may produce a foul odor. If an infection is present, it easily spreads throughout the body, causing **sepsis** (a potentially fatal systemic infection).

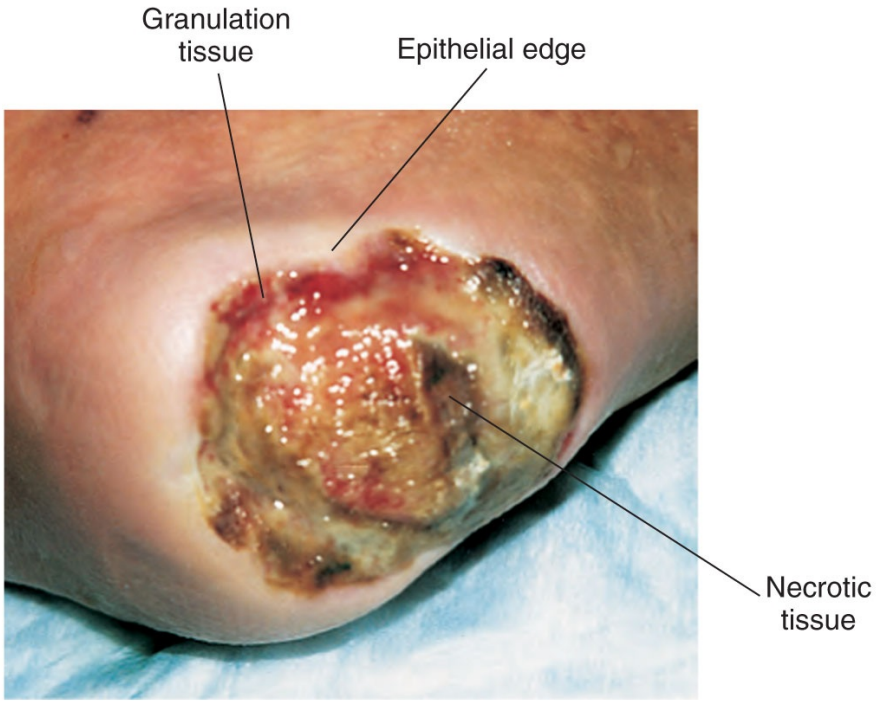


FIGURE 28-24 Example of stage IV pressure sore.

Prevention and Treatment of Pressure Ulcers

The first step in prevention is to identify clients with risk factors for pressure ulcers (Box 28-2). The second step is to implement measures that reduce conditions under which pressure ulcers are likely to form (see Nursing Guidelines 28-2).

BOX 28-2 Risk Factors for Developing Pressure Ulcers

- Inactivity
- Immobility
- Malnutrition
- Emaciation
- Diaphoresis
- Incontinence
- Vascular disease
- Localized edema
- Dehydration
- Sedation

NURSING GUIDELINES 28-2

Preventing Pressure Ulcers

- Change the bedridden client's position frequently. Remind a client who is sitting in a chair to stand and move hourly or at least to shift his or her weight every 15 minutes while sitting. *Changing positions relieves pressure and restores circulation.*
- Lift rather than drag the client during repositioning. *Dragging causes friction, which abrades the skin and damages underlying blood vessels.*
- Avoid using plastic-covered pillows when positioning clients. *Plastic prevents evaporation of perspiration because it is nonporous. It also raises skin temperature, further contributing to the growth of microorganisms.*
- Use positioning devices such as pillows to keep two parts of the body from direct contact with each other. *Such devices absorb perspiration, reduce localized heat, and avoid the compression of tissue between two body parts.*
- Use the lateral oblique position (see Chap. 23) rather than the conventional lateral position for side lying. *The lateral oblique position more effectively reduces the potential for pressure on vulnerable bony prominences.*
- Massage bony prominences only if the skin blanches with pressure relief. *Massage improves circulation to normal tissue but causes further damage to areas where pressure ulcers—even those that are stage I—are already established.*
- Keep the skin clean and dry especially when clients cannot control their bladder or bowel function. *Cleansing removes substances that chemically injure the skin.*
- Use a moisturizing skin cleanser rather than soap, if possible. *A nonsoap cleanser maintains skin hydration and avoids altering the skin's natural acidity, which protects it from bacterial colonization.*
- Rinse and dry the skin well. *Cleansing then drying removes chemical residues and surface moisture.*
- Use pressure-relieving devices such as special beds or mattresses (see Chap. 23). *These special devices maintain capillary blood flow by reducing pressure.*
- Pad body areas such as the heels, ankles, and elbows, which are vulnerable to friction and pressure (Fig. A). *Padding prevents friction and adds a cushioning layer over the bony prominence.*

Heel and ankle protection.

- Use seat cushions such as a commercial gel-filled pad when clients sit for extended periods. *These cushions distribute pressure over a wider area, relieving direct pressure on the coccyx.*
- Keep the head of the bed elevated no more than 30 degrees. *Sliding down in bed can produce a* **shearing force** *(the effect that moves layers of tissue in opposite directions).*
- Provide a balanced diet and adequate fluid intake. *Adequate nutrition maintains and restores cells and keeps tissues hydrated.*

Nutrition Notes

■ Nutritional risk factors linked to the development of pressure ulcers include unintentional weight loss of 5% to 10% of body weight, underweight status (BMI < 18.5), and poor nutritional intake (Thomas, 2015).

■ The ideal protein intake needed to facilitate healing of pressure ulcers has not been determined. An optimal protein intake may be 1.2 to 1.5 g/kg/day, well above the recommended allowance of 0.8 g/kg/day. (National Pressure Ulcer Advisory Panel, European Pressure Ulcer Advisory Panel, Pan Pacific Pressure Injury Alliance, 2014).

■ Clients who may have less than optimal nutritional status may benefit from a routine multivitamin. In well-nourished clients with pressure ulcers, there is no data to support the use of supplements to promote healing, including supplements of vitamin C and zinc. (Thomas, 2015).

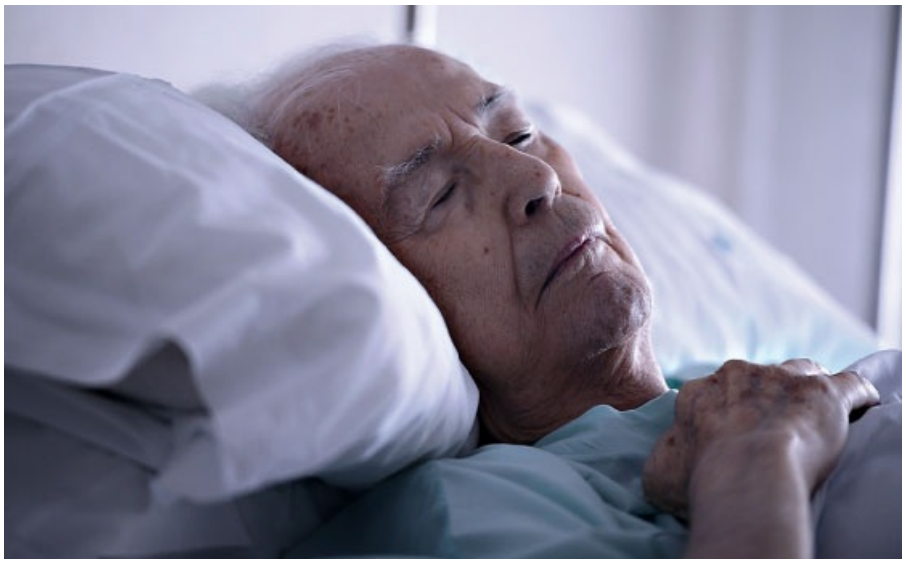
Thomas, D. (2015). The relationship of nutrition and pressure ulcers. In Bales, C, Locher, J., Saltzman, E (eds) *Handbook of Clinical Nutrition and Aging, Nutrition and Health.* Springer Science and Business Media, NY.

NURSING IMPLICATIONS

Clients with a surgical wound, pressure ulcer, or other type of tissue injury are likely to have one or more of the following nursing diagnoses:

- Acute Pain
- Impaired Skin Integrity
- Ineffective Peripheral Tissue Perfusion
- Impaired Tissue Integrity
- Delayed Surgical Recovery
- Risk for Pressure Ulcer
- Risk for Infection

Nursing Care Plan 28-1 is an example of how nurses use the nursing process to care for a client with Impaired Tissue Integrity, defined in the 2015 NANDA-I taxonomy (p. 405) as "damage to mucous membrane, cornea, integumentary system, muscular fascia, muscle, tendon, bone, cartilage, joint capsule, and/or ligament."



Clinical Scenario A A 72-year-old male experienced paralysis below his neck following a fall from a ladder as he was cleaning leaves from the gutters on his home 2 years ago. He has been admitted for treatment of multiple pressure sores that have developed as a consequence of his immobility.

NURSING CARE PLAN 28-1 Impaired Tissue Integrity

Assessment

- Inspect the skin especially over bony prominences.
- Look for skin redness that does not blanch with relief of pressure, evidence of skin tears, or ulceration.
- Observe the client's ability to move and reposition himself or herself independently.
- Assess the status of the client's hydration and nutrition.
- Determine if the client is incontinent or feverish or has other contributing factors to skin and tissue breakdown such as conditions accompanied by edema, those that require the application of devices such as a cast or traction, or treatments that increase the potential for impairment of the integument such as radiation cancer therapy.

Nursing Diagnosis. **Impaired Tissue Integrity** related to unrelieved pressure secondary to immobility from a spinal cord injury at the C7 (seventh cervical vertebrae) level 2 years ago as manifested by stage III pressure ulcer over the coccyx and a stage I ulcer over the bilateral heels and elbows.

Expected Outcome. The tissue integrity in the area of the coccygeal pressure ulcer will be restored as evidenced by the development of granulation tissue around the circumference of the wound by 8/30 and closure by 10/1. The elbows and heels will blanch with pressure relief by 8/18.

Interventions	Rationales
Reposition the client every 2 hours until an air-fluidized bed can be obtained.	Frequent repositioning maintains capillary pressure above 32 mmHg to facilitate the oxygenation of tissue.
Use a 30^{o} lateral oblique (tilted side-lying) position; alternate right and left sides, or prone position if tolerated.	Relief of pressure prevents additional injury to tissues.
Avoid the supine and Fowler's positions as much as possible.	These positions increase the potential for shear forces and pressure over bony prominences on posterior body areas such as the coccyx, shoulders, and heels.
Keep the heels elevated or resting on natural (not synthetic) sheepskin.	Elevation or natural sheepskin padding reduces pressure on tissue and promotes capillary blood flow.
After bathing, spray heels and elbows with Bard Barrier Film.	Skin products, such as Bard Barrier Film, form a clear, breathable film that is impervious to liquids and potential irritants and protects against skin abrasion and friction.
Care for the open coccygeal wound as follows:	
• Cleanse the wound with chlorhexidine antiseptic solution at the time of each dressing change. • Rinse with normal saline. • Pack the wound with foam filler within the package that contains a negative pressure wound closure system. • After sealing the foam with an occlusive dressing, attach the suction tubing set at 125 mmHg. • Change the negative pressure wound closure	• An antiseptic reduces the transient and resident microorganisms that can increase the extent and severity of the pressure ulcer and delay healing. • The vacuum created by a negative pressure wound closure system shrinks the wound and promotes cellular growth to heal the impaired tissue

dressing every 3 days.	
Once healing is evident:	
• Clean, dry, and cover wound with a hydrogel dressing and leave in place for 3 days. • If drainage collects, remove and reapply a new dressing.	• A hydrogel dressing creates a moist environment that continues to accelerate the healing process. Accumulation of fluid beneath the dressing increases the potential for loosening the wound cover. • Changing the dressing restores the occlusive cover over the wound.
Measure the open pressure sore every 3 days (8/18, 8/21, etc.) during the day shift.	Regular assessment of the wound helps to determine the need to continue or revise the plan for wound care.

Evaluation of Expected Outcomes

- Pressure ulcer in area of coccyx measures 2 × 3 × ½ inch on 8/18 with 1/16 inch of granulation tissue around the circumference of the wound.
- Heels and elbows no longer appear red.

CRITICAL THINKING EXERCISES

1. What nursing assessments would be evident to suggest that a wound is healing?
2. A nurse notes that the gauze dressing that covers a wound comes loose repeatedly. What measures could the nurse take?
3. Describe the wound care appropriate for a client with a stage I pressure ulcer, one with an abdominal incision, and one with a peripheral intravenous infusion site.
4. A 75-year-old client is admitted from a nursing home to have surgery to repair a fractured hip. Discuss the factors that may threaten this client's wound healing.

NCLEX-STYLE REVIEW QUESTIONS

1. Which of the following is the best body position the nurse can utilize to promote wound drainage from an abdominal incision with an open drain?
   1. Lithotomy
   2. Fowler's
   3. Recumbent
   4. Trendelenburg

   Test-Taking Strategy: Note the key term "best." Eliminate options to select one that would promote gravity drainage.
2. When the nurse changes a client's dressing, which nursing action is correct?
   1. The nurse removes the soiled dressing with sterile gloves.

2. The nurse frees the tape by pulling it away from the incision.
3. The nurse encloses the soiled dressing within a latex glove.
4. The nurse cleans the wound in circles toward the incision.

Test-Taking Strategy: Use the process of elimination to select the action that supports a principle for managing care of a wound.

3. When a nurse empties the drainage in a Jackson-Pratt reservoir, which nursing action is essential for reestablishing the negative pressure within this drainage device?

1. The nurse compresses the bulb reservoir and closes the vent.
2. The nurse opens the vent, allowing the bulb to fill with air.
3. The nurse fills the bulb reservoir with sterile normal saline.
4. The nurse secures the bulb reservoir to the skin near the wound.

Test-Taking Strategy: Note the key word "essential." Analyze the choices and select the option that must be performed in order to create negative pressure.

4. When the physician considers débriding a wound using an autolytic process, which of the following can the nurse expect to implement? Select all that apply.

1. Removing of necrotic tissue with scissors
2. Applying a chemical enzyme to the wound
3. Applying a dressing with a seaweed component
4. Depositing live fly larvae within the wound
5. Using a dressing that contains gelatin granules
6. Flushing the wound with an irrigating solution.

Test-Taking Strategy: Analyze the choices and select options that promote self-dissolution of devitalized tissue.

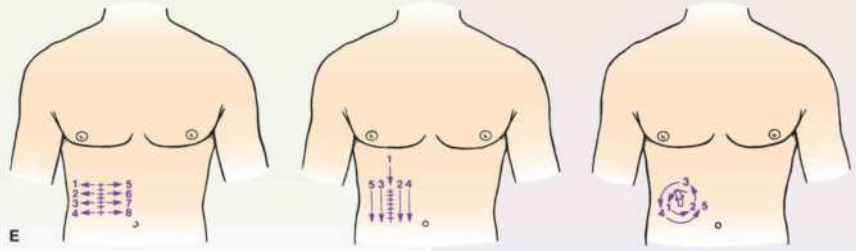
5. Place the physiologic processes that occur during an inflammatory response in the order in which they occur. Use all the options.

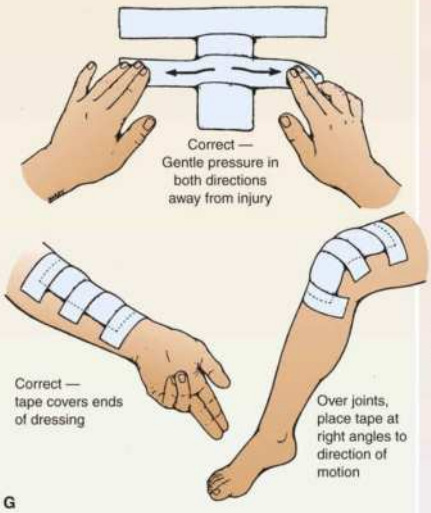
1. Leukocytosis occurs.
2. There is increased cellular permeability.
3. The area becomes red and warm.
4. Local circulation is reduced.

Test-Taking Strategy: Analyze the options and place them in a sequence that begins shortly after tissue is injured through steps leading to wound repair.

SKILL 28-1 Changing a Gauze Dressing

Suggested Action	Reason for Action
ASSESSMENT	
Inspect the current dressing for drainage, integrity, and type of dressing supplies used.	Provides assessments indicating a need to change the dressing and supplies that may be needed.
Check the medical orders for a directive to change the dressing.	Shows collaboration with the prescribed medical treatment.
Determine if the client has allergies to tape or antimicrobial wound agents.	Helps to determine the dressing supplies to use.
Assess the client's level of pain and its characteristics.	Determines if analgesia will be beneficial before changing the dressing.
PLANNING	
Explain the need and technique for changing the dressing.	Relieves anxiety and promotes cooperation.
Consult the client on a preferred time for the dressing change if there is no immediate need for it.	Empowers the client to participate in decision-making.
Give pain medication, if needed, 15–30 minutes before the dressing change.	Allows time for medication absorption and effectiveness.
Gather the necessary supplies, which are likely to include a paper bag for the soiled dressing, clean and sterile gloves, individually packaged gauze dressings, tape, and, in some cases, an antimicrobial agent such as povidone–iodine swabs for wound cleansing.	Facilitates organization and efficient time management.
IMPLEMENTATION	
Wash your hands or use an alcohol-based hand rub (see Chap. 10).	Reduces the transmission of microorganisms.
Pull the privacy curtain.	Shows respect for the client's dignity.
Position the client to allow access to the dressing.	Facilitates comfort and dexterity.
Drape the client to expose the area of the wound.	Ensures modesty but facilitates care.
Loosen the tape securing the dressing; pull the tape toward the wound (Fig. A).	Facilitates removal without separating the healing wound.
A	Loosen the tape. (Photo by B. Proud.)
Put on at least one glove and lift the dressing from the wound (Fig. B).	Provides a barrier against contact with blood and body substances.
B	Remove the dressing.

Moisten the gauze with sterile normal saline if it adheres to the wound. Discard the soiled dressing in a paper bag or other receptacle along with the glove(s) (Fig. C).	Prevents disrupting granulation tissue. Confines the sources of pathogens.
 C	**Dispose of the dressing.**
Wash your hands again or repeat the alcohol-based hand rub. Tear several long strips of tape and fold the ends over, forming tabs (Fig. D).	Removes transient microorganisms. Facilitates handling tape later when wearing gloves and eases tape removal during the next dressing change.
 D	**Prepare the tape.**
Open sterile supplies using the inside wrapper of one of the gauze dressings as a sterile field, if needed. Put on sterile gloves. Inspect the wound. Cleanse the wound with the antimicrobial agent. Use a technique that prevents transferring microorganisms back to a cleaned area (Fig. E).	Ensures an aseptic technique. Ensures sterility. Provides data for description and comparison. Removes drainage and microorganisms. Supports principles of medical asepsis.
 E **Wound cleansing techniques.**	
Use a single swab or a small gauze square for each stroke. Allow the antimicrobial agent to dry.	Prevents transferring microorganisms to clean areas. Ensures that the tape will stay secured when applied.

Cover the wound with the gauze dressing (Fig. F).	Protects the wound.
F	Apply the dressing.
Secure the dressing with tape in the opposite direction of the incision or across a joint. Place a strip of tape at each end of the dressing and in the middle if needed (Fig. G).	Prevents loosening with activity; holds the dressing in place without exposing the wound or incision.
 G	Secure with tape. (From Smeltzer, S. C., & Bare, B. G. [2000]. *Brunner and Suddarth's textbook of medical-surgical nursing* [9th ed.]. Philadelphia, PA: Lippincott Williams & Wilkins.)
Remove and discard gloves. Rewash hands or repeat the alcohol-based hand rub.	Confines the sources of microorganisms. Removes transient microorganisms.

EVALUATION

- Dressing covers the entire wound.
- Dressing is secure, dry, and intact.

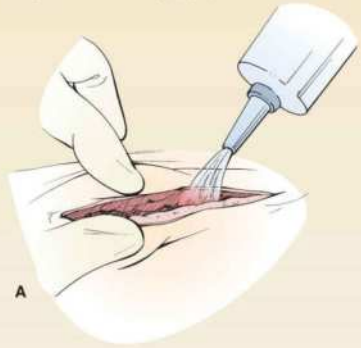
DOCUMENT

- Type of dressing
- Antimicrobial agent used for cleansing
- Assessment data

SAMPLE DOCUMENTATION

Date and Time Gauze dressing changed over abdominal wound. Wound cleansed with povidone iodine. Incision is well approximated with sutures. No drainage, swelling, or tenderness observed. _____ J. Doe, LPN

SKILL 28-2 Irrigating a Wound

Suggested Action	Reason for Action
ASSESSMENT	
Check the medical orders for a directive to irrigate the wound.	Shows collaboration with the prescribed medical treatment.
Determine how much the client understands about the procedure.	Indicates the level of health teaching needed.
PLANNING	
Plan to irrigate the wound at the same time that the dressing requires changing.	Makes efficient use of time.
Gather the equipment required, which is likely to include a container of solution, a basin, a bulb or asepto syringe, gloves, and absorbent material including a towel to dry the skin.	Facilitates organization.
Bring supplies for changing the dressing.	Makes efficient use of time.
Consider additional items for standard precautions such as goggles or face shield and cover apron or gown.	Follows infection control guidelines when there is a potential for being splashed with blood or body substances.
IMPLEMENTATION	
Wash your hands or use an alcohol-based hand rub (see Chap. 10).	Reduces the transmission of microorganisms.
Pull the privacy curtain.	Shows respect for the client's dignity.
Drape the client to expose the area of the wound.	Ensures modesty but facilitates care.
Follow directions in Skill 28-1 for removing the dressing.	Provides access to the wound.
Wash your hands or repeat the alcohol-based hand rub.	Reduces the transmission of microorganisms.
Position the client to facilitate filling the wound cavity with solution.	Ensures contact between the solution and the inner area of the wound.
Pad the bed with absorbent material and place an emesis basin adjacent to and below the wound.	Reduces the potential for saturating the bed linens.
Open and prepare supplies following the principles of surgical asepsis.	Confines and controls the transmission of microorganisms.
Put on gloves and other standard precautions apparel.	Reduces the potential for contact with blood and body substances.
Fill the syringe with solution and instill it into the wound without touching the wound directly (Fig. A).	Dilutes and loosens debris.
 A	**Instill the irrigant.**
Hold the emesis basin close to the client's body to catch the solution as it drains from the wound (Fig. B).	Collects and contains the irrigating solution.
B	**Position the client to drain the irrigant.**
Repeat the process until the draining solution seems clear.	Indicates the evacuation of debris.
Tilt the client toward the basin.	Drains the remaining solution from the wound.
Dry the skin.	Facilitates applying a dressing.
Dispose of the drained solution, soiled equipment, and linens.	Reduces the potential for transmitting microorganisms.
Remove gloves, wash hands, and prepare to change the dressing.	Provides for the absorption of residual solution and coverage of the wound.
EVALUATION	
• Irrigation solution shows evidence of debris removal. • Wound shows evidence of healing.	
DOCUMENT	
• Assessment data • Type and amount of solution • Outcome of procedure	

SAMPLE DOCUMENTATION

Date and Time Dressing removed. Moderate purulent drainage on soiled dressing. Wound is separated 3 inches. Approximately 300 mL of sterile NSS instilled within wound. Drained solution is cloudy with particles of debris. _____ J.Doe, LPN

SKILL 28-3 Providing a Sitz Bath

Suggested Action	Reason for Action
ASSESSMENT	
Check the medical orders for a directive to administer a sitz bath.	Shows collaboration with the prescribed medical treatment.
Determine how much the client understands about the procedure.	Indicates the level of health teaching needed.
Assess the condition of the rectal or perineal wound and the client's level of pain.	Provides baseline data for future comparisons; indicates if pain medication is needed.
PLANNING	
Explain the procedure.	Relieves anxiety and promotes cooperation.
Ask if the client prefers the sitz bath before or after routine hygiene.	Involves the client in the decision-making process.
Obtain disposable equipment unless specially installed tubs are available.	Facilitates organization and efficient time management.
Assemble other supplies such as a bath blanket and towels.	Prepares for maintaining warmth and provides a means for drying the skin.
Inspect and clean the bathroom area or the tub room.	Supports principles of medical asepsis.
Place the basin inside the rim of the raised toilet seat (Fig. A).	Allows for submerging the rectum and the perineum.
A	**Position the sitz bath basin.**
IMPLEMENTATION	
Wash your hands or use an alcohol-based hand rub (see Chap. 10).	Reduces the transmission of microorganisms.
Help the client put on a robe and slippers.	Maintains warmth, safety, and comfort.
Help the client to ambulate to the location where the sitz bath will be administered.	Demonstrates concern for safety.
Shut the door to the bathroom or tub room.	Provides privacy.
Clamp the tubing attached to the water bag.	Prevents a loss of fluid.

29

Gastrointestinal Intubation

Learning Objectives

On completion of this chapter, the reader should be able to:

1. Define intubation and list reasons for gastrointestinal intubation.
2. Identify four general types of gastrointestinal tubes.
3. Name at least four assessments that are necessary before inserting a tube nasally.
4. Explain the purpose of and how to obtain a **N**ose-to-**E**arlobe-to-the-**X**iphoid (NEX) measurement.
5. Describe two methods for determining distal placement in the stomach.
6. Discuss three ways that nasointestinal feeding tubes or their insertion differ from their gastric counterparts.
7. Name four schedules for administering tube feedings.
8. Explain the purpose of assessing gastric residual.
9. Name five nursing activities involved in managing the care of clients who are being tube-fed.
10. Name two nursing responsibilities for assisting with the insertion of a tungsten-weighted intestinal decompression tube.

Words to Know

bolus feeding
continuous feeding
cyclic feeding
decompression
dumping syndrome
enteral nutrition
gastric reflux
gastric residual
gastrostomy tube (G-tube)
gavage
intermittent feeding

intestinal decompression
intubation
jejunostomy tube (J-tube)
lavage
lumen
nasogastric intubation
nasogastric tube
nasointestinal intubation
nasointestinal tubes
Nose-to-**E**arlobe-to-the-**X**iphoid (NEX) measurement
orogastric intubation
orogastric tube
ostomy
percutaneous endoscopic gastrostomy (PEG) tube
percutaneous endoscopic jejunostomy (PEJ) tube
pseudoconfirmatory gurgling
stylet
sump tubes
tamponade
transabdominal tubes

Clients, especially those undergoing abdominal or gastrointestinal (GI) surgery, may require some type of tube placed within their stomach or intestine. The use of a gastric or intestinal tube reduces or eliminates problems associated with surgery or conditions affecting the GI tract, such as impaired peristalsis, vomiting, or gas accumulation. Tubes also can be used to nourish clients who cannot eat. This chapter discusses the multiple uses for gastric and intestinal tubes and the nursing guidelines and skills for managing related client care.

INTUBATION

Intubation generally means the placement of a tube into a body structure; in this chapter, it refers specifically to insertion of a tube into the stomach or intestine by way of the mouth or nose. **Orogastric intubation** (the insertion of a tube through the mouth into the stomach), **nasogastric intubation** (the insertion of a tube through the nose into the stomach), and **nasointestinal intubation** (the insertion of a tube through the nose to the intestine) are performed to remove gas or fluids or to administer liquid nourishment.

A tube also may be inserted within an **ostomy** (a surgically created opening). A prefix identifies the anatomic site of the ostomy; for instance, a *gastrostomy* is an artificial opening into the stomach (Fig. 29-1).

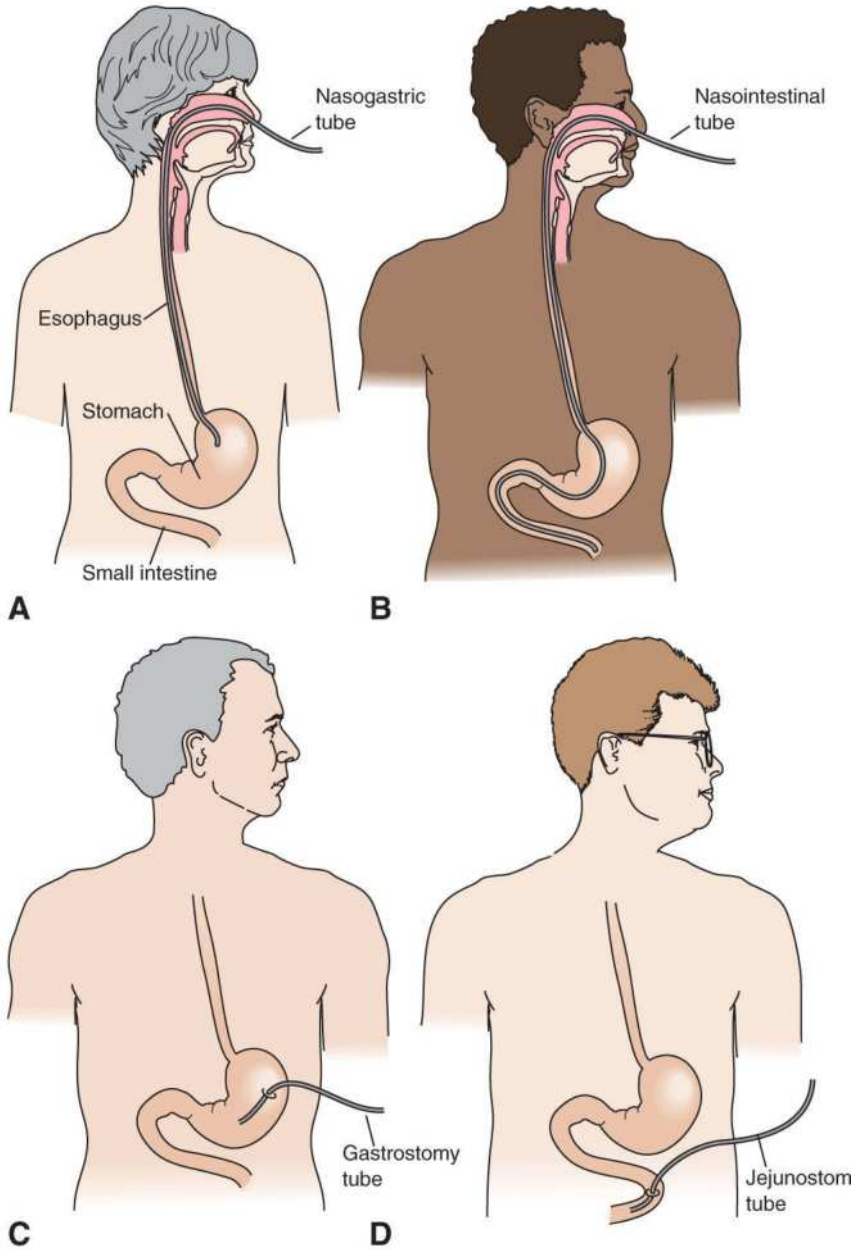


FIGURE 29-1 Examples of various methods of gastric and intestional intubation. [From Carter P (2015). *Lippincott Textbook for Nursing Assistants*, 4th edition. Philadelphia: LWW (PE).]

Gastric or intestinal tubes are used for a variety of reasons, including the following:

- Performing a **gavage** (providing nourishment)
- Administering oral medications that the client cannot swallow
- Obtaining a sample of secretions for diagnostic testing
- Performing a **lavage** (removing substances from the stomach, typically poisons)
- Promoting **decompression** (removing gas and liquid contents from the stomach or bowel)
- Controlling gastric bleeding, a process called compression or **tamponade** (pressure)

TYPES OF TUBES

Although all gastric and intestinal tubes have a proximal and distal end, their size, construction, and composition vary according to their use (Table 29-1). The outside diameter of most tubes is measured using the French scale, indicated by a number followed by the letter F. Each number on the French scale equals approximately 0.33 mm. The larger the number, the larger the diameter of the tube.

Tubes can be identified according to the location of their insertion (mouth, nose, or abdomen) or the location of their distal end (stomach [gastric] or intestinal).

Orogastric Tubes

An **orogastric tube** (a tube inserted through the mouth into the stomach), such as an Ewald tube, is used in an emergency to remove toxic substances that have been ingested. The diameter of the tube is large enough to remove pill fragments and stomach debris. Because of its size, the tube is introduced through the mouth rather than the nose.

Nasogastric Tubes

A **nasogastric tube** (a tube placed through the nose and advanced to the stomach) is smaller in diameter than an orogastric tube but larger and shorter than a nasointestinal tube. Some nasogastric tubes have more than one **lumen** (channel) within the tube.

A Levin tube is a commonly used, single-lumen gastric tube with multiple uses, one of which is decompression. Gastric **sump tubes** (double-lumen tubes) are used almost exclusively to remove fluid and gas from the stomach (Fig. 29-2). The second lumen serves as a vent. The use of sump tubes decreases the possibility that the stomach wall will adhere to and obstruct the distal drainage

openings when suction is applied.

Because nasogastric tubes remain in place for several days or more, many clients complain of nose and throat discomfort. If the tube's diameter is too large or pressure from the tube is prolonged nasopharyngeal tissue irritation or breakdown may occur. Furthermore, gastric tubes tend to dilate the esophageal sphincter, also known as the cardiac valve, a circular muscle between the esophagus and stomach. The stretched opening may contribute to **gastric reflux** (the reverse flow of gastric contents), especially when the tube is used to administer liquid formula. If gastric reflux occurs, the liquid could enter the airway and interfere with respiratory function.

Nasointestinal Tubes

Nasointestinal tubes (tubes inserted through the nose for distal placement below the stomach) are longer than their gastric counterparts. The added length permits them to be placed in the small bowel. They are used to provide nourishment (feeding tubes) or to remove gas and liquid contents from the small intestine (decompression tubes).

Feeding Tubes

Nasointestinal tubes used for nutrition, such as a Dobhoff tube, are usually small in diameter and made of a flexible substance such as polyurethane or silicone. Their narrow width and soft composition allow them to remain in the same nostril for 4 weeks or longer. In addition, they reduce the potential for gastric reflux because they deliver liquid nutrition beyond the stomach.

Narrow tubes are not problem-free. They tend to curl during insertion because they are so flexible. Therefore, some are supplied with a **stylet** (metal guidewire) that helps straighten and support them during insertion. Almost all have a weighted tip that helps them descend past the stomach. Checking the placement of the distal end is more difficult; these tubes also become obstructed more easily. Despite the problems associated with maintenance, small-diameter tubes are preferred for their comfort. They are ideal for providing a continuous infusion of nourishment.

TABLE 29-1 Types of Gastrointestinal Tubes

TUBE	PURPOSE	CHARACTERISTICS
Orogastric		
Ewald	Lavage	Large diameter: 36–40 F Single lumen Multiple distal openings for drainage
Nasogastric		
Levin	Lavage Gavage Decompression Diagnostics	Usual adult size 14–18 F Single lumen 42–50 in (107–127 cm) long Multiple drain openings
Salem sump	Decompression	Same diameter as Levin Double lumen Pigtail vent 48 in. (122 cm) long Marked at increments to indicate depth of insertion Radiopaque
Sengstaken–Blakemore	Compression Drainage	Usual diameter: 20 F 36 in (90 cm) long Triple lumen; two lead to balloons in the esophagus and stomach and the third is for removing gastric drainage; a fourth lumen may be used to remove pharyngeal secretions.
Nasointestinal		
Keofeed	Gavage	Small diameter: 8 F 36 in. (90 cm) long Polyurethane or silicone Weighted tip Extremely flexible and may require the use of a stylet during insertion Radiopaque Bonded lubricant that becomes activated with moisture.
Maxter	Intestinal decompression	Usual size: 18 F 100 in. (250 cm) long Double lumen Tungsten-weighted tip Graduated marks every 10 in. (25 cm)
Transabdominal		
Gastrostomy	Gavage; may be used for decompression while the client is fed through a jejunostomy tube	Sizes 12–24 F for adults Rubber or silicone May have additional side ports for balloon inflation to maintain placement May be capped or plugged between feedings Radiopaque
Jejunostomy	Gavage	Sizes 5–14 F for adults Silicone or polyurethane Radiopaque

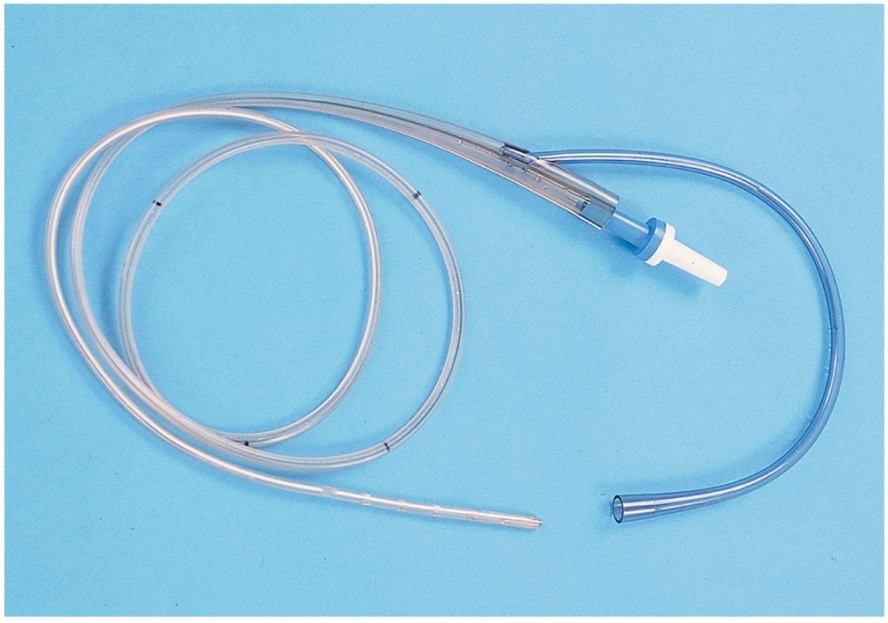


FIGURE 29-2 Vented nasogastric (Salem sump) tube with a one-way valve. (Photo by B. Proud.)

Intestinal Decompression Tubes

Although surgery is often the most common intervention when a client has a

partial or complete bowel obstruction, **intestinal decompression** (the removal of gas and intestinal contents) also may be used. A tube used for intestinal decompression has a double lumen and a weighted tip (Fig. 29-3). One lumen is used to suction the intestinal contents; the other acts as a vent to reduce suction-induced trauma to intestinal tissue. The weighted tip and peristalsis, if present, propel the tube beyond the stomach and into the intestine. The progress of the radiopaque tip through the GI tract is monitored by X-ray.

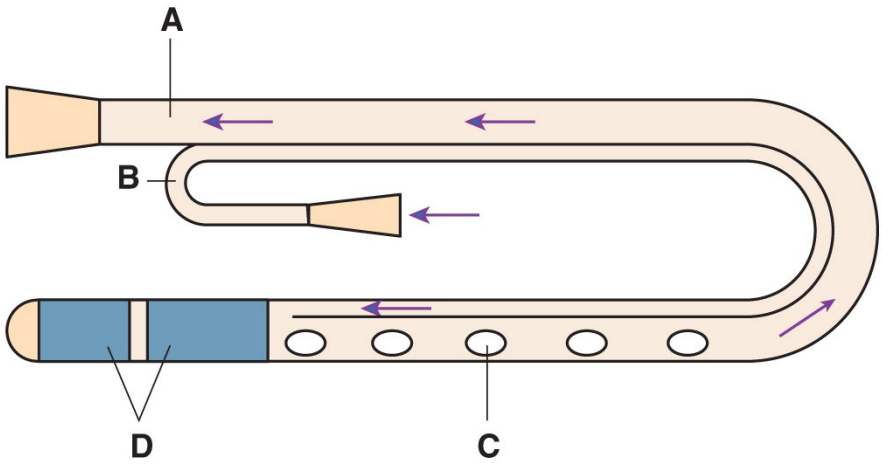


FIGURE 29-3 An intestinal decompression tube, (A) including the suction lumen, (B) the vent lumen, (C) openings for suction, (D) and the radiopaque tungsten tip.

At one time, intestinal tubes, such as the Cantor and Miller–Abbott tubes, were weighted with mercury. Because of mercury's hazards to both the client and the environment, however, mercury-weighted tubes are not used today. Instead, intestinal tubes, like the Maxter tube (see Table 29-1), are now weighted with tungsten.

Transabdominal Tubes

Transabdominal tubes (tubes placed through the abdominal wall) provide access to various parts of the GI tract. Two examples are a **gastrostomy tube** or G-tube (a transabdominal tube located within the stomach) and a **jejunostomy tube (J-tube)** (a transabdominal tube that leads to the jejunum of the small intestine).

A G-tube is placed surgically or with the use of an endoscope. A surgically inserted G-tube resembles a long rubber catheter sutured to the abdomen. A **percutaneous endoscopic gastrostomy (PEG) tube** (a transabdominal tube inserted under endoscopic guidance) is anchored with internal and external crossbars called bumpers (Fig. 29-4A). **A percutaneous endoscopic**

jejunostomy (PEJ) tube (a tube that is passed through a PEG tube into the jejunum) is small in diameter so that it can be inserted through the larger PEG tube (Fig. 29-4B).

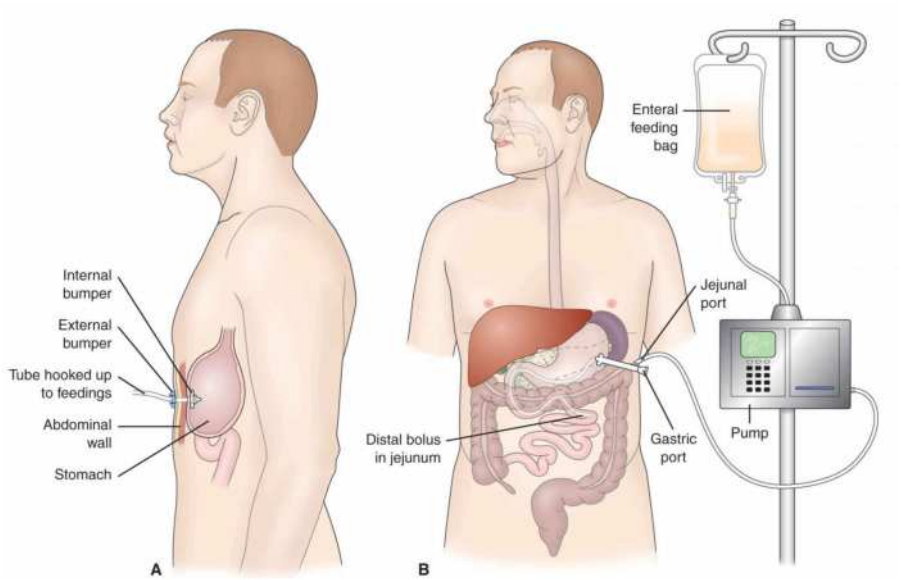


FIGURE 29-4 Transabdominal tubes. (**A)** A percutaneous endoscopic gastrostomy (PEG) tube. (**B)** A percutaneous endoscopic jejunostomy (PEJ) tube. (Courtesy of IVAC Corporation, San Diego, CA.)

Transabdominal tubes are used instead of nasogastric or nasointestinal tubes when clients require an alternative to oral feeding for more than 1 month.

NASOGASTRIC TUBE MANAGEMENT

Usually, nurses insert nasogastric tubes. Additional nursing responsibilities include keeping the tube patent (or unobstructed), implementing the prescribed use, and removing the tube when it has accomplished its therapeutic purpose.

Nasogastric Tube Insertion

Inserting a nasogastric tube involves preparing the client, conducting preintubation assessments, and placing the tube.

Client Preparation

Most clients are anxious about having to swallow a tube. Suggesting that the diameter of the tube is smaller than most pieces of food may foster a positive outcome. Explaining the procedure and giving instructions on how the client can assist while the tube is being passed may further reduce anxiety. One of the most important ways to support clients is to provide them with some means of control. The nurse can establish a signal, such as the client raising a hand, to indicate the

need for a pause during the tube's passage.

Preintubation Assessment

Before insertion, the nurse conducts a focused assessment that includes the client's:

- Level of consciousness
- Weight
- Bowel sounds
- Abdominal distention
- Integrity of nasal and oral mucosa
- Ability to swallow, cough, and gag
- Any nausea and vomiting

Assessment findings serve as a baseline for future comparisons and may suggest a need to modify the procedure or the equipment used. One main goal of the assessment is to determine which nostril is best to use when inserting the tube and the length to which the tube will be inserted.

Nasal Inspection

After the client clears nasal debris by blowing into a paper tissue, the nurse inspects each nostril for size, shape, and patency. The client should exhale while each nostril in turn is occluded. The presence of nasal polyps (small growths of tissue), a deviated septum (nasal cartilage deflected from the midline of the nose), or a narrow nasal passage excludes a nostril for tube insertion.

Nasogastric Tube Measurement

Some tubes are already marked to indicate the approximate length at which the distal tip will be located within the stomach. These markings, however, may not correlate exactly with the client's anatomy. Therefore, before inserting a tube, the nurse obtains the client's **NEX measurement** (length from ***N***ose-to-***E***arlobe-to-the-***X***iphoid process [tip of the sternum]; Fig. 29-5) and marks the tube appropriately.

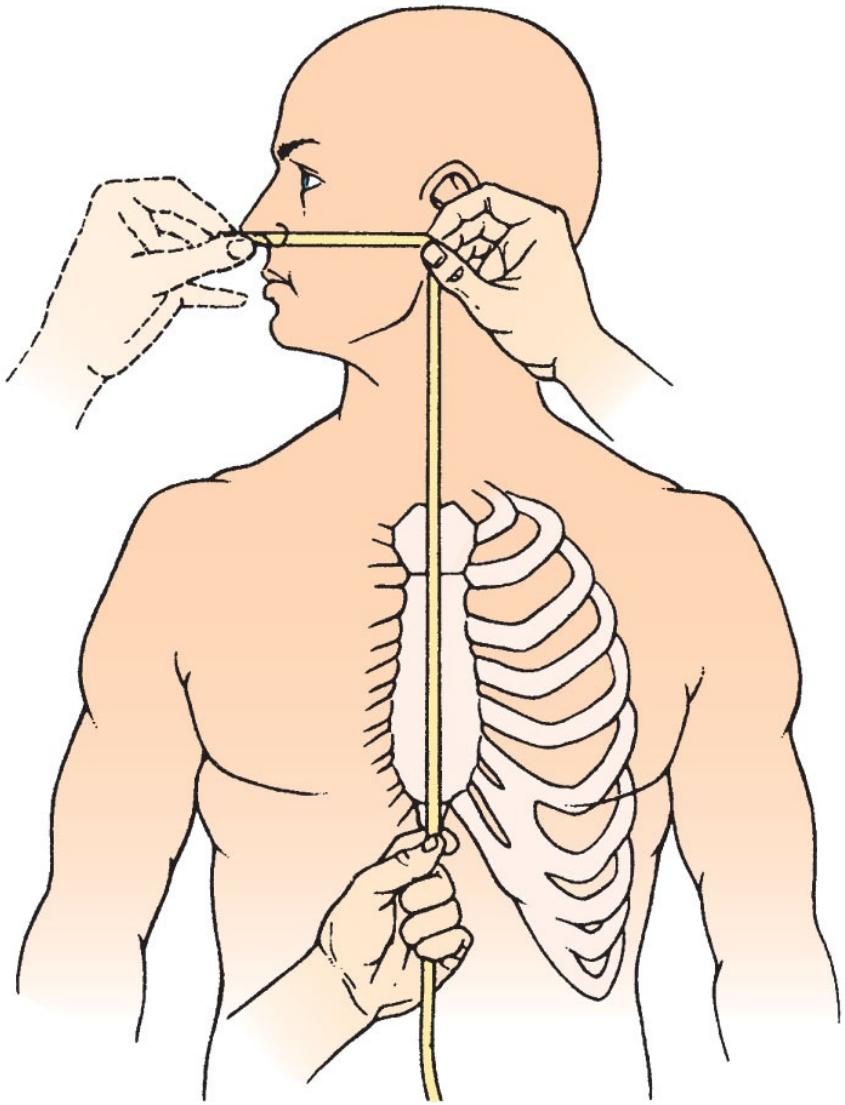


FIGURE 29-5 Obtaining the NEX measurement.

The first mark on the tube is made at the measured distance from the nose to the earlobe. It indicates the distance to the nasal pharynx—a location that places the tip at the back of the throat but above where the gag reflex is stimulated. A second mark is made at the point where the tube reaches the xiphoid process, indicating the depth required to reach the stomach.

Nasogastric Tube Placement

When inserting a nasogastric tube, the nurse's primary concerns are to cause as little discomfort as possible, to preserve the integrity of the nasal tissue, and to locate the tube within the stomach, not in the respiratory passages.

Gerontologic Considerations

■ An age-related reduction in the number of laryngeal nerve endings contributes to diminished efficiency of the gag reflex. Other conditions that can depress the gag reflex include neurologic disorders such as dementia, strokes, and Parkinson's disease.

The placement of weighted-tip feeding tubes may also be confirmed using bedside ultrasonography (Gok et al., 2015; Kim et al., 2012). In a research sample of adults, the technique proved to be 97% accurate in determining distal tube location, could be performed in shorter time than conventional X-rays, but requires someone who is skilled to perform and interpret the results (Kim et al., 2012).

Although several bedside physical assessment methods such as auscultating the abdomen while instilling air and testing the pH of aspirated liquid have been used by nurses and may still be widely practiced to determine the distal location of a nasogastric tube, they do not always provide credible evidence. Current data indicate that testing the pH of aspirated fluid can be unreliable because the pH can be falsely altered by the aspiration of swallowed alkaline saliva, refluxed acidic gastric secretions, and medications that make gastric secretions less acidic. In addition, auscultation over the abdomen while air is instilled is known to cause **pseudoconfirmatory gurgling** when air enters the esophagus or small intestine, thus misinterpreting a gastric location of the tube's tip. The only evidence-based methods for determining the distal location of a nasogastric tube include obtaining an abdominal X-ray after its initial insertion and monitoring the external tube length (the "X" marked component of the NEX measurement to the proximal end of the tube) after radiographic confirmation (Miller et al., 2015).

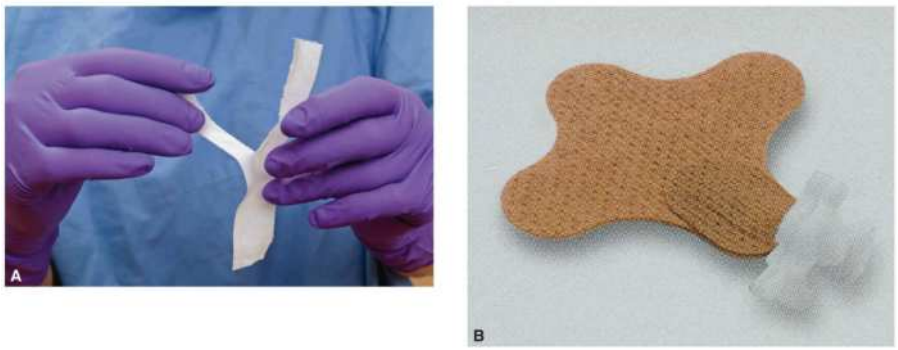


FIGURE 29-6 (**A**) One end of a piece of tape is split, forming two narrower strips, and the opposite end is left intact. The wider intact end of the tape is applied to the nose, and the narrower strips are wound around

the tube in opposite directions to secure the nasogastric tube. (**B)** A commercial adhesive product used to secure a tube's placement.

Once the nurse has confirmed stomach placement, he or she secures the tube to avoid upward or downward migration (Fig. 29-6). The tube is then ready to use for its intended purpose. The steps to follow when inserting a nasogastric tube are outlined in Skill 29-1.

Once the tube is at its final mark, the nurse must verify the location within the stomach.

Checking Placement of Intestinal Feeding Tubes

Initial tube placement is verified with an X-ray and eventually may be identified in real time using an electromagnetic system once the technology becomes more widely used. Other techniques for determining placement of small-diameter nasogastric feeding tubes are less reliable, but may be used by some. Checking the placement by auscultating air may be inconclusive because the air that escapes from the distal tip is less pronounced as a result of the small diameter of the tube. Also, aspiration of stomach contents from small-diameter tubes is not always possible because the negative pressure causes the tube to collapse. Nonetheless, once the feeding tube is inserted and secured to avoid slipping, its continued safe location requires frequent checking and measurement of the external portion of the tube. Repeated X-rays to reassess tube placement are expensive, impractical, and potentially harmful. Nurses may verify the tube's distal placement throughout its use by aspirating fluid from the tube after the initial X-ray using a large-volume (30 to 50 mL) syringe. The large-volume syringe creates less-negative pressure during aspiration and, therefore, provides enough fluid to test the pH (Nursing Guidelines 29-1).

NURSING GUIDELINES 29-1

Inserting a Nasointestinal Feeding Tube

- Wash hands or perform an alcohol-based hand rub (see Chap. 10). *Hand hygiene reduces the transmission of microorganisms.*
- Put on gloves. *Gloves provide a physical barrier between the nurse's hands and body fluids.*
- Obtain a flexible small-gauge feeding tube with a stylet (Fig. A). *Flexible small gauge feeding tubes are designed to remain in place longer and have less potential for nasal trauma.*

A small gauge flexible feeding tube with stylet.

- Follow the manufacturer's suggestions for activating the lubricant bonded to the tube. Two common techniques are to instill water through the tube and to immerse the tip in water (Fig. B). *Activation of the lubricant transforms the dry bond to a gelatinous consistency.*

Activating the bonded lubricant. (From Springhouse. [2009]. *Lippincott's visual encyclopedia of clinical skills.* Philadelphia, PA: Wolters Kluwer Health.)

- Secure the stylet within the tube. *This measure stiffens the tube and facilitates insertion.*
- Insert the tube into the nose (Fig. C) until it reaches the second mark. *Doing so places the tube in the presumed area of the stomach.*

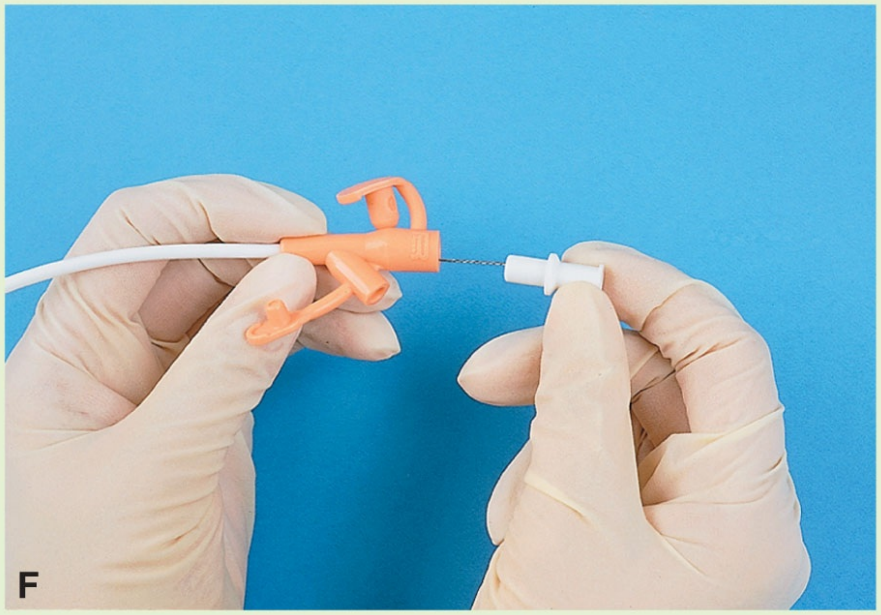
Inserting the feeding tube. (From Springhouse. [2009]. *Lippincott's visual encyclopedia of clinical skills.* Philadelphia, PA: Wolters Kluwer Health.)

- Assess for signs of respiratory distress. *Coughing and difficulty breathing suggest the distal end of the tube is in the respiratory tract.*
- Aspirate fluid using a 30–50 mL syringe (Fig. D) and test the fluid pH (Fig. E).

Aspirating fluid from the tube.

Checking the pH of aspirated fluid.

- Drop a sample of aspirated fluid onto an indicator strip (Fig. F). *This step initiates a chemical reaction on contact and saturation.*



Removing the stylet.

- Compare the color on the test strip with the color guide on the container of reagent strips. *The color of the test strip changes according to the hydrogen ion concentration of the liquid. Stomach fluid usually has a pH of 5 or less even in those receiving gastric–acid inhibitors; secretions from the small intestine have a pH of 6 or greater (*American Association of Critical Care Nurses, 2009).
- Loop the tubing if there is no respiratory distress, and tape it temporarily to the cheek if the desired

distal location is beyond the pyloric valve. *Looping provides slack so the tube can descend into the small intestine.*
- Ambulate or position the client on his or her right side for at least 1 hour or the time specified in the agency's policy. *This duration allows the tube to move by gravity through the pyloric valve.*

Stop, Think, and Respond 29-1

Discuss the consequences of inserting a nasogastric tube into the respiratory passages.

Use and Maintenance of Nasogastric Tubes

Nasogastric tubes are connected to suction for gastric decompression or are used for tube feeding.

Gastric Decompression

Suction is either continuous or intermittent. Continuous suctioning with an unvented tube can cause the tube to adhere to the stomach mucosa, resulting in localized irritation and interfering with drainage. Using a vented tube or intermittent suction prevents or minimizes these effects.

The tube is connected to a wall outlet or portable suction machine (Fig. 29-7). The suction setting is prescribed by the physician or indicated in the agency's standards for care. Usually low pressure (40 to 60 mmHg) is used. The tube is clamped or plugged during ambulation or after instilling medications (see Chap. 32).

FIGURE 29-7 Suction removes liquids and gas from the stomach.

Promoting Patency

Even with intermittent suctioning, the tube may become obstructed. Giving ice chips or occasional sips of water to a client who is otherwise NPO (receives nothing by mouth) promotes tube patency. The fluid helps dilute the gastric secretions. Both must be given sparingly, however, because water is hypotonic and draws electrolytes into the gastric fluid. Because the diluted fluid is ultimately removed, giving the client liberal amounts of water can deplete serum electrolytes (see Chap. 16).

Restoring Patency

The nurse assesses tube patency frequently by monitoring the volume and characteristics of drainage and observing for signs and symptoms suggesting an obstruction (nausea, vomiting, and abdominal distention). Inspection of the equipment helps to identify possible causes for the assessment findings (Table 29-2). Once the cause is identified, a variety of simple nursing interventions can resolve it. Sometimes the nasogastric tube must be irrigated to maintain or restore patency (Skill 29-2). The nurse must obtain a medical order before attempting an irrigation.

TABLE 29-2 Troubleshooting a Poorly Draining Nasogastric Tube

POSSIBLE CAUSES	SOLUTIONS
The drainage holes are adhering to the gastric mucosal wall.	Turn the suction off momentarily. Change the client's position.
The tube is displaced above the esophageal sphincter.	If the measured mark is not at the tip of the nose, remove the tape, advance the tube, check placement, and resecure.
The portable suction machine is disconnected or turned off.	Replace the plug into the electrical outlet or turn on power.
The drainage container is filled beyond capacity.	Empty and record the amount of drainage in the suction container.
The vent is acting as a siphon.	Instill a bolus of air into the vent to restore patency.
The vent is capped or plugged.	Remove the cap and restore the port to atmospheric pressure.
The tubing is kinked or disconnected.	Straighten tubing or reconnect to the suction machine.
The suction is inadequate.	Check that the pressure is 40–60 mmHg.
The cover on the suction container is loose.	Resecure the lid to the container.
A solid particle or thick mucus obstructs the lumen.	Increase suction pressure momentarily. Obtain and implement a medical order for an irrigation.

Enteral Nutrition

Enteral nutrition (nourishment provided through the stomach or small intestine rather than by the oral route) is delivered by instilling formula through a tube. Although a nasogastric tube can be used, it is more likely that liquid formula will be administered through a nasointestinal or transabdominal tube. Both are discussed later in this chapter.

Stop, Think, and Respond 29-2

Explain the reason for using an isotonic saline solution, rather than a hypotonic or hypertonic solution, to irrigate a nasogastric tube.

Removal of a Nasogastric Tube

Nurses remove a nasogastric tube (Skill 29-3) when the client's condition improves, when the tube becomes hopelessly obstructed, or according to the agency's standards for maintaining the integrity of the nasal mucosa. Unobstructed larger diameter tubes usually are removed and changed at least every 2 to 4 weeks for adults (Humphries, 2011–2012). Small-diameter, flexible tubes are removed and changed every 4 weeks to 3 months, depending on the agency's policy. Tubes used for pediatric clients are changed more frequently because the tissue is more fragile and there is a greater potential for infection.

Before permanent removal, some physicians prescribe a trial period during which the tube is clamped and the client is allowed to consume oral fluids. Remaining asymptomatic (i.e., no nausea, vomiting, or gastric distention) is a good indication that the client no longer requires intubation. If symptoms develop, the tube is already in place and can be easily reconnected to suction. This practice avoids subjecting the client to the discomfort associated with tube replacement.

Stop, Think, and Respond 29-3

If the client who has just had a nasogastric tube removed wants something to eat, what nursing actions are appropriate?

NASOINTESTINAL TUBE MANAGEMENT

Nurses also insert nasointestinal tubes used for enteral feeding.

Insertion of a Nasointestinal Tube

The techniques for client preparation, positioning, and advancement of nasointestinal tubes are similar to those for nasogastric tubes. Some modifications are necessary, however, because nasointestinal tubes are constructed differently (see Nursing Guidelines 29-1; Nursing Guidelines 29-2).

To estimate the length of tube required for an intestinal placement, the nurse determines the NEX measurement and adds 9 inches (23 cm). He or she also marks the additional measurement on the tubing.

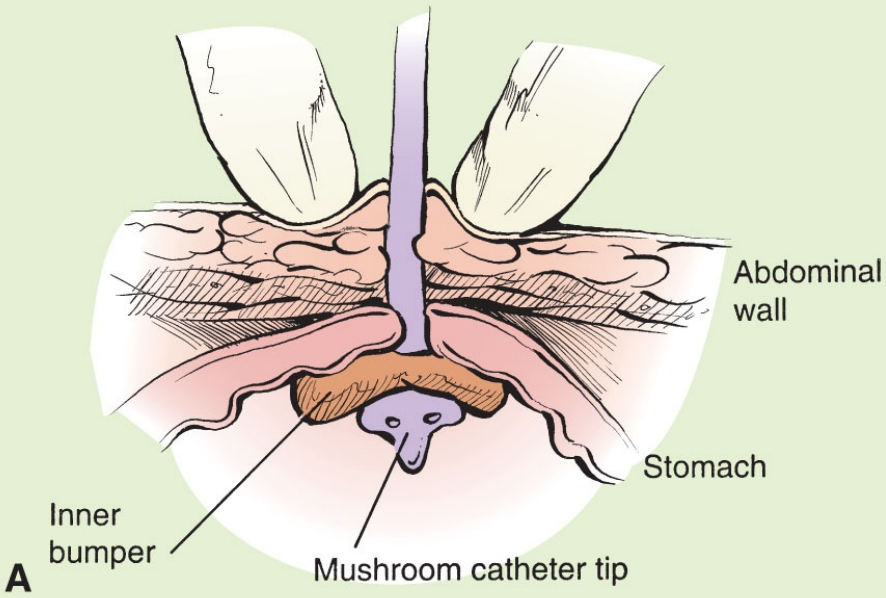
New technologies that promise to promote safety and efficacy in nasoenteric tube placement are becoming available. A computer system that uses electromagnetic technology to direct and locate a feeding tube has also been developed. It consists of an electronically modified feeding tube and a receiver that is placed externally over the mid abdomen. A computer then converts the signal into a graphic display. This helps to identify misplacement immediately,

and subsequent use eliminates the need for repeated radiographic verification of its location (Gerritson et al., 2015; Mathus-Vliegen et al., 2010).

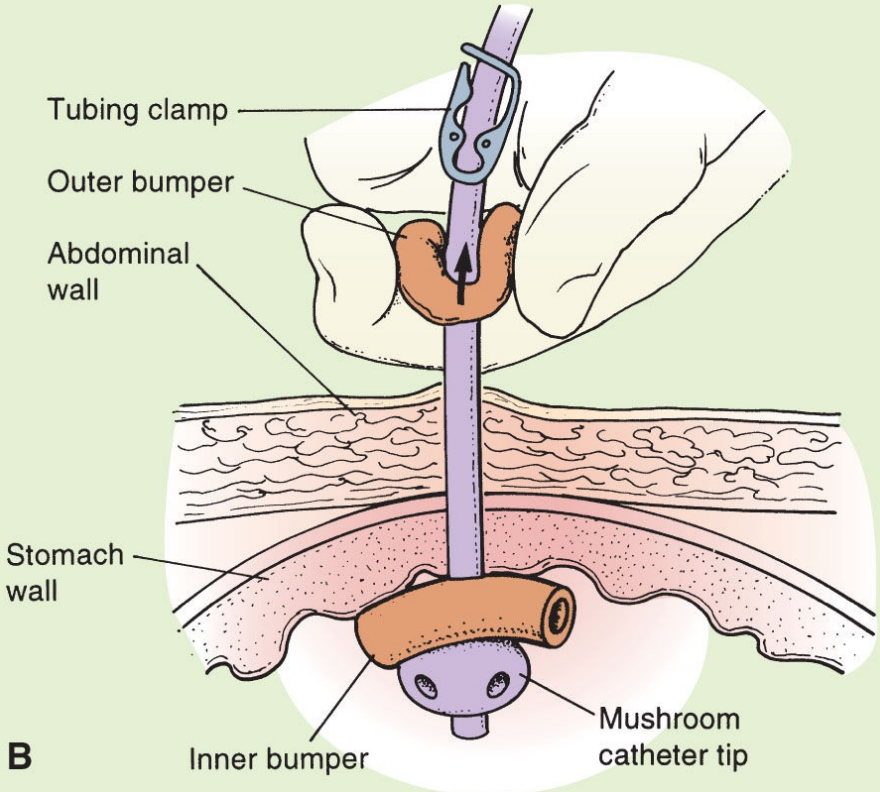
NURSING GUIDELINES 29-2

Managing a Gastrostomy

- Wash hands or perform an alcohol-based hand rub (see Chap. 10). *Hand hygiene reduces the transmission of microorganisms.*
- Put on gloves. *They provide a physical barrier between the nurse's hands and body fluids.*
- Assess and replace the gauze dressing over a new gastrostomy if it becomes moist; slight bleeding or a clear serous drainage from the wound is normal for a few weeks after the procedure. *These measures reduce the conditions that support the growth of microorganisms and maceration of the skin.*
- Remove and discontinue the dressing after the first 24 hours unless the physician orders otherwise. *This facilitates assessment.*
- Inspect the skin around the tube daily. *Regular monitoring provides assessment data about the status of wound repair* (Fig. A).
- Make sure that the sutures holding a surgically placed tube are intact. *Checking prevents tube migration.*
- Report any redness or tissue maceration. *These findings indicate early skin impairment.*
- Apply a skin barrier ointment such as zinc oxide, karaya gum wafer, hydrocolloid dressing, or ostomy pouch if the skin appears irritated (see section on "Ostomy Care," Chap. 31). *Such barriers protect the skin and promote healing.*
- Press down on the skin at the base of the tube. If the client has a PEG tube, compress the arms of the external bumper together and lift them about 1 inch (2.5 cm) (Fig. B). *These steps aid in assessing for drainage, which normally disappears by the end of the first week.*
- Clean the skin with half-strength hydrogen peroxide or 0.9% saline. After 1 week, using soap and water is sufficient. Dry the skin well using air or a blow dryer on a cool or low heat setting. *Appropriate cleaning removes secretions and reduces microorganisms.*
- Rotate the direction of the external bumper 90 degrees or other external retaining device at least once a day. *Doing so relieves pressure and maintains skin integrity.*
- Slide the external bumper down so it is flush with the skin. *Sliding restabilizes the tube.*
- Avoid placing any type of dressing material under the arms of the external bumper. *This helps to avoid creating pressure on the internal bumper and damaging the tissue.*
- Replace the water in the balloon beneath the bumper weekly using a Luer-tip (not Luer-lok) syringe. *This keeps the balloon fully inflated and prevents tube migration.*



Inspecting for drainage.



Inspecting the skin.

- Tape the G-tube to the abdomen or secure it with an abdominal binder or commercial tube stabilizer. *Appropriately securing the tube maintains its position.*

- Make sure the tube is not kinked and the skin is not stretched. *These assessments ensure tube patency and skin integrity.*
- Insert a Foley catheter (see Chap. 30), if the client is not sensitive to latex, 2–5 inches (5–10 cm) within the opening, and inflate the balloon if the tube comes out. *Doing so maintains temporary access to the stomach and, if done within 3 hours of accidental extubation, prevents the site from closing.*
- Use the G-tube in a manner similar to how a nasogastric tube is used for administering feedings. *The tube facilitates nourishment.*

TRANSABDOMINAL TUBE MANAGEMENT

The physician inserts transabdominal tubes, such as G- and J-tubes, but the nurse is responsible for assessing and caring for them and their insertion sites. Conscientious care is necessary because G-tubes may leak (Box 29-1) and cause skin breakdown.

TUBE FEEDINGS

Providing nutrition by the oral route is always best. However, if oral feedings are impossible or jeopardize the client's safety, nourishment is provided enterally or parenterally (see section on Total Parenteral Nutrition, Chap. 16). Tube feedings are used when clients have an intact stomach or intestinal function but are unconscious, have undergone extensive mouth surgery, have difficulty swallowing, or have esophageal or gastric disorders. Skill 29-4 describes the technique for administering tube feedings.

BOX 29-1 Causes of Gastrostomy Leaks

- Disconnection between the feeding delivery tube and G-tube.
- Clamped G-tube while tube feeding is infusing.
- Mismatch between the size of the G-tube and stoma.
- Increased abdominal pressure from formula accumulation, retching, sneezing, and coughing.
- Underinflation of the balloon beneath the skin.
- Less than optimal stoma or stomal location.

Gerontologic Considerations.

■ Long-term use of tube feedings in older adults with dementia or other chronic declining conditions involves many ethical considerations. Refusal to eat, which may be intentional or unintentional, is associated with a variety of conditions, including depression, suicidal behaviors, or advanced cognitive impairment.

■ Caregivers and all health-care providers must comprehensively assess decisions about whether a feeding tube is inserted or removed. In all situations,

nurses should follow the 2011 American Nurses Association position statement which states that adults, or in the event of their incapacity, their surrogates, are in the best position to weigh the harm and benefit of nutrition and hydration, following evaluation and discussion with their health-care team (American Nurses Association, 2011)

- Nurses need up-to-date knowledge about ethical and legal issues related to the use of tube feedings (see Chap. 3).

TABLE 29-3 Comparison of Feeding Tubes

TUBE	ADVANTAGES	DISADVANTAGES
Nasogastric	Low incidence of obstruction	Can damage nasal and pharyngeal mucosa from pressure or friction;
	Accommodates crushed medications	Dilates esophageal sphincter, potentiating gastric reflux
	Facilitates bolus or intermittent feedings	Potential for aspiration
	Easy to check distal placement and gastric residual	Requires frequent replacement to ensure the integrity of nasal tissue
Nasointestinal	Easy to insert	Requires X-ray to verify placement
	Comfortable	Becomes obstructed easily
	Only slight dilation of esophageal sphincter	Best used for continuous feeding
	Reduced danger for aspiration	
	Can remain in place for 4 weeks or longer	
Gastrostomy	No nasal tube	Must wait 24 hours to use after initial placement
	Easily concealed	May leak and cause skin breakdown
	Accommodates long-term use	Increased incidence of infection
	Infrequent tube replacement	Requires skin care at tube site
	Client can be taught self-care	Can migrate or become dislodged if tube is not secured
		Gastric overfill and aspiration possible
Jejunostomy	Same as gastrostomy	Same as gastrostomy
	Reduced potential for reflux and aspiration	

Benefits and Risks of Tube Feedings

Tube feedings are delivered through a nasogastric, nasointestinal, or transabdominal tube. Each has its advantages and disadvantages (Table 29-3).

Instilling nutritional formulas into the stomach uses the body's natural reservoir for food. It also reduces the potential for enteritis (inflammation of the intestine) because the chemicals in the stomach tend to destroy microorganisms. Gastric feedings increase the potential for gastric reflux, however, because of their volume and temporary retention within the stomach.

Although the placement of tubes within the intestine reduces the risk for gastric reflux, it does not eliminate that risk. Additional problems are associated with intestinal tube feedings. For example, an intestinally placed tube may lead to **dumping syndrome** (a cluster of symptoms from the rapid deposition of calorie-dense nourishment into the small intestine). The symptoms, which include weakness, dizziness, sweating, and nausea, are caused by fluid shifts from the circulating blood to the intestine and a low blood glucose level related to a surge of insulin. Diarrhea also may result when administering hypertonic formula solutions.

Formula Considerations

In addition to the type of tube and the access site, the type of formula also is

individualized based on the client's nutritional needs (Table 29-4). Factors include the client's weight, nutritional status, concurrent medical conditions, and the projected length of therapy. The feeding schedule also affects the choice of formula: calories may need to be concentrated if the client is being fed several times a day rather than continuously. Most formulas provide 0.5 to 2.0 kcal/mL with 750 to 800 mL of water per liter (McCarthy & Martindale, 2015).

TABLE 29-4 Tube-Feeding Formulas

TYPE	EXAMPLES	DESCRIPTION
Standard, isotonic	Osmolite Isocal Nutren 1.0	Routine formulas for clients with normal digestion and absorption; do not alter water distribution. Provide approximately 1.0 cal/mL
High calorie	Comply Nutren 1.5 Nutren 2.0 Deliver 2.0	Provide up to double the amount of calories of standard formulas for clients who require a fluid restriction or have high calorie needs.
High protein	Promote Isocal HN Ultracal HN plus	Provide up to double the amount of protein of standard formulas.
Fiber containing	Jevity Compleat Ultracal	Provide fiber to normalize bowel function in clients with diarrhea or constipation.
Partially hydrolyzed	Criticare HN Optimental Vivonex T.E.N.	Provide nutrients in simple form that require little or no digestion for clients with impaired digestion or absorption.

Nutrition Notes

- Specialty formulas are available with altered nutritional profiles for specific disease states, such as for clients with diabetes, renal failure, hepatic failure, respiratory insufficiency, and wound healing. Pediatric formulas are also available.
- Products like BOOST®, Carnation Breakfast Essentials®, and Ensure® are primarily intended as oral supplements, not for tube feedings

Gerontologic Considerations

- Tube-feeding formulas are prescribed based on the older client's condition (i.e., malabsorption syndromes, glucose intolerance). For example, lactose-free tube-feeding formulas may be beneficial to older clients who experience malabsorption syndromes.
- Clients with or who are at risk for pressure sores benefit from formulas fortified with additional zinc, protein, and other nutrients.
- In home and long-term care settings, registered dietitians may be helpful in the ongoing assessment of tube feedings.

Tube-Feeding Schedules

Tube feedings may be administered on bolus, intermittent, cyclic, or continuous schedules.

Bolus Feedings

A **bolus feeding** (the instillation of liquid nourishment in less than 30 minutes four to six times a day) usually involves 250 to 400 mL of formula per administration. This schedule is the least desirable because it distends the stomach rapidly, causing gastric discomfort and an increased risk for reflux. Bolus feedings may be used because they mimic, to some extent, the natural filling and emptying of the stomach. Some clients experience discomfort from the rapid delivery of this quantity of fluid. Clients who are unconscious or who have delayed gastric emptying are at greater risk for regurgitation, vomiting, and aspiration with this method of administration.

Intermittent Feedings

An **intermittent feeding** (the gradual instillation of liquid nourishment four to six times a day) is administered over 30 to 60 minutes, the time most people spend eating a meal. The usual volume is 250 to 400 mL per administration. Intermittent feedings generally are given by gravity drip from a suspended container or with a feeding pump. Gradual filling of the stomach at a slower rate reduces the bloated feeling that can accompany bolus feedings. The container and feeding tube that holds the formula requires thorough flushing after each feeding to reduce the growth of microorganisms. Tube-feeding administration sets are replaced every 24 hours regardless of the feeding schedule.

Cyclic Feedings

A **cyclic feeding** (the continuous instillation of liquid nourishment for 8 to 12 hours) is followed by a 16- to 12-hour pause. This routine often is used to wean clients from tube feedings while continuing to maintain adequate nutrition. The tube feeding is given during the late evening and hours of sleep. During the day, clients eat some food orally. As oral intake increases, the volume and duration of the tube feeding gradually decreases.

Continuous Feedings

A **continuous feeding** (the instillation of liquid nutrition without interruption) is administered at a rate of approximately 1.5 mL/minute. A feeding pump is used to regulate the instillation. Because only a small amount of fluid is instilled at any one time, the formula does not need to be held in the reservoir of the stomach; it can be delivered directly into the small intestine. Instilling small amounts of fluid beyond the stomach reduces the risk of vomiting and aspiration.

Continuous feeding creates some inconvenience, though, because the pump must go wherever the client goes.

Gerontologic Considerations

- Older adults tend to tolerate small, continuous feedings.

Client Assessment

The following daily assessments are standard for almost every client who receives tube feedings: weight, fluid intake and output, bowel sounds, lung sounds, temperature, condition of the nasal and oral mucous membranes, breathing pattern, gastric complaints, status of abdominal distention, vomiting, bowel elimination patterns, and skin condition at the site of a transabdominal tube. Once tube feedings have been initiated, it is also necessary to routinely assess the client's **gastric residual** (the volume of liquid within the stomach). The nurse measures the gastric residual to determine whether the rate or volume of feeding exceeds the client's physiologic capacity. Overfilling the stomach can cause gastric reflux, regurgitation, vomiting, aspiration, and pneumonia. As a rule of thumb, the gastric residual should be no more than 100 mL or no more than 20% of the previous hour's tube-feeding volume. According to McCarthy and Martindale (p. 40, 2015), "Gastric residual volumes (GVR) in the range of 200 to 500 mL should raise concern and lead to the implementation of measures reducing risk of aspiration, but automatic cessation of feeding shouldn't occur for a GVR less than 500 mL in the absence of other signs of intolerance." If the gastric residual is high, the feeding is stopped and the gastric residual is rechecked every 30 minutes until it is within a safe volume for resuming the feeding (Nursing Guidelines 29-3).

NURSING GUIDELINES 29-3

Checking the Gastric Residual

- Wash hands or perform an alcohol-based hand rub (see Chap. 10). *Hand hygiene reduces the transmission of microorganisms.*
- Don gloves. *Gloves provide a physical barrier between the nurse's hands and body fluids.*
- Stop the infusion of the tube-feeding formula. *This measure facilitates assessment.*
- Aspirate fluid from the feeding tube using a 50-mL syringe. *Doing so allows for the collection of a large volume of fluid.*
- Continue aspirating until no more fluid is obtained. *This ensures an accurate assessment.*
- Measure the aspirated fluid and record the amount. *Documentation provides objective data for evaluation.*
- Reinstill the aspirated fluid. *This measure returns partially digested nutrients and electrolytes to the*

client.

- Postpone tube feeding and report residual amounts that exceed agency guidelines or those established by the physician. *Doing so reduces the risk of aspiration.*
- Check gastric residual again in 30 minutes. *This duration allows time for part of the stomach contents to empty into the small intestine.*
- Provide or resume tube feeding if the gastric residual is within an acceptable range. *Doing so prevents overfeeding.*
- Flush a feeding tube with 30 mL of water with a large volume syringe before and after an intermittent feeding or administration of medication, any interruption of feeding, and every 4 hours during a continuous feeding (McCarthy & Martindale, 2015). *Clients require additional water for hydration and to prevent obstructions within the feeding tube.*
- Document the volume of flushes separately from the volume of formula. *Intake and output records facilitate monitoring clients for fluid volume excess or deficit.*

Stop, Think, and Respond 29-4

If a client's nutritional needs are met entirely with tube feedings, what effects might that have on the person physically, emotionally, and socially?

Nursing Management

Caring for clients with feeding tubes generally involves maintaining tube patency, clearing any obstructions, providing adequate hydration, dealing with common formula-related problems, and preparing clients for home care.

Maintaining Tube Patency

Feeding tubes, especially those smaller than 12 Fr are prone to obstruction. Common causes are using formulas with large-molecule nutrients, refeeding partially digested gastric residual, administering formula at a rate less than 50 mL/hour, and instilling crushed or hydrophilic (water-absorbing) medications into the tube. To maintain patency, it is best to flush feeding tubes with 30 to 60 mL of water immediately before and after administering a feeding or medications, after any interruption in a feeding, every 4 hours if the client is being continuously fed, and after refeeding the gastric residual. Exceptions for varying the volume of flush solution would be for clients on fluid restrictions such as those with renal or heart failure.

Clearing an Obstruction

If an obstruction occurs, the nurse consults the physician. Occasionally, it is possible to clear the tube with a solution of warm water in a 30- to 60-mL syringe using a back-and-forth motion, withdrawing the instilled fluid, and repeating the process if necessary (see Nursing Guidelines 29-4). Another option, if declogging with water is unsuccessful, is instilling fluid-activated pancreatic enzyme combined with 1/8 tsp of sodium bicarbonate followed by

letting it dwell for 30 minutes before attempting to flush the tube again (McCarthy & Martindale, 2025). The latter method requires a written medical order. There are other possible mechanical means for clearing obstructions. One is a product called Bionix Feeding Tube Declogger®, which is a stem with a "screw-and-thread design" that when inserted bends and conforms to the shape of the tube freeing the obstruction (Fisher & Blalock, 2014). And similarly, the TubeClear® system provides a stem with a wire encased within a sheath. A power source creates a jackhammer-like motion at the wire tip inside the obstructed tube (Activated Medical, 2015). When an obstruction cannot be cleared, the tube is removed and another is inserted rather than compromising nutrition by the delay.

Providing Adequate Hydration

Although tube feedings are approximately 80% water, clients usually require additional hydration. Adults require 30 mL of water per kilogram of body weight, or 1 mL/kcal, on a daily basis (Dudek, 2013).

To determine whether or not a client's hydration needs are being met, the nurse identifies the amount of water on the label of commercial formula. He or she can then add this amount to the total volume of flush solution and compare it with the recommended amount. If there is a significant deficit, the nurse revises the plan of care to increase either the volume or, preferably, the frequency of flushing the tube. If the fluid volume is excessive, the nurse monitors the client's urine output and lung sounds to determine whether or not the client can excrete comparable amounts (see Chap. 16).

NURSING GUIDELINES 29-4

Clearing an Obstructed Feeding Tube

- Select a syringe with a 30–60 mL capacity. *This capacity reduces negative pressure during aspiration, which could lead to the collapse of the tube walls.*
- Wash hands or perform an alcohol-based hand rub (see Chap. 10). *Hand hygiene reduces the transmission of micro-organisms.*
- Don gloves. *They provide a physical barrier between the nurse's hands and potential contact with body fluids.*
- Aspirate as much as possible from the feeding tube. *Aspiration clears the path above the obstructing debris.*
- Instill 5 to 15 mL of the selected solution. *Instillation allows for direct contact between the irrigating solution and debris.*
- Clamp the tube and wait 15 minutes. *This duration gives the substance in the solution time to physically affect the obstructing debris.*
- Aspirate and flush the tube with water. *Repeat if necessary. Use of negative pressure or positive pressure restores patency.*
- Consult the physician if unable to clear the obstruction with instilled water. *Chemical and mechanical*

measures are alternative options for removing an obstruction in a tube.

Handling Miscellaneous Problems

Clients who require enteral feeding experience several common or potential problems. Many are associated with tube-feeding formulas or the mechanical effects of the tubes themselves (Table 29-5). Nurses must report problems promptly and make necessary adjustments to the plan of care.

Pharmacologic Considerations

- Although liquid medications are preferred for enteral administration, they can cause GI distress. Sorbitol, an inactive ingredient, is used as a sweetener for oral liquid medications. In large amounts, such as multiple medications administered via enteral tube, sorbitol acts as an osmotic laxative. Tablets crushed and mixed with water may reduce diarrhea if it occurs.

Gerontologic Considerations

- Older adults are at increased risk for fluid and electrolyte disturbances and, as a result, may develop hyperglycemia (elevated blood glucose levels) when tube feedings are administered.
- If an older client is receiving tube feedings with full-strength formula concentrations, it is important to check capillary blood glucose levels at intervals until the client's results are within a normal range.
- Monitor older adults for agitation or confusion, which may cause them to pull out feeding tubes inadvertently. Also, a change in behavior or mental status may be an early indicator of a fluid or electrolyte imbalance.

Preparing for Home Care

Because of shortened lengths of stay in hospitals, some clients who continue to need tube feedings are discharged for self-care or a caretaker when home. Before demonstrating the procedure to those who will administer tube feedings, the nurse provides a written instruction sheet that includes the following:

- Places to obtain equipment and formula
- The amount and schedule for each feeding and flush, using household measurements
- Guidelines for delaying a feeding

- Special instructions for skin, nose, or stomal care, including frequency and types of products to use
- Problems to report such as weight loss, reduced urination, weakness, diarrhea, nausea and vomiting, and breathing difficulties
- Names and phone numbers of people to call if questions arise
- Date, time, and place for continued medical follow-up

Depending on the client's or caretaker's self-confidence and competence in self-administering tube feedings, health-care providers often make a referral to a home health agency for postdischarge nursing support.

Gerontologic Considerations

- When teaching older adults or older caregivers how to manage a G-tube or administer tube feedings at home, allow more time for processing information and include several practice sessions.
- A referral for skilled nursing care, which may be covered by Medicare/Medicaid or private health insurance plans, may be appropriate for ongoing teaching and assessments for clients who require tube feedings after being discharged.
- For older adults living on a fixed income, dietitians can suggest ways to prepare less costly, home-blended formulas that meet the client's nutritional needs.

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INTESTINAL DECOMPRESSION

Most nasogastric, nasointestinal, and transabdominal tubes are used for enteral

feeding or gastric decompression. Sometimes, however, clients require intestinal decompression, which is performed with a tungsten-weighted tube (see Table 29-1). Intestinal decompression sometimes makes it possible to avoid surgery.

TABLE 29-5 Common Tube-Feeding Problems

PROBLEM	COMMON CAUSES	SOLUTIONS
Diarrhea	Highly concentrated formula Rapid administration Bacterial contamination Lactose intolerance Inadequate protein content Medication side effects	Dilute initial tube feeding to one-quarter to one-half strength. Start at 25 mL/hour and increase rate by 25 mL q12hour. Hang no more than 4 hours' worth of formula. Wash hands. Change formula bag and tubing q24hour. Refrigerate unused formula. Consult with the physician on using a milk-free formula. Raise serum albumin levels with total parenteral nutrition solutions containing supplemental protein, or administer albumin intravenously. Consult with the physician about adjusting drug therapy or administering an antidiarrheal.
Nausea and vomiting	Rapid feeding Overfeeding Air in stomach Medication side effects	Instill bolus and intermittent feedings by gravity. Delay feeding until gastric residual is <100 mL or <20% of hourly volume. Maintain sitting position for at least 30 minutes after feeding. Consult with the physician about ordering medication that facilitates gastric emptying. Administer continuous feedings. Instill feedings within the small intestine. Keep tubing filled with formula or water. Consult with the physician about adjusting drug therapy or administering drugs to control symptoms.
Aspiration	Incorrect tube placement Vomiting	Check placement before instilling liquids. Keep head elevated at least 30 degrees during feedings and for 30 minutes afterward. Keep cuffed tracheostomy and endotracheal tubes inflated. Refer to measures for controlling vomiting.
Constipation	Lack of fiber Dehydration	Change formula. Increase supplemental water. Consult with the physician on giving a laxative, enema, or suppository.
Elevated blood glucose level	Calorie-concentrated formula	Instill diluted formula and gradually increase concentration. Administer insulin according to medical orders.
Weight loss	Inadequate calories	Increase calories in formula. Increase rate or frequency of feedings.
Elevated electrolytes	Dehydration	Increase supplemental water.
Dry oral and nasal mucous membranes	Mouth breathing Dried nasal mucus	Provide frequent oral and nasal hygiene.
Middle ear inflammation	Narrowing or obstruction of eustachian tube from presence of tube in pharynx	Turn from side to side q2hour. Insert a small-diameter feeding tube.
Sore throat	Pressure and irritation from tube	Use a small-diameter feeding tube.
Plugged feeding tube	Instilling crushed or powdered medications through the tube Formula coagulation from drug-food interactions Kinked tube Large molecules in formula	Use liquid medications. Dilute crushed drugs. Flush the tubing liberally after drug administration. Flush tubing with water before and after drug administration. Follow agency policy for alternative flush solutions. Maintain neck in neutral position or change position frequently. Dilute formula. Flush tubing at least q4hour. Use a larger diameter feeding tube.
Dumping syndrome	Rapid and large instillation of highly concentrated formula into the intestine	Administer small, continuous volume. Adjust glucose content of formula.

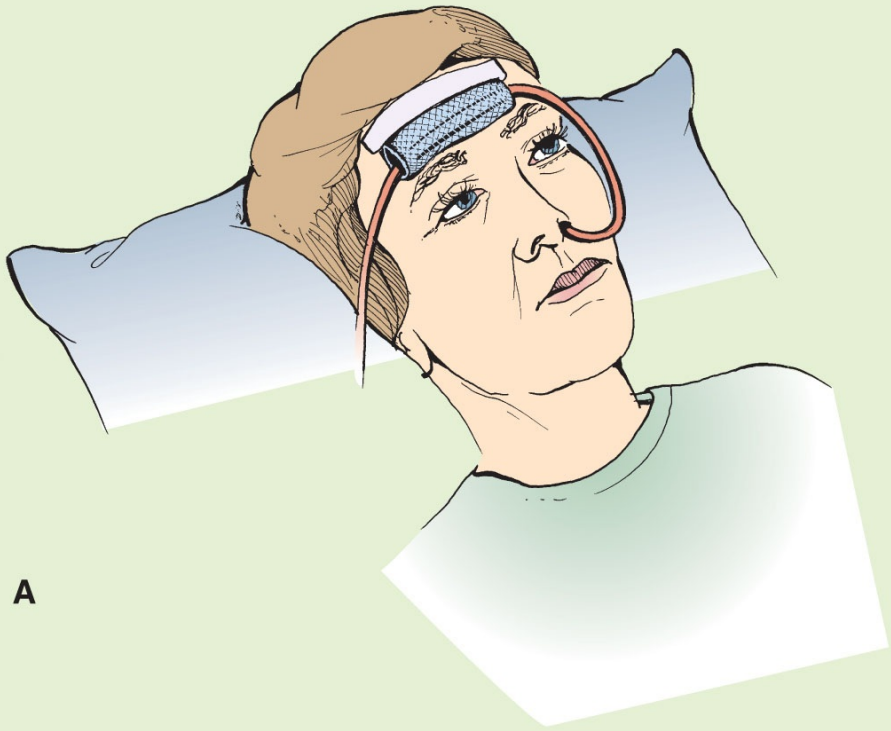
Nasointestinal Decompression Tube Insertion

A nasointestinal decompression tube is inserted in the same manner as a nasogastric tube. The nurse then promotes and monitors its passage into the intestine. In the presence of peristalsis, the weight of the tungsten propels the tip of the tube beyond the stomach. Openings through the distal end provide channels through which the intestinal contents are suctioned. An intestinal decompression tube generally remains in place until the intestinal lumen is patent or until surgical treatment is instituted (see Nursing Guidelines 29-5).

NURSING GUIDELINES 29-5

Inserting an Intestinal Decompression Tube

- Assemble all the necessary equipment as for any nasally inserted tube. *Doing so ensures organization and efficient time management.*
- Follow the techniques in Skill 29-1 for inserting a nasogastric tube. *The same principles are involved during the initial insertion.*
- Thread excess tubing through a sling of folded gauze taped to the forehead (Fig. A) once gastric placement is confirmed. *The sling supports the tube as it advances.*



Fashioning a gauze sling.

- Ambulate the client, if possible. *Ambulation helps the tube to move through the pyloric valve into the small intestine.*
- When the radiograph indicates that the intestinal tube has advanced beyond the stomach, position the client on the right side for 2 hours, then on the back in a Fowler's position for 2 hours, then on the left side for 2 hours. *Gravity and positioning promote movement through intestinal curves.*
- Follow agency policy or physician's instructions for manually advancing the tube several inches each hour. *This advancement supplements the natural peristaltic advancement.*
- Observe the graduated marks on the tube. *The marks provide a means for monitoring the tube's progression and approximate anatomic location.*
- Request an X-ray confirmation when the tube has reached the prescribed distance. *An X-ray provides objective evidence of the terminal location of the distal tip.*
- Secure the tube to the nose once its distal location has been confirmed. *Stabilizes the tube and prevents further migration.*
- Coil the excess tubing and attach it to the client's pajamas or gown. *Coiling and attachment prevent*

accidental extubation.
- Connect the proximal end to a wall or a portable suction source. *Produces negative pressure to pull substances from the intestine.*

Removal of an Intestinal Decompression Tube

Once the intestinal decompression tube has served its purpose, the nurse begins the process of removing it. An intestinal decompression tube is removed slowly because removal is in a reverse direction through the curves of the intestine and the valves of the lower and upper ends of the stomach.

First, the tube is disconnected from the suction source. Next, the tape that secures the tube to the face is removed and the tube is withdrawn 6 to 10 inches (15 to 25 cm) at 10-minute intervals. When the last 18 inches (45 cm) remains, the tube is pulled gently from the nose. Afterward, nasal and oral hygiene measures are provided. The tube cannot be removed nasally if the distal end descends below the ileocecal valve between the small and large intestine. Instead, the proximal end is cut and the tube is gradually removed manually or by peristalsis when it descends through the anus.

NURSING IMPLICATIONS

Depending on data collected during client care, the nurse may identify one or more of the following nursing diagnoses:

- Imbalanced Nutrition: Less Than Body Requirements
- Feeding Self-Care Deficit
- Impaired Swallowing
- Risk for Aspiration
- Impaired Oral Mucous Membrane
- Diarrhea
- Constipation

Clinical Scenario A 56-year-old male with metastatic cancer has been admitted while undergoing chemotherapy. He has been receiving bolus tube feedings due to side effects of his cancer treatment. Currently he has become nauseous, and reports abdominal discomfort and the feeling that vomiting is imminent. Before administering the next bolus feeding, the nurse checks the client's gastric residual volume because overfeeding could lead to aspiration. The nurse obtains a gastric residual volume of 550 mL.

Nursing Care Plan 29-1 is a model for managing the care of a client with a large gastric residual with a nursing diagnosis of Risk for Aspiration, defined by NANDA-I (2015, p. 381) as "vulnerable to entry of gastrointestinal secretions, oropharyngeal secretions, solids, or fluids into tracheobronchial passages, which may compromise health."

NURSING CARE PLAN 29-1 Risk for Aspiration

Assessment

- Measure the gastric residual before each bolus feeding and whenever the client develops gastric symptoms.
- Auscultate bowel sounds.
- Palpate the abdomen and measure abdominal girth for evidence of distention.
- Ask an alert client about feeling full, nauseous, or vomiting.
- Check if any medical orders restrict the positioning of a client in a Fowler's position.

Nursing Diagnosis. **Risk for Aspiration** related to slow gastric emptying as manifested by the measurement of the gastric residual of 550 mL from a 16 F nasogastric tube and the development of feeling nauseous, bloated, and potential for vomiting 2 hours before a scheduled bolus feeding of 400 mL.

Expected Outcome. The client's risk for aspiration will be reduced as evidenced by a gastric residual of

less than 500 mL within 30 minutes of a scheduled feeding.

Interventions	Rationales
Maintain head elevation at no less than 30 degrees at all times.	Elevating the upper body promotes the deposition of the tube-feeding formula within the stomach and movement toward the small intestine.
Monitor bowel sounds; report if absent or fewer than five per minute.	Active bowel sounds suggest that peristalsis is sufficient to facilitate gastric emptying and intestinal absorption and the elimination of liquid nourishment.
Measure the length of the tube as it exits the naris.	Checking the proximal length of a feeding tube provides evidence that the end of the tube has not migrated from its insertion location.
Measure gastric residual before all tube feedings.	This standard of care helps to determine the client's response to liquid nourishment via a gastric tube.
Refeed gastric residual and follow with a 15- to 30-mL tap water flush for a total of no more than 500 mL.	Gastric residual contains partially digested nutrients that should not be discarded; flushing the tube following refeeding helps to prevent obstruction within the tube and provides additional water intake.
Postpone tube feeding for 30 minutes if the gastric residual measures 500 mL.	Distention of the stomach with additional formula predisposes the client to regurgitation and the potential for aspiration.
Report the gastric residual volume to the physician if it remains above the maximum volume after delaying feeding for 30 minutes.	Sharing the assessment findings with the physician facilitates collaboration in modifying the plan of care by changing the type, volume, or frequency of the tube feeding, or administering a medication that promotes gastric emptying.
Keep the client in a high Fowler's position. Maintain a suction machine at the bedside.	A Fowler's position promotes forward movement of gastric contents toward the small intestine. Having equipment for performing oral–pharyngeal suctioning ensures a rapid response for clearing the upper gastrointestinal tract and airway following episodes in which the client vomits.

Evaluation of Expected Outcomes

- Gastric residual measures less than 500 mL.
- Bowel sounds are present and active in all quadrants.
- Head is elevated 30 degrees.
- Tube feeding has resumed on schedule.

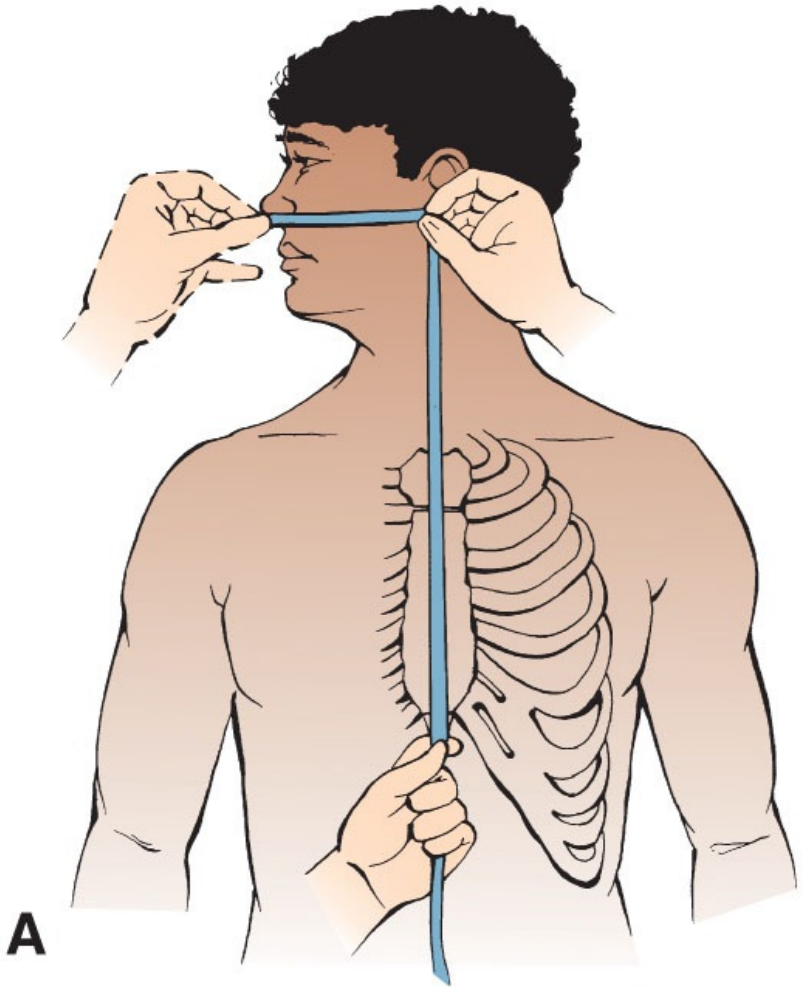
CRITICAL THINKING EXERCISES

1. What nutritional suggestions could a nurse make for a client who has a chronic disease that impairs the ability to swallow food?
2. When a client experiences persistent gagging during attempts to insert a nasogastric tube, what actions can the nurse take?

3. Describe the similarities and differences between inserting a tube for gastric decompression and one for intestinal decompression.
4. What questions would be important to ask if a client receiving tube feedings at home calls to report the onset of diarrhea?

NCLEX-STYLE REVIEW QUESTIONS

1. Place an X on the image where the nurse determines the distal measurement on a nasogastric tube prior to its insertion.



Test-Taking Strategy: Recall the components of the sternum and their locations.

2. When a practical nurse assists with the insertion of a single lumen nasogastric tube, what instruction is correct when the tube is in the client's oropharynx?

1. Breathe deeply as the tube is advanced.
2. Hold your head in a sniffing position.
3. Press your chin to your upper chest.
4. Avoid coughing until the tube is down.

Test-Taking Strategy: Use the process of elimination to select the option that is better than any of the others for facilitating the insertion of a nasogastric tube.

3. What is the most definitive technique for determining whether the distal end of a tube for gastric decompression is in the stomach?

1. Obtaining an abdominal X-ray
2. Checking the pH of aspirated fluid
3. Instilling 100 mL of tap water into the tube
4. Feel for air at the tube's proximal end

Test-Taking Strategy: Note the key term and modifier "most definitive." Select the option that provides the most valid evidence that the nasogastric tube is in the stomach.

4. Immediately after insertion of a transabdominal G-tube, which finding should the nurse consider normal when assessing the gastrostomy site?

1. Milky-appearing drainage
2. Serosanguineous drainage
3. Green-tinged drainage
4. Bright bloody drainage

Test-Taking Strategy: Note the key words "immediately" and "normal." Analyze the choices and select the option that the nurse should consider an appropriate appearance to drainage around a newly inserted G-tube.

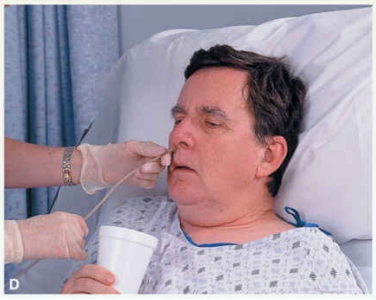
5. When a client with a nasogastric tube for gastric decompression indicates that he or she is very thirsty, which nursing intervention is most appropriate to add to the plan of care?

1. Offer fluids at least every 2 hours.
2. Provide crushed ice in sparse amounts.
3. Increase oral liquids on the dietary tray.
4. Refill the water carafe twice each shift.

Test-Taking Strategy: Note the key word and modifier "most appropriate." Use the process of elimination to select the option that is better than any of the others.

SKILL 29-1 Inserting a Nasogastric Tube

Suggested Action	Reason for Action
ASSESSMENT	
Check that a medical order has been written.	Ensures that care is within the legal scope of practice.
Determine the reason for the nasogastric tube.	Facilitates the evaluation of outcomes.
Identify the client.	Ensures that the procedure will be performed on the correct client.
Assess how much the client understands about the procedure.	Indicates the need for and the level of health teaching.
Inspect the nose after the client blows into a paper tissue (Fig. A).	Provides data that will determine which naris to use.
	Clearing the nose. (Photo by B. Proud.)
Unwrap and uncoil the tube.	Straightens the tube and releases bends from product packaging.
Obtain the NEX measurements (Fig. B).	Determines length for insertion.

Mark the tube at the NE (nose-to-ear) and EX (ear-to-xiphoid) measurements (Fig. C).	Provides a guide during insertion.
	Marking the tube. (Photo by B. Proud.)
PLANNING	
If a plastic tube feels rigid, place it in or flush it with warm water.	Promotes flexibility.
Assemble the following equipment, in addition to the tube: water, straw, towel, lubricant, tissues, tape, emesis basin, flashlight, stethoscope, clean gloves, and 50-mL syringe.	Contributes to organization and efficient time management.
Place a suction machine at the bedside if the client is unresponsive or has difficulty swallowing.	Provides a method for clearing the client's airway of vomitus.
Remove dentures.	Avoids choking should they become loose or displaced.
Establish a hand signal for pausing.	Relieves anxiety by providing the client with some locus for control.
IMPLEMENTATION	
Wash your hands or perform an alcohol-based hand rub (see Chap. 10).	Reduces the transmission of microorganisms.
Pull the privacy curtain.	Demonstrates respect for the client's dignity.
Assist the client to sit in semi-Fowler's or high-Fowler's position and hyperextend the neck as if in a sniffing position.	Ensures the visualization of the nasal passageway to facilitate inserting the tube.
Protect the client, bed clothing, and linen with a towel.	Avoids linen changes.
Put on gloves.	Reduces the transmission of microorganisms.
Lubricate the tube with water-soluble gel over 6–8 inches (15–20 cm) at the distal tip.	Reduces friction and tissue trauma.
Insert the tube into the nostril while pointing the tip backward and downward (Fig. D).	Follows the normal contour of the nasal passage.
	Preparing to insert the tube. (Photo by B. Proud.)

Do not force the tube. Relubricate or rotate it if there is resistance.	Prevents trauma.
Stop when the first mark on the tube is at the tip of the nose.	Places the tip above the area where the gag reflex may be stimulated.
Use a flashlight to inspect the back of the throat.	Confirms that the tube has been maneuvered around the nasal curve.
Instruct the client to lower his or her chin to the chest and swallow sips of water.	Narrows the trachea and opens the esophagus; helps to advance the tube.
Advance the tube 3–5 inches (7.5–12.5 cm) each time the client swallows.	Coordinates insertion; reduces the potential for gagging or vomiting.
Pause if the client gives the preestablished signal.	Demonstrates respect and cooperation.
Discontinue the procedure and raise the tube to the first mark if there are signs of distress such as gasping, coughing, a bluish skin color, or the inability to speak or hum.	Indicates that the tube is possibly in the airway.
Assess placement according to agency policy when the second mark is reached.	Provides data on distal placement.
Withdraw the tube to the first mark and reattempt insertion if the assessment findings are inconclusive, or consult with the physician about obtaining an X-ray.	Ensures safety.
Proceed to secure the tube if the data indicate the tube is in the stomach (Fig. E).	Prevents tube migration.

Securing the tube. (Photo by B. Proud.)

Connect the tube to suction or clamp it while awaiting further orders.	Promotes gastric decompression or potential use.
Remove gloves and wash your hands or use an alcohol-based hand rub.	Reduces the transmission of microorganisms.
Position the client with a minimum head elevation of 30 degrees.	Prevents gastric reflux.
Remove equipment from the bedside.	Restores orderliness and supports the principles of medical asepsis.
Measure and record the volume of drainage at least every 8 hours.	Provides data for evaluating fluid balance.

EVALUATION

- Distal placement within the stomach is confirmed.
- Client exhibits no evidence of respiratory distress.
- Client can speak or hum.
- Lung sounds are present and clear bilaterally.
- No bleeding or pain is noted in the area of nasal mucosa.

DOCUMENT

- Type of tube
- Outcomes of the procedure
- Method for determining placement and outcome of assessment
- Description of drainage
- Type and amount of suction, if the tube is used for decompression

SAMPLE DOCUMENTATION

Date and Time A 16 F Salem sump tube inserted without difficulty. No respiratory distress. Can talk and hum following insertion. Placement verified by radiograph evaluation. Salem sump tube secured to nose and connected to low, intermittent wall suction. Positioned with the head of bed elevated 30 degrees.
J.Doe, LPN

SKILL 29-2 Irrigating a Nasogastric Tube

Suggested Action	Reason for Action
ASSESSMENT	
Monitor the client's symptoms, the volume and rate of drainage, and evidence of abdominal distention.	Provides data for future comparisons.
Check that a medical order has been written, if that is the agency's policy.	Complies with the legal scope of the nursing practice.
Identify the client.	Ensures that the procedure will be performed on the correct client.
Assess how much the client understands about the procedure.	Provides an opportunity for client teaching.
PLANNING	
Assemble the following equipment: Asepto or irrigating syringe, irrigating fluid (isotonic saline solution), container, clean towel or pad, clean gloves, and cover or plug for end of tube.	Contributes to organization and efficient time management.
Turn off the suction.	Facilitates implementation.
IMPLEMENTATION	
Pull the privacy curtain.	Demonstrates respect for the client's dignity.
Wash your hands or perform an alcohol-based hand rub (see Chap. 10).	Reduces the transmission of microorganisms.
Place a clean pad or towel beneath where the tube will be separated.	Avoids changing bed linens and protects the client from soiling.
Put on clean gloves.	Complies with standard precautions.
Disconnect the nasogastric tube from the suction tubing and apply cover or insert plug into suction tubing.	Keeps the connection area clean.
Confirm that the external tube length has not changed.	Ensures safety.
Fill irrigating syringe with 30–60 mL of normal saline solution.	Provides an adequate quantity of isotonic solution to clear tubing.
Insert the tip of the syringe within the proximal end of the tube and allow the solution to flow in by gravity or apply gentle pressure (Fig. A).	Dilutes and mobilizes debris.
A	**Instilling the irrigation solution. (Photo by B. Proud.)**
Aspirate after the fluid has been instilled.	Removes substances that may impair future drainage.
Reconnect the tube to the source of suction.	Resumes therapeutic management.
Observe the characteristics of the aspirated solution; measure and discard.	Provides data for evaluating the effectiveness of the procedure.
Monitor for the flow of drainage through the suction tubing (Fig. B).	Provides evidence that patency is being maintained
B	**Monitoring drainage. (Photo by B. Proud.)**
Remove gloves and perform hand hygiene.	Reduces the transmission of microorganisms.

Record the volume of instilled and drained fluid on the bedside intake and output sheet.	Provides accurate data for determining fluid balance.
EVALUATION	
• Drainage is restored. • Nausea and vomiting are relieved. • Abdominal distention is reduced.	
DOCUMENT	
• Volume and type of fluid instilled • Appearance and volume of returned drainage • Response of client	

SAMPLE DOCUMENTATION

Date and Time Salem sump tube irrigated with 60 mL of normal saline. Solution instilled with slight pressure. A 100 mL of solution returned with several large mucus particles. Reconnected to low, intermittent suction. Gastric tube draining well at the present time. Abdomen is soft. No vomiting. _____ J.Doe, LPN

SKILL 29-3 Removing a Nasogastric Tube

Suggested Action	Reason for Action
ASSESSMENT	
Assess bowel sounds, the condition of the mouth and nasal mucosa, the level of consciousness, and gag reflex.	Provides data for future comparisons and may affect how the procedure is performed.
Check that a medical order has been written.	Complies with the legal scope of nursing practice.
Identify the client.	Ensures that the procedure will be performed on the correct client.
Assess how much the client understands the procedure.	Provides an opportunity for client teaching.
PLANNING	
Assemble the following equipment: towel, emesis basin, cotton-tipped applicator sticks, oral hygiene equipment, and clean gloves.	Contributes to organization and efficient time management.
IMPLEMENTATION	
Pull the privacy curtain.	Demonstrates respect for dignity.
Wash your hands or perform an alcohol-based hand rub (see Chap. 10).	Reduces the transmission of microorganisms.
Place the client in a sitting position, if alert, or in a lateral position if not.	Prevents aspiration of stomach contents.
Cover the chest with a clean towel and place the emesis basin and tissues within easy reach.	Prepares for possible vomiting and protects the client from soiling.
Remove the tape securing the tube to the client's nose.	Facilitates pulling the tube from the stomach.
Put on clean gloves.	Complies with standard precautions.
Turn off the suction and separate the tube.	Prepares for removal.
Instill a bolus of air into the lumen that drains gastric secretions.	Prevents residual fluid from leaking as the tube is withdrawn.

Clamp, plug, or pinch the tube (Fig. A).	Prevents fluid from leaking as the tube is withdrawn.
	Occluding the tube. (Photo by B. Proud.)
Instruct the client to take a deep breath and hold it just before removing the nasogastric tube. Remove the tube from the client's nose gently and slowly. Enclose the tube within the towel or glove and discard the tube in a covered container (Fig. B).	Reduces the risk for aspirating gastric fluid. Lessens the potential for trauma. Provides a transmission barrier against microorganisms.
	Enclosing the tube. (Photo by B. Proud.)
Empty, measure, and record the drainage in the suction container. Remove gloves and perform hand hygiene. Offer an opportunity for oral hygiene. Encourage the client to clear the nose of mucus and debris with paper tissues or cotton-tipped applicators. Discard disposable equipment; rinse and return portable suction equipment.	Provides data for evaluating the client's fluid status. Reduces the transmission of microorganisms. Removes disagreeable tastes from the client's mouth. Promotes the integrity of nasal tissue. Preserves cleanliness and orderliness in the client's unit; demonstrates accountability for equipment.

EVALUATION

- The tube is removed.
- The client resumes eating and taking fluids.
- The client experiences no nausea or vomiting.
- The airway remains clear.
- The nasal mucosa is moist and intact.

DOCUMENT

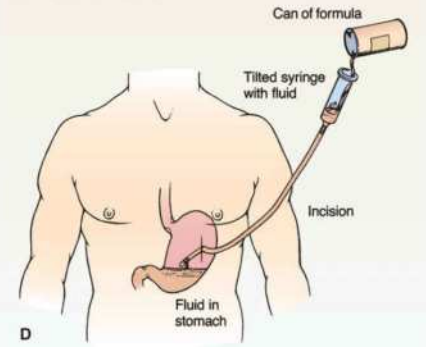
- Type of tube removed
- Response of client
- Appearance and volume of drainage
- Appearance of nose and nasopharynx

SAMPLE DOCUMENTATION

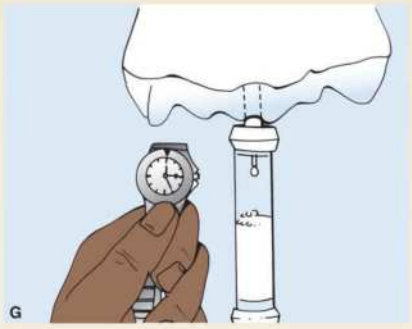
Date and Time Salem sump tube removed. Brief period of retching during removal. Total of 75 mL clear green drainage emptied from suction container. Oral care provided. L. naris swabbed with applicator lubricated with petroleum jelly. Mucosa is red but intact. _____ J.Doe, LPN

SKILL 29-4 Administering Tube Feedings

Suggested Action	Reason for Action
Bolus Feeding	
ASSESSMENT	
Check the medical order for the type of nourishment, volume, and schedule to follow.	Complies with the legal scope of nursing practice.
Check the date and identifying information on the container of tube-feeding formula.	Ensures accurate administration and avoids using outdated formula.
Wash your hands or perform an alcohol-based hand rub (see Chap. 10).	Reduces the transmission of microorganisms.
Identify the client.	Ensures that the procedure will be performed on the correct client.
Distinguish the tubing for gastric or intestinal feeding from the tubing to instill intravenous solutions.	Prevents administering nutritional formula into the vascular system.
Assess bowel sounds.	Provides data indicating the safety for instilling liquids through the tube.
Validate that the external length of the feeding tube has not migrated internally upward or downward.	Supports that the feeding tube remains in the desired anatomic location.
Measure gastric residual if a 12 F or larger tube is in place (Fig. A).	Determines if the stomach has the capacity to manage the next instillation of formula; aspiration of fluid may be impossible with small-lumen tubes.
A	Measuring gastric residual. (Photo by B. Proud.)
Flush the feeding tube with 30 mL of water.	Prevents an obstruction of the tube.
Measure capillary blood glucose or glucose in the urine.	Provides data indicating response to caloric intake.
Assess how much the client understands the procedure.	Provides an opportunity for client teaching.
PLANNING	
Replace any unused formula and feeding bag every 24 hours.	Reduces the potential for bacterial growth.
Wait and recheck gastric residual in 30 minutes if it exceeds 500 mL.	Avoids overfilling the stomach.
Assemble the following equipment: Asepto syringe, formula, tap water.	Contributes to organization and efficient time management.
Warm refrigerated nourishment to room temperature in a basin of warm water.	Prevents chilling and abdominal cramping.
IMPLEMENTATION	
Perform hand hygiene.	Reduces the transmission of microorganisms.
Place the client in a 30- to 90-degree sitting position.	Prevents regurgitation and aspiration.
Reinstill gastric residual by gravity flow if volume is less than 500 mL.	Returns predigested nutrients without excessive pressure.

Pinch the tube just before all the residual has instilled (Fig. B). 	Prevents air from enterin.g the tube Pinch the tubing before it becomes empty.
Add fresh formula to the syringe and adjust the height to allow a slow but gradual instillation (Fig. C). 	Provides nourishment. Administering a bolus feeding.
Continue filling the syringe for a bolus feeding before it becomes empty. If a gastrostomy tube is being used, tilt the barrel of the syringe during the feeding (Fig. D). 	Prevents air from entering the tube. Permits air displacement from the stomach. Bolus feeding through a gastrostomy tube (G-tube).

Flush the tubing with at least 30–60 mL of water after each feeding, or follow the agency's policy for suggested amounts (Fig. E).	Ensures that all nourishment has entered the stomach; prevents fermentation and coagulation of formula in the tube; provides water for fluid balance.
E	Instilling water to flush the tubing.
Plug or clamp the tube as the water leaves the syringe.	Prevents air from entering the tubing; maintains patency.
Keep the head of the bed elevated for at least 30–60 minutes after a feeding.	Prevents gastric reflux.
Wash and dry the feeding equipment. Return items to the bedside.	Supports principles of medical asepsis.
Record the volume of formula and water administered on the bedside intake and output record.	Provides accurate data for assessing fluid balance and caloric value of nourishment.
Provide oral hygiene at least twice daily.	Removes microorganisms and promotes comfort and hygiene of the client.
Intermittent Feeding	
ASSESSMENT	
Follow the previous sequence for assessment.	Principles remain the same.
PLANNING	
In addition to those activities listed for bolus feeding, replace unused formula, feeding containers, and tubing every 24 hours.	Reduces the potential for bacterial growth.
IMPLEMENTATION	
Fill the feeding container with room-temperature formula (Fig. F).	Prevents administration of cold formula, which can cause cramping; room-temperature formula will be instilled before supporting bacterial growth.
F	Filling a formula infusion bag. (From LifeART copyright 2016. Philadelphia, PA: Lippincott Williams & Wilkins. All rights reserved.)

Gradually open the clamp on the tubing. Connect the tubing to the nasogastric or nasoenteral tube. Open the clamp and regulate the drip rate according to the physician's order or the agency's policy. Check at 10-minute intervals (Fig. G).	Purges air from the tube. Provides access to formula. Supports safe administration of liquid nourishment. Ensures early identification of infusion problems.
 G	**Checking the rate of flow.**
Flush the tubing with water after the formula has infused (Fig. H).	Clears the tubing of formula, prevents obstruction, and provides water for fluid balance.
 H	**Flushing the tubing following feeding.**
Pinch the feeding tube just as the last volume of water is administered. Clamp or plug the feeding tube. Record the volume of formula and water instilled. Follow recommendations for postprocedural care as described with a bolus feeding.	Prevents air from entering the tube. Prevents leaking. Provides accurate data for assessing fluid balance and caloric value of nourishment. Principles for care remain the same.

Continuous Feeding

ASSESSMENT

In addition to previously described assessments, check the gastric residual every 4 to 6 hours or according to agency policy.	Principles remain the same. This method ensures a routine pattern for assessment to accommodate the schedule of continuous feedings and prevents inadvertent overfeeding.

PLANNING

In addition to previously described planning activities, obtain equipment for regulating continuous infusion (e.g., tube-feeding pump).	Aids in an accurate administration and sounds an alarm if the infusion is interrupted.
Replace unused formula, feeding containers, and tubing every 24 hours.	Reduces the potential for bacterial growth.
Label the date and attach a time tape on the feeding container.	Facilitates periodic assessment.

IMPLEMENTATION

Flush the new feeding container with water.	Reduces surface tension within the tube and enhances the passage of large protein molecules.
Fill the feeding container with no more than 4 hours' worth of refrigerated formula. *Exception*: Commercially prepared, sterilized containers of formula, or formula that is kept iced while infusing may hang for longer periods.	Prevents growth of bacteria; body heat will warm cold formula when infused at a slow rate.
Purge the tubing of air.	Prevents distention of the stomach or intestine.
Thread the tubing within the feeding pump according to the manufacturer's directions (Fig. I).	Ensures the correct mechanical operation of equipment and accurate administration to the client.

Preparing the pump. (Photo by B. Proud.)

Connect the tubing from the feeding pump to the client's feeding tube (Fig. J).	Provides access to formula.

Connecting the feeding tube to the pump.

Set the prescribed rate on the feeding pump (Fig. K).	Complies with the medical order.
	Programming the pump. (Photo by B. Proud.)
Open the clamp on the feeding tube and start the pump (Fig. L).	Initiates infusion.
	Releasing the clamp. (Photo by B. Proud.)
Keep the client's head elevated at all times.	Prevents reflux and aspiration.
Flush the tubing with 30–60 mL of water or more every 4 hours after checking and refeeding gastric residual and after administering medications.	Promotes patency and contributes to the client's fluid balance.
Record the instilled volume of formula and water.	Provides accurate data for assessing fluid balance and caloric value of nourishment.
Follow recommendations for postprocedural care as described with a bolus feeding.	Principles for care remain the same.

EVALUATION

- The client receives a prescribed volume of formula according to an established feeding schedule.
- The client's weight remains stable or the client reaches the target weight.
- The lungs remain clear.
- Bowel elimination is within normal parameters for client.
- The client has a daily fluid intake between 2,000 and 3,000 mL unless intake is otherwise restricted.

DOCUMENT

- Volume of gastric residual and actions taken, if excessive
- Type and volume of formula
- Rate of infusion, if continuous
- Volume of water used for flushes
- Response of client; if symptomatic, describe actions taken and results

SAMPLE DOCUMENTATION

Date and Time Exit length of tube corresponds with insertion measurement. Gastric residual of 50 mL obtained by aspiration. Residual reinstilled and tube flushed with 30 mL of tap water. A 480 mL of Enrich with Fiber placed in tube-feeding bag. Formula infusing at 120 mL/hour. No diarrhea or gastric complaints at this time. _____ J. Doe, LPN

UNIT 8

Promoting Elimination

30 Urinary Elimination

31 Bowel Elimination

30

Urinary Elimination

Learning Objectives

On completion of this chapter, the reader should be able to:

1. Identify the functions of the urinary system.
2. Describe the physical characteristics of urine and factors that affect urination.
3. Name four types of urine specimens that nurses commonly collect.
4. Identify three alternative devices for urinary elimination.
5. Define continence training.
6. Name three types of urinary catheters.
7. Describe two principles that apply to using a closed drainage system.
8. Explain why catheter care is important in the nursing management of clients with retention catheters.
9. Discuss the purpose for irrigating a catheter and methods for performing this skill.
10. Define urinary diversion.
11. Discuss factors that contribute to impaired skin integrity in clients with a urostomy.

Words to Know

anuria
bedpan
catheter-associated urinary tract infections (CAUTI)
catheter care
catheter irrigation
catheterization
clean-catch specimen
closed drainage system
commode
continence training
continuous irrigation

Credé maneuver
cutaneous triggering
dysuria
external catheter
fenestrated drape
frequency
incontinence
nocturia
oliguria
pelvic floor muscle exercises
peristomal skin
polyuria
residual urine
retention catheter
stasis
straight catheter
24-hour specimen
urgency
urinal
urinary diversion
urinary elimination
urinary retention
urine
urostomy
voided specimen
voiding reflex

This chapter reviews the process of urinary elimination and describes the nursing skills for assessing and maintaining urinary elimination.

OVERVIEW OF URINARY ELIMINATION

The urinary system (Fig. 30-1) consists of the kidneys, ureters, bladder, and urethra. These major components, along with some accessory structures, such as the ring-shaped muscles called the internal and external sphincters, work together to produce **urine** (fluid within the bladder), collect it, and excrete it from the body.

Urinary elimination (the process of releasing excess fluid and metabolic

wastes), or urination, occurs when urine is excreted. Under normal conditions, the average person eliminates approximately 1,500 to 3,000 mL of urine each day. The consequences of impaired urinary elimination can be life-threatening.

Urination takes place several times each day. The need to urinate becomes apparent when the bladder distends with approximately 150 to 300 mL of urine. The distention with urine causes increased fluid pressure, stimulating stretch receptors in the bladder wall and creating a desire to empty it of urine.

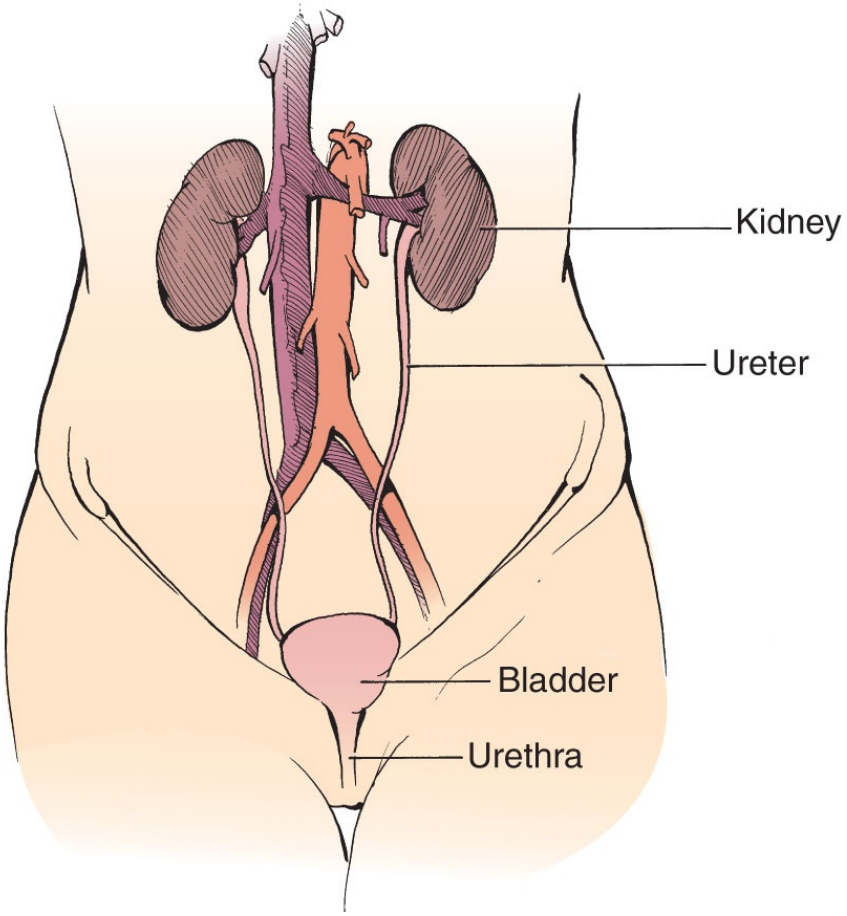


FIGURE 30-1 The major structures of the urinary system.

CHARACTERISTICS OF URINE

The physical characteristics of urine include its volume, color, clarity, and odor. Variations in what is considered normal are wide (Table 30-1).

TABLE 30-1 Characteristics of Urine

CHARACTERISTIC	NORMAL	ABNORMAL	COMMON CAUSES OF VARIATIONS
Volume	500–3,000 mL/day 1,200 mL/day average	<400 mL/day	Low fluid intake Excess fluid loss Kidney dysfunction
		>3,000 mL/day	High fluid intake Diuretic medication Endocrine diseases
Color	Light yellow	Dark amber	Dehydration
		Brown	Liver/gallbladder disease
		Reddish-brown	Blood
		Orange, green, blue	Water-soluble dyes
Clarity	Transparent	Cloudy	Infection Stasis
Odor	Faintly aromatic	Foul	Infection
		Strong	Dehydration
		Pungent	Certain foods

FACTORS AFFECTING URINARY ELIMINATION

Patterns of urinary elimination depend on physiologic, emotional, and social factors. Examples include (1) the degree of neuromuscular development and the integrity of the spinal cord; (2) the volume of fluid intake and the amount of fluid losses, including those from other sources; (3) the amount and type of food consumed; and (4) the person's circadian rhythm, habits, opportunities for urination, and anxiety.

General measures to promote urination include providing privacy, assuming a natural position for urination (sitting for women, standing for men), maintaining an adequate fluid intake, and using stimuli such as running water from a tap to initiate voiding.

Urine Specimen Collection

Health-care providers collect urine specimens, or samples of urine, to identify microscopic or chemical constituents. Common urine specimens that nurses collect include voided specimens, clean-catch specimens, catheter specimens, and 24-hour specimens.

Voided Specimens

A **voided specimen** is a sample of fresh urine collected in a clean container. The first voided specimen of the day is preferred because it is most likely to contain substantial urinary components that have accumulated during the night. Nevertheless, the specimen can be voided and collected at any time it is needed.

The sample of urine is transferred into a specimen container and delivered to the laboratory for testing and analysis. If the specimen cannot be examined in less than 1 hour after collection, it is labeled and refrigerated.

Clean-Catch Specimens

A **clean-catch specimen** is a voided sample of urine considered sterile and is sometimes called a midstream specimen because of the way it is collected. To

avoid contaminating the voided sample with microorganisms or substances other than those in the urine, the external structures through which urine passes (the urinary meatus, which is the opening to the urethra, and the surrounding tissues) are cleansed. The urine is collected after the initial stream has been released.

Clean-catch specimens are preferred to randomly voided specimens. This method of collection is also preferable when a urine specimen is needed during a woman's menstrual period. As soon as the specimen is collected, it is labeled and taken to the laboratory. A clean-catch urine specimen is refrigerated if the analysis will be delayed more than 1 hour.

Research suggests that collecting a specimen in midstream after the use of soap, tap water, and nonsterile gauze for perineal cleansing provides results as reliable and, in some cases, more so than when an antiseptic solution is used, because even a small amount of antiseptic can reduce the bacterial count when a culture is ordered (Latour et al., 2013; ACI Urology Network, 2012). Nurses should follow their agency's policy for collecting midstream voided specimens. When a clean-catch specimen is needed, nurses can instruct clients who are capable of performing the procedure on the collection technique (see Client and Family Teaching 30-1). However, the quality of the results is only as good as the quality of the specimen, making the technique for collection crucial to the reliability of the results (ACI Urology Network, 2012).

Catheter Specimens

A urine specimen can be collected under sterile conditions using a catheter, but this is usually done when clients are catheterized for other reasons such as to control incontinence in an unconscious client. Clean-catch, midstream specimens are a lower-cost alternative to catheterized specimens because of the reduced incidence of cellular and microbial contamination (Preanalytical Resource Center, 2011). For clients who are already catheterized, the nurse clamps the drainage tube for 30 minutes and then aspirates a sample through the lumen of a latex catheter or from a self-sealing port that has been cleansed with an alcohol pad (Fig. 30-2).

Client and Family Teaching 30-1 Collecting a Clean-Catch Specimen

The nurse teaches the female client as follows:

- Wash your hands.
- Remove the lid from the specimen container.
- Rest the lid upside down on its outer surface, taking care not to touch the inside areas.
- Sit on the toilet and spread your legs.
- Separate your labia with your fingers.

- Cleanse each side of the urinary meatus with a separate swab, wiping from front to back toward the vagina.
- Use the final clean, moistened swab to wipe directly down the center of the separated tissue.
- Begin to urinate.
- After releasing a small amount of urine into the toilet, catch a sample of urine in the specimen container.
- Take care not to touch the mouth of the specimen container to your skin.
- Place the specimen container nearby on a flat surface.
- Release your fingers and continue voiding normally.
- Wash your hands.
- Cover the specimen container with the lid.

The male client should follow the same steps as above but should perform the following cleansing routine:

- Retract your foreskin, if you are uncircumcised, or cleanse in a circular direction around the tip of the penis toward its base using a premoistened swab.
- Repeat with another swab.
- Continue retracting the foreskin while initiating the first release of urine and until you have collected the midstream specimen.

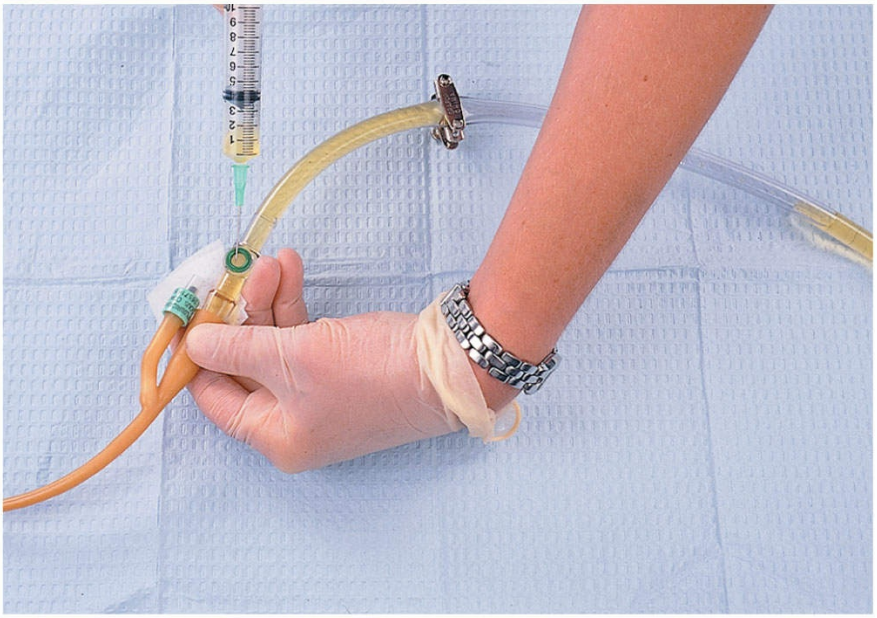


FIGURE 30-2 The location for collecting a urine specimen from a catheter. (Photo by B. Proud.)

24-Hour Specimens

The nurse collects, labels, and delivers a **24-hour specimen** (a collection of all urine produced in a full 24-hour period) to the laboratory for analysis. Because the contents in urine decompose over time, the nurse places the collected urine in a container with a chemical preservative or puts the container in a basin of ice or in a specimen-dedicated refrigerator.

To establish the 24-hour collection period accurately, the nurse instructs the client to urinate just before starting the test and then discards that urine. All urine

voided thereafter becomes part of the collected specimen. Exactly 24 hours later, the nurse asks the client to void one last time to complete the test collection. The final urination and all collected voidings from the preceding 24 hours represent the total specimen, which the nurse labels and takes to the laboratory.

Abnormal Characteristics of Urine

Laboratory analysis is a valuable diagnostic tool for identifying abnormal characteristics of urine. Specific terms describe particular abnormal characteristics of urine and urination. Many terms use the suffix *-uria*, which refers to urine or urination. For example,

- *Hematuria*: urine containing blood
- *Pyuria*: urine containing pus
- *Proteinuria*: urine containing plasma proteins
- *Albuminuria*: urine containing albumin, a plasma protein
- *Glycosuria*: urine containing glucose
- *Ketonuria*: urine containing ketones

ABNORMAL PATTERNS OF URINARY ELIMINATION

Assessment findings may indicate abnormal patterns of urinary elimination. Some common problems include anuria, oliguria, polyuria, nocturia, dysuria, and incontinence.

Anuria

Anuria means the absence of urine or a volume of 100 mL or less in 24 hours. It indicates that the kidneys are not forming sufficient urine. In this case, the term “urinary suppression” is used. In urinary suppression, the bladder is empty; therefore, the client feels no urge to urinate. This distinguishes anuria from **urinary retention**, in which the client produces urine but does not release it from the bladder. A sign of urinary retention is a progressively distending bladder.

Oliguria

Oliguria, urine output less than 400 mL/24 hours, indicates the inadequate elimination of urine. Sometimes oliguria is a sign that the bladder is being only partially emptied during voidings. **Residual urine**, or more than 50 mL of urine that remains in the bladder after voiding, can support the growth of microorganisms, leading to infection. Also, when there is urinary **stasis** (a lack of movement), dissolved substances such as calcium can precipitate, leading to

urinary calculi (stones).

Gerontologic Considerations

- Older adults are more likely to have chronic residual urine (excessive urine in the bladder after urinating), which increases the risk for urinary tract infections.
- Maintenance of good perineal hygiene is one intervention for preventing urinary tract infections. Women should always clean from the urinary area back toward the rectal area to prevent organisms from the stool entering the bladder. In addition, thorough hand washing by the client and caregiver is necessary.
- Also, older adults may benefit from learning double-voiding in which the person voids, then waits a few more minutes to allow any residual urine to be voided (American Urological Association, 2014).

Polyuria

Polyuria means greater-than-normal urinary elimination and may accompany minor dietary variations. For example, consuming higher than normal amounts of fluids, especially those with mild diuretic effects (eg, coffee and tea), or taking certain medications actually can increase urination. Ordinarily, urine output is nearly equal to fluid intake. When the cause of polyuria is not apparent, excessive urination may be the result of a disorder. Common disorders associated with polyuria include *diabetes mellitus*, an endocrine disorder caused by insufficient insulin or insulin resistance, and *diabetes insipidus*, an endocrine disease caused by insufficient antidiuretic hormone.

Nocturia

Nocturia (nighttime urination) is unusual because the rate of urine production is normally reduced at night. Consequently, nocturia suggests an underlying medical problem. In aging men, an enlarging prostate gland, which encircles the urethra, interferes with complete bladder emptying. As a result, there is a need to urinate more frequently, including during the usual hours of sleep.

Pharmacologic Considerations

- Nurses who are pregnant or may become pregnant should not handle crushed or broken finasteride (Propecia, Proscar) or dutasteride (Avodart) tablets or capsules. Absorption of the drug poses substantial risk for abnormal growth to a male fetus.

Dysuria

Dysuria is the difficult or uncomfortable voiding and a common symptom of trauma to the urethra or a bladder infection. **Frequency** (the need to urinate often) and **urgency** (a strong feeling that urine must be eliminated quickly) often accompany dysuria.

Gerontologic Considerations

- Older adults are likely to experience urinary urgency and frequency because of normal physiologic changes such as diminished bladder capacity and degenerative changes in the cerebral cortex. Subsequently, when they perceive the urge to void, they need to access a bathroom as soon as possible.
- Age-related changes, such as a diminished bladder capacity and a relaxation of the pelvic floor muscle tone, increase the risk for incontinence.
- Fluid restriction may be used in an attempt to control urination, but it may actually contribute to incontinence by causing concentrated urine and eliminating the normal perception of a full bladder.

Nutrition Notes

- Observational studies suggest that proanthocyanidins in cranberries may decrease the incidence of recurring urinary tract infections, possibly by impairing the ability of bacteria to adhere to the lining of the urinary tract. However, placebo- controlled trials have failed to confirm a decrease in urinary tract infection risk related to cranberry (Barbosa-Cesnik et al., 2011).
- Problems with assessing cranberry efficacy include the difficulty in standardizing dosing because cranberry is available in various forms, such as juice, capsules, tablets, and extract.

Incontinence

Incontinence means the inability to control either urinary or bowel elimination and is abnormal after a person has achieved earlier continence. The term *urinary incontinence* should not be used indiscriminately: anyone may be incontinent if his or her need for assistance goes unnoticed. Once the bladder becomes extremely distended, spontaneous urination may be more of a personnel problem than a client problem. (The client may not be incontinent if staff members are attentive to his or her need to urinate.)

Pharmacologic Considerations

- Drug therapy can increase the risk for urinary incontinence especially in older adults when cognitive/mobility issues hamper the ability to void in a timely manner. Drugs that increase urinary incontinence include the following:
  - High blood pressure drugs—vasodilate and put more fluid into the vascular system.
  - Antidepressants—impair bladder contraction and worsen overflow incontinence symptoms.
  - Diuretics (water pills)—pull excess water and salt into the bladder.
  - Sleeping pills—clients do not respond and waken to void with a full bladder at night.
- Client care includes more frequent voiding, especially within 2 hours of diuretic administration.

ASSISTING CLIENTS WITH URINARY ELIMINATION

Stable clients who can ambulate are assisted to the bathroom to use the toilet. Clients who are weak or cannot walk to the bathroom may need a commode. Clients confined to the bed use a urinal or bedpan.

Commode

A **commode** (a chair with an opening in the seat under which a receptacle is placed) is located beside or near the bed (Fig. 30-3). It is used for eliminating urine or stool. Immediately afterward, the waste container is removed, emptied, cleaned, and replaced.

FIGURE 30-3 A bedside commode.

Urinal

A **urinal** is a cylindrical container for collecting urine. It is more easily used by males. When given to the client, the urinal should be empty; otherwise, the bed linen may become wet and soiled. If the client needs help placing the urinal,

- Pull the privacy curtain.
- Put on gloves.
- Ask the client to spread his legs.
- Hold the urinal by its handle.
- Direct the urinal at an angle between the client's legs so that the bottom rests on the bed (Fig. 30-4).
- Lift the penis and place it well within the urinal.

FIGURE 30-4 Placement of a urinal.

After use, the nurse promptly empties the urinal. He or she measures and records the volume of urine if the client's intake and output are being monitored (see Chap. 16). The nurse washes his or her hands and always offers the client an opportunity to wash his hands after voiding.

Using a Bedpan

A **bedpan** (a seat-like container for elimination) is used to collect urine or stool. Most bedpans are made of plastic and are several inches deep. A *fracture pan*, a modified version of a conventional bedpan, is flat on the sitting end rather than rounded (Fig. 30-5). Clients with musculoskeletal disorders who cannot elevate

their hips and sit on a bedpan in the usual manner use a fracture pan. When a client confined to bed feels the need to eliminate, the nurse places a bedpan under the buttocks (Skill 30-1).

FIGURE 30-5 Two types of bedpans: a fracture pan (*left*) and a conventional bedpan (*right*). (Photo by B. Proud.)

Stop, Think, and Respond 30-1

Describe measures that may reduce a client's concerns when he or she requires a bedpan.

MANAGING INCONTINENCE

Urinary incontinence, depending on its type, may be permanent or temporary. The six types of urinary incontinence are stress, urge, reflex, functional, total, and overflow (Table 30-2).

TABLE 30-2 Types of Incontinence

TYPE	DESCRIPTION	EXAMPLE	COMMON CAUSES	NURSING APPROACH
Stress	The loss of small amounts of urine when intra-abdominal pressure rises	Dribbling is associated with sneezing, coughing, lifting, laughing, or rising from a bed or chair.	Loss of perineal and sphincter muscle tone secondary to childbirth, menopausal atrophy, prolapsed uterus, or obesity.	Pelvic floor muscle strengthening Weight reduction
Urge	Need to void perceived frequently, with short-lived ability to sustain control of the flow	Voiding commences when there is a delay in accessing a toilet.	Bladder irritation secondary to infection; loss of bladder tone from recent continuous drainage with an indwelling catheter	Restriction of fluid intake of at least 2,000 mL/day Omit bladder irritants, such as caffeine or alcohol. Administration of diuretics in the morning
Reflex	Spontaneous loss of urine when the bladder is stretched with urine, but without prior perception of a need to void	The person automatically releases urine and cannot control it.	Damage to motor and sensory tracts in the lower spinal cord secondary to trauma, a tumor, or other neurologic conditions	Cutaneous triggering Straight intermittent catheterization
Functional	Control over urination lost because of inaccessibility of a toilet or a compromised ability to use one	Voiding occurs while attempting to overcome barriers such as doorways, transferring from a wheelchair, manipulating clothing, acquiring assistance, or making needs known.	Impaired mobility, impaired cognition, physical restraints, inability to communicate	Clothing modification Access to a toilet, commode, or urinal Assistance to a toilet according to a preplanned schedule
Total	Loss of urine without any identifiable pattern or warning	The person passes urine without any ability or effort to control.	Altered consciousness secondary to a head injury, loss of sphincter tone secondary to prostatectomy, anatomic leak through a urethral/vaginal fistula	Absorbent undergarments External catheter Indwelling catheter
Overflow	Urine leakage because the bladder is not completely emptied; bladder distended with retained urine	The person voids small amounts frequently, or urine leaks around a catheter.	Overstretched bladder or weakened muscle tone secondary to obstruction of the urethra by debris within a catheter, an enlarged prostate, distended bowel, or postoperative bladder spasms	Hydration Adequate bowel elimination Patency of catheter Credé maneuver

The management of incontinence is complex because there are so many variations. Treatment is further complicated when clients have more than one type of incontinence; for example, stress incontinence often accompanies urge incontinence.

Some forms of incontinence respond to simple measures such as modifying clothing to make elimination easier. Other forms improve only with a more regimented approach like continence training. Inserting a retention catheter is the least desirable approach to managing incontinence because it is the leading cause of urinary tract infections in hospitals and nursing homes (American Nurses Association, 2015; Centers for Disease Control and Prevention, 2015).

Gerontologic Considerations

- Loss of control over urination often threatens an older adult's independence and self-esteem. It also may cause an older adult to restrict activities, possibly contributing to depression. Teaching older adults to structure activities with planned toileting breaks every 60 to 90 minutes results in less urine in the bladder and thus diminishes urge incontinence.
- Older adults who experience difficulty controlling urine need an evaluation to

identify and treat contributing factors, such as constipation, urinary tract infection, and side effects from medications.

- Older adults need encouragement to discuss urinary incontinence with a knowledgeable, nonjudgmental health care provider. If they understand that urinary incontinence is a condition that frequently responds to medication or behavioral retraining, they are more likely to seek professional help.
- Many resources are available to assist older adults in evaluating and treating incontinence. For example, some health care facilities offer special incontinence clinics and physical therapy departments to teach pelvic muscle exercises. The National Association for Continence (http://www.nafc.org) is an excellent source of information for products, resources, and continence programs. Nurses can encourage older adults to take advantage of these kinds of resources rather than accepting incontinence as an inevitable condition that compromises their quality of life.
- When efforts to restore continence are unsuccessful, nurses can encourage older adults to verbalize their feelings and identify interventions helpful in maintaining dignity, ultimately enabling older adults to participate in meaningful activities.

Continence Training

Continence training to restore the control of urination involves teaching the client to refrain from urinating until an appropriate time and place. This process sometimes is referred to as *bladder retraining*, but this term is inaccurate because the various techniques used involve mechanisms other than those unique to the bladder.

Continence training primarily benefits clients with the cognitive ability and desire to participate in a rehabilitation program. This includes clients with lower body paralysis who wish to facilitate urination without the use of urinary drainage devices such as catheters. Clients who are not candidates for continence training require alternative methods such as absorbent undergarments.

Continence training is often a slow process that requires the combined effort and dedication of the nursing team, client, and family (see Nursing Guidelines 30-1 and Box 30-1).

Gerontologic Considerations

- An individualized toileting schedule should be maintained, for example, at intervals of must be offered every 90 to 120 minutes for clients who have

difficulty maintaining continence.

- Absorbent products may interfere with the person's independence in toileting and may lead to skin breakdown. Incontinence products are never used primarily for staff convenience in institutional settings. In addition, an older person should never be reprimanded for an episode of incontinence.
- When efforts to restore continence are unsuccessful, nurses can encourage older adults to verbalize their feelings and identify interventions helpful in maintaining dignity, ultimately enabling older adults to participate in meaningful activities.

Pharmacologic Considerations

Pharmacologic therapy may be an option for women with urinary incontinence:

- Topical estrogen (creams or Estring vaginal ring) to treat stress incontinence
- Antispasmodics (oxybutynin (Ditropan)) to treat urge incontinence
- Tricyclic antidepressant (imipramine (Tofranil)) to treat mixed incontinence issues

NURSING GUIDELINES 30-1

Providing Continence Training

- Compile a log of the client's urinary elimination patterns. *The data help to reveal the client's type of incontinence.*
- Set realistic, specific, short-term goals with the client. *Short-term goals prevent self-defeating consequences and promote client control.*
- Discourage strict limitation of liquid intake. *Intake maintains fluid balance and ensures adequate urine volume.*
- Plan a trial schedule for voiding that correlates with the times when the client is usually incontinent or experiences bladder distention. *This schedule reduces the potential for accidental voiding or sustained urinary retention.*
- In the absence of any identifiable pattern, plan to assist the client with voiding every 2 hours during the day and every 4 hours at night. *This duration provides time for urine to form.*
- Communicate the plan to nursing personnel, the client, and the family. *Collaboration promotes continuity of care and dedication to reaching goals.*
- Assist the client to a toilet or commode; position the client on a bedpan or place a urinal just before the scheduled time for trial voiding. *These measures prepare the client for releasing urine.*
- Simulate the sound of urination such as by running water from the faucet. *Doing so simulates relaxation of the sphincter muscles, allowing the release of urine.*
- Suggest performing **Credé maneuver** (the act of bending forward and applying hand pressure over the bladder (Fig. A). *Credé maneuver increases abdominal pressure to overcome the resistance of the internal sphincter muscle.*

A Credé maneuver.

- Instruct paralyzed clients to identify any sensation that precedes voiding such as a chill, muscular spasm, restlessness, or spontaneous penile erection. *These cues can help the client anticipate urination.*
- Suggest that paralyzed clients with reflex incontinence use **cutaneous triggering** (lightly massaging or tapping the skin above the pubic area). *Cutaneous triggering initiates urination in clients who have retained a* **voiding reflex** *(the spontaneous relaxation of the urinary sphincter in response to physical stimulation).*
- Teach clients with stress incontinence to perform **pelvic floor muscle exercises** (Kegel exercises) which are isometric exercises to improve the ability to retain urine within the bladder; see Box 30-1. *Pelvic floor muscle exercises (Kegel exercises) strengthen and tone the pubococcygeal and levator ani muscles used voluntarily to hold back urine and intestinal gas or stool.*

- Assist clients with urge incontinence to walk slowly and concentrate on holding their urine when nearing the toilet. *These measures reverse previous mental conditioning in which the urge to urinate becomes stronger and more overpowering close to the toilet.*

BOX 30-1 Technique for Performing Pelvic Floor Muscle Exercises (Kegel Exercises)

- Tighten the internal muscles used to prevent urination or interrupt urination once it has begun.
- Keep the muscles contracted for at least 10 seconds.
- Relax the muscles for the same period.
- Repeat the pattern of contraction and relaxation 10 to 25 times.
- Perform the exercise regimen three or four times a day for 2 weeks to 1 month.

CATHETERIZATION

Catheterization (the act of applying or inserting a hollow tube), in this case, refers to using a device inside the bladder, or externally about the urinary meatus. A urinary catheter is used for various reasons:

- Keeping incontinent clients dry (catheterization is a last resort that is used only when all other continence measures have been exhausted)
- Relieving bladder distention when clients cannot void
- Assessing fluid balance accurately
- Keeping the bladder from becoming distended during procedures such as surgery
- Measuring the residual urine
- Obtaining sterile urine specimens
- Instilling medication within the bladder

Types of Catheters

There are a variety of urinary catheters, but the three most common types are external, straight, and retention. Most catheters are made of latex. For clients who are sensitive or allergic to latex, latex-free catheters such as those made of silicone are used.

External Catheters

An **external catheter**, also known as a condom catheter, (a urine-collecting device applied to the skin) is not inserted within the bladder; instead, it surrounds the penis (Fig. 30-6). An external catheter is obviously more effective for male clients.

Condom catheters are helpful for clients receiving care at home because they

are easy to apply. A condom catheter has a flexible sheath that is unrolled over the penis. The narrow end is connected to tubing that serves as a channel for draining urine. The drainage tube may be attached to a leg bag (Fig. 30-7) or connected to a larger urine-collection device.

Three potential problems accompany the use of condom catheters. First, the sheath may be applied too tightly, restricting blood flow to the skin and tissues of the penis. Second, moisture tends to accumulate beneath the sheath, leading to skin breakdown. Third, condom catheters frequently leak. Applying the catheter correctly and managing care appropriately can prevent these problems (Skill 30-2).

Nurses may apply an external collection device called a urinary bag (U bag). A urinary bag serves as a disposable container for collecting a urine specimen from a male or female infant who has not developed the ability to achieve continence. The upper portion of the bag has a self-adherent pad that covers the pubic area. The bag encircles the penis or labia. A diaper is applied over the U bag to ensure that it does not become loose or removed by the infant. Once the specimen is obtained, the U bag is removed and discarded.

FIGURE 30-6 A condom catheter is an example of an external urine collection device. (Photo by B. Proud.)

Stop, Think, and Respond 30-2

Discuss assessments that indicate common problems associated with the use of a condom catheter and nursing measures that can reduce or eliminate negative outcomes.

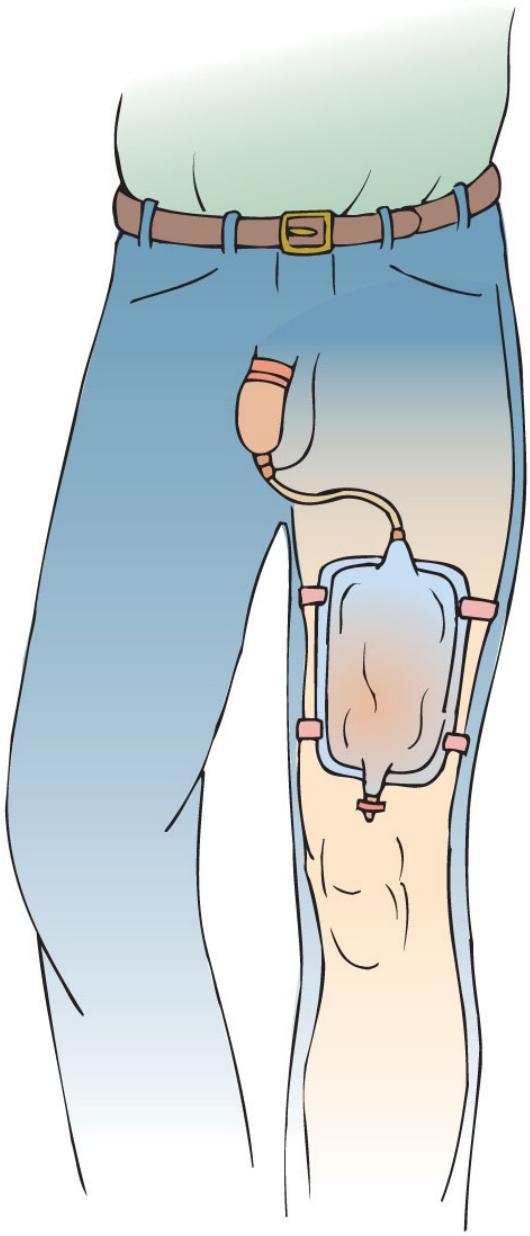


FIGURE 30-7 A leg bag collects urine from a catheter but is concealed under clothing.

Straight Catheters

A **straight catheter** is a urine drainage tube inserted but not left in place. It drains urine temporarily or provides a sterile urine specimen (Fig. 30-8).

Retention Catheters

A **retention catheter**, also called an indwelling catheter, is left in place for a period of time (see Fig. 30-8). The most common type is a Foley catheter.

Unlike straight catheters, retention catheters are secured with a balloon that is inflated once the distal tip is within the bladder. Both straight and retention catheters are available in various diameters, sized according to the French (F) scale (see Chap. 29). For adults, sizes 14, 16, and 18 F are commonly used.

Inserting a Catheter

The techniques for inserting straight and retention catheters are similar, although the steps for inflating the retention balloon do not apply to a straight catheter. When inserting a straight or a retention catheter in a health agency, the nurse uses sterile technique. In the home, nurses and clients who self-catheterize use clean technique because most clients have adapted to the organisms in their own environment. Because of anatomic differences, techniques for insertion differ in men and women and are described in Skills 30-3 and 30-4.

Gerontologic Considerations

■ Enlargement of the prostate, a common problem among older men, can obstruct urinary outflow and make catheterization difficult or impossible. Insertion of a urinary catheter should never be forced. Sometimes a malecot catheter is inserted into the bladder through the abdominal wall (suprapubic catheterization) when it cannot be inserted into a narrowed urethra.

Stop, Think, and Respond 30-3

Discuss factors that predispose a female with a Foley catheter to develop a urinary tract infection.

Connecting a Closed Drainage System

A **closed drainage system** (a device used to collect urine from a catheter) consists of a calibrated bag, which can be opened at the bottom; tubing of sufficient length to accommodate for turning and positioning clients; and a hanger from which to suspend the bag from the bed (Fig. 30-9). The nurse coils excess tubing on the bed but keeps the section from the bed to the collection bag

vertical. Dependent loops in the tubing interfere with gravity flow. The nurse also takes care to avoid compressing the tubing, which can obstruct drainage. Placing the tubing over the client's thigh is acceptable.

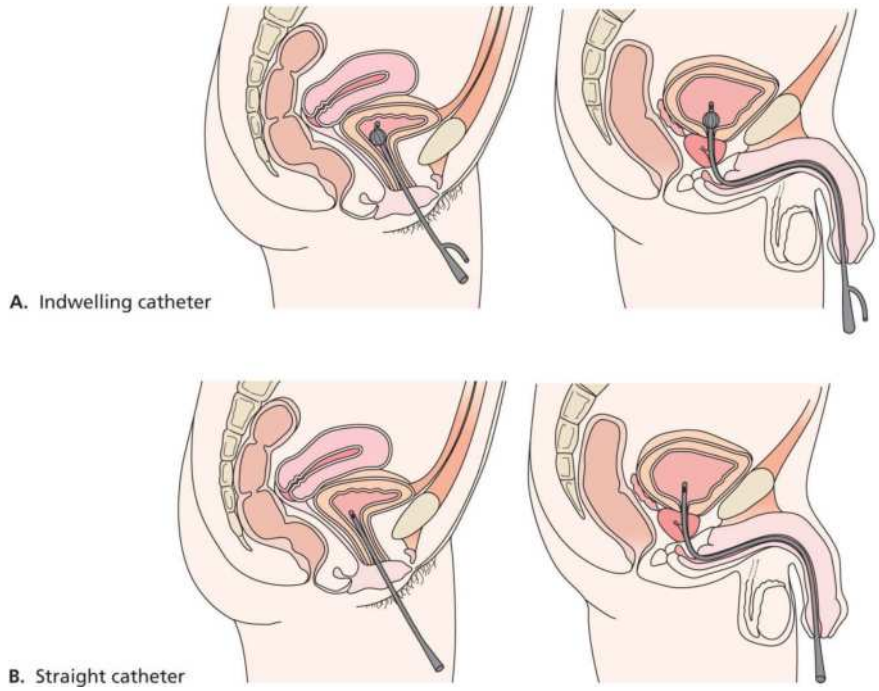


FIGURE 30-8 Types of urinary catheters. (**A)** A retention (Foley) catheter with balloon. (**B)** A straight catheter. (From Carter, P. J. [2011]. *Lippincott textbook for nursing assistants* [3rd ed.]. Philadelphia, PA: Lippincott Williams & Wilkins [PE].)

FIGURE 30-9 A closed urine drainage system.

The nurse always positions the drainage system lower than the bladder to avoid a backflow of urine. When transporting the client in a wheelchair, the nurse suspends the drainage bag from the chair below the level of the bladder. When the client is ambulating, the nurse secures the drainage bag to the lower part of an IV pole or allows the client to carry the bag by hand (Fig. 30-10).

To reduce the potential for the drainage system becoming a reservoir of pathogens, the entire drainage system is replaced whenever the catheter is changed and at least every 2 weeks in clients with a urinary tract infection.

Gerontologic Considerations

- Indwelling catheters should be avoided if at all possible because older people have increased susceptibility to urinary tract infections. Bladder training or other interventions are much more desirable.
- If indwelling catheters are necessary, meticulous daily care is required. The tubing should never be placed higher than the bladder to prevent any backflow of urine into the bladder.

Stop, Think, and Respond 30-4

Discuss possible explanations for why urine may not flow from a catheter.

Catheter-Associated Urinary Tract Infections

Those who are catheterized are at risk for acquiring urinary tract infections. The Centers for Disease Control and Prevention (CDC, 2015) estimates that 75% of all urinary tract infections are acquired by those with indwelling catheters, especially those that are in place for a prolonged period of time. In an effort to prevent **catheter-associated urinary tract infections (CAUTI),** the American Nurses Association (2015) has recommended measures to reduce their incidence:

- Advocate against inappropriate short-term catheter use.
- Secure the catheter appropriately.
- Keep the drainage bag below the level of the bladder at all times.
- Empty the drainage bag regularly.
- Keep the catheter and drainage tube from kinking.
- Maintain a closed drainage system.
- Perform perineal hygiene at least daily.
- Remove urinary catheters in a timely fashion.
- Provide rigorous catheter care.

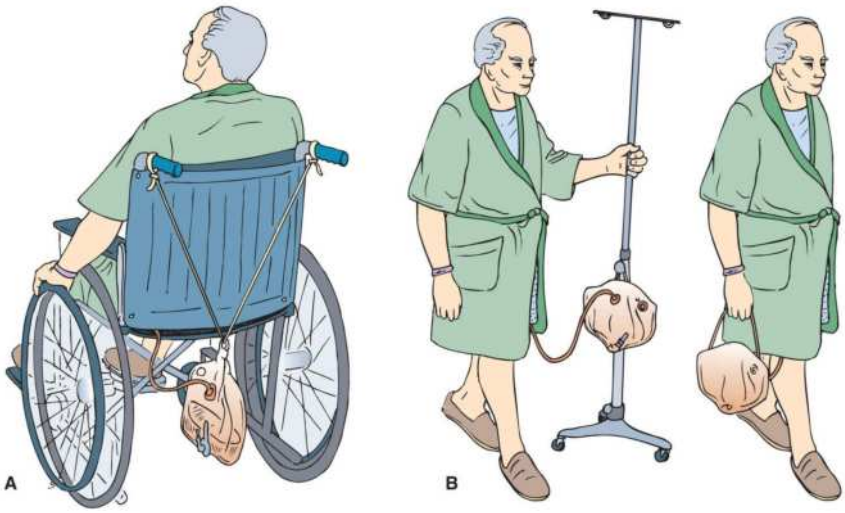


FIGURE 30-10 Techniques for suspending a drainage system below the bladder. **(A)** A wheelchair patient. **(B)** An ambulating patient with and without an IV pole.

Providing Catheter Care

A retention catheter keeps the meatus slightly dilated, providing pathogens with a direct pathway to the bladder where an infection could develop. Bacteria also tend to adhere to the matrix of catheters forming a slimy substance known as biofilm, which supports bacterial growth and subsequent antibiotic resistance (Soto, 2014; Amalaradjou & Venkitanarayanan, 2013).

Nutrition Notes

- Interfering with bacterial adhesion has been shown to repel bacteria such as *Escherichia coli*, a common uropathogen. Currently cranberry products are considered a preventive measure rather than a form of treatment once a urinary tract infection exists (Amalaradjou & Venkitanarayanan, 2013).

Catheter care (hygiene measures used to keep the meatus and adjacent area of the catheter clean) helps to deter the growth and spread of colonizing pathogens. Nursing Guidelines 30-2 describes the technique for providing catheter care. Nurses must follow agency policy for using antiseptic and antimicrobial agents because the use of these substances is not a standard recommendation by the CDC.

Catheter Irrigation

Catheter irrigation (flushing the lumen of a catheter) is a technique for restoring or maintaining catheter patency. A catheter that drains well, however, does not need to be irrigated. A generous oral fluid intake is usually sufficient to produce dilute urine, which thus keeps small shreds of mucus or tissue debris from obstructing the catheter. Occasionally, however, the catheter may need to be irrigated, such as after a surgical procedure that results in bloody urine.

Depending on the type of indwelling catheter, nurses irrigate continuously through a three-way catheter or periodically using an open system or a closed system (Skill 30-5).

Using an Open System

An open system is one in which the retention catheter is separated from the drainage tubing to insert the tip of an irrigating syringe. Opening the system creates the potential for infection because it provides an opportunity for pathogens to enter the exposed connection. Consequently, it is the least desirable of the three methods.

Using a Closed System

A closed system is irrigated without separating the catheter from the drainage tubing. To do so, the catheter or drainage tubing must have a self-sealing port. After cleansing the port with an alcohol swab, the nurse pierces the port with an 18- or 19-gauge, 1.5 in. needle (see Chap. 34). He or she attaches the needle to a 30- to 60-mL syringe containing a sterile irrigation solution. The nurse pinches or clamps the tubing beneath the port and instills the solution, then releases the tubing for drainage. The nurse records the volume of irrigant as fluid intake or subtracts it from the urine output to maintain an accurate intake and output record.

NURSING GUIDELINES 30-2

Providing Catheter Care

- Plan to cleanse the meatus and a nearby section of the catheter at least once a day. *Regular cleansing reduces colonizing microorganisms.*
- Gather clean gloves, soap, water, washcloth, towel, and a disposable pad. *Organization facilitates efficient time management.*
- Wash your hands or perform an alcohol-based hand rub (see Chap. 10). *Hand hygiene reduces the potential for transmitting microorganisms.*
- Place a disposable pad beneath the hips of a female and beneath the penis of a male. *The pad protects the bed linen from becoming wet or soiled.*
- Put on clean gloves and wash the meatus, the catheter where it meets the meatus, the genitalia, and the perineum (in that order) with warm, soapy water. Rinse and dry. *Routine hygiene removes gross secretions and transient microorganisms while following the principles of asepsis.*
- Remove soiled materials and gloves, and repeat hand hygiene measures. *These steps remove colonizing microorganisms.*

Continuous Irrigation

A **continuous irrigation** (the ongoing instillation of solution) instills irrigation solution into a catheter by gravity over a period of days (Fig. 30-11). Continuous irrigations keep a catheter patent after prostate or other urologic surgery in which blood clots and tissue debris collect within the bladder and catheter.

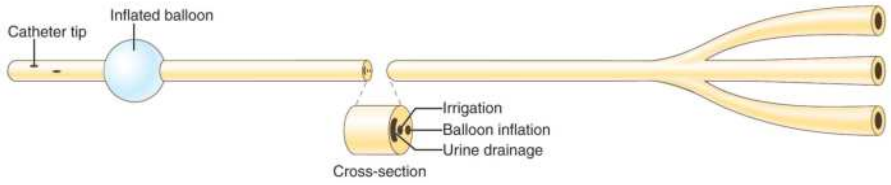


FIGURE 30-12 Components of a three-way catheter.

A three-way catheter is necessary to provide a continuous irrigation. The catheter has three lumens or channels within the catheter, each leading to a

separate port. One port connects the catheter to the drainage system; another provides a means for inflating the balloon in the catheter; and the third instills the irrigation solution (Fig. 30-12).

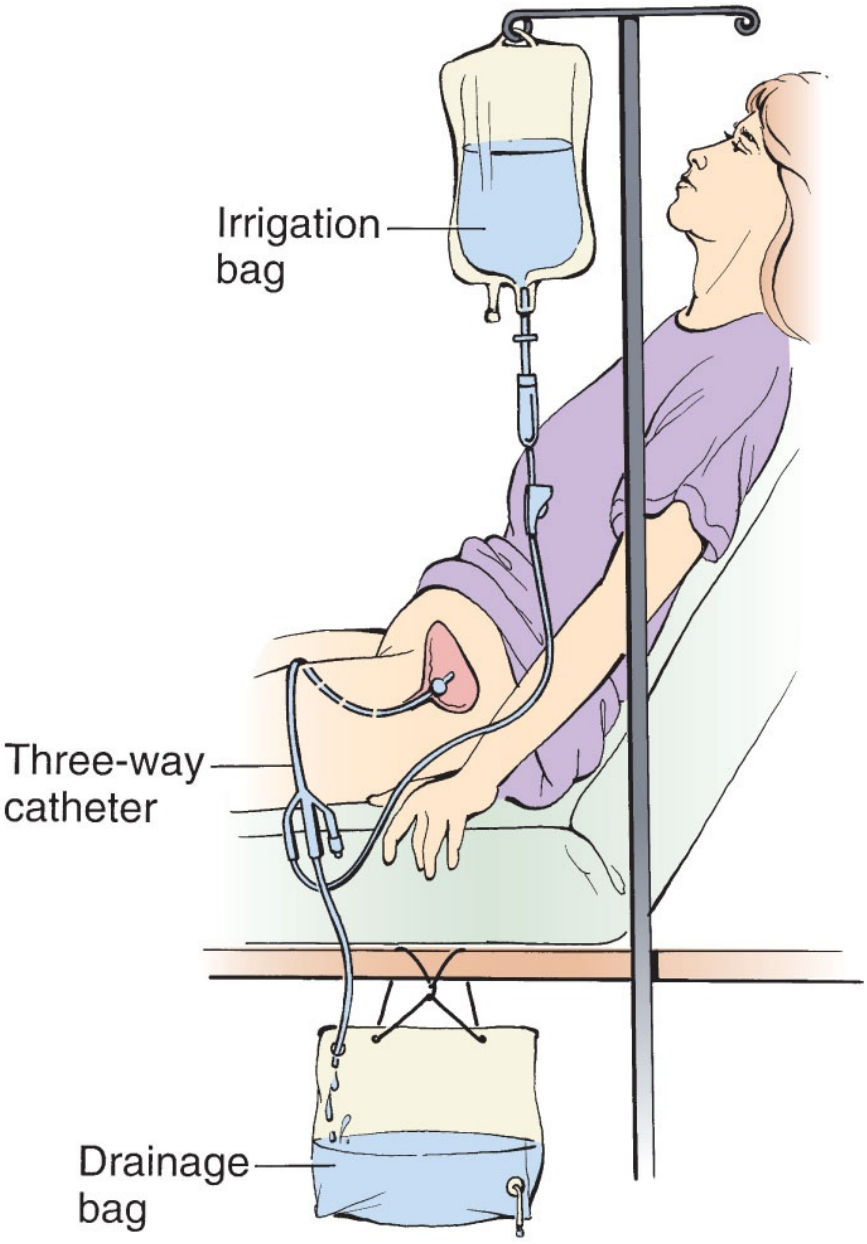


FIGURE 30-11 Bladder irrigation using a three-way catheter.

The steps involved in providing a continuous irrigation are as follows:

- Hang the sterile irrigating solution from an intravenous pole.

- Purge the air from the tubing.
- Connect the tubing to the catheter port for irrigation (Fig. 30-13).
- Regulate the rate of instillation according to the medical order.
- Monitor the appearance of the urine and volume of urinary drainage.

Stop, Think, and Respond 30-5

Discuss what actions might be appropriate if irrigating a catheter is unsuccessful in promoting catheter patency.

FIGURE 30-13 Attaching irrigation tubing to a port on a three-way catheter.

Indwelling Catheter Removal

A catheter is removed when it needs to be replaced or when its use is discontinued. The best time to remove a catheter is in the morning when there is more opportunity to address any urination difficulties without depriving a client of sleep (see Nursing Guidelines 30-3).

URINARY DIVERSIONS

In a **urinary diversion**, one or both ureters are surgically implanted elsewhere. This procedure is done for various life-threatening conditions. The ureters may be brought to and through the skin of the abdomen (Fig. 30-14) or implanted within the bowel (called an ileal conduit). A **urostomy** (a urinary diversion that discharges urine from an opening on the abdomen) is the focus of this discussion.

Care for an ostomy, a surgically created opening, is discussed in more detail in Chapter 31 because those formed for bowel elimination are more common. Chapter 31 also provides a detailed description of an ostomy appliance, the device used for collecting stool or urine, and the manner in which it is applied and removed from the skin.

Caring for a urostomy and changing a urinary appliance are more challenging than the care of intestinal stomas. Urine drains continuously from a urostomy, increasing the risk for skin breakdown. In addition, because moisture and the weight of the collected urine tend to loosen the appliance from the skin, a urinary appliance may need to be emptied and changed more frequently. When changing the appliance, it may help to place a tampon within the stoma to absorb urine temporarily while the skin is cleansed and prepared for another appliance.

It is often difficult to maintain the integrity of the **peristomal skin** (the skin around the stoma) because of the frequent appliance changes and the ammonia in urine. Skin barrier products are used, and sometimes an antibiotic or steroid ointment is applied.

NURSING IMPLICATIONS

Clients with urinary elimination problems may have one or more of the following nursing diagnoses:

- Toileting Self-Care Deficit
- Impaired Urinary Elimination
- Urinary Retention
- Risk for Infection
- Stress Urinary Incontinence
- Urge Urinary Incontinence
- Reflex Urinary Incontinence
- Functional Urinary Incontinence
- Situational Low Self-Esteem
- Risk for Impaired Skin Integrity

NURSING GUIDELINES 30-3

Removing a Foley Catheter

- Wash your hands or perform an alcohol-based hand rub (see Chap. 10) and put on clean gloves. *These measures follow standard precautions.*
- Empty the balloon by aspirating the fluid with a syringe. *This step ensures that all the fluid has been withdrawn.*

- Gently pull the catheter near the point where it exits from the meatus. *Doing so facilitates withdrawal.*
- Inspect the catheter and discard if it appears to be intact. *This ensures safety.*
- Clean the urinary meatus. *This promotes comfort and hygiene.*
- Monitor the client's voiding especially for the next 8 to 10 hours; measure the volume of each voiding. *Findings determine whether or not elimination is normal as well as characteristics of the urine.*

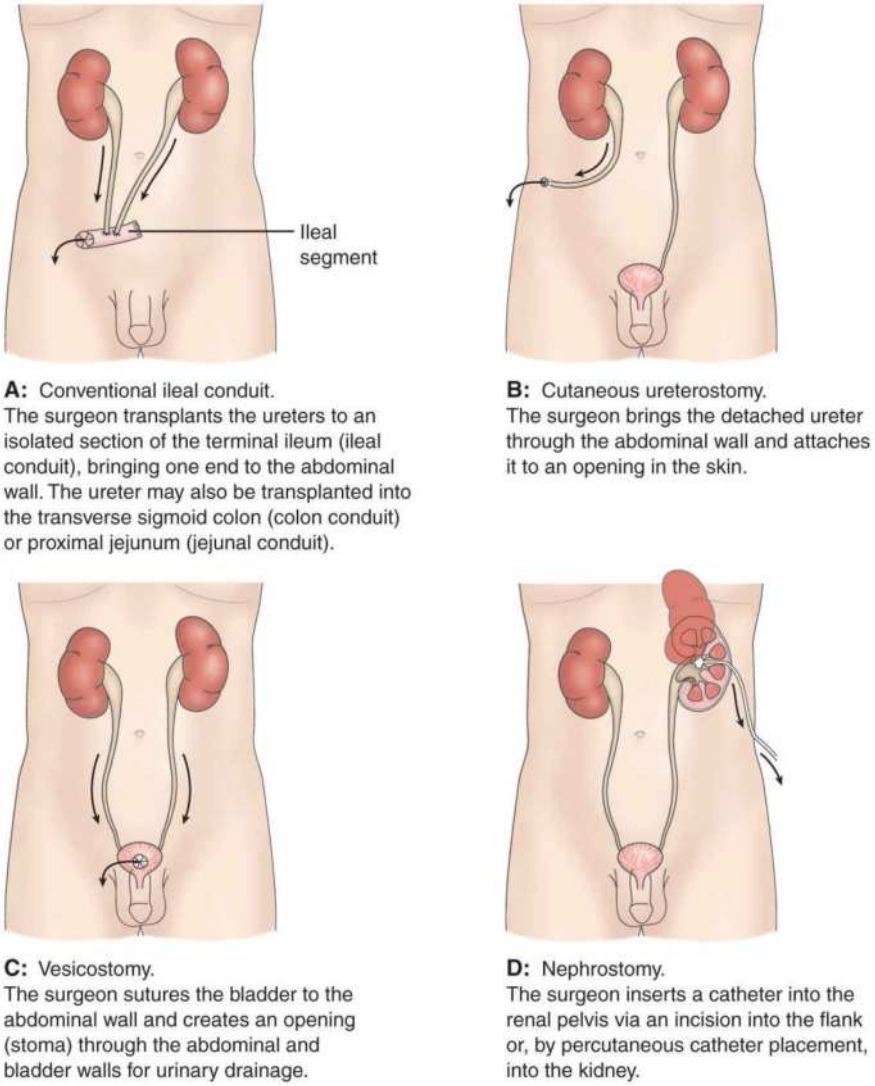


FIGURE 30-14 Examples of urinary diversions. (**A)** An ileal conduit. (**B)** A cutaneous ureterostomy. (**C)** Vesicostomy. (**D)** Nephrostomy. (From Hinkle, J. L., & Cheever, K. H. [2013]. *Brunner and Suddarth's textbook of medical-surgical nursing* [13th ed.]. Philadelphia, PA: Lippincott Williams & Wilkins.)

Clinical Scenario Nursing staff in a nursing home note that an 85-year old female client frequently requires changes in her underclothing because she cannot ambulate to the toilet in time to void. The client is embarrassed because others notice the loss of urine on the floor and her wet outer garments.

Nursing Care Plan 30-1 is developed for a client with Urge Urinary Incontinence, defined by NANDA-I (2015, p. 189) as “the involuntary passage of urine occurring soon after a strong sense of urgency to void.”

NURSING CARE PLAN 30-1 Urge Urinary Incontinence

Assessment

- Inquire about the number of voidings per day; voiding more than eight times in 24 hours or waking up two or more times at night to urinate, or urinating soon after the bladder has been emptied suggests a pattern of urgency or what has also been referred to as an overactive bladder.
- Identify the interim the client can wait to postpone urination following the sensation of a need to empty the bladder, commonly referred to as warning time (Carpenito, 2013).
- Ask the client if the need to urinate is less easily controlled as the person gets nearer to the location of a toilet.
- Determine if the client experiences accidental loss of urine when there is an almost unstoppable need to urinate.

Nursing Diagnosis. **Urge Urinary Incontinence** related to uninhibited bladder muscle contractions as manifested by 14 to 18 voidings per day including awakening three times at night to urinate; daily episodes of urinary incontinence with impaired ability to delay urge to void.

Expected Outcome. The client will report a decrease in the number of daily voidings to <8 per day; absence or limited occasions of nocturia; ability to delay urination by 15 minutes or more when urination seems imminent; and absence of urinary incontinence within 6 to 8 weeks of implementing therapeutic interventions, for example, by 9/15.

Interventions	Rationales
Keep a record of the frequency of voidings and the length of time between the warning sign for voiding and actual voiding for 3 days beginning 8/1 through 8/3.	Documenting the client's unique pattern of urination facilitates appropriate nursing interventions.
Alert all nursing team members to respond as soon as possible to the client's signal for assistance.	Responding promptly reduces episodes of incontinence and demonstrates a united effort to help the client achieve control of urination.
Instruct the client to restrain urination as long as possible after the warning sign is perceived.	Efforts to delay urination help to reverse an established habit of overresponding to an urgent need to void.
Suggest that the client uses a technique such as breathing deeply, singing a song, or talking about family to delay voiding.	Focusing thoughts on something other than urination may provide sufficient distraction to extend the interval between the warning sign and actual voiding.
Encourage the client to eliminate the intake of beverages that contain caffeine or alcohol.	Caffeine promotes urination; alcohol inhibits the antidiuretic hormone, which prevents the reabsorption of water in the nephrons and leads to an increased formation of urine.
Ensure an oral fluid intake of at least 1,500–2,000 mL/day.	An adequate fluid intake reduces the potential for urinary infection or renal stone formation.
Assist the client to the toilet for the purpose of urination at a frequency that corresponds with the client's preconditioning pattern of urination (ie, approximately q1 1/2h), and extend the time by 15 minutes until there is an interval of 2 hours	Increasing the length of time between voidings reduces chronic low-volume voiding, improves bladder muscle tone, and increases bladder capacity, which potentiates achieving continence.

between voidings.	
Continue to extend the intervals between voiding until the client is voiding no more frequently than q4h in a 24-hour period.	Reconditioning the control of urination is facilitated by repetition and gradually extending the efforts to control voiding.
Praise the client every time a short-term goal of delaying or controlling urination is achieved.	Positive reinforcement helps to motivate the client to continue efforts to control incontinence.
Share the client's progress with the physician.	Medical interventions such as prescribing a medication that blocks acetylcholine (anticholinergic agent) may help to inhibit bladder muscle contractions and promote contraction of the urinary sphincter.

Evaluation of Expected Outcomes

- The client is able to gradually delay urination.
- Nocturia is reduced to once per night.
- The client has fewer to no episodes of incontinence.

CRITICAL THINKING EXERCISES

1. During a nursing assessment, a female client reports periodic dribbling of urine. What additional information is important to obtain?
2. An older adult client confides that she would like to participate in activities outside her home, but she is worried that others will notice her problem with urinary incontinence. What response might help this client? What suggestions could you offer?
3. A resident in a nursing home who has had a retention catheter for the last 6 months says, "I'd do anything if I didn't have to have this catheter." What suggestions would be appropriate at this time?
4. The physician orders the removal of a urinary retention catheter. What actions should the nurse take?

NCLEX-STYLE REVIEW QUESTIONS

1. What is the most important nursing assessment before beginning continence retraining?
   1. Recording the time when client is incontinent
   2. Checking the results of a routine urinalysis
   3. Palpating the extent of bladder distention
   4. Observing the characteristics of the client's urine

   Test-Taking Strategy: Note the key words and modifier "most important assessment." Analyze the choices and select the option that is better than any of the others.
2. During continence retraining, what is the best nursing response when a client

wants to restrict fluid intake to remain dry for longer periods?

1. Encourage the practice because it shows evidence of client cooperation.
2. Encourage the practice because it leads to accomplishing the goal.
3. Discourage the practice because it contributes to constipation.
4. Discourage the practice because it predisposes to fluid imbalance.

Test-Taking Strategy: Note the key words and modifier "best nursing response." Consider the pros and cons regarding restricting fluids to achieve continence, then select the option that is better than any of the others.

3. When applying an external condom catheter, which nursing action is correct?

1. Lubricate the penis before applying the catheter.
2. Measure the length and circumference of the penis.
3. Leave space between the penis and bottom of the catheter.
4. Retract the foreskin and roll the catheter over the penis.

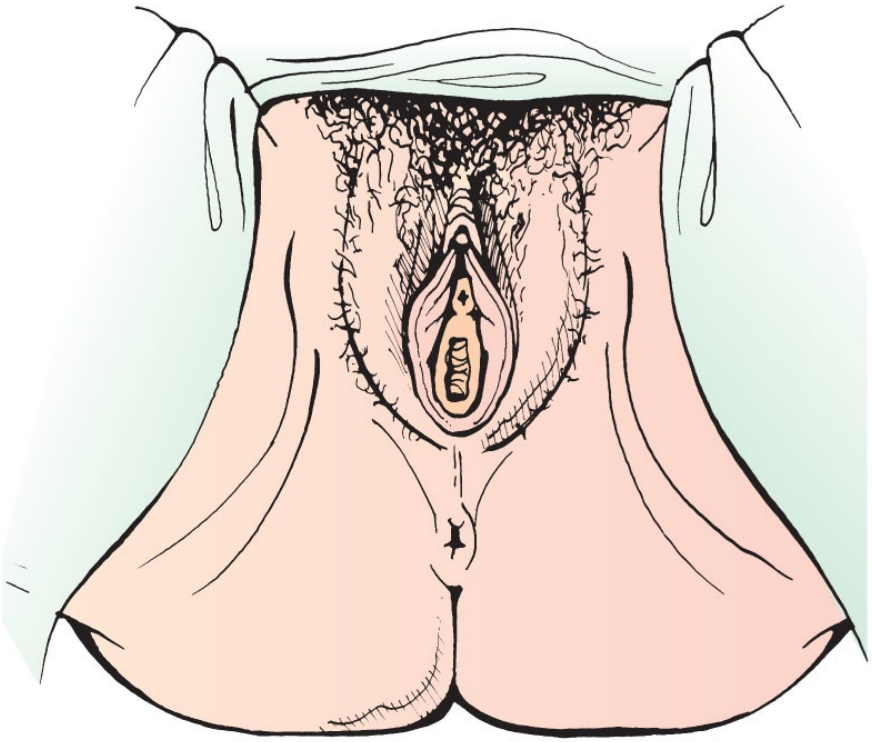
Test-Taking Strategy: Note the key word "correct." Review the choices and eliminate those that are incorrect.

4. When the nurse instructs a female client on the technique for collecting a clean-catch midstream urine specimen for routine urinalysis, which statement is correct?

1. Cleanse the urethral area using several circular motions.
2. Void into the plastic liner that is under the toilet seat.
3. After voiding a small amount, collect a sample of urine.
4. Mix the antimicrobial solution with the collected urine specimen.

Test-Taking Strategy: Note the key word "correct." Review the choices and eliminate those that are incorrect.

5. Identify the female urinary meatus by placing an X on the diagram.



Test-Taking Strategy: Examine the diagram and identify anatomical structures. Mark the location where the nurse inserts a urinary catheter.

SKILL 30-1 Placing and Removing a Bedpan

Suggested Action	Reason for Action
ASSESSMENT	
Ask the client if he or she feels the need to void.	Anticipates elimination needs.
Palpate the lower abdomen for signs of bladder distention.	Indicates bladder fullness.
Determine if a fracture pan is necessary or if there are any restrictions in turning or lifting.	Prevents injury.
PLANNING	
Gather needed supplies such as clean gloves, bedpan, toilet tissue, and a disposable pad.	Promotes organization and efficient time management.
Warm the bedpan by running warm water over it especially if it is made of metal.	Demonstrates concern for the client's comfort.
IMPLEMENTATION	
Wash your hands or perform an alcohol-based hand rub (see Chap. 10); put on clean gloves.	Reduces the transmission of microorganisms.
Place the adjustable bed in high position.	Promotes use of good body mechanics.
Close the door and pull the privacy curtains.	Demonstrates concern for the client's right to privacy and dignity.
Raise the top linen enough to determine the location of the client's hips and buttocks.	Prevents unnecessary exposure.
Instruct the client to bend the knees and press down with the feet.	Helps to elevate the hips.
Place a disposable pad over the bottom sheets, if necessary.	Protects bed linen from becoming wet and soiled.
Slip the bedpan beneath the client's buttocks (Fig. A).	Ensures proper placement.
A	Placing a bedpan.
Or roll the client to the side and position the bedpan (Fig. B).	Reduces work effort and the potential for a work-related injury; aids in placement if client cannot lift buttocks.
B	Placing a bedpan from a side-lying position.

Raise the head of the bed (Fig. C).	Simulates the natural position for elimination.
C	**The position for elimination.**
Ensure that toilet tissue is within the client's reach.	Provides supplies for hygiene.
Identify the location of the signal device and leave the client, if doing so is safe.	Respects privacy yet provides a mechanism for communicating a need for assistance.
Return and remove the bedpan.	Prevents discomfort.
Assist with removing residue of urine from the skin, if necessary.	Prevents offensive odors and skin irritation.
Wrap the gloved hand with toilet tissue and wipe from the meatus of a female toward the anal area.	Supports the principles of medical asepsis.
Place soiled tissue in the bedpan unless it requires measuring.	Contains soiled tissue until the time of disposal.
Help the client to a comfortable position.	Ensures the client's well-being.
Provide supplies for hand hygiene.	Removes residue from elimination and colonizing microorganisms.
Measure the volume of urine if the client's intake and output are being monitored.	Ensures accurate data collection.
Save a sample of urine if it appears abnormal in any way.	Facilitates laboratory examination or further assessment.
Empty the urine into a toilet and flush.	Facilitates disposal.
Clean the bedpan and replace it in a place that is separate from clean supplies.	Supports the principles of asepsis.
Remove gloves and repeat hand hygiene.	Removes colonizing microorganisms.

EVALUATION

- Bedpan is positioned without injury.
- Urine is eliminated.
- Hygiene measures are performed.

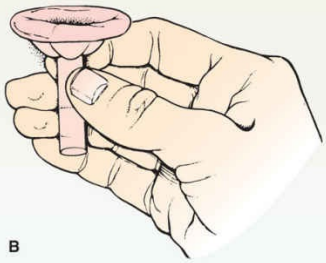
DOCUMENT

- Volume of urine eliminated (for monitoring intake and output)
- Appearance and other characteristics of the urine

SAMPLE DOCUMENTATION

Date and Time Assisted to use the bedpan. Voided 300 mL of clear, amber urine without difficulty._____ J. Doe, LPN

SKILL 30-2 Applying a Condom Catheter

Suggested Action	Reason for Action
ASSESSMENT	
Wash your hands or perform an alcohol-based hand rub (see Chap. 10).	Reduces the potential for transmitting microorganisms.
Assess the penis for swelling or skin breakdown.	Provides data for future comparison or a basis for using some other method for urine collection.
Determine the client's understanding about the application and use of an external catheter.	Provides an opportunity for health teaching.
Verify the client's willingness to use a condom catheter.	Respects the client's right to participate in making decisions.
Check the medical record to determine if the client has a latex allergy.	Maintains client safety and prevents possible allergic reaction.
PLANNING	
Gather supplies such as soap, water, a towel, a condom catheter, drainage tubing, a collection device, and clean gloves. Some devices come packaged with an adhesive strip or Velcro for securing the catheter.	Promotes organization and efficient time management.
Provide privacy.	Demonstrates respect for the client's dignity.
Place the client in a supine position and cover him with a bath blanket.	Facilitates the application of the catheter and maintains privacy.
IMPLEMENTATION	
Wash your hands or perform an alcohol-based hand rub (see Chap. 10) and put on clean gloves.	Reduces the transmission of microorganisms and follows standard precautions.
Wash and dry the penis well.	Promotes skin integrity.
Wind the adhesive strip in an upward spiral around the penis (Fig. A).	Reduces the potential for restricting blood flow.
A	Applying the adhesive strip in a spiral.
Ensure that the wider end of the condom catheter is rolled to the narrower tip (Fig. B).	Facilitates application to the penis.
 B	A rolled condom sheath.

Hold approximately 1–2 in. (2.5–5 cm) of the lower sheath below the tip of the penis and unroll the sheath upward (Fig. C).	Leaves space below the urethra to prevent the irritation of the meatus.
C	Unrolling the condom catheter sheath over the penis.
Secure the upper end of the unrolled sheath to the skin firmly with a second strip of adhesive or a Velcro strap but not so tight as to interfere with circulation (Fig. D).	Ensures that the catheter will remain in place.
D	Securing a condom catheter.
Connect the drainage tip to a drainage collection device (Fig. E).	Allows for urine drainage and collection.
E	Connecting the condom catheter to a drainage collection system.
Keep the penis in a downward position. Assess the penis at least every 2 hours. Check that the catheter has not become twisted. Empty the leg bag, if one is used, as it becomes partially filled with urine. Remove and change the catheter daily or more often if it becomes loose or tight.	Promotes urinary drainage. Ensures prompt attention to signs of impaired circulation. Maintains catheter patency. Ensures that the catheter will not be pulled from the penis by the weight of the collected urine. Maintains skin integrity.
Substitute a waterproof garment during periods of nonuse. Wash the catheter and collection bag with mild soap and water and rinse with a 1:7 solution of vinegar and water.	Provides a mechanism for absorbing urine. Extends the use of the equipment and reduces offensive odors.

EVALUATION

- Catheter remains attached to the penis.
- Penis exhibits no evidence of skin breakdown, swelling, or impaired circulation.
- Linen and clothing remain dry.

DOCUMENT

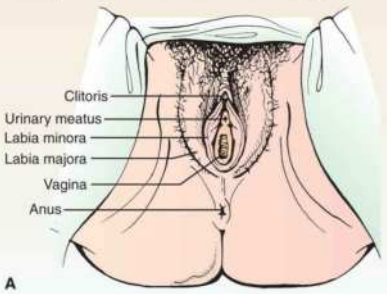
- Preapplication assessment data
- Hygiene measures performed
- Time of catheter application
- Content of teaching
- Postapplication assessment data

SAMPLE DOCUMENTATION

Date and Time Penis washed with soap and water. Penile skin is intact. No discoloration or lesions noted. Condom catheter applied and connected to a leg bag. Instructed to report any swelling or local discomfort.
J. Doe, LPN

SKILL 30-3 Inserting a Foley Catheter in a Female

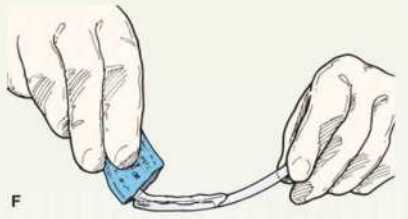
Suggested Action	Reason for Action
ASSESSMENT	
Check the client's record to verify that a medical order has been written.	Demonstrates the legal scope of nursing; catheterization is not an independent measure.
Inspect the medical record to determine if the client has a latex allergy.	Determines if it is safe to use a latex catheter or if a latex-free type is needed.
Determine the type of catheter that has been prescribed.	Ensures the selection of an appropriate catheter.
Review the client's record for documentation of genitourinary problems.	Provides data by which to modify the procedure or equipment.
Assess the client's age, size, and mobility.	Influences the size of the catheter and the need for additional assistance.
Assess the time of the last voiding.	Indicates how full the bladder may be.
Determine how much the client understands about catheterization.	Provides an opportunity for health teaching.
Familiarize yourself with the anatomic landmarks (Fig. A).	Facilitates insertion in the appropriate location.



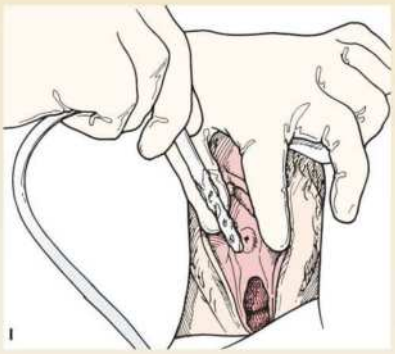
A

Female anatomical landmarks.

PLANNING	
Gather supplies, which include a catheterization kit, a bath blanket, and additional light, if necessary.	Promotes organization and efficient time management.
IMPLEMENTATION	
Close the door and pull the privacy curtain.	Demonstrates concern for the client's dignity.
Raise the bed to a high position.	Prevents back strain.
Wash your hands or perform an alcohol-based hand rub (see Chap. 10).	Reduces the potential for transmitting microorganisms.
Cover the client with a bath blanket and pull the top linen to the bottom of the bed.	Avoids unnecessary exposure.
Position an additional light at the bottom of the bed or ask an assistant to hold a flashlight.	Ensures good visualization.
Use the corners of the bath blanket to cover each leg.	Provides warmth and maintains modesty.
Place the client in a dorsal recumbent position with the feet about 2 feet apart (Fig. B).	Provides access to the female urinary system.
B	**A client draped and placed in a dorsal recumbent position.**
Use a lateral or Sims position for clients who have difficulty maintaining a dorsal recumbent position.	Provides access to the female urinary system, but neither is the preferred position.
If the client is soiled, put on gloves, wash the client, remove gloves, and perform hand hygiene measures again.	Supports the principles of asepsis.
Remove the wrapper from the catheterization kit and position it nearby.	Provides a receptacle for collecting soiled supplies.
Unwrap the sterile cover to maintain the sterility of the supplies inside (see Chap. 10) (Fig. C).	Prevents contamination and the potential for infection.
C	**Opening the sterile catheter tray.**
Remove and put on the packaged sterile gloves (see Chap. 10).	Facilitates handling the remaining equipment without transferring microorganisms.

Remove the sterile towel from the kit and place it beneath the client's hips (Fig. D).	Provides a sterile field.
D	Placing a sterile towel.
Place a **fenestrated drape** (one with an open circle in its center) over the perineum (Fig. E).	Provides a sterile field.
E	Placing a fenestrated drape over the perineum.
Open and pour the packet of antiseptic solution (Betadine) over the cotton balls. Spread lubricant on the tip of the catheter (Fig. F).	Prepares the sterile supplies before contaminating one of two hands later in the procedure.
 F	Lubricating the catheter.
Place the catheterization tray on top of the sterile towel between the client's legs. Pick up a moistened cotton ball with the sterile forceps and wipe one side of the labia majora from an anterior to posterior direction.	Promotes access to supplies and reduces the potential for contamination. Cleanses outer skin before cleansing deeper areas of tissue.

Discard the soiled cotton ball in the outer wrapper of the catheterization kit; repeat cleansing the other side of the labia majora. Separate the labia majora and minora with the thumb and fingers of the nondominant hand, exposing the urinary meatus (Fig. G).	Completes the bilateral outer cleansing. Facilitates the visualization of anatomic landmarks and prevents contaminating the catheter during insertion.
 G	**Separating the labia.**
Consider the hand separating the labia to be contaminated. Clean each side of the labia minora with a separate cotton ball while continuing to retract the tissue with the nondominant hand. Use the last cotton ball to wipe centrally, starting above the meatus down toward the vagina (Fig. H).	Avoids transferring microorganisms to sterile equipment and supplies. Removes colonizing microorganisms. Completes the cleaning of external structures.
 H	**Wiping from above the meatus downward.**
Discard the forceps with the last cotton ball into the wrapper for contaminated supplies. Keep the clean tissue separated.	Follows the principles of asepsis. Prevents recontamination.

Pick up the catheter, holding it approximately 3–4 in. (7.5–10 cm) from its tip (Fig. I).	Facilitates control during insertion.
 I **Preparing to insert the catheter.**	
Insert the tip of the catheter into the meatus approximately 2–3 in. (5–7.5 cm) or until urine begins to flow.	Locates the tip beyond the length of the female urethra, which is approximately 1.5–2.5 in. (4–6.5 cm).
Recheck anatomic landmarks if there is no evidence of urine; remove an incorrectly placed catheter and repeat, using another sterile catheter.	Indicates one of two possibilities: either the bladder is empty or the catheter has been placed within the vagina by mistake; ensures sterility of equipment.
Advance the catheter another 0.5–1 in. (1.3–2.5 cm) after urine begins to flow.	Ensures that the catheter is well within the bladder, where the balloon can be safely inflated.
Direct the end of the catheter so that it drains into the equipment tray or specimen container.	Avoids wetting the linens.
Hold the catheter in place with the fingers and thumb that were separating the labia.	Stabilizes the catheter externally.
Pick up the prefilled syringe with the sterile, dominant hand, insert it into the opening to the balloon, and instill the fluid (Fig. J).	Stabilizes the catheter internally.
 J **Inflating the balloon.**	
Withdraw the fluid from the balloon if the client describes feeling pain or discomfort, advance the catheter a little more, and try again.	Prevents internal injury.

Tug gently on the catheter after the balloon has been filled.	Tests whether or not the catheter is well anchored within the bladder.
Connect the catheter to a urine collection bag. Wipe the meatus and labia of any residual lubricant. Secure the catheter to the leg with tape or other commercial device (Fig. K).	Provides a means of assessing the urine and its volume. Demonstrates concern for the client's comfort. Prevents pulling on the balloon within the catheter.
 K	Securing the catheter to the thigh.
Hang the collection bag below the level of the bladder; coil excess tubing on the mattress.	Ensures gravity drainage.
Discard the catheterization tray and wrapper with soiled supplies. Remove your gloves and perform hand hygiene.	Follows the principles of asepsis. Removes colonizing microorganisms.
Remove the drape, restore the top sheets, make the client comfortable, and lower the bed.	Restores comfort and safety.

EVALUATION

- The catheter is inserted under aseptic conditions.
- The urine is draining from the catheter.
- The client exhibits no evidence of discomfort during or after the insertion.

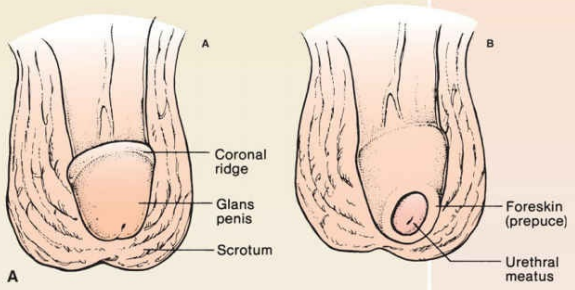
DOCUMENT

- Preassessment data
- Size and type of catheter
- Amount and appearance of urine
- Client's response

SAMPLE DOCUMENTATION

Date and Time Unable to void in past 8 hours. Bladder feels distended. Dr. Peter notified. 16 F Foley catheter inserted per order and connected to gravity drainage. 550 mL of urine drained from bladder at this time. Urine appears light amber. No discomfort reported. _____ J. Doe, LPN

SKILL 30-4 Inserting a Foley Catheter in a Male

Suggested Action	Reason for Action
ASSESSMENT	
Check the client's record to verify that a medical order has been written.	Demonstrates the legal scope of nursing; catheterization is not an independent measure.
Inspect the medical record to determine if the client has a latex allergy.	Determines if it is safe to use a latex catheter or if a latex-free type is needed.
Determine the type of catheter that has been prescribed.	Ensures the selection of the appropriate catheter.
Review the client's record for documentation of genitourinary problems.	Provides data by which to modify the procedure or equipment.
Assess the client's age, size, and mobility.	Influences the size of the catheter and need for additional assistance.
Assess the time of the last voiding.	Indicates the potential fullness of the bladder.
Determine how much the client understands about catheterization.	Provides an opportunity for health teaching.
Familiarize yourself with the anatomic landmarks (Fig. A).	Facilitates insertion.
 Male anatomical landmarks. **A.** Circumcised. **B.** Uncircumcised.	
PLANNING	
Gather supplies, which include a catheterization kit, a bath blanket, and additional light.	Promotes organization and efficient time management.
IMPLEMENTATION	
Close the door and pull the privacy curtain.	Demonstrates concern for the client's dignity.
Raise the bed to a high position.	Prevents back strain.
Perform handwashing or an alcohol-based hand rub (see Chap. 10).	Reduces the potential for transmitting microorganisms.
Place the client in a supine position.	Provides access to the male urinary system.
Cover the client's upper body with a bath blanket and lower the top linen to expose just the penis.	Provides minimal exposure.
Position an additional light at the bottom of the bed or ask an assistant to hold a flashlight.	Ensures good visualization.
If the client is soiled, put on gloves, wash the client, remove gloves, and repeat hand hygiene measures.	Supports the principles of asepsis.
Remove the wrapper from the catheterization kit and position it nearby.	Provides a receptacle for collecting soiled supplies.
Unwrap the sterile inner cover so as to maintain the sterility of the supplies inside (see Chap. 10).	Prevents contamination and the potential for infection.
Remove and put on the packaged sterile gloves (see Chap. 10).	Facilitates handling the remaining equipment without transferring microorganisms.

Place the fenestrated drape over the client's penis without touching the upper surface of the drape (Fig. B).	Provides a sterile field.
	Placing a fenestrated drape.
Open and pour the packet of antiseptic solution (Betadine) over the cotton balls. Place the catheterization tray between the client's thighs.	Prepares sterile supplies before contaminating one of two hands later in the procedure. Promotes ease of access to supplies and reduces the potential for contamination.
Lift the penis at its base with the nondominant hand; retract the foreskin if the client is uncircumcised. Consider the gloved hand holding the penis to be contaminated.	Promotes visualization and support during catheter insertion. Avoids transferring microorganisms to sterile equipment and supplies.
Pick up a moistened cotton ball with the sterile forceps and wipe the penis in a circular manner from the meatus toward the base; repeat using a different cotton ball each time (Fig. C).	Moves microorganisms away from the meatus.
	Cleaning the penis.
Discard the forceps with the last cotton ball into the wrapper for contaminated supplies.	Follows the principles of asepsis.
Apply gentle traction to the penis by pulling it straight up with the nondominant gloved hand.	Straightens the urethra.

Instill the contents of a prefilled syringe containing 5 to 10 mL of water-soluble lubricant or water-soluble 2% lidocaine gel directly through the meatus into the urethra (Fig. D).	Avoids trauma to the urethra caused by insufficient lubrication; this technique replaces the traditional practice of lubricating the outer surface of the catheter, which resulted in its accumulation at the meatus only (Society of Urologic Nurses and Associates, 2015; Gerard & Sueppel, 1997).
D	**Instilling lubricant.**
Insert, but never force the catheter; rather, rotate the catheter, apply more traction to the penis, encourage the client to breathe deeply, or angle the penis toward the toes (Fig. E).	Adjusts for passing the catheter beyond the prostate gland.
E	**Catheter insertion.**
Continue insertion until only the inflation and drainage ports are exposed and urine flows.	Locates the tip beyond the length of the male urethra.
Pick up the prefilled syringe with the sterile, dominant hand, insert it into the opening to the balloon, and instill the fluid (Fig. F).	Stabilizes the catheter internally.
F	**Inflating the balloon.**

Withdraw the fluid from the balloon if the client describes feeling pain or discomfort, advance the catheter a little more, and try again.	Prevents internal injury.
Tug gently on the catheter after the balloon has been filled.	Tests whether or not the catheter is well anchored within the bladder.
Connect the catheter to a urine collection bag.	Provides a means of assessing the urine and its volume.
Wipe the meatus and penis of any residual lubricant.	Demonstrates concern for the client's comfort.
Secure the catheter to the leg or abdomen with tape or other commercial device (Fig. G).	Prevents pulling on the balloon within the catheter.
G	Securing a catheter.
Hang the collection bag below the level of the bladder; coil excess tubing on the mattress.	Ensures gravity drainage.
Discard the catheterization tray and wrapper with soiled supplies.	Follows the principles of asepsis.
Remove your gloves and repeat hand hygiene measures.	Removes colonizing microorganisms.
Remove the drape, restore the top sheets, make the client comfortable, and lower the bed.	Restores comfort and safety.

EVALUATION

- The catheter is inserted under aseptic conditions.
- The urine is draining from the catheter.
- The client demonstrates no evidence of discomfort during or after insertion.

DOCUMENT

- Preassessment data
- Size and type of catheter
- Amount and appearance of urine
- Client's response

SAMPLE DOCUMENTATION

Date and Time 16 F Foley catheter inserted before surgery according to preoperative orders. 350 mL of urine obtained before connecting the catheter to gravity drainage. Urine appears light yellow and clear.
J.Doe, LPN

SKILL 30-5 Irrigating a Foley Catheter

Suggested Action	Reason for Action
ASSESSMENT	
Check the client's record to verify that a medical order has been written.	Demonstrates the legal scope of nursing; a catheter irrigation is not an independent measure.
Verify the type of irrigating solution prescribed, or follow the standard for practice, which usually advises using sterile normal saline solution.	Complies with medical the directives or standards for care.
Assess the urine characteristics.	Provides a baseline for assessing the outcome of the procedure.
Determine how much the client understands about a catheter irrigation.	Provides an opportunity for health teaching.
Locate the port on the drainage tube through which fluid can be instilled (Fig. A).	Ensures a safe procedure and maintains the integrity of the catheter.
A	Identifying the self-sealing irrigation port.
PLANNING	
Gather needed equipment and supplies: an irrigation kit, a flask of sterile irrigating solution, a 30- to 60-mL syringe, and alcohol swabs.	Promotes organization and efficient time management.
IMPLEMENTATION	
Wash hands or perform an alcohol-based hand rub (see Chap. 10).	Follows the principles of asepsis and standards of practice.
Raise the height of the bed.	Reduces back strain.
Pull the privacy curtain.	Demonstrates concern for the client's dignity.
Add 100–200 mL of solution to the irrigating basin.	Avoids contaminating and wasting all the solution in the flask.
Put on gloves kept at the bedside or within the irrigation kit.	Complies with standard precautions.
Attach a needle to the tip of the irrigating syringe found in the irrigation kit. Fill the syringe with 30–60 mL of solution (Fig. B).	Provides a means for penetrating the self-sealing port.
B	Filling the syringe with solution.

Clean the port on the catheter with an alcohol swab (Fig. C).	Removes gross debris and colonizing microorganisms.
C	**Cleaning the irrigation port.**
Clamp or kink the tubing below the port through which the irrigating solution will be instilled (Fig. D).	Ensures that the solution will move forward into the catheter and not into the drainage system.
D	**Clamping the drainage tubing.**
While holding the catheter with one hand, insert the syringe into the port (Fig. E).	Maintains sterility.
E	**Instilling the irrigation solution.**

Gently instill the solution. Remove the syringe. Release the clamp from the drainage tubing and observe the flow of urine through the tubing (Fig. F).	Clears the catheter of debris and dilutes particles within the catheter. Prevents leaking. Facilitates gravity drainage.
 F **Draining the irrigation solution.**	
Repeat the instillation and drainage if the urine appears to contain appreciable debris. Record the volume of instilled solution as fluid intake. Discard or protect the sterility of the irrigating equipment, which may be reused for the next 24 hours as long as it is not contaminated.	Promotes patency. Maintains accurate assessment data. Complies with the principles of infection control.

EVALUATION

- The prescribed amount and type of solution are instilled.
- The principles of asepsis have been maintained.
- The urine continues to drain well through the catheter.
- The client reports no discomfort.

DOCUMENT

- Preassessment data
- Volume and type of solution
- Volume and appearance of drainage

SAMPLE DOCUMENTATION

Date and Time Urine appears amber with some evidence of white particles. 60 mL of sterile normal saline solution instilled into catheter. 120 mL drainage returned. Urine appears to have less sediment. Catheter remains patent. _____ J. Doe, LPN